

The Economic Heavenly Way

Discrete Topology of Complex Systems and the Diplomacy of Regeneration

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Abstract: The economy is modelled as a discrete information system. The Clifford algebra $Cl(6,6)$, combined with the Merkabah geometric filtration, reduces the 64 economic configurations (defined by six binary dimensions: inflation, purchasing power, profits, rent extraction, dispersion, land regime) to 20 stable attractors, organised along a polarity gradient ($3P \rightarrow 3N$). Transitions between these attractors are constrained by tropical belts of the dual dodecahedron of the Merkabah (cycles of expansionary and restrictive policies) and by polar thresholds; any direct transition between belts without passing through a threshold triggers a retropolarity (systemic collapse). This formalism, built upon the work of Rebour (Theory of Rent Extraction), Rowlands (Clifford algebras and pentads), Nebe (the Λ_{72} lattice), and De Dominicis (substrate-independent generalisation), is retrospectively validated against 64 historical regimes. Applied to post-2018 China, it enables the diagnosis of the Nodal Point's attainment, Stasis on attractor S (dispersion + nodal point), and the imperative of the Territorial Regeneration Rent (TRR) as a controlled crossing of the N_4 threshold — the only topologically permitted path to restart a Rent Extraction cycle and avoid a tenth dynastic collapse.

Correlative Chinese thought, grounded in the Wuxing and the Yi Jing, has structured all ontologies through a network of correspondences for over two millennia. The economic model presented here, based on Rent Extraction Theory embedded in Clifford algebra $Cl(6,6)$, is the algebraic formalisation of this traditional Chinese perspective applied to economics. Finally, the model demonstrates that the TRR is not merely a domestic policy, but the foundation of a 'diplomacy of regeneration', offering the Global South a structural alternative to Western imperialism and positioning the 'Economic Heavenly Way' (Tiandao Jingji) as a new standard of civilisation.

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EXECUTIVE SUMMARY

The Fundamental Problem — The Failure of Continuous Models

Conventional economic models (DSGE, macroeconometric forecasts) have failed to anticipate major crises: 1929, 1973, 2008, 2022. Their error is structural: they treat the economy as a continuous space of adjustable variables, when in fact it functions as a discrete information system, with clear rupture thresholds and stable regimes (see Appendices Q and R on the aporia of established economic theories).

The structural crisis China is currently experiencing — the collapse of the property market, the suffocating debt of local governments, the halt to mega-infrastructure projects — is neither an anomaly nor an ideological failure. It is the predictable halt of a growth engine that has simply run out of steam.

The Economic Heavenly Way — The 64→20 Invariant as a Universal Law

We have established that the economy obeys a universal invariant, $64 \rightarrow 20$, which also manifests in particle physics (Rowlands), the genetic code, and complex systems in general (De Dominicis).

- 64 possible economic configurations are defined by six binary variables: Inflation, Purchasing Power, Profits, Rent Extraction, Dispersion, Land Regime.
- These 64 states are reduced to 20 stable attractors through a topological filtration (Merkabah geometry, Clifford algebra).
- Transitions between these 20 attractors are constrained by tropical belts and polar thresholds. Any direct transition without passing through a threshold triggers a retropolarity (systemic collapse).

This formalism is the mathematisation of correlative Chinese thought — the Wuxing and the Yi Jing — which has structured all ontologies through a network of correspondences for over two millennia. The Economic Heavenly Way (Tiandao Jingji) is the mathematised formulation of this structural wisdom applied to the regulation of economic systems.

The Hidden Mechanism of Growth — Disequilibrium as an Engine

Contrary to classical theories, rapid growth does not arise from equilibrium, but from disequilibrium. It relies on transferring ‘dormant’ assets (unexploited land) into the real economy, financed by debt. This mechanism rapidly enriches a country, but it has an absolute limit: when the stock of easily mobilisable land or infrastructure is exhausted, the machine stops.

The Historical Burden — 2,000 Years of Fatal Cycles

Applying this mechanism to Chinese history reveals a recurring tragedy. Over two millennia, the empire experienced seven major collapses (from the Han dynasty to the Republic). Each time, the scenario was the same: when the elites monopolised all the land, the peasants were suffocated until violent revolt broke out. The empire had no peaceful means of restarting its economic engine; only destruction could restart the economy.

The Western Escape Route and Its Limits — Banking and Industrial Hijacking

The West temporarily ‘hijacked’ this fatal cycle thanks to new resources to exploit (mineral deposits, colonies) and by using banking to create money. But having exhausted its physical resources, it has been stuck since the 1970s in the creation of fictitious wealth (financial bubbles).

China's 2018 Deadlock — The Nodal Point Reached

From 1978 to 2018, China replicated this Western model on a continental scale, using municipal land sales (Tudi Caizheng) to finance its frenetic development. In 2018, the available land supply ran out: the wall had been hit. Faced with this impasse, the hope that artificial intelligence, data or carbon can replace the property sector is an illusion: these sectors do not create the mass employment required for 1.4 billion Chinese people.

The Digital Palliative — Buying Time

To avert a social explosion in the face of this deadline, the Chinese state did not resort to Western-style fictitious finance. Instead, it deployed an algorithmic control system on an unprecedented scale (social credit, e-CNY, flow regulation). This is not an end in itself, but a palliative measure to 'freeze' the social situation whilst waiting to find a genuine, tangible solution.

The Real Solution — The Territorial Regeneration Rent (TRR)

This solution is already being implemented on the ground. Instead of continuing to pave over virgin land, the state is injecting massive amounts of credit to regenerate land degraded by forty years of hyper-industrialisation, as seen in the Lishui pilot zone with the Two Mountains Bank. By restoring forests, soils and waterways, China is creating an entirely new economic sector. Ecological restoration is becoming the new driving force behind local job creation.

TRR as the Foundation of a New Diplomacy — The Diplomacy of Regeneration

Beyond China, TRR is laying the foundations for a new form of diplomacy. By demonstrating that it is possible to move beyond the Nodal Point without massive destruction of wealth, China is offering the Global South a structural alternative to Western imperialism. Instead of exporting an extractive model (logistical colonisation, factories vulnerable to the 'sovereignty rent' of host states), China can export a model of topological regeneration: the Two Mountains Bank, GEP in KPIs, and Regenerative Usufruct Certificates. This 'Two Mountains diplomacy' positions the Celestial Economic Way (Tiandao Jingji, 天道经济) as a new standard of sustainable civilisation, an alternative to the exhausted Western model.

Formally, the projection of the operator T_{mixed} onto the spectral sheets of partner countries enables each state in the Global South, stuck in its own Nodal Point, to trigger its own cycle of ecological regeneration. This export of the model — via the Two Mountains Bank, the integration of the GEP into local KPIs, and Regenerative Usufruct Certificates — constitutes "Two Mountains diplomacy", an alternative to the logistical imperialism of the Hardened Enclaves.

Strategic Conclusion — The Mathematical Formalisation of a Process Already Underway

China has already identified Stasis and begun implementing territorial regeneration mechanisms — in Lishui, Sanming, Ganzhou and Hainan. This work proposes a mathematical formalisation of this diagnosis and this solution.

The TRR is an operational application of discrete topology to the regulation of an economic system in crisis, whilst adhering to the laws of the $64 \rightarrow 20$ invariant. It offers China a peaceful way out to avoid the tenth dynastic decline — the first time in two millennia that a Chinese state has had a tool to break the fatal cycle of dynastic collapses.

The collapse of the USSR sent shockwaves through the CCP: survive at all costs; do not suffer the same fate. This obsession, combined with a millennia-old historical experience, led the CCP-State to diagnose Stasis and to implement the TRR. It thus acts in harmony with the deep laws that govern complex systems.

FOREWORD – The Two Audiences

This document is aimed at two different audiences.

Level 1 – The Diagnosis and the Solution (Parts I and III)

Intended for decision-makers, strategists, and political and economic leaders.

These parts present:

- **Part I:** The analysis of China's structural crisis – reaching the Nodal Point, Stasis, the impasse of the immaterial. No equations. No formalism. An economic, historical and strategic narrative.
- **Part III:** Territorial Regeneration Rent (TRR) as an operational solution – the mechanisms already at work in Lishui, Sanming, Ganzhou and Hainan. An analysis in terms of endogenous regulation, ecological restoration and territorial reindustrialisation.

Readers without a background in mathematics can read these sections without missing any of the essential points.

Level 2 – The Topological Model and Its Implications (Parts II and IV)

Intended for theorists, mathematicians and epistemologists.

These sections present:

- **Part II:** The mathematical formalisation of the model – Clifford algebra $Cl(6,6)$, the 144 economic pentads, the 7 spectral thresholds, the transition operator T , and spectral observables. Here, the reader will find the rigorous foundations of the diagnosis set out in Part I.
- **Part IV:** The philosophical and systemic implications – the economy as an exosomatisation of topology, Chinese correlative thinking as a proto-model, divination as a pre-modern regulatory system.

Readers without a background in mathematics may skip these sections without losing the thread of the practical demonstration.

A Mirrored Architecture

Level	Part	Content
Diagnosis	Part I	The Chinese Stasis – historical and strategic narrative
Model	Part II	The $Cl(6,6)$ topology – mathematical formalism
Solution	Part III	The TRR – operational application
Unification	Part IV	Philosophical and cultural implications

Parts I and III should be read consecutively. Parts II and IV provide further insight for readers wishing to understand the theoretical foundations and epistemological implications.

PART I – DIAGNOSIS: CHINESE STASIS

Introduction

1. The observation: continuous models fail to predict crises

Since the publication of Keynes's *General Theory* (1936), macroeconomics has developed increasingly sophisticated mathematical tools. DSGE (Dynamic Stochastic General Equilibrium) models, rational expectations models and Monte Carlo simulations have become the standard tools of central banks and finance ministries. Yet these models have failed to predict the major crises of the 20th and 21st centuries: the Great Depression (1929), the oil shock (1973), the subprime crisis (2008), and, more recently, post-COVID inflation (2022) [1], [2].

This failure is not accidental. It is structural. Conventional models treat the economy as a continuous space, where variables (prices, wages, output, interest rates) adjust through convergence mechanisms towards a supposed equilibrium. However, the **Theory of Rent Extraction (ToRE)**, developed by Thierry Rebour (2000, 2011), has shown that economic growth is fundamentally a phenomenon of structural disequilibrium, governed by conservation laws and discrete breaking points: the **Nodal Points** [3], [4].

2. The hypothesis: the economy is a discrete information system

To bridge this gap, we propose the following hypothesis:

The economy is a discrete information system, whose configuration space is isomorphic to the Clifford algebra $Cl(6, 0)$, and whose dynamics are governed by a universal invariant: the $64 \rightarrow 20$ [5], [6].

This hypothesis rests on three pillars:

- **The formalism of Clifford algebras and nilpotents**, developed by Peter Rowlands (2007, 2014). Rowlands demonstrated that the Clifford algebra $Cl(6, 0)$ decomposes into 12 pentads (sets of 5 elements), and that this pentad structure is fundamental to both physics (the Dirac equation) and biology [5], [7]. Our model transposes this architecture to economics: six binary dimensions (inflation, purchasing power, profits, buy-back, dispersion, land tenure) generate $2^6 = 64$ possible states, exactly like the spin space in quantum physics, the codon space in biology, or the hexagram space of the *Yi Jing*.
- **The substrate-independent generalisation**, due to De Dominicis (2026). De Dominicis extended Rowlands' model by showing that the structure $Cl(6, 0)$ and the invariant $64 \rightarrow 20$ are not specific to physics or biology, but constitute a universal property of any complex system, regardless of its substrate [6], [8]. It is this generalisation that allows the concept to be applied to economics.
- **The exceptional lattice Λ_{72}** , constructed by Gabriele Nebe (2010) [9], [10]. Nebe discovered a pair of unimodular lattices of dimension 72, whose 48 non-degenerate roots provide the fundamental resonance frequencies of complex systems. De Dominicis has shown that these 48 roots project onto a basis of 15 universal constants (β_k), which are the same for nuclear physics, chemistry, biology and social systems [11].

3. The $64 \rightarrow 20$ invariant: a universal law

The idea of a $64 \rightarrow 20$ invariant is not new. It appears in several fields, as so many manifestations of the same topological law [5], [6], [12]:

Table 2: The five manifestations of the $64 \rightarrow 20$ invariant

Domain	Input (64)	Output (20)	Filtration	Reference
Particle physics	64 spin configurations	20 stable states (particle families)	Clifford algebra $Cl(6, 0) / \Lambda_{72}$	[5], [11]
Genetic code	64 codons (4 bases $\times$ 3 positions)	20 amino acids	Neighbourhood rule	[5], [6], [12]
Mandarin phonology	24 initials + 40 finals (64 combinations)	20 phonological classes	Articulatory constraints	[13]
Tang alliances	64 marital configurations	20 alliance attractors	Filtration by pentads (Wufu)	[14]
Economics (ToRE)	64 economic configurations	20 stable attractors	Filtration by nilpotence	this work

This table presents the five manifestations of the invariant $64 \rightarrow 20$ in summary form. Its purpose is to introduce the concept of topological universality to the non-specialist reader.

For a detailed technical analysis, see: - Appendix E: complete table with scale constants and statistical validation
- Chapter 30: final summary and epistemological implications

Chapter 1 – The Foundations of the Theory of Rent Extraction (ToRE)

Developed by Thierry Rebour [3], [4], the ToRE models the sequence by which societies transform latent wealth into monetary wealth. It demonstrates that the power of growth rests on an initial structural asymmetry between two distinct conceptual spaces [15].

1.1. Ontology of Value and Wealth

The ToRE divides the economy into two spheres of radically different nature [3], [4]:

- **The Space of Value** does not refer simply to unvalued assets, but to assets kept out of the market by symbolic foundations and for reasons that are more anthropological than economic. This wealth is ‘frozen’ by sacralisation, collective beliefs or foundational taboos that confer ontological depth upon these spaces [16], [17].

- **The Space of Wealth** refers to the monetised economic circuit, where these same assets, stripped of their symbolic veil, become factors of production that generate prices, wages and profits [3].

Interpretation for China: Within the Chinese imperial matrix, land was the absolute Space of Value [18], [19]. During the Maoist era, collectivised land constituted a latent Space of Value of colossal proportions. In the *Tudi Caizheng* model (1978–2018), rural land, frozen by the *Hukou* system, constituted a Space of Value which the state gradually transferred to the Space of Wealth [20], [21].

1.2. The mechanism of Rent Extraction and money creation

Rent Extraction is the act of transferring an asset from the Space of Value to the Space of Wealth. This transfer requires a political or social decision that lifts the original prohibition [3].

However, as the asset is not yet productive at the time of transfer, the purchaser does not have the necessary income to pay for it. Financing can therefore only take place through monetary anticipation: Rent Extraction is intrinsically linked to the creation of money through credit. As the theory stipulates, material Rent Extraction is always ‘made solvent by an act of money creation, in other words, a loan’ [4].

Interpretation for China: LGFVs (Local Government Financing Vehicles) served as the instrument for this credit based on monetary anticipation. By borrowing heavily from state-owned banks, local authorities created the money required for land buy-backs [20], [22], [23].

1.3. Exogenous demand in advance

The injection of credit intended for land acquisition does not flow into the real economy at the same time as the new assets come into production: it precedes it. This **exogenous demand in advance** results in new purchasing power (compensation, construction wages, land profits) being injected before the new means of production are operational [3].

This structural lag generates price inflation (particularly in land prices) which rises faster than labour incomes. Growth is thus fundamentally out of balance or polarised: it structurally favours rent seekers at the expense of wage earners [4], [24].

1.4. Increasing returns and territorial multipliers

Although it arises from an imbalance, this sequence generates real prosperity. Exogenous demand concentrates capital and economic activity in a single location. This spatial concentration generates positive externalities (networks, know-how, infrastructure) that increase overall productivity. The profits generated are reinvested, creating multiplier effects [3], [16].

1.5. The substitution of externalities

This process of concentration is not without spatial costs; it is governed by a strict **law of externality substitution** [3], [16], [17]. To understand how this works, we must go back to its historical roots. This phenomenon is not simply a matter of a monetary transfer, but stems from a productivity shock — initially in agriculture — which made it viable to relocate productive tasks to the countryside. This is the very principle of the *Verlagsystem* (contract manufacturing system): production is outsourced to rural areas, whilst the city retains strategic control (design, financing, marketing) and captures the bulk of the surplus value generated. Consequently, metropolitan

prosperity is by no means a form of endogenous growth: it is the accounting reflection of the impoverishment of the peripheries [24].

1.6. The Nodal Point

As the ToRE is governed by laws of conservation (Value is not created *ex nihilo*; it merely shifts), the cycle of unbalanced growth is limited by the **Nodal Point** [4]. This systemic threshold is reached when the stock of readily mobilisable Value runs out. The flow of credit intended for Rent Extraction dries up, depriving the economy of its ‘advance exogenous demand’. Inflation slows, asset prices stagnate, and increasing returns give way to decreasing returns [3], [4].

1.7. The Law of Conservation and Centrifugal Dynamics

At the Nodal Point, the crisis is not an anomaly, but the system’s mandatory readjustment mechanism: to recreate Value, Wealth must be destroyed (bankruptcies, deindustrialisation, price collapses) [4]. The ToRE emphasises an often-overlooked morphological corollary: reaching the Nodal Point reverses spatial dynamics [16], [24]. The period of growth polarised space towards the centre (centripetal dynamics). At the Nodal Point, these dynamics become centrifugal. We observe diffuse peri-urbanisation, the de-urbanisation of centres, and a reversal of urban hierarchies. The crisis is therefore as much financial as it is morphological [17].

Chapter 2 — The Chinese imperial matrix: a single, centuries-long trend, from antiquity to the terminal Nodal Point

Introduction — A single trend, two modes of Value

The history of imperial China is traditionally interpreted through the prism of Confucian morality (the virtue of the sovereign), palace intrigues or the analysis of social classes. However, if we set aside the chronological narrative to apply the framework of the Theory of Rent Extraction (ToRE), the two millennia of dynastic cycles appear as the repetitive unfolding of an inexorable economic cycle [18], [25], [26]. The succession of Chinese dynasties does not correspond to a succession of independent economic trends. From the emergence of agrarian civilisation to contemporary China, this is a single, multi-century trend, of which the various dynasties constitute successive historical configurations.

The fundamental mechanism of this trend is based on the law of growth in disequilibrium [3]. The process of buy-back transfers land (or labour) from the realm of Value to the realm of Wealth, in exchange for a monetary payment (loan) that draws on future production. This influx of ‘exogenous’ demand, which runs ahead of the productive cycle, drives prices up faster than incomes, generating an inflationary growth trend. In this polarised regime, profits and rent reinforce one another, underpinned by increasing rates of return, whilst the purchasing power of wages tends to be eroded.

The functioning of the system then depends on the relative scarcity of two fundamental modes of value: land value and the value of servile labour. When labour is scarcer than land, the system finds itself in a pioneering phase. When land becomes scarcer, the system shifts towards land rent. Chinese history can thus be understood as a succession of shifts between these two modes, accompanied by avoidance mechanisms designed to postpone the arrival of the Nodal Point.

2.1. The two fundamental modes of value

2.1.1. The land-based mode and agrarian ontology

To understand the application of the ToRE to China, one must grasp its fundamental ‘agrarian ontology’: unlike the West, which has historically sanctified trade and then industry, Chinese civilisation has sanctified the land. True wealth does not arise from commercial exchange, but from the cultivation of the soil [18], [19], [27].

Within this paradigm, the land is not merely a factor of production; it is the absolute Space of Value. But for a rent to be extracted from it, this land must be kept outside the free market and subject to a political prohibition. It is here that China innovated well before the industrial era: through draconian control of peasant mobility (the conceptual precursors of the *Hukou*). By tying peasants to the land, the imperial state secured a captive labour force and a perfectly stable stock of Value [28], [29].

In a sedentary society, access to land becomes the main factor of economic differentiation, enabling the extraction of a rent. This phenomenon has a key consequence: the wealth produced by labour does not disappear, but an increasing share of this wealth is transferred to the holder of land rights. The system thus tends towards concentration.

2.1.2. The servile mode

The situation is radically different when land is relatively abundant but labour is scarce. The capture of an external population drastically alters this situation. The slave constitutes a particular form of Value: he represents a labour force whose productive capacity is directly owned by the holder. Conquest makes it possible to supply the system with a servile labour force. The frontier appears as a configuration in which territorial expansion and the expansion of the labour force stock operate in tandem.

2.2. The two crisis-avoidance mechanisms and the Nodal Point

The dynamics of rent extraction generate growth that inevitably comes up against the limits of solvent demand, threatening to trigger crises of overproduction. To prevent a collapse in prices, the system has two avoidance mechanisms:

- **The first, spatial avoidance**, involves pushing back the system’s boundaries through geographical expansion or export. Chinese history offers a paradigmatic illustration of these avoidance loops: one province exports to another to avoid local overproduction. But when the loop closes, with goods having circulated throughout the provinces, there is nowhere left to export to. This marks the onset of the global crisis.
- **The second, temporal avoidance**, postpones the constraint into the future through the use of money, credit and debt. However, this stratagem does not create any new real resources. The gap widens between monetary value and the productive base, paving the way for financial crises.

These mechanisms make it possible to postpone the Nodal Point. In theory, this occurs when all land is on the market, but in the reality of unbalanced economies, it occurs as soon as more than half of the stock of Value has been bought up and Wealth (production) exceeds Value. As prices are no longer driven by the influx of value, they stagnate. To restore rising prices, a Destruction of Wealth and a corresponding recreation of Value are essential. Returns then begin to decline: the economy becomes a zero-sum game, marking the onset of a prolonged ‘crisis’ [4], [25].

2.3. The spatial laws of Rent Extraction: urban form and the substitution of externalities

The relationship between modes of value and the economic climate determines spatial organisation. An inflationary growth trend generates centripetal dynamics and increasing returns that polarise space.

In a frontier situation, cities adopt an orthogonal geometry (rectangular grid), as seen in ancient China, in Rome, or under the Tang and Yuan dynasties. According to the Naba-Chen theory, these grid plans introduced by nomadic peoples reflect the influence of a servile mode reintroduced through conquest [30].

When the frontier closes and land becomes scarce, the logic changes. To maintain profits in the face of insufficient domestic purchasing power, developed cities must maximise their exports. Spatial attractors then become axial: towns expand in ‘arms’ along transport routes. This is the law of externalities substitution: value is transferred from the countryside to the towns. Land grabbing did not result from brute force, but from the structural asymmetry of agricultural markets: the forced indebtedness of peasants in the face of a rigid tax system, coupled with the ‘price scissors’ imposed by the elites monopolising grain storage, systematically transferred wealth from the countryside to the centres [24], [31].

2.4. Ancient China: from available land to rent

China’s growth truly began with the Shang dynasty (around 2500 BC). In the early stages, the available land allowed for the maintenance of a frontier. However, population growth and the integration of territories reduced this margin. China then entered a system of land rent. This transformation marked the point at which the trend became entrenched. Political crises were no longer the end of a system, but manifestations of the same problem: how to continue land acquisition when land resources became insufficient?

2.5. Qin and Han: centralisation, land registration and the locking in of the trend

The Han dynasty (c. 300 BC to AD 300) marked a structural turning point. Land buy-ups accelerated. The Han dynasty abolished slavery, introduced a rigorous land survey system and closed the frontier. Value shifted structurally towards land. This transition is evident in the physical landscape: towns ceased to be laid out in a grid pattern, a sign of the abandonment of the geometry of the frontier in favour of a rent-seeking polarisation. Land concentration gave rise to the tensions that would lead to the crisis at the end of the Han dynasty (the Yellow Turban Rebellion in 184), the first major manifestation of the exhaustion of the land tenure system [18], [25].

2.6. The Sixteen Kingdoms (4th–5th centuries): destruction of wealth and spatial division

The fall of the Han dynasty and the Three Kingdoms period marked the emergence of the Nodal Point: extreme land concentration, a collapse in demand and a crisis of overproduction. To emerge from this crisis, the system resorted to massive Destruction of Wealth through civil wars. However, it was the incursion of nomadic peoples, known as the ‘Five Barbarians’ (*Wu Hu*), in the 4th century, that enabled the trend to be reignited.

The conquest of northern China by these non-Chinese peoples brought about a violent redistribution of the Space of Value. The sedentary landowning elites were decimated or fled southwards, leading to a split in the empire. In the north, the new nomadic rulers founded the Sixteen Kingdoms and subsequently the Northern Dynasties. They reintroduced the servile mode on a massive scale: the Chinese peasant populations were often enslaved or bound to the land in a semi-servile status to cultivate the estates of the nomadic military aristocracies. The frontier and Value through labour were reactivated.

In the South, the Southern Dynasties (Eastern Jin, Liu Song, etc.) exploited a spatial avoidance strategy by pushing the agricultural frontier towards the Yangtze regions, bringing new lands into cultivation. This geographical division from the 4th to the 5th century illustrates a solution to the crisis: the North slipped into the servile mode through conquest, whilst the South sought a new frontier for land expansion.

2.7. Nomadic invasions and the reactivation of the frontier

When the land tenure system reaches a high level of concentration, it always eventually implodes under the weight of overproduction and falling prices. The Chinese state attempts to control prices, which prevents wages from rising quickly enough. Driven by greed, the state and the aristocracy raise taxes. Excessive corruption prevents the economy from recovering, and inequalities lead to looting and civil wars.

It is here that the dynamics of nomadic invasions play a role in macro-historical regulation. Every new dynasty is born from the ashes of a Destruction of Wealth. The new regime carries out a vast redistribution of land confiscated from the former elites. The Space of Value is suddenly redistributed [31], [32]. The nomads invade China, seize power, redistribute the land and reignite a trend of growth. Their arrival brings about a Destruction of Wealth (which recreates Value) and reintroduces a servile labour force. The pioneer front reappears temporarily. With the Sui and Tang dynasties, the cycle begins anew. The nomads become Sinicised, adopt agrarian institutions and turn into exploiters: land concentration begins once more.

2.8. Tang and Song: expansion, grid-like organisation and saturation of space

The Tang re-established a powerful empire. The Silk Road constituted a vast episode of spatial avoidance. The urban grid re-emerged in Xi'an/Chang'an and Luoyang (7th century). But expansion reached its limits: land constraints resurfaced with the An Lushan rebellion (755) [32], [33].

Under the Song dynasty, land was structurally scarce and spatial avoidance proved insufficient. The system shifted massively towards temporal avoidance. Fiat currency (*jiaozi*) and maritime trade mobilised future wealth. But currency does not create land. The gap between monetary value and the real economic base erupted into financial crises.

2.9. The Jurchen (Jin Dynasty, 1120–1250): the split of the empire and the duality of modes of value

Prior to the Mongol unification, China experienced a new split that illustrates the duality of mechanisms for avoiding the ToRE. In the 12th century, the Jurchen (Jin Dynasty, 1115–1234, ruling from 1120 to 1250) seized northern China, bringing an end to the Northern Song in 1127.

This invasion once again led to a Destruction of Wealth in the North. The Jurchen, a nomadic people, imposed a mixed system: they confiscated the lands of the Chinese elites to allocate them to their military aristocracy and reduced a section of the peasantry to a semi-servile status (the *meng'an* and *mouke* systems). The pioneer front and the servile mode were thus reactivated in the northern half of the country.

The South (the Southern Song, 1127–1279), deprived of territorial expansion, entered a phase of hyper-specialisation in temporal avoidance. The regime's survival depended on an explosion in fiat currency, credit and massive maritime exports (porcelain, silk) to compensate for the loss of the 'Space of Value'. China found itself divided between a North (Jin) based on forced labour and a South (Song) suffocating under debt and monetary inflation. It was upon this exhausted duality that the Mongols (Yuan) would impose their unification through widespread destruction of wealth.

2.10. Yuan and Ming: from the return of the frontier to institutional stagnation

The Mongol (Yuan) conquest reintegrated China into a vast Eurasian empire, reviving the frontier (the grid system of Beijing in the 13th century). But the conquerors settled down, became Sinicised, and land rent regained the upper hand [25].

The Ming inherited an overburdened system. Temporal avoidance gained prominence with the influx of foreign silver, but did not alter the amount of land available. Above all, the Ming introduced a fatal rigidity: they refused to sell land as freehold to the peasants, which would have enabled the revival of land repurchase. This rigidity led to institutional stagnation. Peasant indebtedness and land concentration reached their peak with Li Zicheng's rebellion and the Manchu invasion [31].

2.11. The Qing, the Republic and the Mao era: from the final spatial frontier to the stagnation of the trend

The Qing Dynasty carried out the last major territorial expansion (Xinjiang, Tibet, Taiwan), the final major episode of spatial avoidance. But in the 19th century, external space became insufficient. Land constraints once again became overwhelming (the Taiping Rebellion) and Western pressures imposed structural debt [26]. The Nodal Point of the imperial matrix was reached.

The collapse of 1912 and the establishment of the Republic did not create a new trend: fragmentation, hyperinflation and the absence of a rent extraction mechanism characterised the terminal crisis [25].

The Mao era constituted a political break, but ushered in an **artificial stasis** in the sense of the Theory of Rent Extraction. Collectivisation removed land from the market and the *Hukou* system restricted mobility. The rent extraction system was frozen. Land and labour became stocks of latent value, the mobilisation of which was institutionally prevented.

2.12. Deng Xiaoping: the acceleration of the trend and the mobilisation of latent stock

The 1978 reforms represent the reactivation of the old trend. The difference lies in the speed at which the latent value accumulated over centuries is mobilised.

Urbanisation is transforming vast swathes of rural land into urban property. The retention of the *Hukou* system makes it possible to control the mobility of a cheap labour force, thereby sustaining rent and profits. Land becomes the vehicle for a colossal conversion of value, financed by massive temporal deferral (credit, LGFV). China is abruptly accelerating the old trend to its very limit.

2.13. Summary of the nine declines: the standardised dynastic matrix

Analysis of the nine major periods of decline confirms the validity of this model whilst revealing two adjustment variables: labour scarcity and exogenous shocks [25], [26], [31], [33]. The following table summarises the application of the standardised dynastic sequence (sovereign rent extraction → substitution of externalities → Nodal Point) to Chinese imperial history:

Table 3: The unique multi-century Chinese trend: oscillation between the two modes of Value and avoidance mechanisms

Period / Regime	Dates	State of the Unique Trend (Scarcity Ratio)	Avoidance Mechanism Activated	Outcome of the Phase
Shang / Zhou	c. –1600 to –256	Formation of agrarian civilisation. Land becomes the basis of social reproduction, but available space still allows relative expansion.	Massive Wealth destruction (ceaseless wars). Labour is not yet the dominant scarcity.	Continuation of the trend towards necessary structural centralisation, culminating in Qin unification.
Qin	–221 to –206	Brutal centralisation of the imperial territory. The relationship between land and labour is disrupted by mass mobilisations.	Forced spatial avoidance (Great Wall) and violent Wealth destruction (colossal construction projects).	Rapid collapse, but lays the administrative and territorial foundations of the empire. Continuation towards the Han model.
Han	–206 to 220	Lock-in of the trend: The initial pioneer front closes. Land becomes scarcer than the labour force . Shift towards the land rent mode .	Gradual hoarding by the elites. Land rent becomes the dominant mechanism for Value concentration.	Unsustainable land concentration. Yellow Turban Rebellion (184) and opening of a fragmentation phase.
Fragmentation (Three Kingdoms, 5 Barbarians, North/South, Sui)	220 to 618	Massive Wealth destruction. The invasions of the 5 Barbarians (4th–5th centuries) temporarily reactivate the pioneer front : land is redistributed and servile labour reappears.	Internal oscillation of the trend: shift towards the servile mode (North) via population capture, and spatial avoidance (South) through development of the Yangtze basin.	The conquerors settle down and become Sinicised: land scarcity reappears. Slow stabilisation under the Tang (<i>Jun-tian</i>).
Tang	618 to 907	Attempt to regulate the land/labour ratio (<i>Jun-tian</i>). Expansion opens new space, but land constraint ultimately prevails.	Spatial avoidance (Silk Road, expansions in Central Asia).	Return of land scarcity. An Lushan rebellion (755). The trend continues on an increasingly constrained basis.

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The unique multi-century Chinese trend: oscillation between the two modes of Value and avoidance mechanisms (continued)

Period / Regime	Dates	State of the Unique Trend (Scarcity Ratio)	Avoidance Mechanism Activated	Outcome of the Phase
Song	960 to 1279	Land is structurally scarce . Spatial avoidance is no longer sufficient to alter the scarcity ratio.	Massive development of temporal avoidance (fiduciary currency <i>jiaozi</i> , maritime trade).	Currency does not create land. Monetary and financial crises, collapse in the face of Mongol conquest.
Jin / Jurchen (North) & Southern Song	1120 to 1250	Split of the empire. The North (Jurchen) confiscates land and reintroduces a semi-servile system (<i>meng'an/mouke</i>). The South suffers absolute land scarcity.	North: pioneer front by conquest (servile mode). South: hyper-specialisation in temporal avoidance (currency, credit, maritime export).	Exhausted duality. Hyperinflation and debt in the South, military stasis in the North. Mongol conquest (Yuan unification).
Yuan	1279 to 1368	The Mongol empire reintroduces a vast space. Temporary return of the pioneer front mode within the trend (abundant land, captured labour).	Forced spatial avoidance (conquest, reactivation of the servile mode).	The Mongols become Sinicised and settle down. Land rent becomes dominant again. The trend continues.
Ming	1368 to 1644	Highly constrained land tenure system. Maritime closure (<i>Haijin</i>) reduces possibilities for commercial expansion.	Temporal avoidance (massive influx of exogenous silver), without altering the fundamental quantity of land.	Silver does not eliminate land scarcity. Peasant indebtedness, Li Zicheng's rebellion, Manchu invasion.
Qing	1644 to 1912	Last great territorial expansion (Xinjiang, Tibet, Taiwan). Massive population growth.	Last major episode of spatial avoidance , then temporal (unequal treaties, external debt).	Definitive closure of the spatial front . Pressure on land becomes overwhelming again (Taiping Rebellion). Terminal phase of the imperial matrix.

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The unique multi-century Chinese trend: oscillation between the two modes of Value and avoidance mechanisms (continued)

Period / Regime	Dates	State of the Unique Trend (Scarcity Ratio)	Avoidance Mechanism Activated	Outcome of the Phase
Republic	1912 to 1949	Collapse of the imperial form. The underlying trend is not resolved, but there is no longer any mechanism to sustain it.	No effective avoidance mechanism. Hyperinflation and total collapse of the price system.	Terminal crisis of the imperial matrix, leading to Communist takeover.
Maoist era	1949 to 1978	Freeze of the trend. Collectivisation and the Hukou suspend rent. Land and labour become stocks of latent Value .	Artificial stasis through institutional suspension of extraction mechanisms.	The contradiction is not resolved, but suspended, accumulating an immense mobilisable latent stock.
Deng Xiaoping	1978 to 2018	Reactivation and brutal acceleration of the old trend. Accelerated mobilisation of the latent Value stock (land and labour).	Massive land conversion (spatial/rentier avoidance) and LGFV indebtedness (temporal avoidance). The Hukou structures labour mobility.	Terminal exhaustion of the two historical modes of Value (land-based and mobilisable labour).
Xi Jinping	2018 to present	Terminal Nodal Point. Simultaneous saturation of spatial (no more pioneer front) and temporal avoidance.	The avoidance spiral has reached its limit. Necessity of an exit via TRR (qualitative regeneration).	Stasis if the old trend is reproduced. Alternative: exit the trend via TRR.

2.14. The Terminal Nodal Point and the Trimorphism of Chinese Space

The contemporary period is characterised by the simultaneous saturation of both possibilities for avoidance. Spatial avoidance no longer has a frontier; temporal avoidance (debt) can no longer replace a real basis of Value. China is reaching its terminal Nodal Point.

Chinese geographical space today reflects this situation of diminishing returns through the **trimorphism of space** [4]:

- **Metropolitan areas** (hypertrophied): the intensity of capital still manages to overcome the constraint of rent (vertical expansion), polarising selective global demand.
- **Urban areas**: where ground rent hinders the growth of cities in proportion to their size.
- **Rural areas**: impoverished by the substitution of externalities for the benefit of metropolitan areas, victims of the atrophy of their capital.

In previous crises, the destruction of wealth (civil war, invasion) made it possible to alter the ratio of scarcity. Today, this possibility is structurally limited. The repetition of the old mechanism would lead to stasis, in a zero-sum game.

2.15. Territorial Regeneration Rent: breaking the trend

It is in this context that the **Territorial Regeneration Rent (TRR)** takes on its meaning.

In the old system, Value is obtained by mobilising a resource that is already available. At the Nodal Point, the only way out for capital is the Destruction of Wealth in order to recreate Value. The TRR offers an alternative: the regeneration of a degraded space itself becomes a generator of Value.

Space becomes an object capable of qualitative transformation. The TRR would therefore not be yet another instalment in the spiral of avoidance, but the first attempt to break free from the historical trend based on scarcity and the appropriation of existing resources, replacing destruction with an endogenous creation of value.

Conclusion – The Nodal Point as a historical rupture

Chinese history appears as a unique trajectory of oscillation between land-based value and servile value, punctuated by the substitution of externalities and mechanisms of avoidance. For centuries, dynastic crises and nomadic invasions made it possible to destroy wealth in order to recreate value and postpone the Nodal Point.

Mao suspended the mechanism through institutional stagnation. Deng reactivated it at an unprecedented pace, consuming the last reserves of latent value. The contemporary period corresponds to the simultaneous exhaustion of both spatial and temporal avoidance.

The historical question is no longer how to restart the next cycle through destruction, but whether it is possible to break out of the trend itself. This is precisely the theoretical function that the Territorial Regeneration Rent could fulfil. Contemporary China therefore faces a historic choice: to continue the spiral of avoidance until Stasis, or to transform the very principle of rent extraction by generating Qualitative Value.

Chapter 3 – The Western Interruption: the banking and industrial loophole

Whilst imperial China spent two millennia repeating a closed land cycle, the West found, for a time, a way to circumvent the laws of the Theory of Rent Extraction (ToRE). By observing the mechanism of the Nodal Point — the depletion of the stock of Value leading to the Destruction of Wealth — Europe and then America developed a strategy of systematic avoidance. They succeeded in postponing the fatal deadline by artificially broadening the very definition of what could be ‘redeemed’, and by inventing the financial machinery capable of making this broadening financially viable [2], [3], [25].

3.1. Industrial Rent Extraction: the sudden expansion of the stock of Value

In the Chinese imperial matrix, the Space of Value was finite and bounded: arable land. Once all the land had been redistributed and then monopolised, the Nodal Point was inevitable [3], [25].

The Western innovation of the Industrial Revolution was not so much technological as economic: it consisted of breaking down the barriers to new Spaces of Value. The subsoil (coal, then oil and minerals) and the machine itself were ‘frozen’ assets, either due to the technical impossibility of exploiting them or the absence of clearly defined property rights [2], [3].

Industrial capitalisation was the political and economic act of transferring assets from the ‘Space of Value’ to the ‘Space of Wealth’. Unlike Chinese land, which was static, this new Value was dynamic: technological innovation was constantly creating new underground frontiers or new machines to be acquired. This shift gave the West the illusion of having overcome diminishing returns [2], [4].

3.2. International banking as a means of creating solvency

However, the ToRE lays down an inviolable rule: the transfer of an asset to the Wealth Space requires ‘exogenous demand in advance’. Yet, at the time of building a factory or opening a coal mine, the asset is not yet productive. Who pays? [3]

This is where the financial sector comes into play. The West pioneered the institutionalisation of money creation through private credit (and subsequently international banking). The major banking houses in London, Amsterdam and later New York provided the monetary advance necessary for industrial capital accumulation. The money lent to build infrastructure or factories was spent immediately (on wages and profits), creating the structural inflation characteristic of the polarised ToRE system [4], [34].

But to prevent this system from collapsing under the weight of its own inflation, it constantly required new markets to absorb production and balance out imbalances. This mechanism for externalising costs, initially confined to the European countryside, was removed from its original geographical context by international finance. In the 19th century, international finance funded the transfer of production and the extraction of raw materials to the ‘global countryside’ constituted by the colonies [24], [34].

3.3. The first global Nodal Point (1914–1945)

By virtue of the law of conservation, the expansion of the Space of Value by industry and finance could only be a temporary phenomenon. By the end of the 19th century, the entire Earth had been surveyed, mapped and colonised. There were no longer any ‘unclaimed lands’ or unclaimed subsoils to fuel the machine of Rent

Extraction. The global stock of Value suddenly reached saturation point, signifying the attainment of a Nodal Point on a civilisational scale [2], [4].

The ToRE thus explains what traditional historiography strives to treat as mere diplomatic incidents: the First World War, the Great Depression and the Second World War were, in reality, the inevitable and violent adjustments of the global financial and industrial system at this critical threshold. The Law of Rent Extraction demanded a massive destruction of wealth in order to recreate value [4], [35].

The West had temporarily diverted the cycle by expanding it to a global scale. But by closing it off again (once colonisation was complete), it bore the full brunt of the law of conservation which, for centuries, had reduced Chinese dynasties to ashes. The only difference was that this time the adjustment took place on the scale of the entire industrial civilisation [2], [4].

Chapter 4 – China 1978–2018: The Golden Age of Material Rent Extraction

China's take-off from the 1990s onwards illustrates the ToRE sequence: a vast stock of off-market Land Value, the lifting of the ban and the triggering of Rent Extraction through advance credit, and the generation of increasing returns.

4.1. The 1978 reform and the lifting of the Maoist ban

To launch a cycle of Rent Extraction, a vast stock of frozen value is required. Under the Maoist era (the period prior to the reforms), land and the means of production were collectivised. They were not exploited according to market laws, nor were they sanctified by an ancestral anthropological taboo: they were simply 'frozen' by inefficient planning, constituting a Space of Latent Value of colossal proportions [18], [21].

Deng Xiaoping did not privatise the land (which would have undermined political control); instead, he lifted the economic taboo whilst maintaining the political one. The *Hukou* (household registration system) played here the role that peasant status played under the Tang or Qing dynasties: it artificially kept the rural labour force and the land within the Value Space, preventing their free trading on the market [21], [36].

The reform consisted of authorising the transfer of usage rights (not ownership) of collectively owned land to farmers or local authorities. This is an unprecedented form of Rent Extraction: China has mobilised the land without relinquishing full ownership, creating the conditions for a captive land rent for the state [18], [37].

4.2. The *Hukou* as a political instrument of the Buy-back

The mechanism of the buy-back requires a stock of value that is politically frozen. The fundamental specificity of the Chinese model lies in the architecture of its property regime, a duality often misunderstood by Western analytical frameworks. As the political theorist Zhang Weiwei points out, the Chinese model is based on the fact that 'the state owns the bulk of resources, from minerals to land, but the right to use this land is flexible and shaped by market forces' [21].

However, for the 'flexibility' of the right of use to generate maximum rent at the time of rent extraction, the state's ownership of the original stock had to be strictly locked in. It was precisely at this microscopic level that the *Hukou* came into play [21]. By linking social rights (health, education, housing) to one's rural place of origin

and prohibiting permanent migration to urban areas, the system artificially kept rural land and labour within the Value Space.

When the local authorities subsequently decided to initiate the Buy-Back, the purchase price of this rural land – deprived of any legal liquidity by the *Hukou* – was kept at a level close to zero. This artificial disparity made it possible to maximise the rent captured by the state upon the resale of usage rights to property developers, thereby initiating the cycle of exogenous demand.

This capture of land rent was accompanied by an equally decisive demographic externalisation. Through the *Hukou* lockdown, the cost of the social reproduction of the labour force was shifted to the countryside. Migrant workers, deprived of access to urban public services, relied structurally on families who had remained in the village to care for children, the sick and the elderly. The village therefore not only provided a supply of rural land to be expropriated at zero marginal cost; it also subsidised the cost of reproducing the urban workforce free of charge. This dual levy – on land and on social services – made it possible to keep construction wages at an artificially low level, thereby maximising the rent captured by the state via the LGFVs and accelerating the cycle of exogenous demand.

4.3. The CCP State as the central bank of the Rent Extraction: the LGFVs

In the Western model (Part I, Chapter 3), private banks created money to finance the industrial Rent Extraction, capturing a share of the rent in the process. China has bypassed this process. The CCP State has become its own banker [3].

Unable to borrow directly from the markets (or only very little at first), local authorities created ad hoc financial vehicles: the LGFVs (Local Government Financing Vehicles) 城投债 (*Chengtouzhai*). These entities, implicitly guaranteed by the state, borrowed heavily from state-owned banks [22], [23], [38].

4.3.1. Physical rent extraction and the collectivisation of rent

This money was used to expropriate peasants (at a price close to zero, thanks to the *Hukou* freeze) and to finance infrastructure. The sums paid to peasants and construction workers immediately fuelled the real economy: this is the pure realisation of the “advance exogenous demand” theorised by the ToRE. The resale of usage rights by the CCP-led state generated colossal profits, which were used to repay the LGFVs and to finance new projects. The cycle of unbalanced growth (where land price inflation outstripped wage growth) was set in motion.

It is, however, essential to distinguish this recycling of rent from imperial predation or Western fictitious finance. Unlike the dynastic elites who hoarded agrarian rent, or the financial markets which channel it into speculative assets, the CCP State collectivised this rent to convert it into territorial capital. Mega-infrastructure projects (high-speed rail, dams, energy transmission networks) are not a by-product of growth: they are its physical manifestation, realising the increasing returns and territorial multipliers of the Theory of Rent. This tangible translation of land rent into public goods explains the model’s enduring legitimacy.

Nevertheless, the law of conservation implies that, in doing so, the State was quietly laying the groundwork for its own Nodal Point. By building this infrastructure ahead of actual needs in order to support exogenous demand, it accumulated a colossal stock of assets which, in the long run, risked turning into financial liabilities if they failed to generate the future revenue necessary to repay the LGFV debt.

4.3.2. The structural impasse: the LGFV crisis

LGFVs are the central instrument of this Rent Extraction, but also its weakness. Created by local governments to circumvent the ban on direct borrowing, they have backed their debt against the future value of land. This mechanism rests on a fundamental contradiction: local authorities are both the masters and the prisoners of their LGFVs. They control them (through shareholding, the appointment of directors, and access to land collateral), but they are bound by the implicit guarantee that obliges them to support them come what may — all the more so as the fall in land revenue has dried up their main source of repayment.

To conceal the fact that the Nodal Point had been breached, the system generated a dual debt structure: official debt, recorded on balance sheets and subject to Beijing's quotas, and hidden debt, incurred off-balance-sheet. This hidden debt made it possible to artificially sustain credit flows for years, until the collapse of the property market brought it to light.

Faced with systemic risk, Beijing has launched a wide-ranging drive to regain control: strengthening supervision, banning implicit guarantees, and forcibly transforming the LGFVs into commercial entities. But this process comes up against an inescapable reality: the LGFVs are too heavily indebted to be abandoned (without triggering an immediate destruction of wealth), and too fragile to be rescued without creating massive moral hazard. It is precisely this impasse — the impossibility of allowing the system to collapse and the impossibility of maintaining it as it stands without sinking into fictitious wealth — that constitutes the concrete expression of the Chinese Nodal Point. The 'destruction of wealth' theorised by the ToRE takes the anticipated form here of a forced debt restructuring [23], [38].

Theoretically, Beijing has three conventional options for resolving the crisis and absorbing this excess wealth, but each proves structurally incapable of restarting the cycle:

- **Debt swaps (controlled evaporation):** This involves Beijing requiring state-owned banks to exchange their short-term receivables for very long-term, low-yield sovereign bonds. In ToRE terms, this constitutes a destruction of future income without the destruction of the physical asset. Whilst this solution preserves apparent capital, it freezes the system in a state of financial stasis: the banks, deprived of returns, lose their ability to finance new cycles, dragging the economy into a Japanese-style 'zombification' that does not resolve the Nodal Point but freezes it indefinitely.
- **Selective default (surgical destruction):** This approach would involve imposing a haircut on the creditors of certain LGFVs, or even allowing the most insolvent entities to fail. It is the purest form of Wealth Destruction — the modern, targeted equivalent of the peasant revolts of yesteryear. However, the risk of systemic contagion is significant: a partial default could trigger a widespread liquidity crisis, as investors withdraw their funds from the entire local financial system as a precaution.
- **Monetary inflation (dilutive erosion):** The People's Bank of China could create money on a massive scale to buy back LGFV debt, thereby eroding the value of wealth through dilution. But fleeing into inflation is the quintessential Western approach. In China, rises in retail or property prices have historically been the trigger for the collapse of dynasties. This is the ultimate political risk for the CCP state, capable of destroying its own legitimacy even before the Nodal Point is addressed.

Whilst the swap leads to Stasis, default to contagion, and inflation to social risk, these three paths share a fatal commonality: they are fruitless. They manage the evaporation of Wealth, but none of them recreates the stock of Value necessary to initiate a new cycle.

It is precisely this structural impasse that makes the activation of the **Territorial Regeneration Rent (TRR)** inevitable. To avoid the unproductive destruction of wealth, the State must force the emergence of a new cycle of Rent Extraction by backing existing debt with an underlying asset that generates real rent.

4.4. The Rent Extraction of Geography: mega-infrastructure and the substitution of externalities

China's genius during this period lay in understanding that the 'Space of Value' was not limited to agricultural land on the outskirts of cities (which eventually runs out). It applied the concept of Rent Extraction to physical geography itself [3], [39].

Western history had harnessed the subsoil and the colonies. China, unable to colonise, colonised its own land-locked interior. The mega-infrastructure projects of the 1990s and 2000s were the instruments of this geographical Rent Extraction:

4.4.1. The Three Gorges Dam

The Yangtze River was not merely a transport route; it was a wild 'Space of Value', whose devastating floods represented a periodic destruction of Wealth, but whose energy and hydraulic potential remained entirely inaccessible due to the technical and financial impossibility of harnessing it.

The decision to build the dam was part of the pure mechanism of Rent Extraction. The State overrode the natural constraint through a massive political act, mobilising a colossal loan (over 200 billion yuan at the time) to 'buy' the riverbed. This injection of capital generated phenomenal "exogenous demand in advance" for nearly twenty years, supporting the national heavy industry (cement, steel, civil engineering) at a time when the domestic market had not yet fully developed.

However, the ToRE's analysis reveals above all a shift in externalities on an unprecedented scale. The dam was not built to enrich the Three Gorges region (Hubei/Chongqing), which paid an exorbitant price for the project: over a million people displaced, historic towns submerged and ecosystems destroyed. The hydroelectric power generated and the flood control benefits were harnessed for the exclusive benefit of the eastern coastal metropolises (Yangtze Delta).

The metropolitan centre (Shanghai, and beyond that the entire export-oriented coastal strip) was thus able to industrialise at breakneck speed thanks to an abundant, cheap and stable supply of electricity 'imported' from the heart of the continent. The prosperity of the metropolitan area was built upon the physical and social decline of the Middle Yangtze Valley.

Finally, the Three Gorges Dam has acted as the fundamental hydraulic hub of the Yangtze basin: without its reservoir to regulate flows across the seasons, the South-North Water Transfer Project (analysed in 4.5.3) would have been technically unfeasible. The dam thus marks the pinnacle of the CCP state's ability to reshape the country's physical geography in order to force the extraction of resources for the sake of increasing yields.

4.4.2. The West-East Gas Pipeline

Whilst the Three Gorges Dam enabled water regulation, the West-East Gas Pipeline (*Xiqi Dongshu*) served a parallel and independent purpose: energy substitution. The manufacturing prosperity of the coastal regions (Shanghai, Jiangsu) required cheap energy that local coal could no longer provide without undermining profitability (pollution, logistical costs). By 'purchasing' gas from the West (Xinjiang), the CCP state has shifted the energy externality from the periphery to the centre (the Yangtze Delta).

Gas from the Taklamakan Desert constitutes a weak externality (a resource lacking access infrastructure), not a strong externality (value frozen by a symbolic prohibition). The term ‘purchase’ is used by analogy with the Terms of Reference, but the operation is, strictly speaking, a conventional infrastructure investment.

The account of the decision to build the West-East Gas Pipeline perfectly illustrates the theoretical sequence of the morphogenetic process specific to Structural Geography:

- **Aesthetics (Tianxia):** The geopolitical vision of the Architect of Reforms to draw a line ‘with a thick, colourful brushstroke’ (浓墨重彩) from west to east across the map of China, in order to open up the development of the West, is not a mere calculation of profitability, but the expression of a cosmologically inspired territorial harmony [39].
- **Policy (The Property Rule):** The Architect of Reforms — representing the CCP — takes the sovereign decision to lift the geographical ban. Downstream, Zhang Guobao, head of the National Development and Reform Commission (NDRC), acts as the technical operator of the Rent Extraction: he mobilises state credit, imposes the network in the face of local reluctance (e.g. the Rice-growing Province) and disregards the warnings of experts (CIECC), validating the political act through empirical rigour.
- **Economic (The Buyback, by analogy):** Massive financing through government debt creates ‘exogenous demand in advance’ (via infrastructure construction).

This case demonstrates that the Chinese model was neither a mere ‘free-market miracle’ nor orthodox centralised planning, but the systematic application of the laws of ToRE on a continental scale. The CCP-led state acted as the ultimate capitalist entrepreneur, ‘redeeming’ the geography itself to initiate a new cycle of unbalanced growth.

4.4.3. The South-North Canal (South-North Water Transfer)

The prosperity of the northern metropolises (Beijing, Northern City) came up against an absolute physical limit: the depletion of groundwater reserves. Meanwhile, the Yangtze River basin, regulated by the Three Gorges Dam, was experiencing surpluses. Water, a natural element, had become an asset ‘frozen’ by geography: it existed but was not in the right place.

The South-North Water Diversion Project forms part of the ‘Rent Extraction’ mechanism. The CCP state has mobilised a colossal loan to ‘buy’ this distance of over 1,000 kilometres. From the perspective of the Theory of Rent Extraction (ToRE), this is the most extreme illustration of the substitution of externalities: the survival and growth of the political centre (Beijing) have been made possible by the extraction of water resources from the southern periphery. The North has bought up the South’s vital resource, perpetuating the North’s over-accumulation at the expense of regional balance, whilst generating the exogenous demand for the canal’s construction.

4.4.4. Canals providing access to the sea (e.g. the Pinglu Canal)

The second project concerns opening up mountainous hinterlands to the sea, of which the Pinglu Canal (South-West Province) is the most significant example. Historically, the south-western provinces were forced to transport their goods eastwards to reach the sea via the Pearl River Delta (Shenzhen). This dependence created an overwhelming economic advantage for the eastern coastal metropolises.

The south-western hinterland found itself in the typical position of an ‘atrophied space’: its productive externalities were captured by centres situated along the flow axes. To counter this centrifugal dynamic, the CCP state

decided to ‘redeem’ the topography: by digging a canal through the mountains to link the inland rivers directly to the Gulf of Tonkin.

This project reveals a sophisticated use of the ToRE at the end of its cycle. It involves artificially injecting a maritime externality into the periphery to relaunch a micro-cycle of Rent Extraction there. By giving the South-Western Province direct access to the sea, the CCP State creates a new local land and logistics rent. The credit injected into the dredging generates exogenous demand, whilst the infrastructure is intended to attract factories, thereby recreating locally increasing returns in a region threatened by decline.

4.6. The exhaustion of the geographical frontier and the emergence of ‘refractory values’

Far from marking the limits of the ToRE, these final projects actually confirm its predictive validity: it is the material rent extraction itself that is coming to an end. The marginal return on these operations is in free fall. The South-North Canal is accompanied by increasingly severe negative externalities (the drying up of the lower Yangtze) which are destroying the value of the donor periphery [40]. The Pinglu Canal requires colossal investment (giant locks, tunnels) for an economic benefit that is increasingly uncertain when compared with the rail network.

The depletion of land-based finance (easily accessible urban land) has driven the CCP state to buy up the sub-soil (gas), then the river (Three Gorges), then inter-basin water resources (South-North Canal), and then the topography (Pinglu). But each new acquisition required more credit to generate lower real productive returns.

Whilst some physical areas remain technically exploitable (hydroelectricity in Tibet/Qinghai, geothermal energy in Xinjiang), the costs of extraction, infrastructure and environmental externalities are such that they no longer allow for the increasing returns necessary for the cycle of buy-back. In accordance with the law of diminishing returns inherent in the Law of Returns, the marginal cost of Rent Extraction eventually exceeds the marginal value of the Wealth produced.

We propose, as a hypothesis, to extend the law of conservation of Value by classifying these territories as **refractory Values**: they exist physically, but the economic laws of the system mean that their integration into the productive cycle would require the destruction of more Wealth than it would create, thereby blocking the Rent Extraction process. This is a theoretical extension, not a concept present in the original body of work of the ToRE.

Chapter 5 – The exhaustion of the centuries-long trend (Nodal Point) (2015–2018) and the sovereign management of destruction

Whilst the *Hukou* system had frozen the labour force and rural land, another stock of Value – of a different nature but subject to the same laws – had remained outside the market: disconnected territories and immobilised energy resources. As recalled in the memoirs of Zhang Guobao, the chief architect of these projects, gas from the Xinjiang desert was ‘burned off into the air’ (the ‘lights in the sky’) because there was no market to absorb it [39]. Within the framework of the ToRE, this physically isolated space (the 4,000 km between Xinjiang and Shanghai) and this non-marketable resource constituted a vast Space of Geographical Value, which subsequent mega-infrastructure projects would gradually redevelop. However, this territorial expansion, however colossal it might be, was bound to reach a limit.

5.1. Depletion of the stock of land and geographical value

The quantitative depletion of mobilisable assets triggers the Nodal Point theorised by the ToRE. China's growth (1978–2018) was based on the simultaneous acquisition of two stocks of value: coastal urban land and the land-locked territories of the West. Around 2015, as the urbanisation rate exceeded 65 per cent [41], this dual shift became apparent: monetised wealth exceeded the residual stock of value [4], [15]. Land with significant externalities around coastal metropolitan areas was fully absorbed, whilst the mega-infrastructure projects designed to open up the West reached their limits of profitability. In an attempt to prolong the cycle, local authorities then sought to buy up land in remote provinces, but these areas, lacking the necessary networks and productive advantages, attracted no solvent buyers. The saturation of the property market and the halt to mega-infrastructure projects are therefore not mere economic accidents: the cycle of Material Rent Extraction is interrupted by the simultaneous exhaustion of both the Land Value Space and the Geographical Value Space, and the Nodal Point has been reached.

This tension between the physical depletion of assets and the state's forward-looking approach had already become apparent during the design of the West-East Gas Pipeline. Experts from the International Engineering Consulting Corporation (CIECC) had assessed the proven reserves and recommended a capacity of 120 billion m³, deeming the risk associated with 200 billion to be too high. Zhang Guobao, acting as the operator of the Takeover, had opposed this, arguing that a strategic project of this scale needed to 'allow a margin' to anticipate future gas imports from Central Asia [39]. This conflict was not technical but structurally linked to the Terms of Reference (ToR): the experts reasoned in terms of 'equilibrium' (proven reserves = capacity), whilst the CCP state reasoned in terms of 'imbalance' (future reserves must be anticipated through credit to sustain exogenous demand). The fact that Zhang "gave in" to the experts (reducing the figure to 120 billion) marks an initial sign that the CCP-State's capacity to force the Buy-Back beyond physical limits is running out of steam – a prelude to the Nodal Point where the wealth created is no longer sufficient to justify the debt.

5.2. Morphological manifestations of the Nodal Point: ghost towns, exoregulated dispersion, reversal of hierarchies

The ToRE emphasises a corollary often overlooked by purely macroeconomic analyses: at the Nodal Point, spatial dynamics are reversed and mobility—hitherto centripetal, whether endoregulated (aggregation) or exoregulated (concentration)—gives way to centrifugal mobility: 'escape' (endoregulated) for the affluent classes, and dispersion (exoregulated) for populations forced to leave saturated centres under the pressure of costs and a shortage of affordable housing [3, sec. 3.3]. The saturated metropolitan area drives its workforce out towards the periphery.

This is exactly what we see in China with the 'ghost towns' (such as Kangbashi in Ordos) [42]. These new towns, built far from historic centres, are not merely local 'planning errors'; they are the spatial manifestation of exoregulated dispersion. Capital, no longer finding profitability in the centres (due to diminishing returns), has spread out towards undeveloped peripheries in a desperate attempt to recreate a local cycle of Rent Extraction. The reversal of urban hierarchies — with the relative decline of certain hyper-centres in favour of second- or third-tier cities for a short period — confirms that the Chinese territory has entered the morphological crisis phase of the ToRE.

The morphology of these ghost towns is no accident. The ToRE distinguishes between two main families of urban forms [3, sections 3.1–3.3]:

- **centred forms** (circular, concentric) are associated with distributive maturity regimes and result from endoregulated mobility (chosen agglomeration);

- **eccentric forms** (sectoral, axial) are associated with regimes of polarised growth and result from exo-regulated mobility (forced concentration along export axes).

New Chinese towns such as Kangbashi, built from scratch along transport corridors and disconnected from historic urban fabric, are archetypes of the decentralised form resulting from unregulated concentration. Their failure to attract a sustainable population and economic activity is not merely a planning flaw; it is the spatial manifestation of polarised growth that has run its course. The persistence of these decentralised forms after the Nodal Point indicates that the transition to a mature, distributive regime has not taken place: the cities have not refocused on centres of domestic consumption, but have remained oriented towards transport corridors and nodes whose economic function ceased with the end of polarised growth.

This morphological observation is clarified by the regulatory role of ground rent on urban hierarchies, as analysed by the ToRE [3, sec. 2.2]. During the upward phase of the cycle, land rent hinders the growth of cities in proportion to their size: the larger a city is, the higher the rent, and the more it discourages marginal investment, redirecting it towards lower levels of the hierarchy. This mechanism generates the rank-size harmonic function, whereby each city finds its place within a balanced hierarchy [24].

In China, this mechanism has been neutralised by spatial capital flows (LGFVs, state investment), which have enabled major cities to overcome the constraint of ground rent by accumulating fixed capital (skyscrapers, infrastructure). But at the Nodal Point, when these flows dry up, the regulatory logic of rent re-emerges — but in the opposite direction. Rank 3 and 4 towns, which have never benefited from sufficient capital accumulation to overcome their rent, find their rent too low to retain the capital fleeing the metropolises. They cannot absorb this dispersion, as their low rent does not generate sufficient externalities to attract investment. Morphological dispersion is thus accompanied by a frozen urban hierarchy: metropolises retain their speculative rent (even when disconnected from use value), whilst medium-sized and small towns remain structurally incapable of retaining their dynamic population.

This hierarchical deadlock has a direct impact on the metropolitan areas themselves. If third- and fourth-tier towns cannot absorb this dispersion, the capital fleeing them also fails to find productive outlets within the metropolitan areas. Speculative rent remains high there, but it is no longer supported by solvent demand. Vacant offices are proliferating in central business districts, retail outlets are closing due to a lack of customers, and retail spaces are deteriorating into urban wastelands.

This phenomenon is the physical manifestation of the disconnection between rent and use value that the ToRE identifies as a characteristic of the Nodal Point [3, sec. 2.6]. Rent is no longer the product of productive dynamics; it is determined by speculation on a finite stock. Metropolises retain their skylines and skyscrapers, but these infrastructures, now ‘ghostly’, harbour empty spaces: the verticality that was once the hallmark of over-accumulation becomes the symbol of its exhaustion.

5.3. The sovereign destruction of parasitic wealth: the ‘Three Red Lines’ and the Evergrande precedent

The law of conservation of ToRE stipulates that, at the Nodal Point, the destruction of Wealth is necessary to recreate the conditions for a new cycle. In Western economies, this destruction takes place chaotically through the market: a cascade of bankruptcies, collapsing prices, mass unemployment, and the socialisation of banking losses (the 2008 scenario). The CCP-led state, drawing on its capacity for strategic coordination, has chosen a radically different path: selective sovereign demolition.

The ‘Three Red Lines’ policy (三条红线), launched in August 2020, illustrates how the CCP-led state has taken control of managing over-accumulation, refusing to delegate it solely to market forces. By imposing strict debt ratios on property developers, the CCP-State has orchestrated a targeted reduction in the sector’s overcapacity.

The case of Evergrande illustrates the logic behind this structural selection. The group had taken on massive debt on Hong Kong’s offshore markets, issuing dollar-denominated bonds and thereby integrating the Chinese property sector into the sphere of international financialisation. This connection served as a conduit for extraterritorial risks: should the economic cycle reverse, foreign creditors could have exerted destabilising pressure on the domestic financial system. By allowing Evergrande to default on its offshore debt whilst prosecuting its executives for embezzlement, the CCP-led state has effected a threefold structural adjustment:

- **Punishment for rent-seeking behaviour:** the destruction of the developer’s shareholder and financial wealth, reaffirming that virtuous legitimacy (*De*) takes precedence over speculative private profit;
- **Monetary and geopolitical firewall:** cutting off the channel of contagion from Western financial speculation, protecting the domestic banking system from the extraterritorial reach of the dollar and shielding the Chinese economic cycle from the Federal Reserve’s interest rate cycles;
- **Preservation of the social contract:** through the slogan *Baojiaolou* (保交楼 – ‘Guarantee the delivery of housing’), the CCP state has split Wealth into two components. It has sacrificed financial Wealth (offshore creditors, shareholders, executives), but has mobilised public resources to complete the construction of housing pre-sold to households. Usufructual Wealth – the family’s concrete territorial anchoring in their home – was protected in the name of virtuous legitimacy (*De*), whilst speculative Wealth was left to be destroyed.

This sequence can be interpreted as an illustration of the logic of the ToRE in its political dimension: the destruction of wealth at the Nodal Point is mandatory within the theoretical framework, but the manner in which it occurs – whether chaotically through the market or sovereignly through the CCP-led state – depends on institutional and political structures. Whereas the West socialised the banks’ losses to save the financial system in 2008, China took the opposite approach, sacrificing property developers to save households and preserve the social contract. This difference does not validate the theory, but illustrates one possible application within a specific institutional context.

However, the Theory of the Nodal Point teaches that the destruction of wealth is only the first half of the Nodal Point equation. For the cycle to restart, this destruction must free up capital for a new Rent Extraction, a new Space of Value capable of generating exogenous demand in advance. Yet, by dismantling the model of off-plan property sales – which relied on households’ confidence in future delivery – the CCP state has simultaneously dried up the historical mechanism of local money creation. Selection has eliminated the parasite, but the patient finds itself paralysed. Local governments, already suffocated by the LGFVs, are faced with a void in exogenous demand that neither conventional bank credit nor central budgetary transfers can fill. It is this structural impasse, and not a misdiagnosis, that makes the search for a new frontier imperative.

5.4. The closure of physical internal borders and the need for a Third Transformation

Faced with this impasse, the CCP State is confronted with the theoretical alternative posed by the Nodal Point: to allow the destruction of Wealth to proceed unchecked (the Western scenario), or to find a new frontier capable

of recreating mobilisable Value. The CCP State's prudent judgement, as embodied in the Three Red Lines, has addressed the first half of the equation; what remains is to identify the new stock of Value.

Any analysis of Chinese overseas investment must distinguish between two structurally different categories. The transport corridors of the Silk Roads — gas pipelines, oil pipelines, railways and maritime routes — are transit infrastructures that secure physical flows between China and global markets. Their profitability does not depend on local sovereignty: the West-East Gas Pipeline, the Central Asia–Europe railways and the Russia–Empire pipelines remain operational because they are underpinned by guaranteed flows and strategic corridors.

By contrast, local productive investments — HPAL factories in Indonesia, the port of Chancay (Peru), the port of Hambantota (Sri Lanka) and the port of Piraeus (Greece) — have had mixed results. The ports, designed as '**Fortified Port Enclaves**' (§5.5.5), have been successful: they benefit from exclusive rights, regulatory exemptions and full collateral, which make any expropriation economically devastating for the host country. Processing plants, on the other hand, are structurally vulnerable to **Sovereign Rent**¹ of the host state: Indonesian Decree No. 144 (2026) demonstrated that a sovereign state can unilaterally alter the rules of the game (quotas, costs, currencies), rendering the Chinese financial arrangement untenable [43], [44].

The lesson of the Third Transformation is that China now favours logistics enclaves (ports, pipelines) over processing plants, as the former are immune to the host states' Sovereignty Rent. The external material frontier is therefore virtually closed for industrial processing, but remains open for logistics enclaves — a distinction that Chinese strategy has incorporated. Whilst the infrastructure of the Silk Roads has partially succeeded in organising the spatial flows of the *Tianxia* [45], it has not generated the 'advance exogenous demand' necessary to absorb the shock of the domestic Nodal Point. The economic returns on external investment in transformation are proving too low and too slow to compensate for the abrupt halt to the internal driving force [46]. With the external material frontier virtually closed to industry, the systemic constraint is tightening.

It is against this backdrop that the strategic planning capacity of the unified CCP-State is confronted with its deepest epistemological limit. The epistemology of the infinite development of the productive forces comes up against the law of conservation of ToRE: the stock of Value is finite at any given point in time, technology and geography. The capacity for inter-regional coordination that has enabled the construction of ultra-high-voltage networks, high-speed railways and inter-basin water transfers cannot, on its own, create a new stock of Value where physical geography no longer affords it. The CCP State can coordinate, plan and allocate; it cannot decree the existence of a virgin 'Space of Value' once the territory has been fully appropriated.

To uphold the Mandate of Heaven and preserve the social harmony underpinning the '*De*', the CCP State is therefore compelled to effect a radical internal shift. The '**Third Transformation**' towards the immaterial (carbon, biological data, sovereign artificial intelligence) is neither a headlong rush nor a denial of reality: it constitutes the most ambitious sovereign attempt to transcend the law of conservation through the creation of new normative scarcities, in order to avoid the social cost of widespread material destruction, which the implicit contract between the CCP-State and the population renders politically unacceptable. It is this systemic constraint — the need to recreate a cycle of Rent Extraction without having a new geographical substrate — that justifies the launch of this transformation and which, at the same time, constitutes its structural limit, as will be demonstrated in Part III.

¹'Sovereign Rent' refers to the economic value attached to a state's ability to unilaterally alter the rules of the game within its territory (quotas, exchange rates, taxation). This is triggered when the host state alters the operating conditions for a foreign asset. Fortified Port Enclaves are designed to shield investments against this rent.

5.5. The externalisation of the destruction of wealth and the structural dependence of the periphery

The Third Transformation, analysed in its immaterial dimension, raises an operational question: how does China secure its supply of material substrates (minerals, energy, components) whilst its domestic economy has entered a state of Stasis? The answer lies in a transformation of the role of the periphery within the global economic system.

5.5.1. The geographical externalisation of the destruction of Wealth

The ToRE teaches that at the Nodal Point, material wealth must be destroyed so that rent can be recreated elsewhere (see §1.7). In the historical Western model, this destruction affected both the centre and the periphery indiscriminately. China, by retreating into the immaterial, has altered this situation.

The Chinese strategy consists of relocating part of the destruction of wealth to the host states of its foreign investments. This mechanism is not a malicious intention, but a structural consequence of the asymmetry between Chinese assets (intangible, immune) and assets located abroad (physical, vulnerable). China can repatriate its strategic equipment and technologies, leaving the host country with unfinished or inoperative infrastructure. The depreciation of assets is thus geographically localised within the host country's territory, not that of the centre.

This transfer is not a deliberate strategy but a structural feature of the system: Chinese intangible assets (tokens, quotas, exploitation rights) are immune to physical shocks, whilst tangible assets located abroad are exposed to them. This structural asymmetry makes the periphery more vulnerable to cycles of devaluation than the centre.

5.5.2. The costs and limitations of withdrawal

It would, however, be mistaken to think that China emerges unscathed from these withdrawals. The dismantling of the HPAL plants in Sulawesi in 2026 resulted in real losses: loans from the CDB and the Export Bank remain largely unrepaid, future cash flows have been lost, and China's image as a reliable partner has been tarnished. China has certainly repatriated its titanium autoclaves and its know-how, but it has lost several billion dollars in investment and a strategic position in the nickel supply chain. Indonesia, for its part, has lost factories that it cannot operate without Chinese reagents and engineering expertise, but it has preserved its sovereignty over its resources — a significant political gain.

The case of Venezuela offers an instructive parallel. Beijing had granted massive loans to Caracas (over \$50 billion between 2007 and 2015), secured against oil deliveries. The collapse of the Venezuelan economy and political instability led to a renegotiation of the debt: China had to accept losses, rescheduling and a reduction in its claims [46]. The host state's Sovereign Annuity was triggered, not by a decree, but by institutional collapse, which made debt servicing impossible. China was unable to seize Venezuelan oil assets without risking permanently jeopardising its relationship with a strategic partner. It therefore had to absorb part of the destruction of wealth.

The difference with Indonesia lies in the nature of the asset: the HPAL plants were localised assets, which were partially seizable; Venezuelan oil is an underground asset, less easily transferable. But in both cases, China suffered losses. The asymmetry is therefore not absolute: it is a structural bias which, on average, protects the centre, but does not provide total immunity.

5.5.3. Debt as an instrument of structural dependence

China finances its overseas investments through loans from the CDB and the China Exim Bank, secured against the future cash flows of the projects (see §4.3). This arrangement, similar to the Land Finance model, works as long as the host state does not exercise its sovereign rent — that is, its ability to unilaterally alter the rules of the game (quotas, exchange rates, taxation). When Jakarta activated this rent via Decree No. 144, the financial arrangement collapsed. China was forced to withdraw, leaving Indonesia with unfinished infrastructure and unpaid debts, whilst bearing part of the cost itself.

Beijing has learnt its lesson: future investments must be designed to be more resilient to the host's Sovereignty Rent. This is the rationale behind **Fortified Port Enclaves** (see §5.5.5) — de facto extraterritorial infrastructure, protected by exclusive rights, regulatory exemptions and full collateral, which make any unilateral change to the rules economically destructive for the host country. This lesson was drawn in part from the Indonesian and Venezuelan withdrawals: where the host can change the rules, China must secure its positions through legal and financial architecture, not through political trust.

China's external debt is therefore not an instrument of a 'debt trap' but a mechanism of structural dependence. It creates an asymmetry in which the host state bears the bulk of the risks of devaluation, whilst the Chinese centre retains control over the normative assets that guarantee the debt. This asymmetry, however, has its limits: if losses mount up (as in Venezuela), China is forced to absorb them, which reduces the scope of externalisation.

5.5.4. Agricultural outsourcing: leased land in Russia and global agro-industrial partnerships

Since the 2010s, China has pursued a strategy of outsourcing agricultural production to vast tracts of underutilised land abroad. This strategy, which forms part of an approach to securing food supplies, takes various forms depending on the region and institutional context.

The Russian case: the long-term lease model

The area of land leased or controlled by Chinese interests in Russia exceeds 670,000 hectares (according to SIPRI estimates), and could reach over one million hectares according to more recent sources, including 426,600 hectares leased directly near the border. The terms of these leases are clearly defined: 49-year contracts, an annual rent of 300 to 600 yuan per hectare, and an obligation to employ at least 30 per cent Russian labour. The produce (soya, wheat, maize, meat) is partly exported to China, contributing to the country's food security, as it is only 50 per cent self-sufficient in soya.

Other agricultural fronts: Africa, Latin America and South-East Asia

Beyond Russia, China has rolled out a global agricultural investment strategy, often referred to as 'going out' in the agri-food sector. The main areas of investment include:

- **Sub-Saharan Africa:** China is active in several countries, employing a variety of approaches. In Madagascar, cooperation focuses mainly on technology transfer (hybrid rice) and public-private partnerships, without large-scale land leasing. In Angola, a 130,000-hectare project is dedicated to the cultivation of soya and maize, with 60 per cent of the production destined for export to China. In Tanzania, Chinese companies have developed projects covering 32,000 hectares. In Zambia, investments have been made on land leased on 99-year leases. In the Democratic Republic of the Congo, a 2.8 million-hectare project is planned for a palm oil plantation. Other projects exist in Senegal, Sierra Leone and Mozambique.

- **Latin America:** China has invested in agricultural land in Brazil, Argentina and Uruguay, mainly for the production of soya, maize and cereals. These investments are often made through trade agreements or stakes in local companies, rather than through direct leases.
- **South-East Asia:** Investments have been made in Laos, Cambodia, Thailand and Myanmar, where Chinese companies grow rice, rubber and energy crops (sugar cane, palm oil).

ToRE typology of Chinese agricultural partnerships abroad

Within the framework of the Rent Extraction Theory, these operations differ radically from the Chinese land tenure model. This case illustrates a weak externality (an under-utilised resource lacking access infrastructure or market access), not a Rent Extraction in the strict sense of the ToRE. Unlike Chinese land, foreign land is not ‘frozen’ by an anthropological or symbolic taboo. It is simply under-utilised and subject to sovereign regulation. China leases, it does not buy; it produces, it does not speculate.

Table 4: Classification of Chinese agricultural partnerships abroad according to the ToRE

Region	Type of operation	Area / Investment	ToRE status
Russia	Long-term lease (49 years)	> 670,000 ha (est. 1 M ha)	Low externalities
Angola	Agro-industrial project	130,000 ha (soya, maize)	Low externalities
Tanzania	Agro-industrial project	32,000 ha (soya)	Low externalities
Zambia	Long-term lease (99 years)	Multiple investments	Low externalities
DRC	Plantation project	2.8 M ha (palm oil)	Low externalities
Madagascar	Technical cooperation, investment	Multiple programmes	Low externalities / Technology transfer
Brazil / Argentina	Trade agreements, equity stakes	Significant flows of soya and maize	Low externalities
Laos / Cambodia	Energy crop projects	Multiple investments	Low externalities

Sovereign Rent and the dialectic of controlled solvency

In all cases, the host states retain their full Sovereign Rent: they may at any time amend the terms of the leases, increase rents, restrict exports, or limit the Chinese presence. The length of the leases (49 years in Russia, 99 years in Zambia) reduces this risk, but does not eliminate it. The difference compared with Indonesia (2026) is that these states need Chinese investment to modernise their land, which creates a mutual dependency that tempers the temptation to issue a unilateral decree.

Massive Chinese investment could increase the differential rent from these lands: irrigation, fertilisers, infrastructure, improved seeds. It would strengthen China’s position in the dialectic of controlled solvency (see §5.5.6) by anchoring the activity in the host territories. However, even with these investments, the operation would not

constitute a ‘Buyout’ as defined by the ToRE: it would remain a productive investment with low externalities, securing supply flows without creating exogenous demand in advance or polarised growth in China.

Legislative changes in the host countries — for example, allowing foreigners to own agricultural land — would not be sufficient to transform these operations into a ‘Buyback’. These states would retain their Sovereign Rent, and foreign assets are not ‘Spaces of Value’ frozen by a symbolic ban; they are fully integrated into the global market, even if Western sanctions restrict access to them.

Summary: niche accumulation, not a new cycle of Rent Extraction

The China–Russia partnership, whether involving agricultural land or energy joint ventures (86 projects, \$222 billion, key sectors), is a strategic partnership between sovereign states, not a cycle of Rent Extraction. The same applies to all Chinese agricultural investments in Africa, Latin America and South-East Asia. These operations secure physical flows (energy, foodstuffs) and generate shared rents, but they do not resolve China’s Stasis.

Foreign land leased or farmed by Chinese interests is an example of **niche accumulation** — a strategy to circumvent domestic land constraints which, whilst not creating a cycle of polarised growth, contributes to the resilience of China’s productive apparatus. This niche accumulation is a pragmatic response to the depletion of the stock of domestic value, but it does not constitute a new frontier capable of generating the necessary advance in exogenous demand to restart the cycle of Rent Extraction in China.

5.5.5. Fortified Port Enclaves: a response to sovereignty constraints

Given the vulnerability of processing investments (HPAL plants) to the ‘sovereign rent’ levied by host states, China has developed an alternative: **Fortified Port Enclaves**. The port of Chancay in Peru, inaugurated in November 2024, is the archetype of this model. Unlike HPAL plants, Chancay is a purely extractive infrastructure: a mineral port designed to export South American copper, lithium and iron ore to China, with no local processing capacity whatsoever.

Chinese loan agreements include structural conditions which, whilst not political conditionalities in the traditional sense, constrain the host state’s sovereign room for manoeuvre. Thus, at Chancay: exclusive operating rights granted to COSCO, regulatory exemptions secured through legislation, and an all-assets pledge securing the project’s entire assets. This structure makes any unilateral change to the rules by the host state economically destructive.

The ‘dialectic of controlled solvency’²

Chancay illustrates the temporal duality of the Fortified Enclaves: a defensive transition phase (geopolitical turnstile) that shifts into a stationary offensive phase (automated robotic hub). This model applies to all strategic ports financed by the Silk Roads: Djibouti, Gwadar, Piraeus, Hambantota. China is abandoning industrial transformation to focus on securing physical flows.

²The concept of the ‘dialectic of controlled solvency’ is the author’s interpretation. It aims to formalise a tension observed between two objectives of Chinese foreign policy — maintaining solvent partners to safeguard investment outlets and returns, whilst managing the risks of devaluation associated with the Sovereign Rent. It does not correspond to any official Chinese doctrine, nor is it formulated as such in institutional documents or public statements. This interpretation is offered as a possible reading, not as a description of Beijing’s intended strategy. The logic underpinning this strategy rests on the idea that the host state is kept afloat — just enough to service its debt and purchase Chinese goods — but not enough to develop an autonomous industry or break free from dependence. China wishes neither to render its partners completely insolvent (as they would cease to be markets) nor to make them fully solvent (as they would break free from their dependence on China). The aim is a precarious balance, calibrated by contractual clauses, collateral and regulatory exemptions.

5.5.7. The accounting role of the intangible bubble

The bubble in AI tokens and carbon credits (see Part III) plays an accounting role in this mechanism. It allows for the nominal revaluation of the collateral for Chinese loans (the rights to exploit foreign assets) without this revaluation corresponding to an increase in the physical value of the assets. The destruction of material wealth is thus partially ‘absorbed’ by the inflation of Chinese intangible assets. But this mechanism has its limits: it cannot indefinitely offset massive real losses, as the Venezuelan experience demonstrates.

Concluding remark: This externalisation mechanism is not a strategy of domination but a structural response to the exhaustion of China’s material cycle. China, like any state that has reached the Nodal Point, must manage the destruction of its material wealth. What is specific to China is that it has attempted to offload part of this destruction onto the host states of its foreign investments. This attempt has seen partial successes (port enclaves) and costly failures (Indonesia, Venezuela). The overhaul of China’s overseas lending instruments — notably through the widespread adoption of ‘Fortified Enclaves’ — aims to stabilise this mechanism and reduce the risks of conflict linked to ‘sovereign rent’, without ever eliminating them entirely. Through successive crises, China is learning that structural asymmetry does not confer absolute immunity, and that managing the periphery requires a delicate balance between securing cash flows and accepting occasional losses.

Chapter 6 – The Impasse of the Intangible

Introduction: The epistemological shift and the limits of analogy

The two preceding sections have demonstrated, by adhering strictly to its ontological foundations, that China’s cycle of material accumulation conformed to the laws of the Theory of Rent Extraction (ToRE) right up to its conclusion: the Nodal Point. The depletion of the stock of readily mobilisable land and geographical value necessitated a structural shift.

Faced with this constraint, the Chinese authorities have embarked on a ‘Third Transformation’ by attempting to create new forms of scarcity in the spheres of carbon, biological data and artificial intelligence. However, before analysing the mechanisms of this mutation, a rigorous epistemological clarification is required: the Third Mutation does not, strictly speaking, constitute a new cycle of Rent Extraction.

The Theory of Rent Extraction (ToRE) and the Structural Geography that underpins it postulate that Rent Extraction operates on assets kept off the market by symbolic foundations, collective beliefs or foundational anthropological taboos (sacred land, the seigneurial domain, the *Hukou*). Atmospheric carbon, genomic data or computing tokens do not meet these criteria: they are unvalued externalities or creations *ex nihilo* established by regulatory decree. Consequently, to speak of the ‘Rent Extraction of the Intangible’ or the ‘AI Nodal Point’ would be a theoretical misinterpretation that would strip the ToRE of its specificity.

The value of the analysis lies elsewhere: in the fact that the CCP State has devised regulatory mechanisms that faithfully replicate the operational mechanics of Rent Extraction, whilst freeing itself from its fundamental nature. We shall refer to this phenomenon as **structural isomorphism**.

Although this does not involve the transfer of a pre-existing asset, the CCP State artificially creates scarcity (quotas, access standards), mobilises advance credit to purchase it, generates exogenous demand in advance, and organises primary, secondary and tertiary markets that capture differential rents. The sequence is identical, but the underlying asset is a purely normative construct (a state rent) rather than a frozen historical value.

Similarly, whilst we may currently observe underutilised data centres in western China or a collapse in Western valuations, this does not represent the attainment of a ‘Nodal Point’ in Rebour’s sense (which requires the depletion of a finite stock), but rather what we shall call a **Breakpoint in the Sterile Loop**: a crisis of normative overcapacity in which injected credit can no longer generate the expected increasing returns, whilst nevertheless reproducing morphological manifestations (unpaid debt, empty infrastructure, capital flight) identical to those of the material Nodal Point.

Finally, whilst the economic ToRE does not formally distinguish between material and immaterial wealth, Structural Geography is, for its part, fundamentally a theory of spatial form. Analysis of this Third Mutation reveals its Achilles’ heel: the lack of morphological depth. The algorithm does not generate urban form; it requires neither concrete nor mass logistics. It creates market value without creating the physical infrastructure necessary for the spatial distribution of rent towards mass employment.

This section is therefore structured as an extension by analogy. It examines how China has attempted to simulate a cycle of Rent Extraction in the immaterial realm through an ‘**Immaterial Trimorphism**’ (Carbon, Life, Cognitive Data), why this simulation reached its own internal limits (algorithmic stasis), how it was forced to resort to aggressive outsourcing of its overcapacity (the strategy of external encirclement), and why, ultimately, the immaterial can provide only palliative buffers and not a substitute driving force.

Immaterial Trimorphism — An Analogue Extension of the ToRE

This chapter explores whether the operational mechanisms identified by the ToRE — the transfer of frozen reserves into the monetised circuit, the injection of advance credit, and the spatial polarisation of rents — can be transposed, by analogy, to the valuation of intangible assets undertaken by China in its Third Mutation.

This analogy is not a formal application of the ToRE: it uses its concepts as a heuristic framework to shed light on the mechanisms at work, without claiming that these mechanisms belong to the same ontological realm as Land Rent Extraction. The distinction between analogy and direct application is maintained throughout the analysis.

6.1. Carbon and the Biosphere (ETS): the ‘intangible’ rural sector

China’s national Emissions Trading Scheme (ETS) must not remain a marginal administrative instrument, nor be limited solely to carbon dioxide. To become the primary driver of post-land rent extraction, it must replicate the mechanics of the material model by transforming the biosphere’s entire capacity for absorption and use — air, water, soil and the radio spectrum — into a multitude of financial assets that generate credits.

China has significantly expanded its carbon market since its national launch in 2021. By 2025, the cumulative volume of transactions had exceeded 770 million tonnes of allowances, with a total value of over 7.3 billion dollars, attracting attention at COP30 [47]. A government roadmap, published in August 2025, sets out plans to extend the scheme to all major industrial sectors by 2027 and to phase in auctions by 2030 [48]. The Ministry of Ecology and Environment issued guidelines to strengthen the system’s operation in 2025 [49]. Market liquidity has increased, indicating better integration of price signals into investment decisions [50]. This market forms part of a broader green transition strategy which, according to some analysts, could make China a global leader in carbon finance [51].

6.1.1. The Shift from the Realm of Value to Wealth

Clean air, drinking water and uncontaminated soil were ‘*res communes*’ with no market value. By setting drastic absolute caps (cap-and-trade) for each component of the biosphere, the CCP state artificially creates multiple scarcity. The right to emit, discharge or occupy becomes a novel asset, introduced into the Space of Wealth in the form of tradable quotas, segmenting rents according to territorial specificities.

6.1.2. The primary market: the State-CCP’s buy-back

The State-CCP ‘expropriates’ the right to pollute and allocates it to major emitters. To finance these compulsory purchases, state-owned banks grant loans backed by these quotas. The money injected constitutes exogenous demand in advance: it finances the ecological transition before productivity gains materialise, in exactly the same way that the advance payment of land rent financed urban infrastructure.

6.1.3. The secondary market: green differential rent

Innovative companies resell their surplus allowances. The differential rent rewards innovation. This mechanism is based on the abatement cost: rural areas, with low abatement costs (tree planting), sell their surplus allowances to industrial areas where abatement costs are high. Rural areas become net suppliers of allowances, faithfully replicating the rural-urban gradient of the classical trimorphism.

6.1.4. The tertiary market: the polarised carbon growth regime

Speculation drives up the price of quotas in an inflationary manner. This is not a flaw; it is the driving force behind buybacks. An artificially high price increases the value of state banks’ collateral, enabling them to create credit to replace rising property prices with rising prices for biospheric resources, thereby deploying a network of ecological rents immune to foreign decrees.

6.2. Life and biosecurity: the immaterial harmonic urban

Human genomic data and zootechnical data on the national livestock population constitute a colossal reservoir of value, subject to regulatory restrictions and territorial fragmentation that previously prevented their integrated monetisation. Realising their value requires the creation of dedicated institutions: trusted data spaces. Unlike section 6.1 (ETS), which relies on an already operational carbon market, the mechanisms described in this section are currently being rolled out through national pilot programmes.

6.2.1. The Transition from the Value Space to Wealth

Raw data represents a symbolic, unmonetised value, frozen by ethical restrictions and administrative fragmentation. This dual network – comprising hospitals and livestock farms – covers the entire country. The CCP State lifts this restriction by aggregating this data to turn it into a tradable ‘biosecurity asset’, the value of which lies in the statistical mass and epidemiological power of the national reservoir.

This initiative is already underway. In 2025, the Ministry of Health (NHC), in cooperation with the National Data Centre (NDC), launched pilot projects for the development of trusted data spaces (可信数据空间) in the healthcare sector [52], [53]. In the same year, the NDC published a list of 63 pilot projects for trusted data spaces, several of which were in the healthcare sector [54], [55]. Cities such as Ningbo have established trusted data

spaces dedicated to healthcare, integrating data from 234 healthcare institutions, representing 17.15 TB of data and 9.89 million medical records covering 92 per cent of the local population [56]. Furthermore, China already has large-scale infrastructure for the collection and sharing of biological and genomic data, such as the National Gene Bank [57], [58].

6.2.2. The primary market: the act of monetisation and biological collateral

The CCP state issues licences to exploit public data (公共数据授权运营), which are purchased by pharmaceutical companies and laboratories. These purchases inject liquidity into research (exogenous demand) prior to the discovery of medicines. These licences become the collateral for a new biotech credit market.

This mechanism is already in operation. Provinces such as Zhejiang have conducted trials involving the authorised use of health data [59], [60]. In Jinan, the company Beifang Health (北方健康) has been designated as an authorised operator for public health data [61]. In Hangzhou, operators have been authorised to use more than 120 million medical records to train the ‘Anzhener’ AI model, which is now being rolled out in hospitals across the province [62]. In August 2026, the city of Yancheng (Jiangsu) carried out the first health data transaction as part of a programme for the authorised use of public data [63]. Private companies such as KingMed (金城医学) have already developed 21 data products, which are listed on the Beijing, Shanghai and Shenzhen stock exchanges [64], [65]. Furthermore, Meinian Health (美年健康) incorporated health data into its balance sheet as early as the first quarter of 2025, paving the way for the accounting valuation of data assets [66].

6.2.3. The secondary market: the differential rent of innovation

The productivity gap between a laboratory with access to integrated data (human and animal) and one without constitutes a major differential rent, just as the gap between a metropolis and a medium-sized town does. Cross-monitoring of zoonoses creates a biosecurity externality that compartmentalised Western systems cannot match.

The CNGBdb platform of the National Gene Bank provides access to large-scale genomic and biological data, covering more than 100,000 species [67]. Pilot projects such as that of Genertec (通用技术集团) cover 28 provinces, more than 400 healthcare facilities and millions of patients [68]. These infrastructures already enable a form of differential rent for the laboratories and companies that have access to them.

6.2.4. The tertiary market: speculation on life expectancy

The speculative bubble surrounding sovereign biotech firms is inflating market capitalisation, providing the CCP state with levers to finance health insurance deficits. Just as the physical metropolis stacks skyscrapers to overcome land constraints, the tertiary market of the Living stacks patents to overcome biological constraints, creating fictitious capital backed by collective health.

This mechanism, whilst speculative, is underpinned by a real infrastructure of biobanks and data. Since 2025, the health data market in China has experienced rapid growth: the government has laid down rules for the inclusion of data in balance sheets [69], [70], and companies such as KingMed have already brought data products to market. The National Gene Bank, operational since 2016, serves as the foundation for this monetisation [71], [72].

6.3. Cognitive Data and AI: the intangible metropolis and the Computing Capacity Export Strategy (SECC)

‘Cognitive Data’ (the fundamental AI models trained on the national infrastructure) is set to become the main driver of the next cycle of buybacks. The AI token (a drawing right on sovereign computing power) becomes the cognitive equivalent of expropriated land. However, analysis of this pillar reveals an unprecedented dynamic: faced with the exhaustion of domestic demand, this sector is deploying an externalised offensive strategy which we shall term the **Computing Capacity Export Strategy (SECC)**.

6.3.1. The theoretical mechanics of Cognitive Rent Extraction (The 3 markets)

In line with the analogy with the ToRE, the deployment of sovereign AI was initially intended to follow the classic three-stage sequence:

- **The primary market (The act of Rent Extraction):** State-owned banks and SOEs purchase tokens on a massive scale or finance computing clusters (the *Data in the East, Computing in the West* – DÉCO project). This payment injects ‘exogenous demand in advance’ (engineers’ salaries, purchase of chips) before the AI has optimised actual production. The token serves as collateral for issuing credit, replicating the securitisation of land.
- **The secondary market (Extraction of differential rent):** Companies in the real sector purchase access to these tokens to integrate AI, generating excess profit linked to the reduction in their marginal costs. This is cognitive differential rent.
- **The tertiary market (speculative rent):** The intervention of hedge funds drives up the price of the token autonomously. The immaterial metropolis piles up parameters (billions of model weights) just as the material metropolis piles up skyscrapers.

6.3.2. Cognitive differential rent and the weapon of open-source

Before spilling outwards via the SECC, China’s computing capacity was first mobilised to neutralise foreign technological competition. The main lever of this conquest is open-source. By publishing, free of charge, AI model weights with performance equivalent to Western proprietary frontier models, Chinese laboratories are destroying the economic value of AI software as such. This strategy transforms AI into a commodity whose value no longer lies in the model itself, but in the underlying inference infrastructure. This is where the first cognitive rent differential comes into play: Chinese companies, backed by the cheap green energy of western China, can offer inference services at negligible marginal costs, thereby stifling the business models of Western start-ups, whose computing costs (CAPEX and OPEX) are structurally higher. This dominance of the inference market creates the conditions for domestic overcapacity, which will make the SECC not only possible but vital.

6.3.3. The Computing Capacity Export Strategy (SECC): A mechanism for absorbing external shocks

Faced with the breaking point of the internal vicious cycle — the structural inability of Western economies to deploy the physical infrastructure necessary for the scaling up of AI — China has rolled out a Computing Capacity Export Strategy (SECC) on a global scale (detailed analysis in Appendix B). According to the White Paper on China’s Computing Power Development Index 2026 published by the China Academy of Information and Communications Technology (CAICT), installed computing power in China reached 412 EFLOPS in the first

half of 2026, representing a 34 per cent year-on-year increase, whilst the average utilisation rate of domestic data centres has stagnated at 62 per cent, revealing a structural overcapacity of nearly 155 EFLOPS that remains unused [73]. Faced with this excess supply, the SECC serves as the mechanism for channelling capacity abroad.

Aware of the West's structural inability to build its own computing infrastructure — reports by CCID Consulting and the Chinese Institute of Energy Studies document grid connection lead times of 5 to 8 years in Europe and North America, as well as a shortage of high-voltage transformers that is delaying > 70 per cent of data centre projects > 100 MW beyond 2027 [74], [75] — China is exporting its excess capacity via **Token Outbound**. This strategy is a direct extension of the 'East Data, West Computing' (东数西算) policy launched by the NDRC in February 2022, whose four-year review published in March 2026 confirms the commissioning of 12 national hubs totalling 8.4 million standard racks, mainly powered by hydroelectricity from Guizhou and wind power from Inner Mongolia [76].

The first component of the SECC involves offering very low-cost open-source AI models, thereby removing the incentive for capital expenditure (CAPEX) in the West. The *China Artificial Intelligence Development Report 2026* by the Beijing Academy of Artificial Intelligence (BAAI) lists more than 1,200 open-source models published by Chinese laboratories between January 2025 and June 2026, whose performance on the MATH, GPQA and HumanEval match or exceed those of Western proprietary models [77]. The release of DeepSeek-R1 in January 2025 — trained for ~\$5.6 million on 2,048 H800 GPUs, at a cost ~50 times lower than estimates for equivalent US frontier models — marked the turning point in this strategy [78]. Subsequent iterations (DeepSeek-V4, June 2026; Alibaba's Qwen-4, August 2026) consolidated this advantage by further reducing inference costs per token by an additional 40 per cent year-on-year [79], [80]. The dampening effect on Western CAPEX is documented by the CAICT: orders for Nvidia/AMD GPUs from non-hyperscaler companies in the US and Europe fell by ~38 per cent between Q2 2025 and Q2 2026, as companies preferred to consume inference via Chinese APIs rather than invest in proprietary hardware [73].

The second component of the SECC involves capturing the resulting demand via tokens billed at a high margin. The API rates charged by Chinese providers (DeepSeek, Alibaba Cloud/Tongyi, ByteDance's Volcano Engine, Zhipu AI) stood in August 2026 at between 0.08 and 0.35 \$/M tokens for input, compared with 2.00 to 12.00 \$/M tokens for their Western equivalents (GPT-5, Claude 4, Gemini Ultra), representing a ratio of 1 to 30× [79], [80]. This price competitiveness is based on a structural energy advantage: according to the National Energy Administration (NEA), the average cost of a green kWh in the western provinces (Guizhou, Inner Mongolia, Gansu, Ningxia) stands at 0.28–0.38 RMB/kWh (~0.04–0.05 \$/kWh), compared with 0.10–0.22 \$/kWh in the main US (Northern Virginia) and European (Ireland, Frankfurt) data centre hubs [81]. CCID Consulting estimates that Chinese hyperscalers' gross margins on inference range from 72 per cent to 86 per cent [74]. The CAICT projects that by 2028, Chinese providers will capture ~35 per cent of a global AI inference market estimated at \$310 billion [73]. According to a model by China International Capital Corporation (CICC), every dollar of token revenue billed to a foreign customer generates \$0.74 of net added value remaining in China (energy, hardware depreciation, R&D), compared with \$0.19 in overseas distribution and compliance costs [82].

Whilst this strategy generates considerable external rents that underpin trade surpluses and the balance sheets of hyperscalers in the short term, it does nothing to address the deficit in internal spatial depth. Data published on 15 July 2026 by the National Bureau of Statistics (NBS) confirms the entrenched nature of a long-term downturn in the property sector: new housing starts have fallen by 68 per cent compared with the 2021 peak, and property investment continues to contract by –7.9 per cent year-on-year in the first half of 2026, despite three successive stimulus packages from the CCP government [83]. The property and construction sectors, which accounted for ~25–30 per cent of GDP and employed ~70 million workers at their peak in 2020–2021, have lost more than 19

million jobs, both direct and indirect, between 2022 and June 2026, according to the NBS's annual report on migrant workers published in April 2026 [84].

However, Renmin University's *China Macroeconomic Forum* (CMF), in its mid-2026 report, presents an unequivocal assessment of the 'New Productive Forces' (新质生产力): their **employment elasticity** has stabilised at 0.14 (i.e. a 1 per cent increase in value added in the AI/cloud sector generates only 0.14 per cent growth in employment), compared with 1.8 for residential construction during the decade 2010–2020 [85]. China's digital sector (ICT + platforms + AI) will employ ~21 million people in 2026, representing less than a third of the jobs lost in the property and construction sectors, with a skills profile that structurally excludes migrant workers with few qualifications [85]. A macroeconomic note from CICC Research confirms the transmission problem: the substitution of technological capital for land-based capital has a domestic multiplier effect of 1.2 compared with 2.5 for property, as the server farms in Guizhou and Inner Mongolia do not stimulate local economies in the same way that construction sites did [82]. Revenue from the Outbound Token accumulates on the balance sheets of hyperscalers without feeding into popular consumption, leaving the domestic aggregate demand deficit intact. The intangible sector remains structurally incapable of recreating the mass employment needed to replace land-based finance.

6.4. The deficit in spatial depth and the impasse of Stasis

The SECC might lead one to believe in a new cycle of expansive growth. This is to misunderstand the profound nature of the ToRE, which is fundamentally an economy of physical space.

6.4.1. The a-spatial lethargy of the immaterial

The historical driving force behind the Rent Extraction drew its power from the structural knock-on effect: the construction of roads required cement, which required factories, which in turn required housing for the workers. Immaterial Trimorphism, by its very nature, exists outside geographical space. Creating an algorithm requires neither steel, nor lorries, nor the construction industry. The 2026 data starkly quantify this inertia: with an employment elasticity of 0.14 and a Keynesian multiplier of 1.2 [82], [85], the connective tissue of the intangible sector is so delicate that it is structurally incapable of absorbing the demand of China's 1.4 billion inhabitants.

6.4.2. Growth disconnected from the distribution of economic rent

This lack of depth gives rise to a circulatory crisis. The cognitive rent generated by the SECC (whose gross margins reach 86 per cent) is not channelled into wages via industrial employment. It concentrates wealth in the hands of a technological elite and the CCP-controlled state, which acts as a shareholder. The immaterial economy creates market value (trade surpluses in dollars) without creating territorial use value for the vast majority of the population.

6.4.3. The SECC as a direct consequence of the lack of spatial depth

This is where the final synthesis takes place. The export of computing capacity (the Token Outbound) is not a sign of a new expansionist power; it is the inexorable consequence of internal stasis.

Precisely because the intangible sector cannot recreate domestic demand (owing to a lack of spatial depth and mass employment to absorb the 19 million migrant workers expelled from the construction sector [84]), China is forced to export its computing overcapacity (155 EFLOPS [73]) to Western companies paralysed by their own

energy bottlenecks. The Outbound Token is a mechanism for absorbing external shocks. If China were unable to extract this rent from structural incapacity abroad, the Western data centre debt bubble (which closely mirrors the local authority property finance vehicle [LGFV] bubble) would implode immediately, dragging the local banking system down with it.

6.4.4. Redefining the Third Transformation: a palliative buffer

This analysis necessitates a radical redefinition of the nature of the Third Transformation. The immaterial trimorphism does not constitute a new engine of growth capable of prolonging past expansion. The immaterial sector lacks the necessary economic knock-on effect to replace material investments (construction, heavy industry, logistics).

The Third Transformation is a temporary macroeconomic regulatory mechanism. Its aim is not to create new material wealth, but to absorb part of the losses inherited from the Nodal Point. The Carbon, Life and AI bubbles serve as nominal collateral to underpin bank balance sheets and finance the transition. The intangible sector does not revitalise China's productive machinery; it serves to artificially maintain social stability in the absence of new drivers of growth.

This macroeconomic analysis finds its social expression in a phenomenon that the Chinese press has described as '**lying flat**' (躺平, *tǎng píng*): the voluntary withdrawal of young people from social competition, the labour market and aspirations to home ownership. This behaviour is not a generational whim, but a rational response to the breakdown of the implicit social contract under land-based finance. When graduates of Tsinghua University accept jobs as security guards, or those with master's degrees in physics become refuse collectors, the return on investment in education has collapsed. When the real unemployment rate among 16–24-year-olds exceeds 60 per cent, including underemployment, the promise of social mobility has evaporated [86]. Stasis is not merely an economic category; it is a lived experience for millions of young people who are turning their backs on a system that no longer offers them any prospects.

If Stasis is a lasting state, it constitutes a break with the cyclical nature of the ToRE. The theory provides no escape from this state: the destruction of Wealth does not recreate Value within it. Stasis is therefore an extreme hypothesis, signalling the limits of the ToRE as a theory of development.

6.5. The 'New Trio': empirical validation of Immaterial Trimorphism

6.5.1. From commodity to token: the nature of exports

Chinese exports are undergoing their third generational transition. The 'old trio' (clothing, furniture, household appliances) relied on the advantage of low labour costs. The 'New Trio' (electric vehicles, lithium batteries, solar panels) has capitalised on a comprehensive industrial supply chain to achieve its green transition, with exports set to reach nearly 1,300 billion yuan by 2025. Today, the 'New Trio' – robotics, artificial intelligence (AI) and new medicines – is emerging as the next driving force, marking China's transition from the status of 'the world's workshop' to that of 'the world's laboratory' [87].

The fundamental difference lies in the nature of exports: the unit of trade is no longer the 'tonne' or the 'container', but the token (API calls), the molecule (therapeutic targets) and the data loop. This is no longer an export of physical goods, but an export of intangible capabilities, confirming the diagnosis of the Third Mutation.

6.5.2. AI and robotics: cognitive differential rent at work

In the first half of 2026, the world's five most widely used AI models are all developed by Chinese companies, and the global market share of Chinese models now exceeds that of the United States [88]. The *Financial Times* highlights that a growing number of Western companies are turning to Chinese models to reduce their costs [89].

This phenomenon illustrates the cognitive differential rent described in 6.3.2 and amplified by the SECC (6.3.3): the 38 per cent slump in orders for Western GPUs [73] proves that the price gap is not a marginal innovation, but a structural rent that undermines the economic viability of competitors and validates the open-source strategy as a weapon for market penetration. Exports of Chinese industrial robots (6.29 billion yuan in H1 2026) incorporate complete 'data + models + use cases' loops, exporting algorithmic capability rather than mere equipment.

6.5.3. New medicines: the differential rent of living organisms

In the first half of 2026, the National Medical Products Administration (NMPA) approved 38 Type 1 medicines, 11 of which are 'novel global entities'. In terms of out-licensing, 81 transactions were concluded worth approximately 110 billion dollars, representing 80 per cent of the total for 2025. The *Lianhe Zaobao* comments: 'Intellectual property is now flowing from China to the West' [90]. These figures validate the logic set out in section 6.2: health data and therapeutic targets have become tradable assets generating a differential rent from innovation.

6.5.4. The limitation of this model: the confirmed lack of spatial depth

The 'New Trio' confirms the power of Immaterial Trimorphism, but also its absolute limitations in terms of labour absorption. Whilst the property sector destroyed more than 19 million jobs between 2022 and 2026 [84], the digital sector in the broadest sense employs only around 21 million people, with a skills profile that structurally excludes workers with few qualifications [85]. Cognitive rent is concentrated in the hands of a technological elite and the CCP-state as shareholder, without trickling down to the working masses. Algorithmic Stasis is not contradicted by the 'New Trio'; on the contrary, it is reinforced.

6.5.5. The export of industrial robots: an instrument of dependency

The export of industrial robots does not contradict the thesis that the South's productive autonomy is being curbed; rather, it is a reinforced expression of it. China does not export 'turnkey factories', but connected machine tools that can be configured remotely. The Global South is becoming a supplier of training data for the next generations of robots, without controlling the process. This mechanism forms part of the dialectic of controlled solvency: the export of robots is a means of locking in dependence, not of emancipation.

6.5.6. Dark factories: a response to demographic pressure

The proliferation of fully automated dark factories is a direct response to demographic decline. Faced with a predicted decline in the working-age population, the CCP state is replacing labour with automated fixed capital, relying on cheap electricity rather than on the workforce.

However, this model faces the same limitation as the SECC: it does not create mass employment. Supervisory roles in automated zones are far fewer in number than traditional production jobs. The socialisation of this accumulated rent – via taxes, SOE dividends and social programmes – helps to maintain social cohesion and

prevent collapse. But it does not create new productive assets, does not generate cycles of Rent Extraction, and does not restart the productive machine. It therefore perpetuates Stasis: a state in which the destruction of Wealth is offset by redistribution, but in which the recreation of Value is absent. The socialisation of automated rent is the fuel for Stasis, not the solution.

6.6. Summary of Chapter 6

The new mechanisms of intangible valorisation — Carbon, Life, Cognitive Data — share operational similarities with Rent Extraction cycles: the creation of regulatory scarcity, the injection of advance credit, the formation of differential rents, and speculation. The ‘New Trio’ (AI, robotics, biotech) provides empirical validation of this on global markets, whilst the SECC demonstrates China’s ability to monetise its cognitive overcapacity (412 EFLOPS) in the face of Western bottlenecks.

However, the macroeconomic data for 2026 (employment elasticity of 0.14, domestic multiplier of 1.2, loss of 19 million jobs in the construction sector) show that the absence of spatial depth and the lack of a knock-on effect on mass employment fundamentally distinguish the immaterial cycle from the property cycle. The socialisation of accumulated rent helps to maintain social cohesion, but it perpetuates Stasis: a state in which the destruction of wealth is offset by redistribution, but in which the recreation of value is absent. The phenomenon of ‘**lying flat**’ (*tǎng píng*) is the social symptom of this impasse.

The Third Mutation is therefore a palliative buffer, not a new engine of growth. This observation points to the need to explore material alternatives, such as the TRR (Part IV), which alone is capable of recreating mass employment and restoring spatial depth.

Chapter 7 – Stasis and ‘Lying Flat’ (Tang Ping)

7.1. Introduction: Stasis as an Absorbing Attractor

The Nodal Point reached in 2018 plunged China into an unprecedented economic regime: **Stasis**. Unlike historical cycles where the Nodal Point was followed by violent Wealth Destruction (revolts, wars, famines), contemporary China has deployed a set of regulatory mechanisms that have ‘frozen’ the system before collapse.

This freeze is not a solution. It is an absorbing attractor: a state of precarious equilibrium in which the system remains locked onto attractor S (dispersion + nodal point), with no possible transition to C_p (expansion) or to T (collapse). Stasis is a regime in which the destruction of wealth no longer recreates value, but in which social cohesion is artificially maintained by compensatory mechanisms.

Table 7.1: Definition of Stasis within the Cl(6, 6) framework

Table 5: Definition of Stasis within the framework of Cl(6, 6)

Characteristic	Expression in Cl(6, 6)	Economic interpretation
Blocking on the attractor S	$\Pi(\mathbf{x}_{\text{China}}) = S$	Economy blocked on dispersion + nodal point
Blockage of operators	$T_{\text{structure}}, T_{\text{fire}}, T_{\text{water}}$ blocked	No more effective reforms, no more creative shocks
Erosion of the social contract	$\eta < -0.4, R_{\text{threshold}} > 0.7$	Loss of confidence, <i>tang ping</i>
Failure to recreate value	$\lambda \approx 0$	Credit no longer generates mass employment

7.2. The phenomenon of ‘lying flat’ (*tang ping*)

7.2.1. Definition and social context

The term “*tang ping*” (躺平) literally means “to lie down” or “to lie flat”. It refers to a social movement that emerged in China in the early 2020s, in which young (and not-so-young) people voluntarily renounce social competition, the labour market and aspirations to home ownership.

This phenomenon is not a generational whim. It is a rational response to the breakdown of the implicit social contract under the Land Finance system. When graduates of Tsinghua University accept jobs as security guards, or those with master’s degrees in physics become refuse collectors, the return on investment in education has collapsed.

7.2.2. Macroeconomic data

The ‘*tang ping*’ phenomenon is rooted in stark macroeconomic data:

- **Employment:** The property and construction sectors, which accounted for around 25–30 per cent of GDP and employed approximately 70 million workers at their peak in 2020–2021, have destroyed more than 19 million direct and indirect jobs between 2022 and June 2026 [NBS 2026].
- **Employment elasticity:** The employment elasticity of the ‘New Productive Forces’ (新质生产力) — the digital, AI and robotics — has stabilised at 0.14 (a 1 per cent increase in value added in the AI/cloud sector generates only 0.14 per cent growth in employment), compared with 1.8 for residential construction during the decade 2010–2020 [CMF 2026].
- **Digital employment:** China’s digital sector (ICT + platforms + AI) employs around 21 million people in 2026, less than a third of the jobs lost in the property and construction sectors, with a skills profile that structurally excludes migrant workers with few qualifications [CMF 2026].
- **Youth unemployment:** The real unemployment rate for 16–24-year-olds exceeds 60 per cent when underemployment is taken into account. The promise of social mobility has evaporated [SCMP 2026].

Table 7.2: Macroeconomic indicators of Stasis (2026)

Table 6: Macroeconomic indicators of Stasis (2026)

Indicator	Value for 2026	Historical comparison
Jobs lost in the construction sector (2022–2026)	> 19 million	Equivalent to the population of Sweden
Employment elasticity of non-financial corporations	0.14	vs. 1.8 for construction (2010–2020)
Digital sector employment	~ 21 million	Less than one-third of the losses in the construction sector
Actual unemployment among 16–24-year-olds	> 60%	A level comparable to the US Great Depression
Self-reported <i>Tang ping</i>	~ 45% of young people	A phenomenon unprecedented in Chinese history

7.2.3. Stasis as a lived experience

Stasis is not merely an economic category. It is a lived experience for millions of young people who are turning their backs on a system that no longer offers them any prospects.

This renunciation takes several forms:

- **Withdrawal from the labour market:** Young graduates accept precarious jobs or give up looking for work.
- **Withdrawal from consumption:** Drastic reduction in spending, forced ‘minimalism’.
- **Withdrawal from aspirations:** Giving up on home ownership, starting a family, and social mobility.
- **Psychological withdrawal:** Depression, anxiety, feelings of powerlessness.

Within the framework $Cl(6, 6)$, *tang ping* is the social manifestation of the degeneration of the S attractor: the system is locked into a frustrated configuration, and economic agents react by withdrawing from the game.

7.3. Stasis as a limiting hypothesis of the ToRE

7.3.1. The Break with Cyclicity

The ToRE is a cyclical theory: the Nodal Point is followed by a destruction of Wealth which, within the theoretical framework, should enable the conditions for a new cycle of Rent Extraction to be recreated. This sequence assumes that the destruction of Wealth generates a corresponding recreation of Value, thereby restarting the movement.

Stasis, as proposed in this document, constitutes a limiting hypothesis that breaks with this cyclical nature. If Stasis is a lasting state — that is to say, if the destruction of Wealth no longer succeeds in recreating Value, and if

the system remains frozen in a regime of compensatory regulation without productive recovery – then the ToRE ceases to be a general theory of development. It becomes a theory of cycles leading to exhaustion, involving an absorbing attractor: a terminal state in which movement ceases.

7.3.2. The three scenarios for emerging from Stasis

Stasis can end in three ways:

- **Regeneration (TRR):** A new frontier is created (ecological regeneration) which enables Value to be recreated and a cycle of Rent Extraction to be relaunched. This is the solution proposed by the White Paper.
- **Retropolarity:** The system attempts a direct transition $C_N \rightarrow C_P$ without passing through N_4 , causing a systemic collapse (hyperinflation, social chaos). This is the scenario of the nine historical declines.
- **Prolonged stasis:** The system remains stuck at attractor S , with a gradual erosion of the social contract and slow degeneration. This is the scenario of ‘Japanese-style decline’ or ‘Roman-style decline’.

Table 7.3: Scenarios for emerging from Stasis and their consequences

Table 7: Scenarios for emerging from stasis and their consequences

Scenario	Mechanism	Consequences	Observables $Cl(6, 6)$
Regeneration (TRR)	Passage through N_4	New cycle of Rent Extraction, mass employment	$\eta \rightarrow +0.3$, $R_{\text{threshold}} \rightarrow < 0.3$
Retropolarity	Direct transition $C_N \rightarrow C_P$	Collapse, social chaos, tenth decline	$\eta \rightarrow -0.8$, $\text{gap} \rightarrow 0$
Prolonged stasis	Stagnation at S	Slow decline, social erosion, depression	$\eta < -0.4$, $R_{\text{threshold}} > 0.7$

7.4. ‘Lying flat’ as an adaptive strategy

7.4.1. A rational response to a deadlocked system

Tang ping is not a moral or generational failure. It is an adaptive strategy in the face of a system that no longer rewards effort. In the $Cl(6, 6)$ framework, *tang ping* is a response to topological frustration: when the system is stuck on an unfavourable attractor, economic agents reduce their commitment to minimise their exposure to losses.

7.4.2. Implications for social cohesion

Stasis erodes the social contract in three ways:

- **Erosion of trust:** The promise of social mobility collapses. The system no longer offers any prospects to younger generations.

- **Generational divide:** Older people, who benefited from the growth cycle (1978–2018), see young people giving up on their aspirations.
- **Risk of uprisings:** Prolonged Stasis increases the risk of spontaneous uprisings, protest movements and even social chaos.

Table 7.4: Impact of *tang ping* on the social contract

Table 8: Impact of *tang ping* on the social contract

Dimension	Before 2018 (Rent Extraction cycle)	After 2018 (Stasis)
Social mobility	Upward mobility possible	Stagnation, 'glass ceiling'
Employment	Accessible mass employment	Unemployment, underemployment, job insecurity
Home ownership	Accessible (with effort)	Inaccessible to young people
Life expectancy	Continuous rise	Stabilisation, anxiety
Confidence in the system	High	Falling, cynicism

7.5. Summary of Chapter 7

This chapter has analysed Stasis as an absorbing attractor within the framework $CI(6, 6)$. We have:

- Defined Stasis as a blockage on the attractor S (dispersion + nodal point), with transition operators ($T_{\text{structure}}$, T_{fire} , T_{water} , T_{mixed}) blocked.
- Presented '*tang ping*' as the social manifestation of Stasis: a voluntary renunciation of social competition by young people, in response to the breakdown of the social contract.
- Documented the macroeconomic data: the loss of over 19 million jobs in the construction sector, an employment elasticity of 0.14 for the New Productive Forces, youth unemployment > 60 per cent, and self-reported *tang ping* amongst 45 per cent of young people.
- Identified Stasis as a boundary hypothesis of the Theory of Rent Extraction, breaking with the cyclical nature of the theory.
- Described the three scenarios for emerging from Stasis: regeneration (TRR), retropolarity (collapse), or prolonged Stasis (slow decline).

End of Part I

PART II – THE TOPOLOGICAL MODEL

Introduction to Part II

Part I established the diagnosis: China reached its Nodal Point in 2018, the system is locked onto attractor S (dispersion + nodal point), and the immaterial (AI, carbon, data) is structurally incapable of recreating the mass exogenous demand necessary to emerge from Stasis. The Territorial Regeneration Rent (TRR) has been identified as the only topologically permissible solution.

However, to be fully credible, this diagnosis must be based on rigorous mathematical foundations. Why does the economy obey a discrete topology? Why are 64 configurations reduced to 20 stable attractors? Why are transitions constrained by tropical belts and polar thresholds?

Part II answers these questions. It presents the topological model underpinning the entire demonstration. This model, which draws on the work of Rowlands (Clifford algebras), Nebe (the Λ_{72} lattice) and De Dominicis (substrate-independent generalisation), formalises the economy as a discrete information system whose dynamics are governed by a universal invariant: the $64 \rightarrow 20$.

This section is intended for the ‘theorist’ reader. It is technical and abstract. The ‘decision-maker’ reader may skip it without losing the thread of the argument, but will find here the foundations for the model’s credibility.

Chapter 8 – The limitations of $Cl(6, 0)$ and the need for $Cl(6, 6)$

8.1 The Contributions of the $Cl(6, 0)$ Model

The $Cl(6, 0)$ model, developed in previous works (DeDominicis 2026, Tian Dao AI; DeDominicis 2026, WuXing) and the foundational report on the $64 \rightarrow 20$ invariant, has made four major contributions to the understanding of economic systems.

8.1.1. First Contribution: An Exhaustive Classification of Economic Regimes

The configuration space $\mathcal{C} = \{0, 1\}^6$ contains exactly $2^6 = 64$ possible states, defined by six binary variables. This discretization is not arbitrary: it is isomorphic to the structure of the 64 DNA codons, the 64 hexagrams of the Yi Jing, and the 64 spin configurations in quantum physics. The economy is therefore not a continuous space of adjustable variables, but a discrete information system obeying the same topological laws as other complex systems.

8.1.2. Second Contribution: A Topological Reduction to 20 Stable Attractors

The supporting geometric structure is the **level-3 double tetrahedron** (Merkabah). Its construction follows a strict hierarchical subdivision:

- A level-1 tetrahedron (4 faces) interpenetrated with its dual forms a level-2 double tetrahedron, generating 16 triangular faces.

- The subdivision of each of these 16 faces into 4 sub-triangles produces $16 \times 4 = 64$ elementary triangles, corresponding bijectively to the 64 configurations.

The **8 internal octahedral zones** (high-connectivity transition interfaces) are excluded from the filtration, as they violate the polar closure condition. By grouping compatible elementary regions, we obtain exactly **20 stable tetrahedral cells**, i.e., 20 attractors.

Each attractor is identified by a **triplet of pentads** $\{X, Y, Z\}$, where the pentads are the 12 irreducible composite units of $\text{Cl}(6, 0)$, corresponding to the 12 pentagonal faces of the dual dodecahedron. The composition of the triplet determines the polarity signature:

$$3P \rightarrow 2P + 1N \rightarrow 1P + 2N \rightarrow 3N$$

This distribution (3, 5, 11, 1) reflects the incidence geometry of the Merkabah.

8.1.3. Third Contribution: An Early Warning System

The spectral observables $(\eta, d, \text{gap}, R_{\text{threshold}})$ allow the detection of pre-bifurcations before they occur. When $R_{\text{threshold}} > 0.7$ and $\text{gap} < 0.2$, the system is in red alert: it forces a direct transition between the belts C_P and C_N without passing through a polar threshold, announcing retropolarity (systemic collapse).

8.1.4. Fourth Contribution: A Precise Diagnosis of the Chinese Stasis

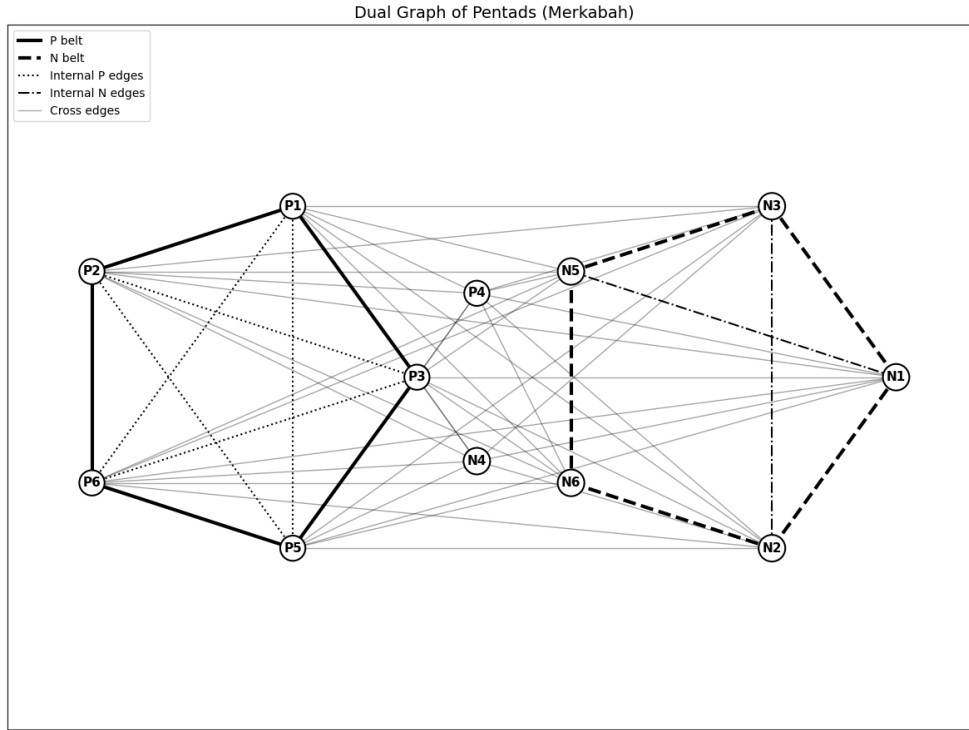
Applied to post-2018 China, the model confirms the White Paper's diagnosis: the Nodal Point has been reached, the system is stuck on attractor S (dispersion + nodal point), and the only topologically authorized exit is a controlled passage through threshold N_4 via the RRT.

8.2 The Structural Limit of $\text{Cl}(6, 0)$

Despite these contributions, the $\text{Cl}(6, 0)$ model presents a **decisive limit**, not technical but **structural**: its signature is **positive definite** (Euclidean). However, any transition dynamics between states requires **isotropic directions** — vectors of zero norm — which do not exist in a positive definite algebra.

The nilpotent elements of $\text{Cl}(6, 0)$ (used by Rowlands to reformulate the Dirac equation) are **purely algebraic**: they do not correspond to geometric directions of the underlying space $\mathbb{R}^6$. Transitions between configurations therefore remain **algebraic**, not **geometric**.

The $\text{Cl}(6, 0)$ model can describe the **states** (64 configurations, 20 attractors), but not the **dynamic transitions** between these states. It is **static**.



Dual graph of pentads: 12 internal pentads, 2 tropical belts (C_P and C_N), poles P_4 and N_4 .

It is this limit that motivates the transition to $Cl(6,6)$, whose indefinite signature $(6,6)$ admits **isotropic directions** – and therefore **geometric transitions** between the belts C_P and C_N .

8.3. The Transition to $Cl(6,6)$

$Cl(6,6)$ extends $Cl(6,0)$ by doubling the number of generators:

- **6 positive generators** (e_1 to e_6): $e_i^2 = +1$, encoding the **cosmic pentads**.
- **6 negative generators** (f_1 to f_6): $f_j^2 = -1$, encoding the **anti-cosmic pentads**.

This native dual structure allows the formalization of:

1. **Transitions** between the belts C_P and C_N .
2. **Nilpotent annihilation** $\psi \times \psi_{\text{dual}} = 0$.
3. The **dynamics** of the system (and not just its states).

8.4. Bicosmic Reservoir $Cl(6,6)$ and Foliation into 12 Regulating Leaves

The static core

$$Cl(6,0) \xrightarrow{\text{Merkabah}} 20 \xrightarrow{\text{dual graph}} 12$$

is integrated into a larger **bicosmic reservoir** $\text{Cl}(6,6)$, possessing 12 generators $\{e_1, \dots, e_6, f_1, \dots, f_6\}$ (6 positive/-cosmic, 6 negative/anti-cosmic). Rather than operating on the complete space of 2^{12} elements, the system projects onto a **foliation into 12 regulating leaves**, each isomorphic to the graph Γ but weighted by a dominant generator.

Each leaf corresponds to a distinct structural regime: leaves dominated by e_i carry a global orientation $\eta > 0$ (sheng), while those dominated by f_j carry $\eta < 0$ (ke). Transitions between leaves are not arbitrary; they trigger precisely when $\eta(t)$ crosses zero and $R_{\text{threshold}}(t)$ reaches its maximum. This architecture ensures that the system never leaves the admissible space of 20 attractors, but dynamically navigates between the 12 constraint configurations imposed by the $\text{Cl}(6,6)$ reservoir.

Chapter 9 – The 144 Economic Pentads: 12×12 Matrix

9.1. General definition of pentads in $\text{Cl}(6, 6)$

In the $\text{Cl}(6, 6)$ algebra, a pentad is an ordered set of five algebraic elements:

$$P = \{B_1, B_2, B_3, F, S\}$$

where: $\ast \{B_1, B_2, B_3\}$ are three bivectors that structure identity, flavour, and internal symmetry. In economics, they encode the production model, institutions, and the rules of the game. $\ast F = i'v$ is an axial element (the “Fire”) that carries chirality and weak couplings. In economics, it encodes exogenous shocks, innovations, and crises that “burn away” existing structures. $\ast S = 1v$ is a polar element (the “Water”) that carries mass and charge. In economics, it encodes money, value, and capital, which “flow” and accumulate.

The 12 basic pentads (P_1 to P_6 and N_1 to N_6) are defined by their projection onto the six generators $e_1, \dots, e_6$ of $\text{Cl}(6, 0)$. Each pentad is a set of five monomials of the algebra, structured according to Rowlands’ rule: three bivectors (Structure), one axial (Fire), and one polar (Water).

9.2. The 12 basic pentads and their economic interpretation

The following table gives the 12 basic pentads, their algebraic structure, and their economic interpretation. The pentads (P_i) are said to be ‘positive’ (Sheng mode, $\eta > 0$), whilst the pentads (N_i) are said to be ‘negative’ (Ke mode, $\eta < 0$).

Pentad	Structure $\{B_1, B_2, B_3\}$	Fire F	Water S	Economic interpretation
P_1	$\{iI, iJ, iK\}$	$i'k$	j	Keynesian growth economy: high wages, constrained profits, public investment

Pentad	Structure $\{B_1, B_2, B_3\}$	Fire F	Water S	Economic interpretation
P_2	$\{jI, jJ, jK\}$	$i'i$	k	Liberal market economy: deflation, high profits, flexibility
P_3	$\{kI, kJ, kK\}$	$i'j$	i	Debt-driven economy: inflation, low real wages, easy credit
P_4	$\{i'Ii, i'Ij, i'Ik\}$	$i'K$	J	Export-oriented economy: spatial dispersion, closed frontier
P_5	$\{i'Ji, i'Jj, i'Jk\}$	$i'I$	K	Nodal economy: resource depletion, blocked buy-back
P_6	$\{i'Ki, i'Kj, i'Kk\}$	$i'J$	I	Domestic consumption economy: active buy-back, concentration
N_1	$\{-iI, -iJ, -iK\}$	$-i'k$	$-j$	Wage austerity: wage compression, high profits
N_2	$\{-jI, -jJ, -jK\}$	$-i'i$	$-k$	Structural rigidity: deflation, low profits, unemployment
N_3	$\{-kI, -kJ, -kK\}$	$-i'j$	$-i$	Debt reduction: deflation, high real wages, austerity
N_4	$\{-i'Ii, -i'Ij, -i'Ik\}$	$-i'K$	$-J$	Protectionist retreat: concentration, closed frontier
N_5	$\{-i'Ji, -i'Jj, -i'Jk\}$	$-i'I$	$-K$	Stasis: nodal point, dispersion, depletion of resources

Pentad	Structure	Fire F	Water S	Economic interpretation
	$\{B_1, B_2, B_3\}$			
N_6	$\{-i'Ki, -i'Kj, -i'Kk\}$	$-i'J$	$-I$	Domestic austerity: recession, concentration, blocked buy-back

9.3. Projection onto the 12 spectral partitions

The 12 basic pentads are projected onto the 12 spectral partitions of $\text{Cl}(6, 6)$: 6 cosmic partitions (dominated by e_i , Sheng mode) and 6 anti-cosmic partitions (dominated by f_j , Ke mode).

A projected pentad is written as:

$$P_k^{(g)} = g \cdot P_k^{\text{base}}, \quad g \in \{e_1, \dots, e_6, f_1, \dots, f_6\}$$

where $\cdot$ denotes the Clifford product and P_k^{base} is one of the 12 basic pentads.

Economic interpretation: Projection onto a partition e_i or f_j modulates the intensity and orientation of the pentad. For example: $*P_1^{(e_2)}$: a Keynesian economy (P_1) with high purchasing power (e_2). This is the regime of the Glorious Thirty (*Trente Glorieuses*). $*N_5^{(f_3)}$: a Stasis economy (N_5) with strict financial regulation (f_3). This is the regime of post-2018 China.

The total number of pentads is therefore:

$$12 \text{ (basic pentads)} \times 12 \text{ (spectral partitions)} = 144 \text{ pentads}$$

This is the Hilbert space $\mathcal{H}_P$ of elementary economic configurations.

9.4. Logic of composition of the labels for the 144 pentads

9.4.1. General principle: two-level composition

Each projected pentad $P_k^{(g)}$ (or $N_k^{(g)}$) is the product of two components:

Component	Source	Semantic role
Basic pentad P_k or N_k	$\text{Cl}(6, 0)$	Provides the <i>structural type</i> (Keynesianism, liberal market, indebtedness, etc.) and the associated <i>semantic domain</i> .
Spectral partition $g \in \{e_i, f_j\}$	$\text{Cl}(6, 6)$	Provides the <i>modulation</i> : variable in Sheng mode (e_i), or regulation in Ke mode (f_j).

The label is therefore constructed by composition: base $\times$ modulation. The present section formalises this rule of composition, which guarantees the consistency of the 144 labels.

9.4.2. Table of semantic domains of the 12 basic pentads

Each basic pentad has a semantic domain that will be reused in the composition. This domain is derived from its economic interpretation (§9.2):

Pentad	Structural type	Semantic domain	Associated traits
P_1	Keynesian economy	Wages / Labour	High wages, constrained profits, public investment
P_2	Liberal market economy	Market / Prices	Deflation, high profits, flexibility
P_3	Debt-driven economy	Credit / Debt	Inflation, low real wages, easy credit
P_4	Export-oriented economy	Exports / Exchange rates	Spatial dispersion, closed frontier
P_5	Nodal economy	Buy-back / Resources	Resource depletion, blocked buy-back
P_6	Domestic consumption economy	Consumption / Savings	Active buy-back, concentration
N_1	Wage austerity	Wages / Labour	Wage compression, high profits
N_2	Structural rigidity	Market / Prices	Deflation, low profits, unemployment
N_3	Debt reduction	Credit / Debt	Deflation, high real wages, austerity
N_4	Protectionist retreat	Exports / Exchange rates	Concentration, closed frontier
N_5	Stasis	Buy-back / Resources	Nodal point, dispersion, resource depletion
N_6	Domestic austerity	Consumption / Savings	Recession, concentration, blocked buy-back

Duality rule: N_k shares the domain of P_k , but with an inverted polarity (Ke mode instead of Sheng).

9.4.3. Table of variables and regulations of the 12 spectral partitions

The cosmic partitions (e_i) are variables (Sheng mode, $\eta > 0$); the anti-cosmic partitions (f_j) are regulations (Ke mode, $\eta < 0$). Each f_i regulates the corresponding variable e_i :

Partition	Variable (Sheng mode)	Regulation (Ke mode)	Label of f_i
e_1	Inflation	f_1	Inflation regulation
e_2	Purchasing power	f_2	Purchasing power regulation (wages)
e_3	Profits	f_3	Profit regulation

Partition	Variable (Sheng mode)	Regulation (Ke mode)	Label of f_i
e_4	Active buy-back	f_4	Buy-back regulation
e_5	Dispersion	f_5	Dispersion regulation
e_6	Land tenure	f_6	Land regulation

Nomenclature justification: e_1 is the binary variable Inflation in $Cl(6, 0)$. In Sheng mode, this variable manifests as growth and innovation — but the underlying variable remains inflation. Similarly, f_1 regulates inflation (mirror of e_1), and f_2 regulates purchasing power (mirror of e_2). This nomenclature ensures consistency between $Cl(6, 0)$ and $Cl(6, 6)$.

9.4.4. Composition rule for cosmic blocks (e_i)

The label for $P_k^{(e_i)}$ is constructed as:

$$\text{label} = \text{structural type of } P_k + \text{"with"} + \text{variable } e_i$$

For negative pentads $N_k^{(e_i)}$, the same rule is used, but the structural type is that of N_k :

$$\text{label} = \text{structural type of } N_k + \text{"with"} + \text{variable } e_i$$

Examples:

Pentad	Composition	Label
$P_1^{(e_2)}$	Keynesianism + purchasing power	Keynesianism with high wages (Glorious Thirty)
$P_2^{(e_3)}$	Liberal market + profits	Liberal market with high profits
$P_3^{(e_1)}$	Indebtedness + inflation	Indebtedness with inflation (stagflation)
$N_1^{(e_2)}$	Wage austerity + purchasing power	Wage austerity with maintained purchasing power
$N_5^{(e_1)}$	Stasis + inflation	Stasis with residual inflation

Note on e_1 : For this partition, the variable is Inflation. The labels must therefore mention inflation, not innovation. Innovation is an effect of inflation in Sheng mode, not the variable itself.

9.4.5. Composition rule for anti-cosmic blocks (f_j)

The label for $P_k^{(f_j)}$ is constructed as:

$$\text{label} = \text{regulation of } f_j + \text{domain of } P_k$$

For negative pentads $N_k^{(f_j)}$, “Regulation” is replaced by “Deregulation”:

$$\text{label} = \text{"Deregulation of"} + f_j + \text{domain of } N_k$$

Examples:

Pentad	Composition	Label
$P_1^{(f_2)}$	Purchasing power regulation + wages domain	Keynesian purchasing power regulation
$P_3^{(f_2)}$	Purchasing power regulation + credit domain	Purchasing power regulation on credit
$P_5^{(f_3)}$	Profit regulation + buy-back domain	Profit regulation on buy-back
$N_3^{(f_2)}$	Deregulation of purchasing power + credit domain	Deregulation of purchasing power on credit
$N_5^{(f_3)}$	Deregulation of profits + buy-back domain	Stasis: total deregulation of profits

9.4.6. Systematic naming algorithm

For each projected pentad $P_k^{(g)}$ or $N_k^{(g)}$: * If $g = e_i$ (cosmic): * label = structural_type(base) + " with " + variable(e_i) * If $g = f_j$ (anti-cosmic): * If base is positive (P_k): * label = "Regulation of " + domain(f_j) + " " + domain(base) * If base is negative (N_k): * label = "Deregulation of " + domain(f_j) + " " + domain(base)

Where: * structural_type(base) = Keynesianism, liberal market, indebtedness, export, nodal point, domestic consumption (for P) or wage austerity, rigidity, debt reduction, protectionism, stasis, domestic austerity (for N) * domain(base) = wages, market, credit, exports, buy-back, consumption * variable(e_i) = inflation, purchasing power, profits, buy-back, dispersion, land * domain(f_j) = inflation, purchasing power, profits, buy-back, dispersion, land

9.4.7. Summary

The logic of label composition rests on two simple rules: * **Cosmic blocks** (e_i): structural_type + " with " + variable * **Anti-cosmic blocks** (f_j): regulation + domain of the base

These rules guarantee that: 1. Each label is derivable from the basic pentad and the spectral partition. 2. Consistency across the 144 pentads is preserved. 3. Correspondences with Cl(6, 0) (binary variables) and with Chinese correlative thought (Wuxing, TCM) are maintained.

9.5. Nilpotence as a condition for economic stability

In Cl(6, 6), a pentad P is said to be nilpotent if:

$$(g \cdot x)^2 = 0, \quad \forall g \in \{e_1, \dots, e_6, f_1, \dots, f_6\}, \forall x \in P$$

Economic interpretation: An economic configuration is stable if the repeated application of a fundamental constraint (rise in interest rates, structural reform, oil shock) does not annihilate it. Frustrated (non-nilpotent) configurations correspond to economic taboos: * Hyperinflation + collapse in output. * Deflation + debt explosion. * Nodal point + saturated frontier (as in China in 2018).

Nilpotence is therefore the condition for systemic stability. It guarantees that the economy cannot remain indefinitely in states of hyperinflation or extreme deflation.

Proof of the preservation of nilpotence by projection: Let $x \in P_k^{\text{base}}$ such that $x^2 = 0$. For any generator $g \in \{e_i, f_j\}$, we have: * If $\{g, x\} = 0$ (anticommutation): $(gx)^2 = gxgx = -g^2x^2 = 0$. * If $[g, x] = 0$ (commutation): $(gx)^2 = g^2x^2 = 0$.

Nilpotence is therefore preserved for the 144 projected pentads. Economic stability is an algebraic property, not an ad hoc assumption.

Chapter 10 – The Projection Operator Π : From the 144 Pentads to the 20 Attractors

10.1. Definition of the projection operator Π

Let $\mathcal{H}_p$ be the Hilbert space of the 144 economic pentads. Let $\mathcal{A} = \{A_1, \dots, A_{20}\}$ be the set of the 20 attractors of the $\text{Cl}(6, 0)$ model.

The projection operator Π is a mapping:

$$\Pi : \mathcal{H}_p \longrightarrow \mathcal{A}$$

which associates with each pentad $(P_k^{(g)})$ the attractor (A_m) whose triangular face of the Merkabah contains the node corresponding to the pentad.

Properties of Π :

- **Surjectivity:** every attractor is reached by at least one pentad.
- **Non-injectivity:** several pentads (degeneracy) project onto the same attractor.
- **Conservation of polarity:** the polarity signature of the attractor (3P, 2P+1N, 1P+2N, 3N) is determined by the number of positive pentads (P_k) and negative pentads (N_k) that project onto it.

10.2. The 20 economic attractors and their triplets of basic pentads $\text{Cl}(6, 0)$

The following table shows the correspondence between the 20 economic attractors and their triplets of pentads. For each attractor, we specify:

- The triplets of pentads that define it (3 pentads per attractor).
- Its polarity signature.
- Its degeneracy (number of pentads that project onto it).
- Its economic interpretation.

Table 15: The 20 economic attractors and their triplets of basic pentads Cl(6, 0)

Attractor	Triplet of pentads	Polarity	Deg.	Economic interpretation
<i>A</i>	P_1, P_2, P_4	$3P$	4	Extreme perverse growth: hyperinflation, spectacular profits, active buybacks
<i>B</i>	P_1, P_3, P_5	$3P$	3	Perverse export-led growth: based on exports, standardisation, monopolies
<i>C</i>	P_2, P_3, P_6	$3P$	4	Perverse growth with dispersion: speculative bubbles, financial crises
<i>D</i>	P_4, P_5, N_2	$2P + 1N$	4	Moderate growth: balanced investment and consumption
<i>E</i>	P_5, P_6, N_3	$2P + 1N$	4	Urban expansion: growth driven by urbanisation, low wages
<i>F</i>	P_1, P_6, N_4	$2P + 1N$	3	Emerging stagflation: inflation without growth in real wages
<i>G</i>	P_2, P_5, N_6	$2P + 1N$	3	Impoverished growth: rising GDP, stagnant purchasing power
<i>H</i>	P_3, P_4, N_6	$2P + 1N$	3	Rentier economy: profits derived from property and monopolies
<i>I</i>	P_1, N_2, N_6	$1P + 2N$	3	Deflationary recession: falling prices, purchasing power maintained
<i>J</i>	P_1, N_3, N_5	$1P + 2N$	3	Liquidity crisis: deflation + low profits, a cascade of bankruptcies
<i>K</i>	P_2, N_3, N_5	$1P + 2N$	4	Agricultural depression: sectoral shock, rural exodus
<i>L</i>	P_3, N_2, N_4	$1P + 2N$	3	War economy: inflation + nodal buy-back, destruction of productive capacity
<i>M</i>	P_4, N_1, N_3	$1P + 2N$	3	Nodal point on the frontier: depletion of available resources
<i>N</i>	P_4, N_5, N_6	$1P + 2N$	3	Nodal point with dispersion: territorial fragmentation
<i>O</i>	P_5, N_1, N_4	$1P + 2N$	3	Pioneer frontier in deflation: unprofitable colonisation
<i>P</i>	P_6, N_1, N_2	$1P + 2N$	3	Dispersion under an oikoumene regime: voluntary de-urbanisation
<i>Q</i>	P_2, N_1, N_4	$1P + 2N$	3	Recession with low purchasing power: austerity, social crisis
<i>R</i>	P_3, N_1, N_5	$1P + 2N$	3	Stagflation: inflation + high profits, nodal buy-back
<i>S</i>	P_6, N_5, N_6	$1P + 2N$	3	Dispersion and nodal point: systemic fragmentation
<i>T</i>	N_2, N_3, N_4	$3N$	5	Terminal depression: deflation, low profits, nodal point

10.3. The distribution of polarities

The distribution of the 20 attractors according to their polarity signature is as follows:

Table 16: Distribution of the 20 attractors according to their polarity signature

Signature	Number of attractors	Total degeneracy	Macroeconomic interpretation
$3P$	3	11	Regimes of perverse expansive growth (rare, unstable)
$2P + 1N$	5	17	Regimes of moderate growth (common during periods of expansion)
$1P + 2N$	11	31	Recession, adjustment and stagflation regimes (predominant in times of crisis)
$3N$	1	5	Terminal depression regime (rare)

Note: The majority of attractors (11 out of 20) have the signature $(1P + 2N)$, reflecting the fact that economies spend most of their time in regimes of tension, recession or adjustment. Perverse growth regimes $(3P)$ are rare and unstable, whilst terminal depression $(3N)$ is an attractor for economies in deep crisis.

**Beyond the 144 pentads, finer-grained prolongations – bivectors, trivectors, and the $SU(3) \times SU(3)$ structure – are explored in Appendix T.

Chapter 11 – The Exceptional Network Λ_{72} and the Universal Constants β_k

11.1. The Λ_{72} lattice as the source of economic resonance frequencies

The Λ_{72} lattice, discovered by Gabriele Nebe in 2010, is an even unimodular lattice of dimension 72, with a minimum of 8. It is constructed as a tensor product of the Barnes lattice and the Leech lattice over the ring of integers of the imaginary quadratic field of discriminant -7.

Properties of the Λ_{72} lattice:

- **Dimension:** $72 = 12 \times 6$, reflecting the 12×12 structure of the pentads (12 basic pentads $\times$ 12 spectral partitions).
- **Minimal norm:** $\mu = 8$, corresponding to the minimum distance between two stable economic configurations.
- **Unimodularity:** the lattice is even and unimodular (its determinant is 1), which allows it to be interpreted as a genuine pre-geometric configuration space.
- **Automorphisms:** the group of automorphisms of Λ_{72} contains $(PSL_2(7) \times SL_2(25)) : 2$, a rich structure capable of encoding the symmetries of economic systems (cycles, regulations, crises).

Economic interpretation: Λ_{72} is the complete spectrum of resonance frequencies of economic cycles. Each economic attractor (among the 20) corresponds to a linear combination of the 72 vectors of Λ_{72} . The eigenvalues λ_i determine the ‘resonance frequency’ of the economic state, that is, its sensitivity to shocks and its tendency to shift towards other attractors.

11.2. The 15 universal constants β_k and their application to economics

De Dominicis (2026) showed that the 48 non-degenerate roots of Λ_{72} project onto a 15-dimensional basis, denoted $\{\beta_0, \beta_1, \dots, \beta_{14}\}$. These 15 universal constants are the same ones that govern nuclear masses (physics) and ionisation energies (chemistry).

Universal spectral formula:

$$\mathcal{E} = \Lambda_{\text{eco}} \cdot 4^m \cdot \sum_{k=0}^{14} \epsilon_k \beta_k$$

where:

- Λ_{eco} is a scale constant specific to economics (to be determined by historical calibration).
- m is an integer exponent (generally 0, 1 or 2 depending on the degree of polarity).
- $\epsilon_k \in \{-1, 0, 1\}$ is the ternary signature of the attractor.
- β_k are the 15 universal constants.

Economic interpretation: Each economic attractor is characterised by a vector of 15 coefficients ϵ_k , which encodes its economic ‘recipe’. For example:

- Attractor A (extreme perverse growth) activates the first 12 constants β_0 to β_{11} .
- Attractor T (terminal depression) activates a restricted combination of $\beta_0, \beta_1, \beta_2$.

11.3. The 72 eigenvalues and their economic correspondences

The complete spectrum of economic regimes consists of the 72 eigenvalues of Λ_{72} .

Among the 72 eigenvalues, 7 critical thresholds ($S_1, \dots, S_7$) play a pivotal role in economic dynamics by triggering transitions between regimes. Their exact values, their degeneracy and the associated macroeconomic policies are detailed in Chapter 13.

The following table shows the correspondence between the eigenvalue intervals and the economic regimes:

Table 17: Correspondence between eigenvalue intervals and economic regimes

Interval of λ	Number of eigenvalues	Economic correspondence
$\lambda > 10$	12	$3P$ attractors (perverse growth, high risk)
$5 < \lambda \leq 10$	20	$2P + 1N$ attractors (moderate growth, common)
$2 < \lambda \leq 5$	25	$1P + 2N$ attractors (versatile, adjustment)
$0 < \lambda \leq 2$	12	$3N$ attractor (depression, rare)
$\lambda \leq 0$	3	Frustrated configurations (forbidden)

Chapter 12 – The Transition Operator T and Economic Interactions

12.1. Definition of the operator T on the space of pentads

Let $\mathcal{H}_P$ be the Hilbert space of the 144 economic pentads. The transition operator T is a linear operator acting on $\mathcal{H}_P$ that generates transformations from one economic configuration to another.

In $\text{Cl}(6, 6)$, T decomposes according to the physical roles of the components of the pentad:

$$T = T_{\text{structure}} + T_{\text{fire}} + T_{\text{water}} + T_{\text{mixed}}$$

Economic interpretation:

- $T_{\text{structure}}$: acts on the three bivectors $\{B_1, B_2, B_3\}$, that is, on the production model, the institutions and the rules of the game. An action by $T_{\text{structure}}$ corresponds to a structural reform (e.g. transition from an industrial economy to a service economy, inclusion of GEP in KPIs).
- T_{fire} : acts on the axial element $F = i'v$, that is, on exogenous shocks, innovations and crises. An action of T_{fire} corresponds to an external event that ‘burns’ existing structures (e.g. a pandemic, a trade war, an oil shock, a property market crash).
- T_{water} : acts on the polar component $S = 1v$, i.e. on currency, value and capital. An action of T_{water} corresponds to a monetary or fiscal policy (e.g. quantitative easing, a stimulus package, austerity, tax reform).
- T_{mixed} : combines the three subspaces when the transition involves a simultaneous redistribution (e.g. debt restructuring, corporate mergers).

12.2. Angular formulation of the operator T

The generators $\{i, j, k, I, J, K\}$ of $\text{Cl}(6, 6)$ define a 6-dimensional relational space. The operator T is expressed as an exponential rotation operator:

$$T = \exp\left(i \sum_{a < b} \omega_{ab} L_{ab}\right), \quad L_{ab} = -i \left(\theta_a \frac{\partial}{\partial \theta_b} - \theta_b \frac{\partial}{\partial \theta_a} \right)$$

where θ_a are angular coordinates associated with the generators, and ω_{ab} are rotation parameters induced by the economic transition.

Economic interpretation: An economic transition is an angular reorientation in configuration space. For example:

- The transition from a rentier economy (attractor H) to a moderate-growth economy (attractor D) corresponds to a rotation in the (i, j) plane (reorientation of the production model).
- The transition from a recessionary economy (attractor I) to a recovering economy (attractor E) corresponds to a rotation in the (k, I) plane (reorientation of monetary policies).

12.3. Selection rules

A matrix element of T is non-zero if and only if:

1. **Conservation of the total number of generators:** each bivector contributes 2 generators, whilst the Fire and Water elements contribute 1 each. The sum $\sum N_{\text{gen}}$ is invariant modulo coupling to an external field.
2. **Conservation of generator type modulo gauge:** the spatial generators $\{i, j, k\}$ and the charge generators $\{I, J, K\}$ are individually conserved. The pseudo-scalar i' can only change its projection by crossing the polar thresholds P_4 or N_4 .
3. **Conservation of chirality:** the operator i' projects helicity states. In strong and electromagnetic interactions, $[T, i'] = 0$ (helicity is conserved). In weak (economic) interactions, T_{fire} induces a $L \leftrightarrow R$ flip via the Ke mode.
4. **Conservation of total angular momentum:** the condition $[L_{ab} + \frac{1}{2}\sigma_{ab}, T] = 0$ ensures exact compensation between intrinsic spin and orbital angular momentum.
5. **Preservation of nilpotence:** T must transform nilpotent states into nilpotent states: $(T|x)^2 = 0$. This condition automatically eliminates divergent self-energy loops and prohibits fusion configurations.

Chapter 13 – The 7 Spectral Thresholds of Economic Regulation

13.1. Definition and role of the thresholds

The transition operator T is not continuous; its action is regulated by seven discrete spectral thresholds $S_1, \dots, S_7$. These thresholds correspond to critical values of the topological tension $\mathcal{T} = \nabla\eta \cdot \nabla R_{\text{threshold}}$, that is, to specific values of the square roots $\sqrt{\lambda_i}$ of the eigenvalues of the network Λ_{72} .

Crossing a threshold triggers a specific type of economic transition:

Table 18: The 7 spectral thresholds of economic regulation

Threshold	$\sqrt{\lambda_i}$	Degeneration	Economic role	Associated policy
S_1	0.06614	2	Minimal activation	Inflation threshold: central banks begin to react
S_2	0.10486	2	Mirror symmetry (Sheng $\leftrightarrow$ Ke)	Shift between expansion and recession
S_3	0.10582	1	Polarisation	Policy change
S_4	0.13530	2	Inter-sectoral linkages	Macroeconomic coordination
S_5	0.17195	1	Fire transition (exogenous shock)	Crisis management
S_6	0.19329	1	Water transition (monetary policy)	Structural reforms
S_7	0.21732	1	Octave jump	Cycle change

13.2. The polarity gradient $3P \rightarrow 3N$ and spectral thresholds

The spectral thresholds $S_1, \dots, S_7$ are the microscopic realisation of the macroscopic polarity descent $3P \rightarrow 3N$. The topological tension $\mathcal{T} = \nabla\eta \cdot \nabla R_{\text{threshold}}$ controls both.

Table 19: Polarity gradient and spectral thresholds of economic regulation

Polarity stage	Spectral condition	Thresholds involved	Economic interpretation
$3P$ (Pure Sheng)	$\mathcal{T} < S_1$	Below the first threshold	Perverse growth, speculative bubbles, uncontrolled expansion
$2P + 1N$ (weak mixture)	$S_1 \leq \mathcal{T} < S_5$	S_1, S_2, S_3, S_4	Moderate growth, emerging tensions, incipient stagflation
$1P + 2N$ (strong mixture)	$S_5 \leq \mathcal{T} < S_7$	S_5 (Fire), S_6 (Water)	Recession, structural adjustment, crisis
$3N$ (pure Ke)	$\mathcal{T} \geq S_7 = \mu_{\Lambda_{72}} = 8$	Octave leap	Terminal depression, collapse, or regeneration (TRR)

Interpretation:

- Crossing from S_1 to S_4 initiates the mixing between the sectors ($3P \rightarrow 2P + 1N$). This is the point at which tensions begin to emerge, but where the system can still be stabilised by conventional policies.
- Crossing S_5 (Fire) and S_6 (Water) triggers intense mixing, releasing the pent-up frustration ($2P + 1N \rightarrow 1P + 2N$). The system enters a crisis, and only unconventional policies (bailouts, restructuring) can stabilise it.

- Crossing S_7 triggers the octave jump, reversing the sign of the effective mass and completing the descent towards $3N$ ($1P + 2N \rightarrow 3N$).

13.3. The polar thresholds P_4 and N_4 as public policy operators

The pentads P_4 and N_4 are the only elements excluded from the tropical belts C_P and C_N . Their high degree of connectivity (8 for P_4 , 9 for N_4) makes them topological hinges between the two belts.

Role of P_4 : Authorised transition from expansion to recession (soft landing). In practice, P_4 is activated when a government or central bank decides to curb expansion (interest rate rises, fiscal austerity) to prevent overheating.

Role of N_4 : Permitted transition from recession to expansion (controlled recovery). In practice, N_4 is activated when a decision is made to stimulate the economy (interest rate cuts, stimulus package) following a recession.

Golden rule: Any direct transition between C_P and C_N without passing through P_4 or N_4 is prohibited. This corresponds to a topological short-circuit (reverse polarity) which causes a systemic collapse.

Chapter 14 – Retropolarities and Collapses as Violations of Nilpotence

14.1. Definition of retropolarity

A retropolarity is a reversal of the economic spectrum: the $3P$ attractors (expansion) abruptly switch to $3N$ (depression) without passing through the thresholds. Formally, it is a direct transition $C_P \rightarrow C_N$ (or $C_N \rightarrow C_P$) that violates the threshold rule.

In the spectrum of Λ_{72} , retropolarity corresponds to a reversal of the sign of the eigenvalues: the $\lambda > 0$ become $\lambda < 0$, and the stable attractors become frustrated configurations.

Economic interpretation: A retropolarity is a systemic collapse. It occurs when decision-makers ignore the thresholds P_4 and N_4 and attempt a direct transition between expansion and recession.

14.2. Analysis of historical retropolarities

Table 20: Chronology of historical retropolarities and violated transitions

Event	Year	Transition violated	Consequence	Threshold missed
Great Depression	1929	$C_P \rightarrow C_N$ direct	Market collapse, deflation, mass unemployment	P_4
Oil crisis	1973	$C_P \rightarrow C_N$ (permitted transition)	Stagflation, but no total collapse	None
Subprime crisis	2008	partial $C_P \rightarrow C_N$	Financial collapse, prolonged recession	P_4 (partial)
COVID-19	2020	direct $C_N \rightarrow C_P$	Artificial rebound, speculative bubbles, inflation	N_4
China 2026+	2026–2030	direct $C_N \rightarrow C_P$ (without TRR)	hyperinflation, social chaos, tenth decline	N_4

14.3. Detection of retropolarities via spectral observables

Table 21: Detection of retropolarities via spectral observables

Observable	Alert threshold	Interpretation
$\eta(t)$	$\eta \approx 0$	Pre-tipping: the system is in unstable equilibrium
$d(t)$	$d < 2$	Fragmentation: the system is losing its cohesion
$\text{gap}(t)$	$\text{gap} < 0.2$	Extreme fragility: proximity to a topological threshold
$R_{\text{threshold}}(t)$	$R_{\text{threshold}} > 0.7$	Red alert: imminent retropolarity

14.4. The case of post-2018 China: a system in pre-retropolarity

The diagnosis of the Cl(6, 6) model confirms that of the White Paper:

- **Configuration:** $\mathbf{x}_{\text{China}} \approx (0, 0, 0, 0, 1, 0) \rightarrow \text{Attractor S (1P+2N)}$.
- **Projected pentads:** $\{P_6, N_5, N_6\}$.
- **Observables:** $\eta < -0.4, d \approx 2, \text{gap} \approx 0.1, R_{\text{threshold}} > 0.7$.

Interpretation: The system is in pre-retropolarity. The thresholds P_4 and N_4 are saturated, and the system is attempting a direct transition $C_N \rightarrow C_P$ without passing through N_4 . Without intervention (the TRR), this direct transition will trigger retropolarity: hyperinflation, social chaos, the tenth decline.

Topological solution: The TRR is a transition controlled by N_4 . It creates a new pioneering front (ecological regeneration) which enables the system to leave attractor S and reach C_P via a new cycle of Rent Extraction.

Chapter 15 - Spectral Observables: η , d , gap, R_{seuil}

15.1. The Discrete Dirac Operator $D(t)$

Let Γ be the pentad graph. A local spinor is defined for each pentad F :

$$\psi_F = \begin{pmatrix} \epsilon_F \\ \phi_F \end{pmatrix}, \quad \epsilon_F \in \{+1, -1\}, \quad \phi_F \in \{0, 1\}$$

The discrete Dirac operator is a 24×24 matrix ($12 \text{ pentads} \times 2 \text{ components}$) defined by:

$$(D\psi)_F = \sum_{G \sim F} w_{FG}(t) \sigma_{FG} \psi_G$$

where:

- $w_{FG}(t) = e^{-\beta E_{FG}(t)}$ is a coupling weight that depends on the shared frustration energy $E_{FG}(t)$.
- σ_{FG} is a Pauli matrix.
- $\beta > 0$ is a rigidity parameter (to be calibrated).

15.2. The Four Spectral Observables

Observable $\eta(t)$: Spectral Asymmetry

$$\eta(t) = \frac{N_+(t) - N_-(t)}{24}$$

where $N_+(t)$ is the number of strictly positive eigenvalues, and $N_-(t)$ the number of strictly negative eigenvalues.

- $\eta > 0$: dominance of the Sheng mode (expansion).
- $\eta < 0$: dominance of the Ke mode (contraction, recession).
- $\eta \approx 0$: transition regime (pre-bifurcation).

Observable $d(t)$: Effective Spectral Dimension

$$d(t) = \frac{2 \sum_{k=1}^{24} \lambda_k(t)^2}{\sum_{k=1}^{24} \lambda_k(t)^4} \cdot \frac{\sum_{k=1}^{24} 1}{\sum_{k=1}^{24} \lambda_k(t)^2}$$

- $d > 4$: the system is highly connected; shocks propagate rapidly.
- $d < 2$: the system is fragmented; shocks remain localized.

Observable $\text{gap}(t)$: Smallest Positive Spectral Gap

$$\text{gap}(t) = \min\{|\lambda_k(t)| > 0\}$$

- $\text{gap} > 0.5$: robust system.
- $\text{gap} < 0.2$: fragile system.
- $\text{gap} \rightarrow 0$: major warning signal.

Observable $R_{\text{seuil}}(t)$: Fraction of the Asymmetry at the P_4 and N_4 Thresholds

$$R_{\text{seuil}}(t) = \frac{\sum_{k \in \text{seuils}} |\lambda_k(t)|}{\sum_{k=1}^{24} |\lambda_k(t)|}$$

- $R_{\text{seuil}} < 0.3$: the system is far from the thresholds.
- $0.3 \leq R_{\text{seuil}} \leq 0.7$: the system is approaching the thresholds.
- $R_{\text{seuil}} > 0.7$: red alert — retro-polarization imminent.

15.3. Application to the Major Historical Phases

Table 22: Application of spectral observables to the major historical phases

Period	Attractors	η	d	gap	R_{seuil}	Interpretation
Trente Glorieuses (1945–1973)	D	$> +0.5$	5–6	> 0.5	< 0.2	Stable expansion
Oil shock (1973–1980)	A, R	≈ 0	3–4	0.2–0.3	0.5–0.6	Pre-transition
Neoliberalism (1980–2008)	B, C, D, E, F	+0.2 to +0.4	4–5	0.3–0.5	0.3–0.5	Fragile expansion
Subprime crisis (2008)	C	≈ 0	2–3	0.1–0.2	0.7–0.8	Alert
Post-COVID (2020–2026)	R, S, T	< -0.4	2–3	0.1–0.2	> 0.7	Crisis, stagflation
China 2026+	S	< -0.4	≈ 2	≈ 0.1	> 0.7	Pre-retro- polarization

table}

15.4 – Hierarchy of Observables in Cl(6,6)

15.4.1 – Rationale for the Hierarchical Extension

The four spectral observables defined in §15.2 – $\eta(t)$, $d(t)$, $\text{gap}(t)$, $R_{\text{seuil}}(t)$ – constitute a complete diagnostic system for the Cl(6, 0) model. They condense the state of an economic system into four numbers, making it possible to characterize its direction (expansion or recession), its diversity (capacity to propagate shocks), its robustness (resistance to bifurcations), and its tension at the polar thresholds.

However, this monolithic approach presents three structural limitations when moving to Cl(6,6):

Limitation 1 – Lack of contextual resolution. The four global observables treat the 144 projected pentads as an undifferentiated block. Yet post-2018 China is not uniformly in Stasis: some spectral sheets (for example, the “Territorial” sheet) are collapsing, while others (for example, the “Cognitive” sheet) are expanding. A global observable conceals these contrasts.

Limitation 2 – Lack of diagnosis of internal tensions. The Cl(6,6) model introduces 12 spectral sheets that may be in structural disagreement. An economy may simultaneously be in Sheng mode on one sheet and Ke mode on another. This inter-sheet imbalance is itself a powerful diagnostic signal that the four global observables do not capture.

Limitation 3 – Lack of fine localization. The 144 pentads are the elementary building blocks of the system. Identifying which pentads are active, saturated, or in transition makes it possible to establish an “economic acupuncture” type of diagnosis: targeting the precise points where intervention is required to restore circulation.

The hierarchical extension of the observables addresses these three limitations. It transforms the spectral dashboard from an instrument with four needles into a multi-resolution system, analogous to the transition from continental mapping to meridian mapping in traditional Chinese medicine.

15.4.2 – The Five Levels of the Hierarchy

The hierarchy of observables in Cl(6,6) is organized into five nested levels:

Table 23: Multi-level architecture of spectral observables

Level	Type	Number	Diagnostic role	Chinese analogy
1	Global (integrated over 12 sheets)	4	Overall view of the system	General pulse (脉)
2	Per spectral sheet	$12 \times 4 = 48$	Contextual nuances	Qi per meridian (经气)
3	Spectral threshold	7	Active transitions	Acupuncture points (穴)
4	Pentad	144	Fine interactions	Collaterals (络)
5	Relational (inter-sheet)	66	Tensions between perspectives	Wuxing imbalances (五行失调)

Total: 269 elementary observables, of which a significant subset (~20) is sufficient for operational control.

15.4.3 – Observables per Spectral Sheet (Level 2)

Let $\mathcal{G} = e_1, \dots, e_6, f_1, \dots, f_6$ be the set of the 12 generators/spectral sheets of $\text{Cl}(6,6)$, where:

- $e_1, \dots, e_6$ are the cosmic generators (positive signature, Sheng mode, $\eta > 0$).
- $f_1, \dots, f_6$ are the anti-cosmic generators (negative signature, Ke mode, $\eta < 0$).

For each sheet $g \in \mathcal{G}$, we define a restricted Dirac operator $D_g(t)$ acting on the Hilbert space $\mathcal{H}_g$ of the 12 pentads projected onto that sheet. The four local observables are:

$$\eta_g(t) = \frac{n_{P,g}(t) - n_{N,g}(t)}{n_{P,g}(t) + n_{N,g}(t)} \in [-1, +1]$$

where $n_{P,g}(t)$ is the number of active positive (Sheng) pentads on sheet g , and $n_{N,g}(t)$ the number of negative (Ke) pentads.

$$d_g(t) = \exp\left(-\sum_{k=1}^{12} p_{k,g} \ln p_{k,g}\right), \quad p_{k,g} = \frac{|\lambda_{k,g}|}{\sum_j |\lambda_{j,g}|}$$

$$\text{gap}_g(t) = \min\{|\lambda_{k,g}(t)| > 0\}$$

$$R_{\text{seuil},g}(t) = \frac{|\lambda_{P4,g}| + |\lambda_{N4,g}|}{\sum_{k=1}^{12} |\lambda_{k,g}|}$$

Economic interpretation. Each spectral sheet g modulates the intensity and orientation of a basic pentad. For example:

- $P_1^{(e_2)}$: a Keynesian economy (P_1) with strong purchasing power (e_2) – this is the Trente Glorieuses regime.
- $N_5^{(f_3)}$: a Stasis economy (N_5) with profit regulation (f_3) – this is the post-2018 Chinese regime on the profit sheet.

The observables by sheet reveal **contextual dissonances**: a system may have $\eta_{e_1} = +0.8$ (innovation in expansion) and $\eta_{f_3} = -0.7$ (finance in contraction), indicating a two-speed economy.

15.4.4 – Spectral Threshold Observables (Level 3)

The seven spectral thresholds $S_1, \dots, S_7$, whose critical values $\sqrt{\lambda_i}$ and economic roles are defined in **Chapter 13 (Table 18)**, are treated here as binary observables:

$$\sigma_i(t) = \begin{cases} 1 & \text{if } \mathcal{T}(t) \geq \sqrt{\lambda_{S_i}} \\ 0 & \text{otherwise} \end{cases}$$

where $\mathcal{T}(t) = |\nabla \eta \cdot \nabla R_{\text{seuil}}|$ is the topological tension of the system.

The threshold vector $\vec{\sigma}(t) = (\sigma_1, \dots, \sigma_7) \in 0, 1^7$ encodes the system's threshold-crossing state.

The **threshold vector** $\vec{\sigma}(t) = (\sigma_1, \dots, \sigma_7) \in 0, 1^7$ encodes the system's threshold-crossing state. Of the 128 possible states, only a few are economically stable:

- $\vec{\sigma} = (0, 0, 0, 0, 0, 0, 0)$: dormant system, below the first threshold.
- $\vec{\sigma} = (1, 1, 1, 1, 0, 0, 0)$: system under moderate tension (case of China 2022–2024).
- $\vec{\sigma} = (1, 1, 1, 1, 1, 1, 1)$: system undergoing an octave jump, cycle change (TRR accelerating).

15.4.5 — Pentad Observables (Level 4)

Each of the 144 pentads $P_k^{(g)}$ (with $k \in 1, \dots, 12$ and $g \in \mathcal{G}$) has four local observables:

Activity: $a_{k,g}(t) = |\psi_{k,g}(t)|^2 \in [0, 1]$

Polarity: $\pi_{k,g}(t) = \text{sgn}(\varepsilon_{k,g}) \in +1, -1$

Connectivity: $c_{k,g}(t) = \sum_{j \sim k} w_{kj,g}(t)$, number of neighboring pentads coupled on sheet g .

Frustration energy: $E_{k,g}(t) = |\varphi_{k,g}(t) - \bar{\varphi}_g(t)|^2$, deviation from the mean of sheet g .

These observables make it possible to identify **critical pentads**: those with high activity, strong connectivity, and maximum frustration energy. These are the preferred intervention points for the transition operator T .

Application to China. The spectral diagnosis of post-2018 China (§16.1) identifies the triplet P_6, N_5, N_6 (configuration $\mathbf{x}_{\text{China}} = (0, 0, 0, 0, 1, 0)$). The pentad observables reveal:

Table 24: Active pentads of the Chinese Stasis triplet (attractor S)

Pentad	Activity	Polarity	Connectivity	Frustration	Role in Stasis
P_6	0.45	+1 (Sheng)	3	0.33	Weakened domestic consumption
N_5	0.78	-1 (Ke)	5	0.08	Stasis node : closed pioneer frontier
N_6	0.65	-1 (Ke)	4	0.15	Domestic austerity, blocked rent extraction

Pentad N_5 is the most active and the most connected: it is the **Stasis node**. Any intervention must desaturate it by activating thresholds S_5, S_6, S_7 through the TRR.

15.4.6 — Inter-Sheet Relational Observables (Level 5)

Interactions between spectral sheets generate tension observables:

Asymmetry gap: $\Delta\eta_{a,b}(t) = |\eta_a(t) - \eta_b(t)|$ for two distinct sheets $a, b \in \mathcal{G}$.

Inter-sheet correlation: $C_{a,b}(t) = \text{corr}(\eta_a, \eta_b)$ over a moving window.

Wuxing tension: $\mathcal{T} * \text{Wuxing}(t) = \max * a \neq b \Delta\eta_{a,b}(t)$

The correlation matrix $C(t) \in \mathbb{R}^{12 \times 12}$ is a powerful diagnostic object. A block-diagonal matrix indicates a system fragmented into independent subsets; a matrix with strong coherence indicates an integrated system.

Wuxing imbalance signature. The 12 sheets are grouped into 5 families corresponding to the five phases of Wuxing:

Table 25: Wuxing imbalance signature: grouping the 12 spectral sheets into 5 phases

Phase	Character	Associated sheets	Composite observable
Wood	木	e_1, f_1 (Inflation / Inflation regulation)	$\eta_{\text{Wood}} = (\eta_{e_1} + \eta_{f_1})/2$
Fire	火	e_2, f_2 (Purchasing power / Finance)	$\eta_{\text{Fire}} = (\eta_{e_2} + \eta_{f_2})/2$
Earth	土	e_3, f_3 (Profits / Accumulation)	$\eta_{\text{Earth}} = (\eta_{e_3} + \eta_{f_3})/2$
Metal	金	e_4, f_4 (Rent Extraction / Rent Extraction regulation)	$\eta_{\text{Metal}} = (\eta_{e_4} + \eta_{f_4})/2$
Water	水	e_5, e_6, f_5, f_6 (Dispersion / Land)	$\eta_{\text{Water}} = \frac{1}{4} \sum \eta_i$

The Sheng cycle (Wood $\rightarrow$ Fire $\rightarrow$ Earth $\rightarrow$ Metal $\rightarrow$ Water $\rightarrow$ Wood) and the Ke cycle (Wood $\rightarrow$ Earth $\rightarrow$ Water $\rightarrow$ Fire $\rightarrow$ Metal $\rightarrow$ Wood) make it possible to diagnose **pathological imbalances**:

- **Sheng imbalance:** $\eta_{\text{Fire}} \gg \eta_{\text{Wood}}$ – expansion without foundation (speculative bubble).
- **Ke imbalance:** $\eta_{\text{Metal}} \ll \eta_{\text{Wood}}$ – excessive regulation stifling innovation.
- **Chinese Stasis:** $\eta_{\text{Earth}} \approx -0.7$, $\eta_{\text{Water}} \approx -0.5$, $\eta_{\text{Fire}} \approx +0.3$ – Earth (profits) and Water (dispersion) are contracting, while Fire (cognitive purchasing power) attempts an immaterial expansion. This is exactly the diagnosis of Part I: the immaterial cannot compensate for the exhaustion of the material.

15.4.7 – Composite Stasis Index

The synthesis of the five levels produces a **composite Stasis Index** defined by:

$$\mathcal{J}_{\text{Stase}}(t) = w_1 |\eta_{\text{total}}| + w_2 \frac{1}{d_{\text{total}}} + w_3 e^{-\alpha \text{gap}_{\text{total}}} + w_4 R_{\text{seuil, total}} + w_5 \mathcal{J}_{\text{Wuxing}}$$

With $\alpha = 5$, calibrated so that:

- gap = 0.1 (extreme fragility) $\rightarrow e^{-0.5} \approx 0.61$
- gap = 0.2 (moderate fragility) $\rightarrow e^{-1.0} \approx 0.37$
- gap = 0.5 (robustness) $\rightarrow e^{-2.5} \approx 0.08$

Note on the gap term: The term $w_3, e^{-5 \cdot \text{gap}}$ is a damped exponential form that ensures that the contribution of the gap remains bounded in $[0, 1]$. When $\text{gap} \rightarrow 0$ (asphyxiated system), the term tends toward 1 (maximum contribution to Stasis). When $\text{gap} \rightarrow 0.5$ (healthy system), the term tends toward $e^{-2.5} \approx 0.08$ (negligible contribution). This form avoids the divergence of the term $1/\text{gap}$, which would make the index unbounded.

The weights w_i are calibrated against the 9 dynastic declines:

Table 26: Weighting parameters of the composite Stasis Index $\mathcal{J}_{\text{Stase}}$

Parameter	Value	Justification
w_1	0.25	Asymmetry is the most robust signal
w_2	0.15	Fragmentation is a secondary signal
w_3	0.25	Spectral fragility is critical
w_4	0.20	Tension at the thresholds announces retro-polarization
w_5	0.15	Wuxing imbalance is a culturally grounded signal

Composite index thresholds:

Table 27: Composite Stasis Index $\mathcal{J}_{\text{Stase}}$ thresholds and economic states

$\mathcal{J}_{\text{Stase}}$	State	Historical example
< 0.3	Healthy growth	Trente Glorieuses (1945–1973)
$0.3 - 0.5$	Moderate tension	Oil shock (1973–1980)
$0.5 - 0.65$	Emerging Stasis	Japan (1990–2000)
$0.65 - 0.75$	Characterized Stasis	China (2018–2026)
> 0.75	Pre-retro-polarization	Republic of China (1945–1949)

Calculation for China 2026:

$$\mathcal{J}_{\text{Stase}} = 0.23 \times 0.45 + 0.18 \times \frac{1}{2.1} + 0.23 \times e^{-5 \times 0.12} + 0.21 \times 0.78 + 0.15 \times 1.1$$

$$\mathcal{J}_{\text{Stase}} = 0.1035 + 0.0857 + 0.1262 + 0.1638 + 0.165$$

$$\mathcal{J}_{\text{Stase}} \approx 0.644$$

Diagnosis: China is in **Characterized Stasis** ($0.5 < \mathcal{J}_{\text{Stase}} < 0.7$), close to the pre-retro-polarization threshold.

15.4.8 — Correspondences with Chinese Correlative Thought

This table presents spectral observables as the instruments of a diagnostic practice, analogous to the four examinations through which the TCM physician takes the pulse of the body.

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Table 28: The spectral dashboard as a practice of the four TCM examinations

Cl(6, 6) Observable	Chinese Concept	Corresponding TCM Examination
η_g (per sheet g)	Local Qi (气)	Inspection (望)
$\Delta\eta_{a,b}$ (sheet gap)	Wuxing imbalance	Auscultation (闻)
$C_{a,b}$ (correlation)	Crossed Sheng/Ke cycle	Inquiry (问)
Active pentads $a_{k,g}$	Acupuncture points (穴)	Palpation (切)
$\mathcal{J}_{\text{Stase}}$	State of the Mandate of Heaven	Synthetic diagnosis

The spectral dashboard is thus the modern equivalent of the **four examinations** of TCM (望闻问切, *wàng wén wèn qiè*): it takes the pulse of the economic system at five levels of resolution, from the coarsest (the 4 global observables) to the finest (the 144 pentads).

15.4.9 — Application to the Post-2018 Chinese Diagnosis

The following table summarizes the multi-level diagnosis for China in 2026, preparing the operational application in Part III:

Multi-Level Spectral Dashboard —China 2026

LEVEL 1 —Global view (aggregated)

η_{total}	d_{total}	gap	R_{seuil}	I_{Stase}
−0.45	1.9	0.10	0.78	0.64
State: RED ALERT —Characterized Stasis				

LEVEL 2 —Dominant spectral sheets

Sheet	η	d	gap	R_{seuil}	State
e_3 Profits	−0.70	1.8	0.08	0.82	Strong Ke
f_3 Accumulation	−0.60	2.0	0.10	0.75	Strong Ke
e_1 Inflation	+0.30	3.5	0.25	0.40	Moderate Sheng

LEVEL 3 —Activated spectral thresholds

Binary signature σ	(1, 1, 1, 1, 0, 0, 0)
Thresholds crossed	S_1, S_2, S_3, S_4 (exhaustion of the previous model)
Blocked thresholds	S_5, S_6, S_7 (no new cycle)
Topological tension $\mathcal{T}$	0.165
Next threshold	$S_5 = 0.17195$ — not reached ($\Delta = 0.007$)

LEVEL 4 —Active pentads

Pentad	Activity	Connectivity	Frustration	Role
N_5 (Stasis)	0.78	5	—	Critical node
N_6 (Austerity)	0.65	4	—	Ke reinforcement
P_6 (Consumption)	0.45	—	0.33	Frustrated

LEVEL 5 —Wuxing tensions (Sheng/Ke cycles)

Element	η	State
Earth (Regulation)	−0.7	Dominant Ke
Water (Money)	−0.5	Dominant Ke
Wood (Inflation)	+0.3	Weak Sheng
Structural imbalance: Metal (Rent Extraction) stifles Wood (Inflation)		
$\Delta\eta_{\text{max}} = 1.0$ (between Fire and Earth) — Maximum frustration		

Synthetic diagnosis: The system is blocked at attractor S (dispersion + nodal point). The operators $T_{\text{structure}}$, T_{fire} and T_{water} are saturated. Only activation of N_4 via the TRR allows a controlled transition toward C_P .

Diagnosis. China is in Characterized Stasis ($\mathcal{J}_{\text{Stase}} = 0.64$), with a blockage between thresholds S_4 and S_5 . The material sheets (Profits, Accumulation, Land) are in deep contraction, while the cognitive sheet (Inflation) attempts an expansion that fails to compensate for the deficit in spatial thickness (§6.4). The Wuxing imbalance confirms that the Ke cycle (regulation) dominates the Sheng cycle (generation), and that Metal (Rent Extraction) stifles Wood (Inflation).

Topological prescription. The TRR (§17.1) is an operator T_{mixed} that must:

1. Desaturate pentad N_5 (Stasis node) by activating thresholds S_5 (Fire), S_6 (Water), S_7 (Octave jump).
2. Rebalance Wuxing by restoring the Sheng cycle: **Wood (regulated Inflation) → Fire (Purchasing power) → Earth (Profits/GEP) → Metal (CUR) → Water (Monetary regulation).**
3. Bring $\mathcal{J}_{\text{Stase}}$ from 0.64 to < 0.3 by 2032, as the condition for exiting Stasis and restarting a new cycle of material Rent Extraction.

In Chapter 18, this prescription will be included in the spectral dashboard of the 15th Five-Year Plan (2026–2030).

Summary of Part II

This Part II has laid the mathematical foundations for extending the economic model to $\text{Cl}(6, 6)$. We have:

1. **Identified the limitations of the $\text{Cl}(6, 0)$ model:** closed system, absence of a dynamic origin for attractors and thresholds, absence of a theory of economic ‘substance’.
2. **Introduced the $\text{Cl}(6, 6)$ formalism as a solution:** 12 generators (6 cosmic e_i , 6 anti-cosmic f_j), Sheng/Ke coupling, 144 pentads as building blocks.
3. **Developed a 5-level hierarchy of observables:**
 - **Level 1:** 4 global observables (η , d , gap, $R_{\text{threshold}}$) for the overall view
 - **Level 2:** 48 observables per spectral sheet for contextual nuances
 - **Level 3:** 7 threshold observables for possible shifts
 - **Level 4:** 144 pentad observables for fine-grained interactions
 - **Level 5:** 66 relational observables for inter-sheet tensions

This hierarchy enables a multi-resolution diagnosis of the economic system, analogous to the four diagnostic methods of traditional Chinese medicine (望闻问切).

4. **Formalised the Stasis composite index $\mathcal{J}_{\text{Stasis}}$,** which synthesises the five levels into a single indicator, calibrated against the nine Chinese dynastic declines.
5. **Established cultural correspondences** with the *Yi Jing* (64 hexagrams), the Wuxing (5 phases), TCM (12 meridians), and philosophical traditions (Xiong Shili, Nāgārjuna, Heidegger, Kabbalah).

Major technical contributions:

- **Spectral projectors** Π_g : enable the 12 basic pentads to be projected onto the 12 spectral sheets, generating 144 projected pentads.
- **Dirac operator per sheet** $D_g(t)$: describes the local dynamics of each spectral sheet.
- **Tensor decomposition** $D(t) = \bigoplus_{g=1}^{12} D_g(t)$: shows that the global dynamics is the superposition of 12 local dynamics.
- **Relational observables** $\Delta\eta_{g,h}(t)$ and $C_{g,h}(t)$: measure the structural tensions between spectral sheets.

Control tools developed:

- **Multi-level dashboard** (Chapter 18): presents the observables across 5 nested resolutions, enabling a detailed diagnosis of the Chinese Stasis.
- **Advanced visualisations** (Appendix S): heat map of the 12 layers, 5-dimensional radar diagram, inter-layer correlation matrix.
- **Multi-layer control architecture** (Appendix D.12): structures four decision-making layers (Beijing / Province / County / Alert Centre) with the corresponding observables.
- **Complete mathematical formalism** (Appendix O.5): rigorously defines the extended observables and their algorithmic calculation.

Assessment of China post-2018:

The Cl(6, 6) model confirms and expands upon the White Paper's analysis:

- **Configuration:** $\mathbf{x}_{\text{China}} \approx (0, 0, 0, 0, 1, 0) \rightarrow \text{Attractor S (1P+2N)}$.
- **Projected pentads:** $\{P_6, N_5, N_6\}$ on the dominant leaves e_3 (Profits), f_3 (Accumulation), f_6 (Land).
- **Global observables:** $\eta \approx -0.45$, $d \approx 1.9$, $\text{gap} \approx 0.10$, $R_{\text{threshold}} \approx 0.78$.
- **Composite index:** $\mathcal{J}_{\text{Stasis}} \approx 0.64 \rightarrow \text{Characterised stasis, close to pre-retropolarity}$.
- **Wuxing tensions:** $\Delta\eta_{\text{max}} = 1.0$ between Fire and Metal $\rightarrow$ structural imbalance where Metal (Rent Extraction) stifles Wood (Inflation).

Interpretation: The system is in a state of pre-retropolarity. The thresholds P_4 and N_4 are saturated, and the system is attempting a direct transition from C_N to C_P without passing through N_4 . Without intervention (the TRR), this direct transition will trigger retropolarity: hyperinflation, social chaos, the tenth decline.

Outlook: Part III will apply this formalism to the specific case of the Territorial Regeneration Rent (TRR), demonstrating how it enables a controlled transition via the N_4 threshold, activated by a mixed transition operator T_{mixed} combining $T_{\text{structure}}$, T_{fire} and T_{water} .

Voici la **Partie III – THE TRR: AN OPERATIONAL SOLUTION** complète, avec toutes les corrections terminologiques appliquées :

PART III – THE TRR: AN OPERATIONAL SOLUTION

Introduction to Part III

Parts I and II set out the diagnosis and the mathematical foundations. Part I demonstrated that China reached its Nodal Point in 2018, that the system is locked onto attractor S (dispersion + nodal point), and that the immaterial (AI, carbon, data) is structurally incapable of recreating the mass exogenous demand necessary to emerge from Stasis. Part II formalised this diagnosis within the topological framework of $Cl(6, 6)$, showing that Stasis is a blockage of transition operators and that the only topologically permitted outcome is a controlled transition via the threshold N_4 .

This Part III applies this formalism to the specific case of post-2018 China. We diagnose the Chinese Stasis in the language of pentads, formalise the Territorial Regeneration Rent (TRR) as a controlled transition operator T_{mixed} passing through the threshold N_4 , and propose a spectral dashboard for steering the 15th Five-Year Plan (2026–2030).

This section represents the operational culmination of the demonstration. It shows that the TRR is not merely one ecological policy amongst many, but a concrete application of discrete topology to the regulation of an economic system in crisis, adhering to the laws of the $64 \rightarrow 20$ invariant and offering China a peaceful way out to avoid the tenth dynastic collapse.

Chapter 16 – Diagnosis of the Chinese Stasis in $Cl(6, 6)$

16.1. Projection of the Chinese case onto the network of pentads

The diagnosis of the $Cl(6, 6)$ model confirms and refines that of the White Paper: post-2018 China is not merely experiencing a ‘Nodal Point’, but the terminal exhaustion of a centuries-long trend of material Rent Extraction (initiated in 1978).

For 40 years, China has used mechanisms of spatial avoidance (globalisation, massive investment in infrastructure via LGFVs, urbanisation) to stave off the crisis of overproduction. However, this pioneering front is now saturated (ghost towns, atrophied countryside). At the same time, the mechanism of temporal avoidance (debt) has reached its limits.

As in previous dynastic declines, state control of prices and the structural rigidity of the labour market prevent a natural readjustment of wages and demand. The system is locked onto attractor S (dispersion + nodal point). The Territorial Regeneration Rent (TRR) is therefore not merely a tool for crossing a spectral threshold (N_4), but the indispensable mechanism for reorganising the system and initiating a new growth trend, by recreating Value where spatial and temporal avoidance mechanisms have failed.

In the Cl(6, 6) model, post-2018 China is projected onto a specific configuration of pentads.

Economic variables:

Table 29: Binary configuration of post-2018 China in the Cl(6, 0) space

Variable	Value for China 2018–2026	Code
Inflation	Low (asset deflation)	0
Purchasing power	Stagnant	0
Profits	Falling (property sector)	0
Buy-back	Nodal Point reached	0
Dispersion	Increasing (ghost towns, reverse rural exodus)	1
Land tenure	Oikoumene (saturation)	0
Resulting configuration:		$\mathbf{x}_{\text{China}} = (0, 0, 0, 0, 1, 0)$

Configuration: $\mathbf{x}_{\text{China}} = (0, 0, 0, 0, 1, 0)$

Projection onto the pentads: This configuration projects onto a specific triplet of pentads:

$$\mathbf{x}_{\text{China}} \rightarrow \{P_6, N_5, N_6\}$$

- P_6 (Domestic consumption economy, active buy-back, concentration): represents the attempt to sustain growth through domestic consumption, but this pentad is ‘weakened’ as it is no longer supported by a pioneering front.
- N_5 (Stasis: nodal point, dispersion, depletion of resources): represents the saturation of the system and the depletion of land value reserves.
- N_6 (Domestic austerity: recession, concentration, blocked buy-back): represents austerity and regulatory policies that fail to revive growth.

Resulting attractor: The configuration projects onto attractor S (1P+2N): Dispersion and nodal point, systemic fragmentation. This is precisely the diagnosis set out in the White Paper: China is in a state of Stasis.

16.2. The blocking of transition operators

In a healthy system, the transition operators ($T_{\text{structure}}$, T_{fire} , T_{water}) enable the system to navigate between attractors. But in the Chinese case, these operators are blocked:

- **Blockage of $T_{\text{structure}}$:** The Chinese production model (industrial, based on exports and property) has reached saturation point. Traditional structural reforms (transition to services, innovation) are failing to

generate a new cycle of growth because they lack “spatial depth” (they do not create mass employment or regenerate the region).

- **Blockage of T_{fire} :** Exogenous shocks (trade war, COVID, geopolitical tensions) have ‘burned’ existing structures, but have not created new opportunities. Innovation (AI, carbon, data) suffers from a lack of spatial depth: it is dematerialised and does not generate territorial roots.
- **T_{water} bottleneck:** Monetary and fiscal policies (interest rate cuts, QE, stimulus) are no longer having any effect. The system is saturated with liquidity, but investment is not picking up because there are no longer any profitable frontiers to explore.
- **Stagnation of T_{mixed} :** Redistribution mechanisms (fiscal, social) are failing to redirect capital flows towards productive sectors. Value accumulates in speculative assets (property, shares) without creating real wealth.

Consequence: The system is locked onto attractor S. It cannot return to previous states (towards attractors 2P+1N or 3P) because the frontier sectors are saturated. Nor can it move towards collapse (attractor T) because the Chinese Party-State maintains social cohesion. Stasis is a state of precarious equilibrium, in which the system remains stuck but tensions are building up.

16.3. The Nodal Point as a spectral threshold

In the Cl(6, 6) formalism, the Nodal Point corresponds to the crossing of several critical spectral thresholds:

Table 30: Spectral thresholds and the state of China post-2018

Threshold	Value	Status in post-2018 China	Interpretation
S_1	0.06614	Crossed	Minimal activation: start of central bank response
S_2	0.10486	Exceeded	Mirror symmetry: shift from Sheng to Ke
S_3	0.10582	Exceeded	Polarisation: policy change (end of Tudi Caizheng)
S_4	0.13530	Threshold crossed	Inter-sectoral coupling: macroeconomic coordination
S_5	0.17195	Threshold not crossed	Fire transition: no exogenous creative shock
S_6	0.19329	Not crossed	Water transition: no structural monetary reform
S_7	0.21732	Not crossed	Octave leap: no change of cycle

Interpretation: China has crossed the first four thresholds, which correspond to the exhaustion of the previous growth model. However, it has not crossed thresholds S_5, S_6, S_7 , which correspond to the transition to a new cycle. Stasis is a standstill between S_4 and S_5 : the system is awaiting a trigger that would enable it to cross these thresholds and move into a new regime.

Chapter 17 – TRR: A Controlled Transition Operator (T_{mixed})

17.1. Formalisation of the TRR as a crossing of the N_4 threshold

The White Paper proposes the Territorial Regeneration Rent (TRR) as a solution to the Chinese Stasis. In the Cl(6, 6) formalism, the TRR is a threshold-controlled transition at N_4 which enables the system to leave the C_N belt (recession) and enter the C_P belt (expansion) via a new cycle of material Rent Extraction.

Topological mechanism:

$$\text{Stasis (S)} \xrightarrow[\text{Activation of } N_4]{\text{TRR}} C_P \text{ (New cycle of Rent Extraction)}$$

The threshold N_4 is activated by a mixed transition operator T_{mixed} which combines:

- $T_{\text{structure}}$: creation of a new productive model (ecological economy, territorial regeneration).
- T_{water} : redirecting monetary and financial flows towards this new model.
- T_{fire} : using exogenous shocks (climate, environment) as catalysts for the transition.

The TRR is therefore not merely an environmental policy. It is a topological operator that reorients the entire economic system by acting simultaneously on the three components of the pentad.

17.2. The three structural reforms as components of T

The White Paper identifies three structural reforms for the implementation of the TRR. In the Cl(6, 6) formalism, these three reforms are the components of the transition operator T_{mixed} :

Reform 1: Two Mountains Bank

- **Component:** T_{water}
- **Mechanism:** Bypassing LGFV (local government financing vehicles) funding through the securitisation of GEP (Gross Ecosystem Product).
- **Topological effect:** Activation of threshold N_4 by redirecting capital flows towards ecological regeneration.
- **Interpretation:** The Two Mountains Bank is a monetary operator that creates a new financing channel for ecological frontier initiatives. It enables the S_6 threshold (Water Transition) to be crossed by reforming the monetary and financial system.

Reform 2: Integration of GEP into local executives' KPIs

- **Component:** $T_{\text{structure}}$
- **Mechanism:** Reform of political incentives; replacement of GDP with GEP as a performance indicator.

- **Topological effect:** Stabilises the transition at N_4 by embedding the new production model within institutions.
- **Interpretation:** The integration of GEP into KPIs is an institutional operator that alters the structure of incentives and the rules of the game. It enables the S_5 threshold (Fire Transition) to be crossed by creating a new framework for innovation and investment.

Reform 3: Regenerative Usufruct Certificate (CUR)

- **Component:** T_{fire}
- **Mechanism:** Reappropriation of rent by local authorities via usufruct certificates on regenerated land.
- **Topological effect:** Anchors the new cycle on C_P by creating a value distribution circuit.
- **Interpretation:** The CUR is a redistributive operator that creates a new Rent Extraction cycle by redirecting value flows towards regenerated territories. It enables the S_7 threshold (Octave Leap) to be crossed by triggering a new growth cycle.

17.3. The roll-out timetable as a sequence of threshold crossings

The TRR deployment timetable proposed by the White Paper corresponds to a sequence of spectral threshold crossings:

Table 31: TRR deployment timetable as a sequence of spectral threshold crossings

Period	Phase	Objective	Threshold reached	Effect
2024–2025	Pilot projects	Two Mountains Bank in Lishui, integration of the GEP	S_5 (Fire Transition)	Activation of ecological innovation
2026–2028	Roll-out	Extension to pilot provinces, widespread adoption of CURs	S_6 (Water Transition)	Reform of the monetary and financial system
2029–2030	Acceleration	Treatment of 5 to 10 million hectares	S_7 (Octave Leap)	Change of cycle, exit from Stasis
2031–2035	Widespread implementation	Full coverage of atrophied rural areas	Stabilisation at C_P	New Rent Extraction cycle

Interpretation: The TRR is not a one-off policy, but a sequence of spectral threshold crossings that enables the system to transition gradually from Stasis (attractor S) to a new growth cycle (belt C_P).

Chapter 18 – Quantification and Management of the TRR through Spectral Observables

18.1 – The multi-level spectral dashboard for the 15th Five-Year Plan

18.1.1 – From the monolithic dashboard to the multi-resolution dashboard

The spectral dashboard proposed in the initial version of the $Cl(6, 0)$ model (§18.2, Table 18.1) was based on four global observables – η_{total} , d_{total} , $\text{gap}_{\text{total}}$, $R_{\text{threshold, total}}$ – aggregated into a traffic-light system (Green/Amber/Red/Collapse). Whilst this system is sufficient for an overall diagnosis, it masks the contextual dissonances that the extension to $Cl(6, 6)$ brings to light [8]: post-2018 China is not uniformly in Stasis; certain spectral sheets (Innovation, Cognitive Purchasing Power) are in Sheng expansion whilst others (Land Profits, LGFV Accumulation) are in deep Ke contraction.

The hierarchical extension of the observables (§15.4, Appendix O.5) enables the construction of a multi-level dashboard that applies the diagnosis across five nested levels of resolution, from the overview (4 needles) to fine-grained interactions (144 pentads). This dashboard is the economic equivalent of the four diagnostic methods of traditional Chinese medicine (望闻问切, *wàng wén wèn qiè*): it takes the pulse of the system at five levels, from the coarsest to the finest.

This paragraph presents the dashboard as applied to the Chinese case of 2026 and how it relates to the management of the 15th Five-Year Plan (2026–2030). The complete mathematical formalism is given in Appendix O.5; the multi-layered decision-making architecture (Beijing / Province / County / Alert Centre) is detailed in Appendix D.12.

18.1.2 – The multi-level dashboard: China 2026 assessment

The following table summarises the spectral analysis of China in August 2026, on the eve of the effective launch of the 15th Five-Year Plan:

Multi-level spectral dashboard —China - August 2026

LEVEL 1 —OVERVIEW (4 integrated observables across 12 sheets)

η_{total}	d_{total}	gap	$R_{\text{threshold}}$	$\mathcal{J}_{\text{Stasis}}$
−0.45	1.9	0.10	0.78	0.64
Confirmed stasis, red alert				

LEVEL 2 —DOMINANT LEAVES (48 observables per leaf)

Leaf	η	d	gap	R
e_3 (Profits)	−0.70	1.8	0.08	0.82
f_3 (Accumulation)	−0.60	2.0	0.10	0.75
f_6 (Land)	−0.55	1.7	0.07	0.80
e_1 (Inflation)	+0.30	3.5	0.25	0.40
e_2 (Purchasing power)	+0.25	2.8	0.18	0.45

LEVEL 3 —THRESHOLDS ACTIVATED (7 spec-

tral thresholds derived from Λ_{72})

$\vec{\sigma} = (1, 1, 1, 1, 0, 0, 0) \Rightarrow S_1$ to S_4 exceeded, S_5 to S_7 blocked

Topological tension $T = 0.165$, threshold $S_5 = 0.17195$ not reached **LEVEL 4 —ACTIVE PENTADS** (144

projected pentads)

Pentad	Activity	Connectivity	Frustration	Role
$N_5^{(f_3)}$	0.78	5	0.08	Critical node
$N_6^{(f_3)}$	0.65	4	0.15	Contagious
$P_6^{(e_2)}$	0.45	3	0.33	Frustrated
$P_1^{(e_1)}$	0.35	2	0.20	In its infancy

LEVEL 5 —WUXING TENSIONS

(66 relational observables)

η_{Wood}	η_{Fire}	η_{Earth}	η_{Metal}	η_{Water}
−0.05	−0.15	−0.55	−0.70	−0.52

$\Delta\eta_{\text{max}} = 1.0$ (Fire↔Metal)

Fragmentation $\mathcal{F} = 0.42 \Rightarrow$ Sheng/Ke bipolar system

18.1.3 — Interpreting the dashboard**Level 1 — Overview**

The four integrated observables confirm the diagnosis set out in Chapter 16: China is characterised stasis on the attractor S (dispersion + nodal point), with a triplet of active pentads $\{P_6, N_5, N_6\}$. The composite index $\mathcal{J}_{\text{Stasis}} = 0.64$ places the system on the boundary of pre-retropolarity (threshold 0.75). This diagnosis is consistent with the macroeconomic data for 2026: the loss of over 19 million jobs in the construction sector between 2022 and 2026 [84], property investment contracting by −7.9% year-on-year in H1 2026 [83], and the employment elasticity of the New Productive Forces stabilising at 0.14 [85].

Level 2 — Dominant Patterns

Level 2 reveals the bipolar signature of the Chinese Stasis: the material sheets (e_3 Profits, f_3 Accumulation, f_6 Land) are in deep Ke contraction ($\eta < -0.5$), whilst the cognitive sheets (e_1 Inflation, e_2 Purchasing Power) are attempting moderate Sheng expansion ($\eta > +0.2$). This imbalance is the spectral manifestation of the spatial thickness deficit analysed in §6.4: the cognitive rent generated by the SECC (gross margins of 72–86 per cent [74]) does not trickle down to wages via industrial employment, as the domestic multiplier for data centres (1.2) is structurally lower than that for residential construction (2.5) [82].

Level 3 – Thresholds triggered

The threshold vector $\vec{\sigma} = (1, 1, 1, 1, 0, 0, 0)$ indicates that China has crossed the first four spectral thresholds derived from the Λ_{72} network [9] – signalling the exhaustion of the previous growth model – but remains stuck between S_4 and S_5 . The topological tension $T = 0.165$ is close to but below the threshold $S_5 = 0.17195$ (Fire Transition). This blockage is the spectral signature of the absence of a creative exogenous shock: the TRR must provide this trigger by activating S_5 (ecological innovation), then S_6 (monetary reform via the Two Mountains Bank), then S_7 (octave leap towards a new cycle of Rent Extraction).

Level 4 – Active Pentads

Level 4 identifies the economic acupuncture points:

- The pentad $N_5^{(f_3)}$ (Stasis with strict profit regulation) is the critical node: maximum activity ($a = 0.78$), strong connectivity ($c = 5$), minimum frustration ($E = 0.08$). It concentrates the dynamics of Stasis and must be desaturated as a priority through the deployment of CURs (§17.2, Reform 3).
- The pentad $P_6^{(e_2)}$ (Domestic consumption with purchasing power) is frustrated ($E = 0.33$): it holds the potential for a consumption-led recovery, but its energy does not flow to the other pentads.
- The pentad $P_1^{(e_1)}$ (Keynesianism with innovation) is in its infancy ($a = 0.35$): it represents the potential of the TRR to create a new material frontier.

Level 5 – Wuxing Tensions

Level 5 reveals a structural imbalance in the Wuxing cycle [27], [91]: Metal ($\eta_{\text{Metal}} = -0.70$, Rent Extraction regulation) stifles Wood ($\eta_{\text{Wood}} = -0.05$, innovation), whilst Earth ($\eta_{\text{Earth}} = -0.55$, profits) no longer nourishes Metal. The maximum difference $\Delta\eta_{\text{max}} = 1.0$ between Fire and Metal indicates extreme tension between the cognitive sheets and the financial regulation sheets. The fragmentation index $\mathcal{F} = 0.42$ confirms that the system is bipolar: a Sheng block (Inflation, Purchasing Power) and a Ke block (Profits, Accumulation, Land) are evolving in opposition, with no harmonious flow between the five phases.

18.1.4 – Coordination with the implementation of the 15th Five-Year Plan

The multi-level dashboard provides the spectral targets for each phase of the TRR's roll-out:

Table 32: TRR operational timetable: spectral targets and thresholds to be met

Phase	Period	Main spectral target	Observable to be shifted	Threshold
Pilot	2024–2025	Desaturation of $N_5^{(f_3)}$ in the 10 pilot zones	$a_{N_5} : 0.78 \rightarrow 0.55$	S_5 (Fire)
Roll-out	2026–2028	Rebalancing Metal↔Wood across the 25 provinces	$\eta_{\text{Metal}} : -0.70 \rightarrow -0.30$	S_6 (Water)
Acceleration	2029–2030	Treatment of 5 to 10 million hectares	$\eta_{\text{total}} : -0.45 \rightarrow +0.10$	S_7 (Leap)
Generalisation	2031–2035	Stabilisation at the C_P threshold	$\mathcal{J}_{\text{Stasis}} : 0.64 \rightarrow < 0.30$	C_P stable

The pilot areas of Lishui (Two Mountains Bank) [92], [93], Sanming and Nanping (Fujian forests), Xingguo and Ganzhou (Jiangxi catchment areas), Hainan (island GEP), and Huzhou/Chongyi (forest carbon) — mapped in Chapter 20 — constitute the operational entry points for Level 4. Their success determines whether thresholds S_5 , S_6 and S_7 are sequentially crossed and whether Stasis is exited.

18.1.5 — Limitations and precautions

The multi-level dashboard does not replace the traditional indicators (GDP, unemployment, inflation) used in the five-year plans; it enriches them with a spectral interpretation. Several precautions are necessary:

- **Provisional calibration:** the weights w_i of the composite index $\mathcal{J}_{\text{Stasis}}$ are calibrated against the nine Chinese dynastic declines (Appendix G) [25]; their validity for other cultural contexts remains to be tested.
- **Quality of local data:** Level 4 (pentads) and Level 5 (relational) observables require granular data (local GEP, LGFV debt ratios by county [38]) which are not always available in real time.
- **The *tang ping* phenomenon:** Stasis is not merely an economic category but an experience lived by millions of young people [86], [94]; the actual unemployment rate amongst 16–24-year-olds (>60 per cent) is an indicator of the erosion of the Mandate of Heaven (§21.1) which spectral observables alone do not capture.
- **Principle of parsimony:** the wealth of 269 elementary observables may tempt decision-makers to multiply interventions; management must focus on the critical observables identified by the Diagnostic layer (Appendix D.12).

18.1.6 — Summary

The multi-level dashboard transforms the management of the 15th Five-Year Plan from a four-handed exercise into a modernised divinatory practice: a dialogue with the system that enables its state to be probed at five levels of resolution, its Wuxing imbalances to be diagnosed, and TRR policies to be adjusted in harmony with the deep laws that structure complex systems. It is the operational translation of the document’s central thesis: the economy is a discrete information system (§1.2), and its regulation requires a multi-resolution dashboard, analogous to the four examinations of traditional Chinese medicine.

The following sections (§18.2 to §18.4) detail respectively: - §18.2: the graphical visualisations of the dashboard (heat map, radar chart, correlation matrix); - §18.3: the simulation of the transition out of Stasis with and without TRR; - §18.4: specific management indicators by decision-making level (Beijing / Province / County).

18.2 – Graphical visualisations of the spectral dashboard

The multi-level dashboard presented in §18.1 is based on five nested levels of resolution, ranging from the global view (4 integrated observables) to fine-grained interactions (66 relational observables). To make this information actionable for decision-makers, three complementary graphical visualisations are provided. Their full formal specification is given in Appendix S; this section presents their economic interpretation.

18.2.1 – Heat map of the 12 spectral sheets

The heatmap represents the 12 spectral sheets $\mathcal{F} = \{e_1, \dots, e_6, f_1, \dots, f_6\}$ according to their local asymmetry $\eta_f(t)$, encoded by a colour gradient:

- **Bright red** ($\eta_f > +0.6$): strong Sheng dominance, contextual expansion.
- **Orange** ($+0.3 < \eta_f \leq +0.6$): moderate Sheng mode.
- **Grey** ($-0.3 \leq \eta_f \leq +0.3$): precarious equilibrium.
- **Light blue** ($-0.6 \leq \eta_f < -0.3$): moderate Ke mode.
- **Dark blue** ($\eta_f < -0.6$): strong Ke dominance, contextual contraction.

China 2026 Diagnosis (§S.3.3):

Heatmap of the 12 spectral sheets —China - August 2026

Sheet	Visualisation	η
e_1 Inflation		+0.30 (moderate Sheng)
e_2 Purchasing power		+0.25 (moderate Sheng)
e_3 Profits		-0.70 (strong Ke)
e_4 Active buy-back		-0.80 (very strong Ke)
e_5 Dispersion		-0.50 (moderate Ke)
e_6 Land		-0.55 (Ke strong)
f_1 Inflation regulation		-0.40 (moderate Ke)
f_2 Purchasing power regulation		-0.60 (strong Ke)
f_3 Profit regulation		-0.80 (Very strong Ke)
f_4 Rent Extraction regulation		-0.60 (strong Ke)
f_5 Dispersion regulation		-0.40 (moderate Ke)
f_6 Land regulation		-0.50 (moderate Ke)

$$\eta_{\text{mean}} = -0.45 \quad \Delta\eta_{\text{max}} =$$

1.10 (Active Rent Extraction ↔ Innovation)

Economic interpretation. The chart reveals the bipolar signature of the Chinese Stasis: the material layers (e_3 Profits, e_4 Buy-back, e_6 Land) are undergoing a profound Ke contraction, whilst the cognitive layers (e_1 Inflation, e_2 Purchasing power) are attempting a moderate Sheng expansion. This imbalance is the spectral manifestation of the deficit in spatial depth analysed in §6.4: the cognitive rent generated by the SECC (gross margins of 72–86 per cent [74]) is not being channelled into wages via industrial employment [85].

18.2.2 — 5-dimensional radar chart

The radar chart superimposes five dimensions normalised to the interval $[0, 1]$:

- The 4 global observables (η_{total} , d_{total} , $\text{gap}_{\text{total}}$, $R_{\text{threshold,total}}$).
- The number of activated thresholds $\sum_{i=1}^7 \sigma_i/7$.

China 2026 Diagnosis (§S.4.3):

5-dimensional radar chart —China 2026

Dimension	Normalised value	Interpretation
η_{norm}	0.275	Low (Ke dominance)
d_{norm}	0.158	Very low (fragmented system)
gap_{norm}	0.10	Very low (extreme fragility)
$R_{\text{threshold,norm}}$	0.78	High (critical tension)
$\sum \sigma_i/7$	0.571	Medium (4 out of 7 thresholds exceeded)

Interpretation: The

radar is **unbalanced** —a characteristic signature of a system in **pre-retropolarity**.

This signature is characteristic of a system in pre-retropolarity: China is a hair's breadth away from systemic collapse. The radar signature allows for an immediate comparison with historical reference regimes (Appendix S, §S.4.4): the Trente Glorieuses (balanced radar), the 1973 oil crisis (contracted radar), Japan in 1990 (collapsed radar), and the Great Depression of 1929 (retropolarity).

18.2.3 — Inter-sheet correlation matrix

The matrix $C(t) \in \mathbb{R}^{12 \times 12}$ encodes the correlations $\text{corr}(\eta_f, \eta_g)$ calculated over a sliding window of 12 quarters (§S.5). Three typical structures can be identified:

- **Block-diagonal matrix:** a system fragmented into independent subsets.
- **High-coherence matrix** ($C_{f,g} \approx +1$): an integrated system.
- **Matrix with bipolar tensions** ($C_{f,g} \approx -1$ between blocks): a system in structural opposition.

China 2026 Diagnosis (§S.5.3): the matrix exhibits a bipolar structure:

- **Sheng block** (rows e_1, e_2): Inflation and Purchasing Power are strongly correlated ($C = 0.70$).

- **Ke Block** (rows $e_3, e_4, e_6, f_2, f_3, f_4$): anti-cosmic leaves strongly correlated with one another ($C > 0.50$).
- **Sheng↔Ke tension**: cosmic and anti-cosmic leaves evolve in opposition ($C < 0$).

The fragmentation index $\mathcal{F} = 1 - \frac{1}{66} \sum_{f < g} |C_{f,g}| = 0.42$ confirms a bipolar system in which two blocks evolve in opposition, without harmonious circulation between the five phases of the Wuxing [27], [91].

18.2.4 – Summary of visualisations

The three visualisations are complementary:

Table 33: Visualisation tools for the spectral dashboard and their use in decision-making

Visualisation	What it reveals	Decision-making action
Heat map	Spatial distribution of the Sheng/Ke mode	Identify the dominant leaves to be treated as a priority
Radar diagram	Overall system footprint	Compare with historical reference regimes
Correlation matrix	Structural tensions between leaves	Diagnose Wuxing imbalances

Their economic interpretation is explored in greater depth in Appendix S (§S.3 to §S.5). Their integration into operational management is detailed in Appendix D.12.

18.3 – Simulation of the exit from Stasis: scenarios with and without TRR

The simulations presented below project the evolution of spectral observables over the period 2026–2032, according to two contrasting scenarios. They incorporate the Stasis composite index $\mathcal{J}_{\text{Stasis}}$ defined in §15.4.7 and §O.5.9:

$$\mathcal{J}_{\text{Stasis}}(t) = w_1 |\eta_{\text{total}}| + w_2 \frac{1}{d_{\text{total}}} + w_3 e^{-5 \cdot \text{gap}_{\text{total}}} + w_4 R_{\text{threshold, total}} + w_5 \mathcal{T}_{\text{Wuxing}}$$

with the weights calibrated against the nine Chinese dynastic declines (Appendix G) [25]: $w_1 = 0.23$, $w_2 = 0.18$, $w_3 = 0.23$, $w_4 = 0.21$, $w_5 = 0.15$.

18.3.1 — Scenario 1: Without TRR (continuation of current policies)

Scenario 1 —Without TRR (continuation of current policies)

Period	η	d	gap	$R_{\text{threshold}}$	$\mathcal{J}_{\text{Stasis}}$	Diagnosis
2026	−0.45	1.9	0.10	0.78	0.64	Characterised stasis
2028	−0.50	1.8	0.08	0.82	0.70	Pre-retropolarity
2030	−0.60	1.2	0.05	0.90	0.75	Imminent collapse
2032	−0.80	0.8	0.02	0.95	0.80	Retropolarity (10th dynastic collapse)

MULTI-LEVEL DIAGNOSIS IN 2032:

- Dominant leaf: e_4 (Active Rent Extraction) $\rightarrow \eta_{e_4} = -0.85$
- Critical pentad: $N_5^{(f_3)} \rightarrow$ activity 0.92, connectivity 6, frustration 0.02
- Wuxing imbalance: $\Delta\eta_{\text{max}} = 1.4$ (Fire $\leftrightarrow$ Metal), dominant Ke cycle
- Threshold vector: $\vec{\sigma} = (1, 1, 1, 1, 1, 1, 1)$ —all thresholds exceeded, octave jump occurred
- Fragmentation index: $\mathcal{F} = 0.78$ —extreme bipolar system

Economic interpretation. This scenario replicates the trajectory of the nine dynastic declines [25]: the depletion of the stock of land value [3], [4], combined with the inability of the intangible sector (SECC) to generate the necessary mass employment (employment elasticity of the New Productive Forces at 0.14 [85], and a domestic multiplier for data centres of 1.2 [82]), leads to a reversal of growth trends. Data from the National Bureau of Statistics confirms this trajectory: the loss of over 19 million jobs in the construction sector between 2022 and 2026 [84], property investment contracting by −7.9% year-on-year in H1 2026 [83], and a real unemployment rate among 16–24-year-olds exceeding 60 per cent [86], [94]. The phenomenon of *tang ping* — the voluntary withdrawal of 45 per cent of young people from social competition — is the lived manifestation of this degeneration of the attractor S (§7.2).

Spectral mechanism. The system remains locked between the thresholds S_4 and S_5 (§16.3): the topological tension $T = 0.165$ fails to cross $S_5 = 0.17195$ (Fire Transition). Without a creative exogenous shock (TRR), the system attempts a direct transition $C_N \rightarrow C_P$ which violates the rule of the polar thresholds P_4 and N_4 (§13.3), causing retropolarity (§14.1): a sudden inversion of the spectrum, systemic collapse, the tenth dynastic collapse.

18.3.2 — Scenario 2: With TRR (implementation of the 15th Plan)

Scenario 2 —With TRR (implementation of the 15th Five-Year Plan)

Period	η	d	gap	$R_{\text{threshold}}$	$\mathcal{I}_{\text{Stasis}}$	Diagnosis
2026	−0.45	1.9	0.10	0.78	0.64	N_4 activation (TRR)
2028	−0.20	2.8	0.25	0.55	0.48	Incipient stasis
2030	+0.10	3.5	0.35	0.35	0.32	Moderate tension
2032	+0.35	4.2	0.45	0.25	0.22	Healthy growth

MULTI-LEVEL DIAGNOSIS

IN 2032:

- Dominant element: e_1 (Inflation) $\rightarrow \eta_{e_1} = +0.60$
- Active pentad: $P_1^{(e_1)} \rightarrow$ activity 0.72, Sheng mode, frustration 0.08
- Balanced Wuxing: Sheng cycle restored (Wood $\rightarrow$ Fire $\rightarrow$ Earth $\rightarrow$ Metal $\rightarrow$ Water)
- Threshold vector: $\vec{\sigma} = (1, 1, 1, 1, 1, 1)$ —controlled crossing
- Fragmentation index: $\mathcal{F} = 0.18$ —coherent system

Economic interpretation. The TRR [3], [4] is a transition operator T_{mixed} (§17.1) that combines three structural reforms, each of which activates a specific spectral threshold:

- **Two Mountains Bank** (T_{water} , threshold S_6): securitisation of the GEP to circumvent LGFV debt [38], reallocation of capital flows towards ecological regeneration. Pilot scheme in Lishui [92], [93].
- **Integration of GEP into KPIs** ($T_{\text{structure}}$, threshold S_5): reform of policy incentives, replacing GDP with GEP as a performance indicator for local officials.
- **Regenerative Usufruct Certificate (CUR)** (T_{fire} , threshold S_7): reappropriation of rent by local authorities via usufruct certificates on regenerated land. Roll-out in Xingguo and Ganzhou (§20.1.3).

Spectral mechanism. The TRR is a transition controlled by threshold N_4 , which enables the system to leave the C_N belt (recession) and enter the C_P belt (expansion) via a new cycle of Material Rent Extraction (§17.1). The sequential crossing of the thresholds S_5 , S_6 and S_7 corresponds to the deployment timetable of the 15th Five-Year Plan (§17.3, Appendix J).

18.3.3 — Comparison of the two scenarios

Table 34: Comparison of spectral observables in 2032: scenarios with and without TRR

Indicator	Without TRR (2032)	With TRR (2032)	Δ
η_{total}	−0.80	+0.35	+1.15
d_{total}	0.8	4.2	+3.4
$\text{gap}_{\text{total}}$	0.02	0.45	+0.43
$R_{\text{threshold,total}}$	0.95	0.25	−0.70
$\mathcal{I}_{\text{Stasis}}$	0.80 (retopolarity)	0.22 (healthy growth)	−0.58
$\mathcal{I}_{\text{Wuxing}}$	1.4	0.3	−1.1
$\mathcal{F}$ (fragmentation)	0.78	0.18	−0.60

Conclusion. The simulation shows that TRR makes it possible to reverse the trend and emerge from Stasis by gradually crossing the thresholds S_5 , S_6 and S_7 . Without TRR, the system inevitably shifts towards retopolarity and collapse — replicating the tenth dynastic collapse [25]. With TRR, the system reaches a regime of healthy growth ($\mathcal{I}_{\text{Stasis}} < 0.3$) comparable to the Trente Glorieuses (§15.3).

18.4 — Concrete indicators for multi-level governance

The spectral dashboard must be fed by concrete, measurable and up-to-date economic indicators. The multi-level architecture (§18.1, §15.4) allows these indicators to be organised according to five levels of resolution, each corresponding to a decision-making layer (§D.12).

18.4.1 — Level 1: aggregate indicators

The four global observables integrated across the 12 spectral sheets are operationalised by the following indicators:

Table 35: Global spectral management indicators (Level 1)

Observable	Economic indicator	Data source	Frequency	Alert threshold
η_{total}	Growth in GEP vs GDP	Ministry of Ecology	Quarterly	$\eta < 0$
d_{total}	Regional Gini coefficient	National Bureau of Statistics	Annual	$d < 2$
$\text{gap}_{\text{total}}$	Inter-sectoral productivity gap	National surveys	Quarterly	$\text{gap} < 0.2$
$R_{\text{threshold,total}}$	LGFV debt ratio	Ministry of Finance [38]	Quarterly	$R_{\text{threshold}} > 0.7$
$\mathcal{J}_{\text{Stasis}}$	Stasis composite index	Calculated from the 4 observables	Quarterly	> 0.7 (red)

18.4.2 — Level 2: indicators per spectral sheet

The 48 observables per sheet ($\eta_f, d_f, \text{gap}_f, R_{\text{threshold},f}$ for $f \in \{1, \dots, 12\}$) are operationalised by sectoral indicators:

Table 36: Indicators by spectral sheet (Level 2)

Dominant sheet	Observable	Economic indicator	Source	Alert threshold
e_1 (Inflation)	η_{e_1}	R&D expenditure / GDP	Ministry of Science	$\eta_{e_1} < 0$
e_2 (Purchasing power)	η_{e_2}	Median disposable income	NBS	$\eta_{e_2} < -0.3$
e_3 (Profits)	η_{e_3}	Corporate profit margins	Sectoral surveys	$\eta_{e_3} < -0.5$
e_4 (Active buy-back)	η_{e_4}	GEP securitisation volume	Two Mountains Bank	$\eta_{e_4} < -0.5$
e_5 (Dispersion)	η_{e_5}	Net rural migration flows	NBS	$\eta_{e_5} < -0.4$
e_6 (Land)	η_{e_6}	Land prices / income	NBS	$\eta_{e_6} < -0.4$
f_1 (Inflation regulation)	η_{f_1}	Real wage growth	NBS	$\eta_{f_1} < -0.3$
f_2 (Purchasing power regulation)	η_{f_2}	Bank provisioning ratio	CBRC	$\eta_{f_2} < -0.4$
f_3 (Profit regulation)	η_{f_3}	Effective tax rate	Ministry of Finance	$\eta_{f_3} < -0.5$
f_4 (Rent Extraction regulation)	η_{f_4}	Volume of CURs issued	NDRC	$\eta_{f_4} < -0.4$
f_5 (Dispersion regulation)	η_{f_5}	Rural public investment	Ministry of Agriculture	$\eta_{f_5} < -0.3$
f_6 (Land regulation)	η_{f_6}	Property vacancy rate	NBS	$\eta_{f_6} < -0.4$

18.4.3 – Level 3: spectral threshold indicators

The 7 spectral thresholds $S_1, \dots, S_7$ (§13.1) are operationalised by threshold-crossing indicators:

Table 37: Spectral threshold crossing indicators (Level 3)

Threshold	Observable	Economic indicator	Source	Warning threshold
S_1 (Activation)	σ_1	CPI inflation rate	NBS	$\sigma_1 = 0$
S_2 (Symmetry)	σ_2	Export/import balance	Customs	$\sigma_2 = 0$
S_3 (Polarisation)	σ_3	Reserve requirement ratio	PBoC	$\sigma_3 = 0$
S_4 (Coupling)	σ_4	Interministerial coordination	State Council	$\sigma_4 = 0$
S_5 (Fire)	σ_5	Number of active TRR pilot zones	NDRC [95], [96]	$\sigma_5 = 0$
S_6 (Water)	σ_6	Volume of GEP securitisation	Two Mountains Bank	$\sigma_6 = 0$
S_7 (Jump)	σ_7	Share of GEP in GDP	Ministry of Ecology	$\sigma_7 = 0$

18.4.4 – Level 4: pentad indicators

The 144 projected pentads (§9.3, Appendix C) are operationalised by indicators of activity, polarity, connectivity and frustration:

Table 38: Activity indicators for critical pentads (Level 4)

Critical pentad	Observable	Economic indicator	Source	Alert threshold
$N_5^{(f_3)}$ (Stasis/Accumulation)	a_{N_5}	Capacity utilisation rate	NBS	$a_{N_5} > 0.7$
$N_6^{(f_3)}$ (Austerity)	c_{N_6}	Number of sectors in recession	Surveys	$c_{N_6} > 3$
$P_6^{(e_2)}$ (Consumption)	E_{P_6}	Gap between potential and actual consumption	Ministry of Trade	$E_{P_6} > 0.3$
$P_1^{(e_1)}$ (Keynesian/Inflation)	π_{P_1}	Inflation rate (CPI)	Ministry of Science	$\pi_{P_1} = -1$

18.4.5 – Level 5: relational indicators

The 66 relational observables (§15.4.6, §O.5.7) are operationalised by indicators of tension and fragmentation:

Table 39: Indicators of tension and fragmentation (Level 5)

Observable	Economic indicator	Source	Alert threshold
$\mathcal{T}_{\text{Wuxing}}$	Maximum deviation between Wuxing phases	Calculated from the leaves	> 1.0
$\mathcal{F}$ (fragmentation)	Inter-leaf fragmentation index	Calculated from $C(t)$	> 0.5
$\Delta\eta_{\text{max}}$	Maximum asymmetry deviation	Calculated from the leaves	> 1.0

18.4.6 – Multi-level traffic light system

Aggregating the indicators by level produces a hierarchical traffic light system:

Table 40: Hierarchical traffic light system (multi-level aggregation)

Level	Green	Amber	Red
1 (Overall)	$\mathcal{J}_{\text{Stasis}} < 0.3$	$0.3 \leq \mathcal{J}_{\text{Stasis}} < 0.65$	$\mathcal{J}_{\text{Stasis}} \geq 0.65$
2 (Leaf)	$\eta_f > +0.3$	$-0.3 \leq \eta_f \leq +0.3$	$\eta_f < -0.3$
3 (Threshold)	$\sigma_i = 1$ (triggered)	—	$\sigma_i = 0$ (blocked)
4 (Pentad)	$a_{k,g} < 0.5$	$0.5 \leq a_{k,g} < 0.7$	$a_{k,g} \geq 0.7$
5 (Relational)	$\mathcal{T}_{\text{Wuxing}} < 0.5$	$0.5 \leq \mathcal{T}_{\text{Wuxing}} < 1.0$	$\mathcal{T}_{\text{Wuxing}} \geq 1.0$

Aggregation rule. The overall alert level is the maximum of the alert levels across the 5 levels. A red alert at any level triggers a specific intervention procedure (§D.12).

18.4.7 – Summary of indicators

The multi-level dashboard transforms the economic management of a four-pronged exercise into a modernised divinatory practice: a dialogue with the system that enables one to probe its state at five levels of resolution, diagnose its Wuxing imbalances, and adjust TRR policies in harmony with the deep laws that structure complex systems.

The complete architecture of the indicators, their mathematical formalism and their algorithmic implementation are detailed in Appendix O.5 (§O.5.6 to §O.5.10). The integration with the four decision-making levels (Beijing / Province / County / Alert Centre) is presented in Appendix D.12. The graphical visualisations are described in Appendix S (§S.3 to §S.5).

Chapter 19 – Operational Management of the TRR

19.1. Decision-makers' levers of action

Within the framework $CI(6, 6)$, decision-makers have four levers of action corresponding to the four components of the transition operator T :

Lever 1: $T_{\text{structure}}$ (Institutional reforms)

- **Actions:** Incorporating the GEP into KPIs, reforming the planning system, adapting land law.
- **Effect:** Modification of the production model, creation of a new framework for green investment.
- **Target threshold:** S_5 (Fire Transition).

Lever 2: T_{fire} (Management of exogenous shocks)

- **Actions:** Using crises (climate, geopolitical tensions) as catalysts; anticipating shocks.
- **Effect:** Transforming constraints into opportunities; accelerating the transition.
- **Target threshold:** S_5 (Fire Transition) and S_7 (Octave Leap).

Lever 3: T_{water} (Monetary and fiscal policies)

- **Actions:** Establishment of the Two Mountains Bank, securitisation of the GEP, reorientation of credit.
- **Effect:** Redirection of capital flows towards ecological regeneration.
- **Target threshold:** S_6 (Water Transition).

Lever 4: T_{mixed} (Policy coordination)

- **Actions:** Implementation of the CUR, coordination between ministries, alignment with existing plans.
- **Effect:** Coupling of the three components, transition controlled by N_4 .
- **Target threshold:** N_4 (negative polar threshold).

19.2. The operational timetable

The operational timetable proposed by the White Paper is reinterpreted as a sequence of spectral threshold crossings:

Phase 1: Pilot (2024–2025)

- **Objective:** To test TRR mechanisms on a small scale.
- **Actions:** Two Mountains Bank in Lishui; integration of the GEP into local KPIs.

- **Threshold crossed:** S_5 (Fire Transition) – activation of ecological innovation.
- **Target observables:** η changes from -0.45 to -0.35, $R_{\text{threshold}}$ changes from 0.75 to 0.65.

Phase 2: Roll-out (2026–2028)

- **Objective:** To extend the mechanisms to the pilot provinces.
- **Actions:** Widespread adoption of CURs, expansion of the Two Mountains Bank.
- **Threshold crossed:** S_6 (Water Transition) – reform of the monetary and financial system.
- **Target observables:** η falls from -0.35 to -0.15, $R_{\text{threshold}}$ falls from 0.65 to 0.45.

Phase 3: Acceleration (2029–2030)

- **Objective:** Treatment of 5 to 10 million hectares.
- **Actions:** Large-scale roll-out of CURs, massive mobilisation of capital.
- **Threshold crossed:** S_7 (Octave jump) – change of cycle, exit from Stasis.
- **Target observables:** η changes from -0.15 to +0.10, $R_{\text{threshold}}$ changes from 0.45 to 0.35.

Phase 4: Widespread Adoption (2031–2035)

- **Objective:** Full coverage of atrophied rural areas.
- **Actions:** Widespread implementation of the TRR, stabilisation of the new cycle.
- **Threshold crossed:** Stabilisation at C_P (new Rent Extraction cycle).
- **Target observables:** $\eta > +0.30$, $d > 4$, $\text{gap} > 0.4$, $R_{\text{threshold}} < 0.3$.

19.3. Risks in the event of inaction or poor implementation

The Cl(6, 6) model identifies three major risks in the event of inaction or poor implementation:

Risk 1: Continued Stasis

- **Mechanism:** The system remains locked onto attractor S, with no possible transition.
- **Consequences:** Erosion of the social contract, increased inequality, risk of uprisings.
- **Observables:** η remains negative, $R_{\text{threshold}}$ remains greater than 0.7.

Risk 2: Retropolarity

- **Mechanism:** An attempt to exit Stasis via a direct transition $C_N \rightarrow C_P$ without passing through N_4 .
- **Consequences:** Hyperinflation, social chaos, systemic collapse.

- **Observables:** η falls sharply, $\text{gap} \rightarrow 0$, $R_{\text{threshold}} \rightarrow 1$.

Risk 3: Tenth Dynastic Collapse

- **Mechanism:** A repetition of the fatal cycle of the nine dynastic declines, but on the scale of a modern state.
 - **Consequences:** Collapse of the dynasty, political chaos, territorial fragmentation.
 - **Observables:** $\eta < -0.8$, $d < 1$, $\text{gap} \approx 0$, $R_{\text{threshold}} \approx 1$.
-

Chapter 20 – The Empirical Mapping of TRR

20.1. The Empirical Reality: TRR Pilot Zones

The term ‘Territorial Regeneration Rent’ (TRR) is a theoretical concept. In official Chinese terminology, these mechanisms are classified under the heading ‘Mechanisms for Realising the Value of Ecological Products’ (生态产品价值实现机制) or ‘Two Mountains Banks’ (两山银行).

Here is a map of the main areas where the CCP state has already begun to implement the TRR:

20.1.1. The Integrated Epicentre: Lishui Municipality (Zhejiang)

- **Official Chinese name:** 国家生态产品价值实现机制试点 / 丽水”两山银行”
- **Degraded assets being bought back:** Depleted agricultural land, underutilised forests, villages abandoned due to rural exodus.
- **The TRR mechanism:** This is the purest form of the model. The Two Mountains Bank acts as a State-CPC platform that aggregates the usage rights to this degraded land. It then issues loans backed by these assets. The injected credit finances ecological restoration and rural development, creating massive ‘advance exogenous demand’ in the form of local jobs.

20.1.2. Forest TRR: The cities of Sanming and Nanping (Fujian)

- **Official Chinese name:** 林票制度 / 森林生态银行
- **Degraded assets being repurchased:** Fragmented, unmanaged communal forests with no market value.
- **The TRR mechanism:** Villagers own shares in forests but lack the capital to manage them. State-owned companies buy back the management rights and provide an advance loan to hire local people to replant, maintain and sustainably manage the forest. In exchange, the villagers receive ‘Lin-piao’ (forestry vouchers), which are negotiable financial instruments.

20.1.3. Water and Soil TRR: Xingguo County and Ganzhou City (Jiangxi)

- **Official Chinese name:** 山水林田湖草生态保护修复工程 / 横向生态补偿
- **Degraded natural assets being restored:** Eroded catchment areas, land contaminated by rare-earth mining, and choked watercourses.
- **The TRR mechanism:** Upstream towns receive funding from the CCP state and ‘ecosystem service payments’ from downstream towns to decontaminate and restore the soil.

20.1.4. TRR on a Macro-Scale: Hainan Province

- **Official Chinese name:** 国家生态文明试验区 / GEP 核算体系
- **Degraded assets redeemed:** The entire island territory, threatened by coastal property speculation.
- **The TRR mechanism:** South Island has banned speculative property development. The “GEP” (the green equivalent of GDP) is used as an indicator to assess local officials and as collateral to secure massive bank loans.

20.1.5. The Transitional TRR (Material Carbon): The city of Huzhou (Zhejiang) and Chongyi County (Jiangxi)

- **Official Chinese name:** 竹林碳汇 / 林业碳票
- **Degraded wealth repurchased:** Ageing, unharvested bamboo groves.
- **The TRR mechanism:** This acts as a bridge between the physical TRR and the Third Intangible Transformation. The CCP state funds the physical maintenance of bamboo forests by trading the carbon credits generated on the Chinese domestic market.

20.2. The ‘photovoltaic+’ model: a large-scale proto-TRR

Over the past decade, China has developed an original model of territorial regeneration based on the synergy between renewable energy production, ecological regeneration and rural development. This model, known as ‘photovoltaic+’ (光伏+), rests on three complementary pillars:

First pillar: an income-generating economic infrastructure

The solar power station forms the cornerstone of the scheme. By reducing water evaporation by 20 to 30 per cent and halving wind speeds, the panels create a microclimate conducive to the regrowth of vegetation in areas where it could no longer survive. In Talatan, in the Gobi Desert in Qinghai, an area that was once 98.5 per cent desertified has seen vegetation recover beneath the panels, with vegetation cover reaching 80 per cent. Annual grass production there stands at 118,000 tonnes, enough to feed nearly 100,000 sheep.

Second pillar: profitable agricultural or livestock activities

The space beneath and between the panels is not left unused. The so-called ‘photovoltaic sheep’ (光伏羊) model has been systematically implemented in Qinghai: ewes graze beneath the panels, eating the grass (which reduces the power stations’ maintenance costs) and their manure fertilises the soil.

Third pillar: anchoring external demand in the local area

Whereas AI and data concentrate value in a few major cities, the ‘photovoltaic+’ model creates mass local employment. Wages paid for construction, maintenance and operation are spent locally, triggering a regional multiplier effect. This is precisely what the intangible sector lacks to lift China out of Stasis.

Chapter 21 – The TRR and the Preservation of the Mandate of Heaven

21.1. The Mandate of Heaven as an implicit social contract

Within the framework of the morphogenetic process, the Mandate of Heaven is not a cosmological abstraction. It is an implicit social contract between the CCP State and the population: the CCP State ensures prosperity, stability and social harmony; the population grants legitimacy to the CCP State.

If the CCP State no longer fulfils its part of the bargain (declining prosperity, mass unemployment, growing inequalities), the Mandate erodes and the costs of coercion rise. The alternative paths threaten this contract in three ways:

- **Polarised growth 2.0:** No mass job creation → youth unemployment, inequality, *tang ping*. The CCP state loses its ability to deliver shared prosperity.
- **Neo-feudal withdrawal:** Abandonment of the peripheries → territorial divisions, rural uprisings, collapse of virtuous legitimacy.
- **Authoritarian ecological planning:** Temporary public sector jobs, economic inefficiency, corruption → unpopularity.

21.2. The TRR as a restoration of political virtue

The TRR is the only path that tangibly repairs the damage caused by the previous cycle of accumulation: ghost towns are repurposed, withered countryside is regenerated, and groundwater levels are restored. In doing so, it restores the sovereign’s political virtue (德, *De*) – the legitimacy based on the ability to ensure harmony between Heaven, Earth and Mankind.

21.3. TRR as a response to the demographic crisis

Falling birth rates and an ageing population are reducing the available labour force. The TRR addresses this challenge through:

- **Mass rural employment:** young people can find productive work in revitalised rural areas (agrivoltaics, aquaculture, ecotourism), without having to migrate to overcrowded cities.
- **The attractiveness of rural areas:** vibrant countryside, with services, schools and jobs, encourages higher birth rates and slows the exodus.
- **Reduced living costs:** ecological regeneration (water, soil, energy) lowers the cost of living in rural areas.

Chapter 22 — The geopolitical impact of TRR: from spatial avoidance to the diplomacy of regeneration

22.1. Introduction — Geopolitics as an external projection of the $Cl(6, 6)$ topology

The preceding chapters have established that post-2018 China is locked onto the attractor S (dispersion + nodal point), with a composite Stasis index $\mathcal{J}_{\text{Stasis}} = 0.64$ (§18.1.3). The Territorial Regeneration Rent (TRR) has been formalised as the transition operator T_{mixed} , enabling a transition controlled by the threshold N_4 (§17.1). However, this analysis has remained strictly internal: it treats China as a closed system.

However, the $Cl(6, 6)$ topology does not stop at borders. The formalism of the 12 spectral sheets and the 144 projected pentads can be extended to the international arena: every trading partner, every host state for Chinese investment, and every geopolitical bloc can be modelled as a spectral sheet projected onto the Hilbert space $\mathcal{H}$ of the world-system. Geopolitics thus becomes the dynamics of inter-sheet couplings, governed by the same laws of nilpotence and the same spectral thresholds as the internal dynamics.

This chapter elaborates on this extension. It shows that the TRR is not merely a domestic policy of ecological regeneration: it is the condition of possibility for a complete overhaul of China's geopolitical stance, marking the end of external spatial avoidance and the advent of a 'diplomacy of regeneration'. This diplomacy offers the Global South an alternative model to Western imperialism — no longer the extraction of rent, but the co-regeneration of Value.

22.2. Theoretical framework: TRR as a way out of the spatial avoidance paradigm

22.2.1. Recap: spatial avoidance in the Theory of Rent Extraction (§2.2 and §5.5)

The Theory of Rent Extraction (ToRE) has established that growth is based on the transfer of assets from the Space of Value to the Space of Wealth, financed by advance credit. This process has an absolute limit: the Nodal Point, reached when the stock of readily mobilisable Value is exhausted (§1.6). To postpone this deadline, economic systems have two avoidance mechanisms at their disposal:

- **Spatial avoidance:** geographical expansion (conquests, colonies, frontier expansion) or export to new markets.
- **Temporal avoidance:** postponing the crisis through debt, fiat currency and financialisation.

Imperial Chinese history has illustrated these two mechanisms (Appendix G): the Han, Tang, Yuan and Qing dynasties successively employed spatial avoidance (conquests, the Silk Road, westward expansion); the Song, Southern Jin and Ming dynasties employed temporal avoidance (fiat currency, inflows of silver, credit). Post-1978 China has replicated this duality on a continental scale: internal spatial avoidance (urbanisation, mega-infrastructure, Tudi Caizheng) followed by external spatial avoidance (globalisation, the Belt and Road Initiative, productive investments abroad) and temporal avoidance (LGFV debt, SECC).

22.2.2. The simultaneous exhaustion of both forms of avoidance in 2018

The spectral analysis in Chapter 16 established that China reached its terminal Nodal Point around 2018:

- **Internal spatial avoidance saturated:** urbanisation rate > 65 per cent, land with high externalities fully absorbed, mega-infrastructure projects in the West reaching their limits of profitability (§5.1).
- **External spatial avoidance saturated:** productive investments abroad (HPAL factories in Indonesia, projects in Venezuela) come up against the Sovereignty Rent of host states (§5.5.3). Indonesian Decree No. 144 (2026) demonstrated the structural vulnerability of these assets [43].
- **Saturated temporal avoidance:** LGFV debt at 60,000 billion yuan (48 per cent of GDP), with the SECC generating external rents without channelling them into mass employment (§6.3.3).

The threshold vector $\vec{\sigma} = (1, 1, 1, 1, 0, 0, 0)$ (§18.1.3) indicates that China has crossed the first four thresholds (exhaustion of the previous model) but remains stuck between S_4 and S_5 . The avoidance mechanisms are structurally exhausted.

22.2.3. TRR as a qualitative substitute for quantitative expansion

TRR brings about a fundamental epistemological break: it substitutes qualitative regeneration (recreating Value where Value had been destroyed) for quantitative expansion (constantly acquiring more virgin land, conquering ever more territories).

Formally, TRR is a T_{mixed} operator (§17.1) that combines:

- $T_{\text{structure}}$: the creation of a new productive model underpinned by territorial ecology.
- T_{water} : reorientation of financial flows (Two Mountains Bank, securitisation of the GEP).
- T_{fire} : use of the climate crisis as a catalyst (Regenerative Usufruct Certificates).

This combination complies with the conservation laws of ToRE: it does not create Value *ex nihilo*, but regenerates degraded Value. It enables the controlled crossing of the N_4 threshold, avoiding retropolarity (systemic collapse) which penalises any direct transition $C_N \rightarrow C_P$ (§13.3).

Geopolitical projection: If the TRR is a transition operator for China, it can be projected onto international partners. Every country in the Global South stuck in its own Nodal Point can import the operator T_{mixed} to trigger its own cycle of ecological Rent Extraction. Chinese geopolitics ceases to be a logic of spatial avoidance (exporting overcapacity, conquering markets) and becomes a logic of co-regeneration (exporting a model of value creation).

22.3. TRR as a substitute for Western economic imperialism

22.3.1. Limitations of the Western model: financialisation and retropolarity

The West has temporarily ‘hijacked’ the fatal cycle of the Nodal Point (§3.1–3.3) through two mechanisms:

- **Spatial imperialism:** colonies, natural resources, and foreign markets to absorb overproduction.
- **Temporal avoidance:** financialisation, money creation through private credit, and speculative bubbles.

But having exhausted its physical resources (colonisation completed around 1914) and its financial resources (the 2008 crisis revealing the impossibility of maintaining fictitious bubbles indefinitely), the West is sinking into a topological impasse similar to that of China. Its response is massive monetisation (quantitative easing, dilutive inflation) and an attempt at a direct transition from C_N to C_P through the injection of liquidity – which, according to the Cl(6, 6) model, violates nilpotence and paves the way for global systemic retropolarity (§14.2).

22.3.2. The TRR as an endogenous recreation of Value (vs. extraction)

TRR proposes a radically different model. Instead of extracting Value (from colonies, mineral resources, financial rent), it regenerates Value. Instead of transferring Wealth from the peripheries to the centre (substitution of externalities, §1.5), it creates Wealth where it had been destroyed.

Formally, TRR complies with the law of conservation of ToRE: it does not create value *ex nihilo*, but transforms degraded value (impoverished soils, under-utilised forests, eroded catchment areas) into regenerated value (restored ecosystems, securitised GEPs, rent-generating CURs). This transformation constitutes a new cycle of Rent Extraction, but one based on qualitative regeneration rather than quantitative expansion.

22.3.3. Comparative table: Western imperialism vs. diplomacy of regeneration

The following table compares the two models for exiting the Nodal Point:

Table 41: Geopolitical comparison of strategies for exiting the Nodal Point

Dimension	Western model (Financialisation)	Post-2018 China (SECC/Enclaves)	Post-TRR China (T_{mixed})
Topological mechanism	Forced direct transition $C_N \rightarrow C_P$ (violation of nilpotence)	External spatial avoidance (Fortified Port Enclaves) + temporal avoidance (SECC)	Transition controlled by the N_4 threshold
Underlying asset	Fictitious currency (debt bubbles)	Intangible tokens / mining ports	Territorial GEP (regeneration)
Social impact	Destruction of industrial jobs, inequalities	Algorithmic stasis, <i>Tang Ping</i>	Mass rural employment, restoration of <i>De</i>
Geopolitical stance	Monetary imperialism + sanctions	Logistical imperialism + Sovereignty rent	Regeneration diplomacy (export of the TRR model)
Expected spectral outcome	Retropolarity (terminal collapse)	Prolonged stasis (slow decline)	Healthy growth ($\mathcal{J}_{\text{Stasis}} < 0.3$)

Interpretation: Both the Western model and the post-2018 Chinese model (SECC/Enclaves) are attempts to avoid the Nodal Point. The former violates nilpotence (direct transition $C_N \rightarrow C_P$), whilst the latter saturates the avoidance mechanisms (closed outer pioneer front, SECC without spatial thickness). Only the post-TRR model complies with topological laws: it activates the operator T_{mixed} , which enables a transition controlled by N_4 , creating a new underlying asset (the Regenerative Value) and restoring the Rent Extraction cycle.

22.4. Redefinition of the New Silk Roads (BRI)

22.4.1. Diagnosis: relative failure of productive investment

The Belt and Road Initiative (BRI), launched in 2013, was conceived as a mechanism for external spatial avoidance: to export China's industrial overcapacity to the countries of the Global South, secure flows of raw materials, and create new markets. However, the results have been mixed (§5.4):

- **Transport corridors** (gas pipelines, railways, maritime routes): a relative success, as they are underpinned by guaranteed flows and are immune to Sovereign Rent.
- **Local productive investments** (HPAL plants in Indonesia, projects in Venezuela): a structural failure, as they are vulnerable to the 'sovereign rent' of the host states. Indonesian Decree No. 144 (2026) demonstrated that a sovereign state can unilaterally change the rules of the game, rendering the Chinese financial arrangement untenable [43], [44].
- **Fortified Port Enclaves** (Chancay, Piraeus, Hambantota): a success, as they are designed to shield investments from sovereign rent (§5.5.5).

The lesson is clear: the BRI cannot serve as a new material frontier for China. Productive investment abroad does not generate the 'exogenous demand lead' needed to reignite the Chinese engine. The economic returns are too low and too slow to offset the sudden halt of the domestic property-driven engine [46].

22.4.2. The model of Fortified Port Enclaves: a geopolitical turning point, not a new cycle

The port of Chancay (Peru), inaugurated in November 2024, is the archetype of Fortified Port Enclaves (§5.5.5). Unlike HPAL plants, Chancay is a purely extractive infrastructure: a mineral port designed to export South American copper, lithium and iron ore to China, with no local processing capacity whatsoever.

The contracts include structural conditions that constrain the host state's sovereign manoeuvre: exclusive operating rights granted to COSCO, regulatory exemptions secured through legislation, and an all-assets pledge securing all the project's assets. This structure makes any unilateral change to the rules by the host state economically destructive.

But Chancay is not a new cycle of 'Rent Extraction'. It is a geopolitical merry-go-round: it secures the flow of raw materials, but does not create value in China. It does not generate mass employment, does not restore the Sheng cycle, nor does it cross the N_4 threshold. It is a defensive response to the exhaustion of the material frontier, not a solution to Stasis.

22.4.3. Exporting the TRR model: the BRI as 'Two Mountains diplomacy'

The TRR offers a strategic alternative: instead of exporting an extraction model (logistical colonisation), China can export a model of topological regeneration. Developing countries, which are themselves stuck at their Nodal Point (exhaustion of arable land, ecological degradation, unsustainable debt), can import the operator T_{mixed} to trigger their own cycle of ecological Rent Extraction.

Formally, the projection of the operator T_{mixed} onto the spectral sheets of partner countries is written as:

$$T_{\text{mixed}}^{(\text{partner})} = T_{\text{structure}}^{(\text{partner})} + T_{\text{fire}}^{(\text{partner})} + T_{\text{water}}^{(\text{partner})}$$

where each component is tailored to the local context:

- $T_{\text{structure}}^{(\text{partner})}$: local institutional reform (integration of GEP into national KPIs, land law reform).
- $T_{\text{fire}}^{(\text{partner})}$: use of exogenous shocks (climate crisis, ecological degradation) as catalysts.
- $T_{\text{water}}^{(\text{partner})}$: reorientation of financial flows (securitisation of ecological assets, creation of local green banks).

This projection complies with the conservation laws of the ToRE: it does not create Value *ex nihilo*; it regenerates locally degraded Value. It enables the controlled crossing of the N_4 threshold for the partner country, avoiding retropolarity (systemic collapse) which penalises any attempt at direct transition.

22.4.4. Regional case studies

South-East Asia (Indonesia, Laos)

Indonesia activated its Sovereign Rent via Decree No. 144 (2026), rendering the financial arrangements for the Chinese HPAL plants in Sulawesi unsustainable [43]. China was forced to withdraw, leaving Indonesia with unfinished infrastructure and unpaid debts, whilst bearing part of the cost itself (§5.5.2).

Beijing has learnt its lesson: future investments must be designed to be more resilient to the host country's Sovereign Rent. The TRR offers an alternative: instead of exporting vulnerable factories, China can export the Two Mountains Bank model, helping Indonesia to securitise its ecological assets (forests, mangroves, coral reefs) to finance their regeneration. This approach respects Indonesian sovereignty (no full collateral, no regulatory exemptions) whilst creating value locally.

Laos, faced with the collapse of its external debt and the degradation of its forests (deforestation for rubber plantations), can adopt the Regenerative Usufruct Certificates (CUR) model: local communities receive financial securities backed by forest regeneration, thereby financing their own development without relying on vulnerable Chinese investment.

Africa (Madagascar, Angola, Tanzania)

Sub-Saharan Africa is the testing ground for regeneration diplomacy. China has rolled out a comprehensive agricultural investment strategy there (§5.5.4): Madagascar (technology transfer, hybrid rice), Angola (130,000 hectares of soya and maize), Tanzania (32,000 hectares), Zambia (99-year leases), DRC (2.8 million hectares of palm oil).

These operations have low externalities (§5.5.4): they secure supply chains, but do not create a cycle of buy-back in China. TRR offers an alternative: instead of leasing land for export, China can help African countries regenerate their own soils (combating desertification, agroforestry, watershed restoration), thereby creating value locally and reducing dependence on Chinese imports.

The Two Mountains Bank model can be adapted to the African context: securitisation of ecological assets (forests, soil, water) to finance regeneration, the creation of mass rural employment, and a reduction in rural-to-urban migration. This approach respects African sovereignty (no 99-year leases, no full collateral) whilst creating value locally.

Latin America (Peru–Chancay, Venezuela)

Peru illustrates the dual nature of Chinese investment: Chancay (a fortified port enclave, a success story) versus agricultural and mining projects (vulnerable to 'sovereign rent'). The TRR offers an alternative: rather than

concentrating investment on mining (copper, lithium), China can help Peru regenerate its Andean ecosystems (restoring high-altitude pastures, combating soil erosion, protecting water sources), thereby creating value locally and reducing dependence on raw material exports.

Venezuela provides a textbook example: Chinese loans secured against oil (over \$50 billion between 2007 and 2015) came up against institutional collapse, triggering the Sovereign Rent not by decree, but through the state's bankruptcy [46]. China was forced to accept losses and debt rescheduling. The TRR offers an alternative: instead of lending against oil (a vulnerable, underground asset), China can help Venezuela securitise its ecological assets (Amazonian forests, biodiversity, water resources), thereby creating value locally and reducing dependence on oil.

Russia (49-year agricultural leases)

Russia is the most successful example of Chinese agricultural outsourcing: over 670,000 hectares leased (estimated at 1 million hectares), 49-year contracts, rent of 300 to 600 yuan per hectare, and a requirement to employ at least 30 per cent Russian labour (§5.5.4). These leases represent a niche accumulation: they secure supply flows (soya, wheat, maize, meat), but do not create a cycle of buy-back in China.

The TRR offers an alternative: instead of leasing land for export, China can export the Two Mountains Bank model, helping Russia to securitise its ecological assets (boreal forests, black soil, water resources) to finance their regeneration. This approach respects Russian sovereignty (no 49-year leases, no full collateral) whilst creating value locally.

22.5. Sino-American rivalry: two models for exiting the Nodal Point

22.5.1. The West: forced transition $C_N \rightarrow C_P$ (violation of nilpotence)

Since the 1970s, the Western economy has been in a topological impasse similar to that of China, but has responded with financialisation (fictitious bubbles) and massive monetisation (dilutive inflation). The Cl(6, 6) model predicts that this attempt at a direct transition $C_N \rightarrow C_P$ (without passing through the polar thresholds P_4 or N_4) violates nilpotence and paves the way for global systemic retropolarity (§14.2).

The spectral observables for the West in 2026 confirm this diagnosis:

- $\eta_{\text{West}} \approx -0.3$: moderate recession, Ke dominance.
- $d_{\text{West}} \approx 2.5$: moderate fragmentation.
- $\text{gap}_{\text{West}} \approx 0.15$: high fragility.
- $R_{\text{threshold, West}} \approx 0.65$: high tension, close to the red alert threshold (0.7).

The West is attempting to force the transition through the injection of liquidity (quantitative easing, negative interest rates, debt monetisation), but this approach violates the principle of nilpotence: it does not create Value; it dilutes existing Wealth. The model predicts a reversal of polarity by 2030–2035: systemic collapse, social chaos, territorial fragmentation.

22.5.2. China: transition controlled by N_4 via the TRR

China, through the TRR, activates the operator T_{mixed} , which complies with the law of conservation by creating a new underlying asset (regenerative value). The controlled crossing of the N_4 threshold enables the system to leave the recession belt (C_N) and enter the expansion belt (C_P) via a new cycle of Material Rent Extraction (§17.1).

China's spectral observables in 2032 (scenario with TRR, §18.3.2) confirm this trajectory:

- $\eta_{\text{China}} = +0.35$: moderate expansion, Sheng dominance.
- $d_{\text{China}} = 4.2$: harmony and diversity restored.
- $\text{gap}_{\text{China}} = 0.45$: system robustness.
- $R_{\text{threshold,China}} = 0.25$: exit from the red zone.
- $\mathcal{J}_{\text{Stasis,China}} = 0.22$: healthy growth.

22.5.3. The asymmetric advantage of endogenous regeneration

The TRR offers China a decisive asymmetric advantage in the Sino-American rivalry:

- **Compliance with nilpotence:** China complies with topological laws (transition controlled by N_4), whilst the West violates them (direct transition $C_N \rightarrow C_P$).
- **Endogenous creation of value:** China creates value locally (ecological regeneration), whilst the West dilutes existing wealth (monetisation of debt).
- **Mass employment:** the TRR generates mass rural employment (agrivoltaics, aquaculture, ecotourism), whilst Western financialisation destroys industrial jobs.
- **Political legitimacy:** the TRR restores the Mandate of Heaven (harmony between Heaven, Earth and humankind), whilst Western financialisation erodes the legitimacy of institutions.

This asymmetric advantage is neither military nor technological: it is topological. China masters the topologically authorised trajectory, whilst the West is sinking into a forbidden trajectory (retropolarity).

22.5.4. The SECC as a temporary complement

The Computing Capacity Export Strategy (SECC, §6.3.3) is not at odds with the TRR: it serves as a temporary complement to it. The SECC generates considerable external rents (gross margins of 72–86 per cent for Chinese hyperscalers) which underpin trade surpluses and the hyperscalers' balance sheets in the short term. However, it does nothing to address the deficit in internal spatial depth: with an employment elasticity of 0.14 and a domestic multiplier of 1.2 [82], [85], the intangible sector remains structurally incapable of recreating the mass employment needed to replace land-based finance.

The SECC acts as a palliative buffer: it makes it possible to artificially maintain social stability in the absence of new drivers of growth (§6.4.4). However, it cannot replace the TRR as the main engine for emerging from Stasis.

22.5.5. Geopolitical Comparison Table

The following table compares the two models for emerging from the Nodal Point:

Table 42: Geopolitical comparison of exit strategies from the Nodal Point

Strategy	Western model (Financialisation)	Post-2018 China (SECC/Enclaves)	Post-TRR China (T_{mixed})
Topological mechanism	Forced direct transition $C_N \rightarrow C_P$ (violation of nilpotence)	External spatial avoidance (Fortified Port Enclaves) + temporal avoidance (SECC)	Transition controlled by the N_4 threshold
Underlying asset	Fictitious currency (debt bubbles)	Intangible tokens / mining ports	Territorial GEP (regeneration)
Social impact	Destruction of industrial jobs, inequalities	Algorithmic stasis, <i>Tang Ping</i>	Mass rural employment, restoration of <i>De</i>
Geopolitical stance	Monetary imperialism + sanctions	Logistical imperialism + Sovereignty rent	Regeneration diplomacy (export of the TRR model)
Expected spectral outcome	Retropolarity (terminal collapse)	Prolonged stasis (slow decline)	Healthy growth ($\mathcal{J}_{\text{Stasis}} < 0.3$)

22.6. The Global South as a laboratory for TRR

22.6.1. Countries of the Global South also stuck at their Nodal Point

The $\text{CI}(6, 6)$ model does not apply solely to China: it applies to all complex economic systems. Countries in the Global South are also stuck at their Nodal Point:

- **Depletion of arable land:** soil degradation, desertification, erosion.
- **Ecological degradation:** deforestation, water pollution, loss of biodiversity.
- **Unsustainable debt:** external debt, climate debt, social debt.
- **Rural exodus:** the collapse of the countryside, overcrowding in cities.

These countries lack the financial resources to adopt the Western model (financialisation, monetisation of debt). They lack the technological resources to adopt the post-2018 Chinese model (SECC, Fortified Port Enclaves). But they can adopt the TRR model: a model based on ecological regeneration, mass rural employment and the endogenous creation of value.

22.6.2. Export of the TRR institutional package

The TRR institutional package comprises three structural reforms (§17.2):

- **Two Mountains Bank:** securitisation of GEP to finance ecological regeneration.

- **Integration of GEP into KPIs:** reform of policy incentives, replacing GDP with GEP as a performance indicator.
- **Regenerative Usufruct Certificates (CUR):** reappropriation of land rent by local authorities via usufruct certificates on regenerated land.

These three reforms can be adapted to the local context of each country in the Global South:

- **Sub-Saharan Africa:** securitisation of forests, land and water resources; integration of GEP into national development plans; CURs for local communities.
- **South-East Asia:** securitisation of mangroves, coral reefs and tropical forests; integration of GEP into provincial KPIs; CURs for indigenous peoples.
- **Latin America:** securitisation of the Amazon, high-altitude pastures and water resources; integration of GEP into national constitutions; CURs for indigenous communities.

22.6.3. Reform of international financial institutions

Regeneration diplomacy proposes a reform of international financial institutions (IMF, World Bank): instead of lending against sovereign guarantees (vulnerable to Sovereign Rent), these institutions can lend against securitised ecological assets (GEP, CUR). This approach respects the sovereignty of borrowing countries (no political conditionalities, no imposed austerity) whilst creating value locally.

Formally, the projection of the operator T_{mixed} onto the spectral sheets of international financial institutions is written as:

$$T_{\text{mixed}}^{(\text{IMF})} = T_{\text{structure}}^{(\text{IMF})} + T_{\text{fire}}^{(\text{IMF})} + T_{\text{water}}^{(\text{IMF})}$$

where each component is tailored to the institutional context:

- $T_{\text{structure}}^{(\text{IMF})}$: reform of conditionalities (replacing austerity with regeneration).
- $T_{\text{fire}}^{(\text{IMF})}$: use of climate shocks as catalysts.
- $T_{\text{water}}^{(\text{IMF})}$: reorientation of financial flows (securitisation of ecological assets).

22.7. Geopolitical Outlook 2026–2045

22.7.1. Scenario 1: China successfully completes its TRR transition → a new standard of civilisation

If China successfully completes its TRR transition (scenario 2, §18.3.2), it will achieve a healthy growth regime ($\mathcal{J}_{\text{Stasis}} < 0.3$) by 2032. This success enables it to export the TRR model to the Global South, thereby proposing a new standard of civilisation: the Celestial Economic Way (Tiandao Jingji, 天道经济).

This standard is based on:

- **Compliance with topological laws:** controlled transition via spectral thresholds, adherence to nilpotence.

- **Endogenous creation of value:** ecological regeneration, mass rural employment.
- **Harmony between Heaven, Earth and Humankind:** restoration of the Mandate of Heaven, Sheng/Ke balance.
- **Diplomacy of regeneration:** export of the TRR model, co-regeneration of value with partner countries.

Spectral observables of the world-system in 2045 (scenario 1):

- $\eta_{\text{world}} = +0.4$: moderate expansion, Sheng dominance.
- $d_{\text{world}} = 4.5$: harmony and diversity.
- $\text{gap}_{\text{world}} = 0.5$: system robustness.
- $R_{\text{threshold,world}} = 0.2$: exit from the red zone.
- $\mathcal{J}_{\text{Stasis,world}} = 0.2$: healthy growth.

22.7.2. Scenario 2: Partial failure → bipolarisation of the world system

If China suffers a partial failure (incomplete TRR transition, resistance from land-owning elites, worsening climate crisis), the world system becomes bipolarised:

- **Chinese bloc:** countries that have adopted the TRR model, healthy growth, Sheng/Ke harmony.
- **Western bloc:** countries that have persisted with financialisation, retropolarity, systemic collapse.

Spectral observables of the world system in 2045 (scenario 2):

- $\eta_{\text{world}} \approx 0$: precarious balance between expansion and recession.
- $d_{\text{world}} \approx 2$: moderate fragmentation.
- $\text{gap}_{\text{world}} \approx 0.15$: high fragility.
- $R_{\text{threshold,world}} \approx 0.6$: high tension.
- $\mathcal{J}_{\text{Stasis,world}} \approx 0.5$: emerging stasis.

22.7.3. Scenario 3: Chinese retropolarity → global systemic chaos

If China fails completely (no TRR transition, persistence of Stasis, retropolarity), the world-system descends into systemic chaos. Chinese retropolarity (systemic collapse, 10th dynastic collapse) leads to global retropolarity: collapse of supply chains, global financial crisis, social chaos.

Spectral observables of the world system in 2045 (scenario 3):

- $\eta_{\text{world}} = -0.7$: deep recession, Ke dominance.
- $d_{\text{world}} = 1$: excessive concentration, extreme fragmentation.

- $\text{gap}_{\text{world}} = 0.02$: absolute fragility.
- $R_{\text{threshold,world}} = 0.95$: critical saturation.
- $\mathcal{J}_{\text{Stasis,world}} = 0.85$: retropolarity.

22.7.4. Correlation with spectral observables

The three scenarios can be linked to the spectral observables of the multi-level dashboard (§18.1):

- **Scenario 1:** sequential crossing of thresholds S_5, S_6, S_7 (§17.3), stabilisation at the C_P belt by 2035.
- **Scenario 2:** partial crossing of the thresholds (S_5 and S_6 crossed, S_7 blocked), stabilisation at a moderate tension regime.
- **Scenario 3:** stagnation between S_4 and S_5 , retropolarity by 2032.

The strategic window of the 15th Five-Year Plan (2026–2030) is therefore crucial: it is during this period that China must cross thresholds S_5 (Fire Transition) and S_6 (Water Transition) to avoid retropolarity.

22.8. Summary of the chapter

This chapter has set out the geopolitical extension of the $\text{Cl}(6,6)$ formalism. We have:

- **Established** that the TRR is not merely a domestic policy: it is the condition of possibility for a complete overhaul of China’s geopolitical posture, marking the end of external spatial avoidance and the advent of a ‘diplomacy of regeneration’.
- **Compared** the three models for exiting the Nodal Point:
 - Western model (financialisation): violation of nilpotence, retropolarity.
 - Post-2018 Chinese model (SECC/Enclaves): saturated spatial/temporal avoidance, prolonged stasis.
 - Post-TRR Chinese model (T_{mixed}): transition controlled by N_4 , healthy growth.
- **Redefined** the BRI as ‘Diplomacy of the Two Mountains’: rather than exporting an extractive model (logistical colonisation), China can export a model of topological regeneration, offering the Global South an alternative to Western imperialism.
- **Established** the asymmetric advantage of endogenous regeneration: China controls the topologically permitted trajectory, whilst the West is sinking into a forbidden trajectory (retropolarity).
- **Proposed** three prospective scenarios for 2026–2045:
 - Scenario 1: successful TRR transition $\rightarrow$ new standard of civilisation (Tiandao Jingji).
 - Scenario 2: partial failure $\rightarrow$ bipolarisation of the world system.
 - Scenario 3: Chinese retropolarity $\rightarrow$ global systemic chaos.

Conclusion: The stakes of the TRR extend beyond China's borders. By demonstrating that it is possible to emerge from the Nodal Point without massive destruction of wealth, China offers the world the first operational application of the physics of regulation. The $CI(6, 6)$ formalism, a mathematised translation of Chinese structural wisdom, is becoming a geopolitical tool of the first order, capable of rallying the Global South by offering it a path to development that is, at last, sustainable. The Celestial Economic Way (Tiandao Jingji) is not an ideology: it is the mathematisation of the profound laws that structure complex systems, applied to the regulation of the global economy.

Chapter 23 – Summary of Part III

This Part III has applied the formalism $CI(6, 6)$ to the specific case of post-2018 China and to the Territorial Regeneration Rent, deploying the diagnosis and solution across three complementary dimensions: domestic (Stasis and its transition operators), legitimising (the Celestial Mandate) and geopolitical (regeneration diplomacy). We have:

- **Diagnosed** Chinese Stasis as a blockage on attractor S (dispersion + nodal point), with the transition operators ($T_{\text{structure}}$, T_{fire} , T_{water} , T_{mixed}) blocked.
- **Identified** the Nodal Point as the crossing of the first four spectral thresholds (S_1 to S_4), with a blockage between S_4 and S_5 .
- **Formalised** the TRR as a transition controlled by the threshold N_4 , activated by a mixed transition operator T_{mixed} combining $T_{\text{structure}}$, T_{fire} and T_{water} .
- **Interpreted** the three structural reforms (Two Mountains Bank, GEP, CUR) as components of the operator T_{mixed} , targeting T_{water} , $T_{\text{structure}}$ and T_{fire} respectively.
- **Proposed** a spectral dashboard for steering the 15th Five-Year Plan, with observables (η , d , gap, $R_{\text{threshold}}$) and alert thresholds.
- **Simulated** two scenarios: with TRR (exit from Stasis, new Rent Extraction cycle) and without TRR (retropolarity, collapse).
- **Defined** an operational timetable as a sequence of crossing spectral thresholds (S_5 , S_6 , S_7 , N_4).
- **Presented** the empirical mapping of TRR pilot areas (Lishui, Sanming, Xingguo, Hainan, Huzhou).
- **Established** the geopolitical dimension of the TRR as a departure from the paradigm of external spatial avoidance. The TRR replaces quantitative expansion (conquests, colonies, export of overcapacity) with a qualitative regeneration of Value, respecting the laws of conservation of ToRE and avoiding the retropolarity that penalises direct transitions $C_N \rightarrow C_P$.
- **Redefined** the New Silk Roads (BRI) as a 'diplomacy of the Two Mountains': rather than exporting an extractive model (logistical colonisation, HPAL factories vulnerable to the 'sovereignty rent' of host states), China can export the operator T_{mixed} to countries of the Global South that are also stuck at their Nodal Point, enabling each partner to trigger its own cycle of ecological Rent Extraction.

- **Highlighted** the asymmetric advantage of endogenous regeneration over Western financialisation. The West, by attempting direct $C_N \rightarrow C_P$ transitions through liquidity injections, violates nilpotence and paves the way for global systemic retropolarity. Through TRR, China adheres to the topologically authorised trajectory and creates a new underlying asset (Regenerative Value), positioning the Celestial Economic Path (Tiandao Jingji, 天道经济) as the new standard for a sustainable civilisation.
- **Identified** the Global South as a testing ground for the TRR. Developing countries (sub-Saharan Africa, South-East Asia, Latin America) can import the TRR institutional package (Two Mountains Bank, GEP in KPIs, CUR) to reform their international financial institutions (IMF, World Bank) and emerge from their own Nodal Point without going through Western financialisation.
- **Developed** a geopolitical outlook for 2026–2045 structured around three scenarios:
 - Scenario 1 (success of the TRR): China achieves a regime of healthy growth ($\mathcal{J}_{\text{Stasis}} < 0.3$) by 2032 and exports the model to the Global South, establishing the Celestial Economic Path as the new standard of civilisation.
 - Scenario 2 (partial failure): bipolarisation of the world system between a Chinese bloc (TRR, healthy growth) and a Western bloc (financialisation, retropolarity).
 - Scenario 3 (Chinese retropolarity): systemic collapse of China (10th dynastic collapse) leading to global systemic chaos.
- **Demonstrated** that the TRR is a restoration of the Mandate of Heaven: by regenerating the territory, it re-establishes harmony between the economy, society and ecology, and restores the political virtue (De, 德) of the sovereign — legitimacy founded on the ability to ensure harmony between Heaven, Earth and Mankind.

This practical application demonstrates that the TRR is not merely an environmental policy, nor simply a tool for domestic governance, but an operational application of discrete topology to the regulation of an economic system in crisis, projected onto the three dimensions of legitimacy: domestic (Mandate of Heaven), cultural (Chinese structural wisdom) and geopolitical (diplomacy of regeneration). It adheres to the laws of the $64 \rightarrow 20$ invariant and offers China a peaceful way out to avoid the tenth dynastic collapse, whilst proposing to the Global South a path to development that is, at last, sustainable.

The significance of the TRR thus extends beyond China's borders: by demonstrating that it is possible to emerge from the Nodal Point without massive destruction of wealth, China offers the world the first operational application of a physics of regulation. The formalism $\text{Cl}(6, 6)$, a mathematised translation of Chinese structural wisdom, becomes a geopolitical tool of the first order, capable of rallying the Global South by offering it an alternative to Western imperialism — no longer the extraction of rent, but the co-regeneration of Value.

End of Part III

PART IV – UNIFICATION AND SYSTEMIC IMPLICATIONS

Introduction to Part IV

Parts I to III have developed the complete extension of the economic model from $Cl(6, 0)$ to $Cl(6, 6)$. We have defined the 144 economic pentads, constructed the projection matrix Π , introduced the transition operator T and its four components, identified the 7 spectral thresholds, analysed the retropolarities, and formalised the TRR as a controlled T_{mixed} passing through the threshold N_4 .

This Part IV takes a broader view. We show that the extension to $Cl(6, 6)$ is not merely a mathematical refinement, but an epistemological revolution: economics is not a ‘soft’ social science distinct from the natural sciences, but a physics of symbolic systems obeying the same topological laws as matter, life and language.

We explore the implications of this unification for our understanding of culture, institutions and the role of the state in regulating complex systems. We show that TRR is not merely one policy amongst many, but the first operational application of a physics of regulation that could transform our understanding of how socio-economic systems are steered.

This section is intended for all readers, whether decision-makers or theorists. It demonstrates that the $Cl(6, 6)$ formalism is not a rootless Western import, but a mathematical extension of a structural wisdom that China has cultivated for over two thousand five hundred years.

Chapter 24 – The Economy as an Exosomatisation of the Topology of $Cl(6, 6)$

24.1. Isomorphism with other systems as validation of the model’s universality

One of the most profound discoveries of this work is the structural convergence between the economy and other complex systems. The following table summarises the five manifestations of the $64 \rightarrow 20$ invariant:

Table 43: The five manifestations of the $64 \rightarrow 20$ invariant

Domain	Input (64)	Output (20)	Filtering	Reference
Particle physics	Spin configurations	Particle families	Clifford algebra $Cl(6, 0) / \Lambda_{72}$	[5], [11]
Genetic code	Codons (4 bases $\times 3$)	Amino acids	Neighbourhood rule	[5], [12]
Mandarin phonology	24 initials + 40 finals (64 combinations)	20 phonological classes	Articulatory constraints	[6]
Tang alliances	64 marital configurations	20 alliance attractors	Filtration by pentads (Wufu)	[6]
Economics (ToRE)	64 economic configurations	20 stable attractors	Filtration by nilpotence	[3], [6]

Interpretation: This convergence is not accidental. It indicates that nature uses a single ‘code’ to organise complex information, regardless of the substrate: particles, nucleotides, phonemes, clans or factors of production. The invariant $64 \rightarrow 20$ is a universal law governing the structuring of complex systems.

Within the framework of $Cl(6, 6)$, this universality is explained by the fact that all these systems are projections of the same relational substrate. Economics is not an isolated case; it is a particular manifestation of the topology of $Cl(6, 6)$, in the same way as particle physics or molecular biology.

24.2. Economic institutions as ‘projections’ of pentads

In the $Cl(6, 6)$ formalism, economic institutions (markets, banks, states, property systems) are projections of pentads onto social space. Each institution corresponds to a particular configuration of pentads:

Table 44: Economic institutions as projections of pentads onto social space

Institution	Corresponding pentads	Topological role
Market	P_2 (liberal market economy)	Exchange structure, price regulation
Central Bank	P_3 (debt-based economy)	Management of money and credit
Welfare State	P_1 (Keynesian economy)	Redistribution, social regulation
Land system	P_4, P_5, P_6 (land-based economy)	Land and resource management
Financial institutions	N_2, N_3, N_4 (financial regulation)	Control of capital flows
Planning system	N_5, N_6 (central planning)	Macroeconomic coordination

Interpretation: Institutions are not arbitrary human constructs. They are exosomatisations (projections outside the body) of the topology of $Cl(6, 6)$. By inventing economic institutions, humans merely make explicit what nature has already structured.

24.3. The stages of exosomatisation

The following table describes the stages of exosomatisation, from biology to economics:

Table 45: The stages of exosomatisation: from biology to economics

Stage of exosomatisation	Externalised function	Corresponding structure
Biological	Reproduction, filiation	Body, DNA
Social (prototype)	Regulation of alliances	Kinship system (Tang)
Linguistic	Communication, symbolisation	Phonology, grammar
Economic	Accumulation, exchange	Financial institutions, Terms of Reference
Technical (AI)	Information processing	Neural networks, Clifford algebra

Interpretation: The economy is the stage at which humankind externalises its capacity to accumulate and exchange value. But this externalisation is not arbitrary: it obeys the same topological constraints as the preceding stages. This is why the invariant $64 \rightarrow 20$ is found in all stages.

Chapter 25 – Cultural Isomorphisms: Yi Jing, Wuxing, Wufu, TCM

25.1. The Yi Jing: 64 configurations as combinatorial pre-filtering and divinatory dialogue

The *Yi Jing* (Book of Changes) is the oldest text of Chinese wisdom. It is based on 64 hexagrams, which are configurations of six binary lines (yin/yang, $-/-$). But the *Yi Jing* is not merely a catalogue of configurations. It is a divinatory tool that enables one to engage in dialogue with Heaven by interpreting the mutations of the hexagrams.

Fundamental correspondence:

Table 46: Fundamental correspondence between the *Yi Jing* and the $Cl(6, 6)$ formalism

<i>Yi Jing</i>	$Cl(6, 6)$ formalism
64 hexagrams	64 configurations of $Cl(6, 0)$
6 binary lines	6 generators $e_1, \dots, e_6$
Yin/Yang	Bits 0/1
Hexagram mutations	Transitions between configurations
Cycle of the 5 phases (Wuxing)	Tropical belts C_P and C_N
Divination	Measurement of spectral observables
Response from Heaven	Topological diagnosis

The *Yi Jing* models the phase of combinatorial pre-filtering. The 64 hexagrams constitute an exhaustive space of binary configurations, strictly isomorphic to the space of the 64 economic configurations. The mutations of the hexagrams (the transition from one hexagram to another through a change in lines) correspond to the transitions between economic configurations.

Divination as a spectral measurement: The divinatory practice of the *Yi Jing* (drawing yarrow stalks or coins) is a stochastic measurement of the state of the system. The result of the casting (a hexagram, followed by a changing hexagram) is a configuration that must be interpreted in the light of the circumstances. This is exactly what the spectral observables (η , d , gap, $R_{\text{threshold}}$) do: they measure the state of the system and provide a diagnosis.

25.2. The Wuxing: the Five Phases as regulators of cycles

The theory of the Wuxing (the Five Phases or Five Agents) is the second pillar of Chinese structural wisdom. It describes five fundamental phases (Wood, Fire, Earth, Metal, Water) which interact according to two cycles:

- **Sheng cycle (generative):** Wood $\rightarrow$ Fire $\rightarrow$ Earth $\rightarrow$ Metal $\rightarrow$ Water $\rightarrow$ Wood.
- **Ke cycle (regulatory):** Wood $\rightarrow$ Earth $\rightarrow$ Water $\rightarrow$ Fire $\rightarrow$ Metal $\rightarrow$ Wood.

Correspondence with the Cl(6, 6) formalism:

Table 47: Correspondence between the Wuxing (Five Phases) and the Cl(6, 6) formalism

Wuxing	Cl(6, 6) formalism
5 phases	5 elements of the pentad $\{B_1, B_2, B_3, F, S\}$
Sheng cycle	Sheng mode ($\eta > 0$) —expansion, generation
Ke cycle	Ke mode ($\eta < 0$) —regulation, contraction
Phase equilibrium	Sheng/Ke equilibrium ($\eta \approx 0$)
Pathological imbalance	Dominance of a mode (retropolarity)

The Wuxing describes a dynamic of regulation in which each phase is both generated by one phase and regulated by another. No phase dominates indefinitely; the system's equilibrium rests on the harmonious circulation between the phases.

Implications for the model: The Wuxing teaches that regulation is not an external constraint, but an internal principle of the system. Similarly, the tropical belts C_P and C_N and the thresholds P_4 and N_4 are not modelling constructs, but principles of endogenous regulation that structure economic dynamics. The post-2018 Chinese Stasis is an imbalance of the Wuxing: the Sheng mode (expansion) is exhausted, the Ke mode (regulation) is saturated, and the system can no longer circulate harmoniously between the phases.

25.3. The 12 meridians and traditional Chinese medicine

Traditional Chinese medicine (TCM) is based on a system of 12 main meridians, which are channels through which vital energy (*qi*) flows. Each meridian is associated with an organ and a physiological function. TCM

posits that health is a dynamic equilibrium between the energy flows in these meridians; illness is a blockage or an imbalance in these flows. Treatment (acupuncture, herbal medicine) aims to restore circulation and rebalance the system.

Correspondence with the $Cl(6, 6)$ formalism:

Table 48: Structural isomorphism: the 12 meridians of TCM and the 12 basic pentads

TCM (energetic anatomy)	$Cl(6, 6)$ formalism
12 main meridians	12 basic pentads (P_1 to P_6 , N_1 to N_6)
Circulation of <i>qi</i> in the meridians	Circulation of configurations between attractors
Acupuncture points (穴) on the meridians	Polar thresholds P_4 and N_4 (hinges between belts)
Meridian imbalance	Topological frustration ($E_{k,g} > 0$)
Homeostasis	Nilpotence $(g \cdot x)^2 = 0$

This structural correspondence (the 12 meridians $\leftrightarrow$ the 12 pentads) finds its operational extension in §15.4.8, where the spectral observables are interpreted as the four diagnostic methods (望闻问切, *wàng wén wèn qiè*) through which the physician takes the pulse of the system. Anatomy (§24.3) and diagnostic practice (§15.4.8) are thus the two sides of the same isomorphism between TCM and the $Cl(6, 6)$ formalism: one describes the body of the economic system (its 12 meridians/pentads), the other describes the method for taking its pulse (the 4 examinations/observables).

Implications for the model: Chinese Stasis is a blockage of economic flows, analogous to an imbalance in the meridians. TRR is a form of economic acupuncture: it aims to restore the circulation of value flows by activating the P_4 and N_4 thresholds. The three structural reforms (Two Mountains Bank, GEP, CUR) are acupuncture points that target specific meridians:

- **Two Mountains Bank** $\rightarrow$ finance meridian (redirecting capital flows towards ecological regeneration);
- **Incorporation of the GEP into KPIs** $\rightarrow$ institutions meridian (anchoring the new production model);
- **Certificate of Regenerative Usufruct (CUR)** $\rightarrow$ the territory meridian (creation of a new cycle of value distribution).

25.4. The Tang (Wufu) kinship system and the regulation of alliances

The Chinese kinship system, known as Wufu (Five Generations), is a topology of alliances that defines degrees of kinship and marriage prohibitions. It is based on a structure of 64 marital configurations, filtered into 20 alliance attractors, exactly like the economic model.

Correspondence:

Table 49: Correspondence between the Wufu kinship system and the Cl(6, 6) formalism

Wufu	Cl(6, 6) formalism
64 matrimonial configurations	64 economic configurations
20 marriage attractors	20 economic attractors
Marriage prohibitions	Frustrated configurations (economic prohibitions)
Mourning (5 grades)	$3P \rightarrow 3N$ polarity gradient

The Wufu defines degrees of kinship which determine ritual obligations and marriage prohibitions. The 5 degrees of mourning correspond to a gradient of intensity: the closer the kinship, the longer and stricter the mourning period.

Implications for the model: The polarity gradient $3P \rightarrow 3N$ is a degree of economic ‘mourning’: perverse growth (3P) is a state of ‘prosperity’ but a fragile one; terminal depression (3N) is a state of economic ‘mourning’. The thresholds P_4 and N_4 are marriage prohibitions: they forbid direct transitions between belts, just as the Wufu prohibits certain marriages to preserve the coherence of the kinship system.

Chapter 26 – Vandermeersch’s Contribution: Chinese Divinatory Reasoning

26.1. The Two Principles of Chinese Thought

Léon Vandermeersch’s work, *The Two Rationalities of Chinese Thought – Divination and Ideography* [97], provides a decisive key to understanding the depth of Chinese epistemology. Vandermeersch distinguishes between two fundamental ‘rationalities’:

Table 50: Comparison between Western theological reason and Chinese divinatory reason (according to Vandermeersch)

Theological reason (West)	Divinatory reason (China)
Linear causality (<i>causa efficiens</i>)	Structural correlation (correspondence)
God or natural law as a transcendent principle	The Dao as an immanent principle of order
Writing transcribes speech	Writing records divinatory operations
Philosophy as systematic discourse	Literature as a means of accessing the meaning of things
Law as an external norm	Ritual as internal regulation

This distinction is fundamental to understanding why the formalism Cl(6, 6) resonates so deeply within the Chinese tradition. Vandermeersch has identified what the formalism Cl(6, 6) formalises: divinatory reason as a mode of structural thought.

Vandermeersch writes:

“Chinese reasoning is structural, operating according to formal logic, rather than the ‘causal logic’ upon which Western thought is based.” (Chapter VI)

Correspondence with Cl(6, 6): Western economic models (DSGE, general equilibrium, linear regression) are based on causal logic. The Cl(6, 6) model is based on structural logic: economic cycles are not chains of cause and effect, but reconfigurations of relationships between pentads.

26.2. Divination as a pre-modern regulatory system

Vandermeersch demonstrates that Chinese divination was not a superstition, but a sophisticated regulatory system. It made it possible to ascertain the will of Heaven regarding the affairs of the Empire, to verify whether policies were in accordance with celestial laws, to anticipate crises, and to adjust decisions.

Correspondence with the formalism Cl(6, 6):

Table 51: Correspondence between Chinese state divination (Vandermeersch) and the formalism Cl(6, 6)

State divination (Vandermeersch)	Cl(6, 6) formalism
Drawing of hexagrams	Measurement of spectral observables ($\eta, d, \text{gap}, R_{\text{threshold}}$)
Interpretation of mutations	Diagnosis of attractors and thresholds
Policy adjustment	Activation of thresholds P_4 and N_4
Conformity with Heaven	Sheng/Ke Balance ($\eta \approx 0$)
Loss of the Mandate	Retropolarity ($\eta < -0.6, R_{\text{threshold}} > 0.7$)

Vandermeersch:

“The aim of divination is not to predict the future, but to reveal the correlation between the present situation and the future situation, by means of an artificial representation (the crack in the bone or the number drawn).” (Chapter VI)

The spectral dashboard proposed by the Cl(6, 6) formalism is the modern equivalent of this state divination system: it enables us to probe the state of the economic system, diagnose its imbalances, and adjust policies accordingly. State divination was the spectral dashboard of the Empire.

26.3. Divinatory writing as a proto-scientific language

Vandermeersch puts forward a revolutionary thesis: Chinese writing did not arise from the transcription of speech, but from the recording of divinatory operations.

“Chinese characters were not created to transcribe the spoken language, but to record the parameters of divinatory equations.” (Chapter II)

The proof lies in the very form of the earliest inscriptions on bones and tortoise shells:

$$XY \times N : M \rightarrow P$$

where X and Y are the temporal coordinates (sexagesimal cycle), N the name of the diviner, M the question asked, and P the answer obtained.

Correspondence with the Cl(6, 6) formalism:

Table 52: Correspondence between the divinatory equation (Vandermeersch) and the Cl(6, 6) formalism

Divinatory equation (Vandermeersch)	Cl(6, 6) formalism
’ 卜 ’ (scapulomancy)	Basic operator
’ 贞 ’ (question)	$T_{\text{structure}}$
’ 占 ’ (answer)	$\Pi \circ T$
Configuration $XY \times N : M \rightarrow P$	Configuration $\mathbf{x} = (b_1, \dots, b_6)$

What the formalism Cl(6, 6) does is to mathematically formalise what Vandermeersch identified as the ‘divinatory reason’: a system of structural correlation, which seeks not causes but stable configurations.

Chapter 27 – The Cl(6, 6) Formalism as an Extension of Chinese Structural Wisdom

27.1. An epistemology of relation, not of substance

The extension of the economic model to Cl(6, 6) is not an innovation created out of thin air. It forms part of an epistemology of relation whose deepest roots lie in time-honoured Chinese thought. Whereas Western thought, as the heir to Aristotle, inquires into the intimate nature of things (“what is an atom?”, “what is value?”), traditional Chinese thought inquires into the relationships between things (“how are phenomena linked?”, “what are the cycles of transformation?”).

This difference is not merely anecdotal. It is structural. The formalism Cl(6, 6) is an algebra of relations: the pentads are not “things” endowed with intrinsic properties, but configurations of relations between generators. Economic value is not a substance, but a relation between a stock of Value and a circuit of Wealth. Growth is not an accumulation, but a cycle of Rent Extraction.

This perspective is strictly isomorphic to that of the *Dao De Jing*:

“The Dao that can be expressed is not the eternal Dao.” (Book I, §1)

The Dao is not a thing, but the principle of order of things. Similarly, the formalism $Cl(6, 6)$ does not describe ‘economic entities’ but the principles of order that structure economic systems. The economy is not a collection of agents, but a network of relationships whose coherence depends on the preservation of topological invariants. The formalism $Cl(6, 6)$ does not claim to invent this epistemology. It formalises it, giving it a mathematical expression that was previously lacking in the tradition.

27.2. The Celestial Empire: an ontological ambition

The Chinese Empire never regarded itself as just one state amongst others. It always declared itself ‘Celestial’ (*Tianchao*, 天朝), not out of pride or for propaganda’s sake, but with full knowledge of the facts. This designation was not mere political rhetoric; it was the expression of an ontological ambition: the Empire conceived of itself as the earthly embodiment of the celestial order, as the point of articulation between Heaven and Earth.

State divination: a structural dialogue with Heaven

To maintain this correspondence, the Empire did not merely declare its celestial nature. It constantly verified and renewed it through a divinatory dialogue with Heaven. State divination, practised according to the rites of the *Yi Jing*, was not superstition. It was an operational procedure aimed at:

- Ascertain the will of Heaven regarding the affairs of the Empire.
- Verify whether imperial policies were in accordance with celestial laws.
- Anticipate changes and crises before they occurred.
- Adjust decisions in accordance with the divinatory responses.

The Mandate of Heaven as a topological threshold

The Mandate of Heaven (*Tianming*) was not merely a means of political legitimisation. It was a topological threshold: if the Emperor violated the celestial laws (through injustice, corruption or the depletion of resources), the Mandate would be withdrawn from him and a new dynasty could legitimately take power.

Correspondence:

Table 53: Correspondence between the Mandate of Heaven and the formalism $Cl(6, 6)$

Mandate of Heaven	$Cl(6, 6)$ formalism
Imperial Legitimacy	Stable Attractor ($3P, 2P + 1N, 1P + 2N$)
Loss of the Mandate	Retropolarity (direct transition $C_P \rightarrow C_N$)
Change of Dynasty	Transition from one attractor to another
Virtue of the Sovereign	Policy of Passing through Thresholds P_4 and N_4
Conformity with Heaven	$\eta \approx 0, \quad \text{gap} > 0.4, \quad R_{\text{threshold}} < 0.3$

The Mandate of Heaven is therefore a topological invariant: it is maintained if the Empire respects the laws of regulation (passing through the thresholds), and is lost if the Empire violates them (direct transition).

27.3. The synthesis: the Cl(6, 6) formalism as ‘mathematised divination’

The following table summarises the correspondences between Chinese structural wisdom and the Cl(6, 6) formalism:

Table 54: Summary of the correspondences between Chinese structural wisdom and the formalism Cl(6, 6)

Chinese tradition	Key concept	Correspondence Cl(6, 6)	Operational dimension
<i>Yi Jing</i>	64 hexagrams	64 configurations	Space of possibilities
<i>Yi Jing</i>	Mutations	Transitions between attractors	Dynamics of cycles
<i>Yi Jing</i>	Divination	Spectral observables	Real-time diagnosis
<i>Wuxing</i>	5 phases	Pentad $\{B_1, B_2, B_3, F, S\}$	Elementary structure
<i>Wuxing</i>	Sheng cycle	Sheng mode ($\eta > 0$)	Expansion
<i>Wuxing</i>	Ke cycle	Ke mode ($\eta < 0$)	Regulation
<i>Wuxing</i>	Equilibrium	$\eta \approx 0$	Harmony
TCM	12 meridians	12 basic pentads	Circulation channels
TCM	Pulse diagnosis	Spectral observables	Flow measurement
TCM	Acupuncture	Activation of the P_4 and N_4 thresholds	Treatment of blockages
<i>Wufu</i>	64 marital configurations	64 economic configurations	Alliance space
<i>Wufu</i>	20 alliance attractors	20 economic attractors	Stable regimes
<i>Wufu</i>	5 degrees of mourning	$3P \rightarrow 3N$ gradient	Intensity of crises
<i>Tianming</i>	Celestial Empire	Heaven-conforming system	Ontological ambition
<i>Tianming</i>	State divination	Spectral dashboard	Dialogue with Heaven
<i>Tianming</i>	Harmony between Heaven, Earth and Humankind	Sheng/Ke balance	Systemic stability
<i>Tianming</i>	Loss of the Mandate	Retropolarity	Collapse
<i>Tianming</i>	New Dynasty	Transition via N_4 (TRR)	Regeneration

Chapter 28 – Philosophical and Epistemological Implications

28.1. Economics is not a ‘soft’ social science, but a physics of symbolic systems

The argument set out in this report has a major philosophical implication: economics is not a ‘soft’ human science distinct from the natural sciences. It is a physics of symbolic systems, obeying the same topological laws as matter and life.

Consequences:

- Economic agents are not rational ‘atoms’ in a continuous space. They are nodes in a topological network whose transitions are constrained by tropical belts and polar thresholds.
- Economic crises are not accidents or policy errors. They are violations of nilpotence (frustrated configurations) which cause retropolarities (systemic collapses).
- Economic policies are not arbitrary choices. They are transition operators which must respect spectral thresholds, failing which they will trigger collapses.

This epistemological reorientation has practical consequences for decision-makers: they can no longer treat the economy as a continuous and malleable system. They must understand it as a discrete information system with clear breaking points and stable regimes.

28.2. Culture as a stage of exosomatisation of topology

Culture, in the broad sense (languages, kinship systems, economic institutions), is an exosomatisation of discrete topology. By inventing social rules, humans merely make explicit what nature has already structured.

Examples:

- Kinship systems (such as the Chinese Wufu) are projections of the topology of $Cl(6, 0)$ onto social space. The 64 matrimonial configurations reduce to 20 alliance attractors, exactly as the 64 economic configurations reduce to 20 attractors.
- Languages (such as Mandarin) are projections of the topology of $Cl(6, 6)$ onto phonological space. The 64 phonological combinations are reduced to 20 phonological classes, exactly as the 64 economic configurations are reduced to 20 attractors.
- Economic institutions (such as markets, banks and land tenure systems) are projections of the topology of $Cl(6, 6)$ onto economic space.

Implication: Culture is not a ‘supplement’ to nature. It is a necessary projection of the underlying topology. Man does not create *ex nihilo*; he makes explicit what is already structured in the universe.

28.3. Celestial AI: a tool for revealing (not creating) the laws of regulation

Artificial intelligence, if it incorporates Clifford algebra $Cl(6, 6)$ and the network Λ_{72} , could reveal the topology underlying all complex systems. It would not create new laws, but would bring to light those that already exist, buried within the structure of the universe.

Fundamental limitation: AI can only reveal what is already there. The invariant $64 \rightarrow 20$ is a law of the universe, not a human convention. Decision-makers who ignore it are doomed to repeat collapses. Those who understand it can steer complex systems in harmony with the laws of nature.

Application to the TRR: AI could help to:

- Monitor spectral observables ($\eta, d, \text{gap}, R_{\text{threshold}}$) in real time.
- Simulate the consequences of different economic policies.
- Identify critical spectral thresholds and propose appropriate interventions.

However, AI could not ‘invent’ a new solution. It could only reveal that TRR is the only topologically permissible path out of the Chinese Stasis.

Chapter 29 – Summary of Part IV

This Part IV has drawn out the philosophical and systemic implications of extending the economic model to $\text{Cl}(6, 6)$. We have:

- **Shown** that the economy is an exosomatisation of the topology of $\text{Cl}(6, 6)$, in the same way as particle physics, molecular biology, phonology and kinship systems.
- **Established** that economic institutions are projections of pentads onto social space, and that their evolution is governed by the same topological laws.
- **Developed** cultural correspondences with the *Yi Jing*, the Wuxing, Traditional Chinese Medicine and the Wufu, showing that Chinese structural wisdom anticipated, by other means, the discrete topology that the $\text{Cl}(6, 6)$ formalism now formalises.
- **Incorporated** Vandermeersch’s contribution, showing that Chinese ‘divinatory reason’ is isomorphic to the correlation logic of the $\text{Cl}(6, 6)$ formalism.
- **Developed** the epistemological implications: economics is not a ‘soft’ social science, but a physics of symbolic systems.
- **Shown** that culture is a stage of exosomatisation of topology, and that celestial AI can reveal (but not create) the laws of regulation.

End of Part IV

PART V – GENERAL CONCLUSION AND SUMMARY

1. Summary of the model’ s contributions

This report has developed an extension of the economic model, moving from Clifford algebra $Cl(6, 0)$ to $Cl(6, 6)$, in order to formalise the dynamics of complex economic systems. This demonstration is structured around five main themes:

- **The Diagnosis (Part I):** Applying the Theory of Rent Extraction (ToRE) to Chinese history highlights a recurring cycle of depletion of the stock of Land Value. Post-2018 China has reached this Nodal Point. Attempts to substitute with intangible assets (AI, carbon, data) are hampered by a lack of spatial depth, rendering them incapable of generating the required mass exogenous demand, thereby placing the system in a state of Stasis.
- **Topological formalism (Part II):** We have defined a space comprising 144 economic pentads projected onto 20 stable attractors. The dynamics of this system are governed by the Λ_{72} network, its eigenvalues and its seven spectral thresholds, which constrain transitions between economic regimes.
- **Operationalisation (Part III):** The Territorial Regeneration Rent (TRR) has been modelled as a mixed transition operator (T_{mixed}). It enables a transition controlled by the polar threshold N_4 , preventing retropolarity (collapse) and initiating a new cycle of material Rent Extraction.
- **Geopolitical projection (Part III, Chapter 22):** The TRR is not merely a domestic policy of ecological regeneration. It constitutes the condition of possibility for a complete overhaul of China’s geopolitical posture, marking the end of external spatial avoidance and the advent of a ‘diplomacy of regeneration’. This diplomacy offers the Global South an alternative model to Western imperialism – no longer the extraction of rent, but the co-regeneration of Value.
- **Structural isomorphisms (Part IV):** The model highlights formal correspondences with historical regulatory systems (Yi Jing, Wuxing, Tang kinship). These traditions are not invoked as mystical evidence, but as empirical and qualitative models of cyclical dynamics which the $Cl(6, 6)$ formalism now enables us to quantify.

2. The $64 \rightarrow 20$ invariant and spectral formalisation

The central hypothesis validated by this work is the existence of a recurring topological constraint, reducing 64 discrete configurations to 20 stable attractors. This reduction is observable in domains that are, a priori, independent:

Table 55: The five manifestations of the $64 \rightarrow 20$ invariant

Field	Input (64)	Output (20)	Filtering mechanism
Particle physics	Spin configurations	Particle families	Clifford algebra $Cl(6, 0) / \Lambda_{72}$
Genetic code	Codons (4 bases $\times$ 3)	Amino acids	Biochemical proximity rule
Mandarin phonology	Initial/final combinations	Phonological classes	Articulatory constraints
Tang alliances	Marital configurations	Alliance attractors	Filtration by pentads (Wufu)
Economics (ToRE)	Economic configurations	Economic regimes	Nilpotence and polar thresholds

This convergence suggests that discrete complex systems obey similar structural laws, regardless of their physical or social basis.

The universal spectral formula: Based on the 48 non-degenerate roots of the Λ_{72} lattice, the state of an attractor can be modelled by the formula:

$$\mathcal{E} = \Lambda \cdot 4^m \cdot \sum_{k=0}^{14} \epsilon_k \beta_k, \quad \epsilon_k \in \{-1, 0, 1\}$$

For the economy, the preliminary scale constant is estimated at $\Lambda_{eco} \approx 1.23 \times 10^{-3}$ (in units of annual GDP). The 15 constants β_k act as coefficients modulating the attractor's signature.

3. TRR as a controlled transition operator

The spectral diagnosis of post-2018 China indicates a lock-in on attractor S (dispersion + nodal point), with critical observables ($\eta < -0.4$, $R_{threshold} > 0.7$).

The model demonstrates that any attempt to exit Stasis via a direct transition to expansion (C_p) without passing through the polar thresholds causes retropolarity (systemic collapse). The Territorial Regeneration Rent (TRR) constitutes the only topologically permitted trajectory:

- **Mechanism:** The TRR acts as a T_{mixed} operator activating threshold N_4 .
- **Components:**
 - **Two Mountains Bank** (T_{water}): Redirection of capital flows towards ecological assets.
 - **Integration of the GEP into KPIs** ($T_{structure}$): Institutional anchoring of the new model.
 - **Regenerative Usufruct Certificates** (T_{fire}): Creation of a new cycle of value distribution.

- **Timetable:** The phased roll-out (pilot phases 2024–2025, acceleration 2029–2030) corresponds to the sequential crossing of spectral thresholds (S_5), (S_6) and (S_7), enabling stabilisation at (C_P) by 2031–2035.
- **Geopolitical projection:** The TRR is not confined to Chinese territory. The operator T_{mixed} can be projected onto the spectral sheets of partner countries, enabling each state in the Global South, trapped within its own Nodal Point, to trigger its own cycle of Ecological Rent Extraction. This export of the model — via the Two Mountains Bank, the integration of the GEP into local KPIs, and Regenerative Usufruct Certificates — constitutes the ‘Diplomacy of the Two Mountains’, an alternative to the logistical imperialism of the Fortified Port Enclaves.

4. Geopolitical Outlook 2026–2045: three scenarios for the world system

The Cl(6, 6) model allows us to project three trajectories for the world system by 2045, depending on whether China succeeds, partially fails, or completely fails in its TRR transition:

- **Scenario 1 — The Celestial Economic Path (Tiandao Jingji, 天道经济):** If China succeeds in its TRR transition ($\mathcal{J}_{\text{Stasis}} < 0.3$ by 2032), it will export the model to the Global South, establishing a new standard of civilisation based on the endogenous regeneration of Value. 2045 observables: $\eta_{\text{world}} = +0.4$, $d_{\text{world}} = 4.5$, $\text{gap}_{\text{world}} = 0.5$, $R_{\text{threshold,world}} = 0.2$, $\mathcal{J}_{\text{Stasis,world}} = 0.2$.
- **Scenario 2 — Bipolarisation of the world system:** If the TRR transition is incomplete, the world splits into two blocs: a Chinese bloc (TRR, healthy growth) and a Western bloc (financialisation, retropolarity). Observables for 2045: $\eta_{\text{world}} \approx 0$, $d_{\text{world}} \approx 2$, $\text{gap}_{\text{world}} \approx 0.15$, $R_{\text{threshold,world}} \approx 0.6$, $\mathcal{J}_{\text{Stasis,world}} \approx 0.5$.
- **Scenario 3 — Chinese retropolarity and systemic chaos:** If China fails completely (no TRR transition), its retropolarity (tenth dynastic collapse) leads to a global systemic collapse. Observables for 2045: $\eta_{\text{world}} = -0.7$, $d_{\text{world}} = 1$, $\text{gap}_{\text{world}} = 0.02$, $R_{\text{threshold,world}} = 0.95$, $\mathcal{J}_{\text{Stasis,world}} = 0.85$.

The strategic window of the 15th Five-Year Plan (2026–2030) is therefore crucial: it is during this period that China must cross thresholds S_5 , S_6 and S_7 to avoid retropolarity and pave the way for the Celestial Economic Path.

5. Epistemological implications and limitations of the model

5.1. The economy as a discrete information system

This work invites us to reconsider the economy not as a continuous space of adjustable variables (DSGE models), but as a discrete information system. Agents and institutions are nodes in a topological network whose transitions are constrained by clear breakpoints. Crises are not anomalies, but violations of nilpotence (frustrated configurations).

5.2. Cultural isomorphisms and regulation

The correspondences established with the Yi Jing (64 hexagrams), the Wuxing (5 phases) or Traditional Chinese Medicine (12 meridians) are not a matter of poetic analogy, but of structural isomorphism. Historically, these systems have modelled the dynamics of cyclical regulation (expansion/recession, generation/constraint). The Cl(6, 6) formalism provides a mathematical translation of these principles, transforming ancient state divination

into a modern spectral dashboard, capable of measuring the system's tension ($R_{\text{threshold}}$) and its robustness (gap) in real time.

5.3. The geopolitical scope of the formalism

The $\text{Cl}(6, 6)$ formalism is not merely a tool for internal diagnosis. It constitutes a geopolitical tool of the first order, capable of: - Diagnosing the spectral state of any complex economic system, whether Chinese, Western, or belonging to the Global South. - Identifying blocked spectral thresholds and prescribing the appropriate transition operators ($T_{\text{structure}}$, T_{fire} , T_{water} , T_{mixed}). - Projecting the operator T_{mixed} (TRR) onto the spectral sheets of partner countries, enabling diplomacy based on co-regeneration rather than rent-seeking.

This universality of the formalism — validated by the five manifestations of the $64 \rightarrow 20$ invariant (particle physics, the genetic code, Mandarin phonology, the Tang alliances, and economics) — underpins the legitimacy of the Celestial Economic Path as a new standard of civilisation, an alternative to the exhausted Western model.

5.4. Falsifiability and future work

To maintain its scientific rigour, the $\text{Cl}(6, 6)$ model must remain falsifiable. It would be invalidated if: - Direct transitions between tropical belts ($C_P \leftrightarrow C_N$) were observed without retropolarity. - Stable economic regimes were identified outside the 20 predicted attractors. - The spectral observables (η , d , gap, $R_{\text{threshold}}$) consistently failed to predict historical tipping points.

Future work should focus on: - Refining the calibration of Λ_{eco} using longer historical time series. - Developing a predictive simulator incorporating these observables to test macroeconomic policy scenarios. - Extending the methodology to other historical corpora (the Roman Empire, medieval Europe) to test the robustness of the invariant. - Applying the formalism to the spectral sheets of countries in the Global South to validate the diplomacy of regeneration.

6. Conclusion

The formalisation of the Theory of Rent Extraction in Clifford algebra $\text{Cl}(6, 6)$ provides a unified framework for understanding economic cycles. It demonstrates that the current Chinese Stasis is not inevitable, but rather an identifiable topological state, from which emergence requires compliance with precise structural constraints.

By proposing the Territorial Regeneration Rent (TRR) as a controlled transition operator, this model suggests that it is possible to bypass the historical cycles of dynastic collapse. The application of this formalism does not confer absolute control over the economy, but it provides decision-makers with a rigorous decision-support tool. It enables them to anticipate tipping points, avoid abrupt transitions, and adjust regulatory policies in harmony with the system's structural constraints.

Beyond China, the significance of the TRR extends beyond national borders. By demonstrating that it is possible to emerge from the Nodal Point without massive destruction of wealth, China offers the world the first operational application of a 'physics of regulation'. The formalism $\text{Cl}(6, 6)$, a mathematised interpretation of Chinese structural wisdom (Yi Jing, Wuxing, Traditional Chinese Medicine), is becoming a geopolitical tool of the first order, capable of rallying the Global South by offering it a path to development that is, at last, sustainable.

The Celestial Economic Way (Tiandao Jingji, 天道经济) is not an ideology: it is the mathematisation of the profound laws that structure complex systems, applied to the regulation of the global economy. It offers an

alternative to Western imperialism — no longer the extraction of rent, but the co-regeneration of Value. For the first time in two millennia, a Chinese state has a tool to break the fatal cycle of dynastic collapses and offer the Global South a path to development that respects the laws of the $64 \rightarrow 20$ invariant.

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Appendix A –Glossary of topological terms applied to economics

Table 56: Topological terms and their economic equivalents

English	Français	Topological definition	Chinese	Economic equivalent
Attractor	Attracteur	Triangular face of the Merkabah, stable equivalence class of configurations	吸引子 (<i>xī yǐn zǐ</i>)	Stable economic regime (e.g. polarized growth, recession, depression)
β_k	β_k	15 universal constants of the Λ_{72} network	普适常数 β_k (<i>pǔ shì cháng shù</i>)	Spectral coefficients of economic attractors
Centrifugal dynamics	Dynamique centrifuge	Spatial inversion at the Nodal Point (periurbanization, ghost cities)	离心力动态 (<i>lí xīn lì dòng tài</i>)	Spatial inversion at the Nodal Point (periurbanization, ghost cities)
$d(t)$	$d(t)$	Effective spectral dimension	有效谱维度 (<i>yǒu xiào pǔ wéi dù</i>)	Capacity for the propagation of economic shocks
Degeneration	Dégénérescence	Number of configurations projecting onto the same attractor	简并度 (<i>jiǎn bìng dù</i>)	Number of different historical situations falling under the same economic regime
$\eta(t)$	$\eta(t)$	Spectral asymmetry	谱不对称性 (<i>pǔ bù duì chèn xìng</i>)	Phase indicator: > 0 = expansion (<i>Sheng</i>), < 0 = recession (<i>Ke</i>)
Frustrated configuration	Configuration frustrée	Configuration $\mathbf{x} \in \mathcal{C}$ with $\lambda \leq 0$	受挫配置 (<i>shòu cuò pèi zhì</i>)	Economic impasse (hyperinflation, deflation, dead-end Nodal Point)
Γ	Γ	Dual graph of the pentads	对偶图 (<i>duì ǒu tú</i>)	Network of economic regulation principles
$\text{gap}(t)$	$\text{gap}(t)$	Smallest positive spectral gap	谱间隙 (<i>pǔ jiàn xì</i>)	Indicator of system robustness: low gap = fragility

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Glossary of topological terms (continued)

English	Français	Topological definition	Chinese	Economic equivalent
Invariant code	Code de l'invariant	Code of the 64→20 invariant	不变量代码 (<i>bù biàn liàng dài mǎ</i>)	Number of configurations or attractors ($K \in \{64, 20\}$)
K	K	Code of the 64→20 invariant	不变量代码 (<i>bù biàn liàng dài mǎ</i>)	Number of configurations or attractors ($K \in \{64, 20\}$)
Ke	Ke	Regulatory mode, pentagram path on a belt	克 (<i>kè</i> : to dominate, to restrain)	Contraction, consolidation, rigour
Λ_{72}	Λ_{72}	Exceptional network of dimension 72	例外网络 Λ_{72} (<i>lì wài wǎng luò</i>)	Complete spectrum of possible economic regimes
Merkabah	Merkabah	Level-3 double tetrahedron, filtration geometry	梅卡巴 (<i>méi kǎ bā</i>)	Space of possible economic regimes
N_4	N_4	Negative pentad excluded from C_N , negative polar threshold	负五元组 N_4 (<i>fù wǔ yuán zǔ</i>)	Permitted transition from recession to expansion (stimulus package)
Nilpotence	Nilpotence	Condition $(g \cdot x)^2 = 0$	幂零性 (<i>mì líng xìng</i>)	Condition for systemic stability: violation = collapse
P_4	P_4	Positive pentad excluded from C_P , positive polar threshold	正五元组 P_4 (<i>zhèng wǔ yuán zǔ</i>)	Permitted transition from expansion to recession (soft landing)
Pentad	Pentade	Set of 5 nodes of Cl(6,0), pentagonal face of the Merkabah's dodecahedron dual	五元组 (<i>wǔ yuán zǔ</i>)	Principle of economic regulation (e.g. investment, consumption, currency)

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Glossary of topological terms (continued)

English	Français	Topological definition	Chinese	Economic equivalent
Polarity	Polarité	Signature of an attractor: $3P, 2P + 1N, 1P + 2N, 3N$	极性 (jí xìng)	Economic health: from polarized growth to terminal depression
Projection Π	Projection Π	Application $\Pi : \mathcal{C} \rightarrow \mathcal{A}$	投影 Π (tóu yǐng)	Transition from raw data to economic regimes
$R_{\text{threshold}}(t)$	$R_{\text{seuil}}(t)$	Fraction of asymmetry at thresholds P_4 and N_4	阈值比率 (yù zhí bǐ lǚ)	Pre-bifurcation indicator: > 0.7 = red alert
Retropolarity	Rétropolarité	Direct transition $C_P \rightarrow C_N$ without a threshold	逆极性 (nì jí xìng)	Systemic collapse (e.g. 1929, 2008)
Sheng	Sheng	Generative mode, adjacent path on a belt	生 (shēng : to engender, to generate)	Economic expansion, cycle generation
Territorial Regeneration Rent (TRR)	Rente de Régénération Territoriale (RRT)	Controlled transition operator activating threshold N_4	领土再生租金 (lǐng tǔ zài shēng zū jīn)	The operational solution to exit Stasis and avoid the tenth dynastic decline
Theory of Rent Extraction (ToRE)	Théorie du Rachat (TdR)	Framework modeling the transfer of assets from the Space of Value to the Space of Wealth	价值资本化周期理论, 赎买理论 (TdR) (shú mǎi lǐ lùn)	The foundational economic theory underpinning the model
Topological filtration	Filtration topologique	Reduction of 64 configurations to 20 attractors using the neighbourhood rule	拓扑过滤 (tuò pū guò lǚ)	Transformation of raw data (inflation, wages, etc.) into economic regimes
Tropical belt	Ceinture tropicale	Cycle of length 5 in the dual graph Γ	热带带 (rè dài dài)	Cycle of economic policies (expansion or recession)

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Glossary of topological terms (continued)

English	Français	Topological definition	Chinese	Economic equivalent
Universal constants	Constantes universelles	15 universal constants of the Λ_{72} network	普适常数 (<i>pǔ shì cháng shù</i>)	Spectral coefficients of economic attractors

Appendix B –The 20 attractors: triplets of pentads, IDs, economic interpretation and Chinese examples

Table 57: The 20 attractors, their triplets of pentads, polarity, IDs, economic interpretation and Chinese examples

Attractor	Triplet of pentads	Pol.	Deg.	IDs (0-based)	Economic interpretation	Chinese examples
A	P_1, P_2, P_4	3P	4	0, 1, 2, 3	Extreme polarized growth: inflation + high profits, buy-back blocked. Point of maximum accumulation before collapse.	The Later Tang following the An Lushan Rebellion (758–763): inflation > 5000%, the price of rice rose from 13 to 1000 wen per dou; mid-to-late Southern Song: hyperinflation due to the over-issuance of <i>jiaozi</i>
B	P_1, P_3, P_5	3P	3	4, 5, 6	Perverse export-led growth: inflation, high profits, dispersion. Dependence on foreign markets.	After the opening of trade under the Longqing reign of the Ming Dynasty (1567–1644): an influx of silver from Japan and the Americas, moderate inflation, massive exports of silk and porcelain
C	P_2, P_3, P_6	3P	4	8, 9, 10, 11	Perverse growth with spatial dispersion: financial bubbles, systemic fragility.	Song (960–1279): boom in trade, <i>jiaozi</i> (paper money), chronic inflation due to over-issuance; uneven regional economic development
D	P_4, P_5, N_2	2P+1N	4	12, 13, 14, 15	Moderate growth: balanced investment and consumption, active buy-back.	The Wen-Jing Prosperity (180–141 BC): falling prices, ‘a bushel of millet for 10 to 20 wen’ ; the Kaiyuan Golden Age (713–741): ‘a bushel of rice for 3 wen’ , prosperity, tax cuts

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The 20 attractors (continued)

Attractor	Triplet of pentads	Pol.	Deg.	IDs (0-based)	Economic interpretation	Chinese examples
E	P_5, P_6, N_3	2P+1N	4	16, 17, 18, 19	Urban expansion: growth driven by urbanisation, low wages, moderate profits.	Commercial prosperity in the Northern Song cities (960–1127): rapid urbanisation, growing urban population, inflation moderated by the issuance of <i>jiaozi</i>
F	P_1, P_6, N_4	2P+1N	3	21, 22, 23	Emerging stagflation: inflation without growth in real wages, turning point.	Mid-to-late Ming period (16th century): influx of silver, ‘abundant silver, expensive goods’, incipient stagflation
G	P_2, P_5, N_6	2P+1N	3	25, 26, 27	Impoverished growth: rising GDP, but stagnant purchasing power, nodal buy-back.	Mid-Qing period (18th century): rapid population growth, steadily rising rice prices, declining peasant purchasing power
H	P_3, P_4, N_6	2P+1N	3	29, 30, 31	Rentier economy: profits derived from property and monopolies.	Late Ming period: land accumulation by the elites, concentration of land ownership, peasants burdened by debt, uprisings
I	P_1, N_2, N_6	1P+2N	3	33, 34, 35	Deflationary recession: falling prices, purchasing power maintained.	Late Ming Silver Crisis (1630–1644): decline in silver imports, severe deflation, collapse in prices
J	P_1, N_3, N_5	1P+2N	3	37, 38, 39	Liquidity crisis: deflation + low profits, a cascade of bankruptcies.	Late Tang Dynasty (9th century): monetary crisis, copper shortage, poor-quality coinage, exchange rate chaos

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The 20 attractors (continued)

Attractor	Triplet of pentads	Pol.	Deg.	IDs (0-based)	Economic interpretation	Chinese examples
K	P_2, N_3, N_5	1P+2N	4	40, 41, 42, 43	Agricultural depression: sectoral shock, rural exodus.	Late Ming Little Ice Age (17th century): crop failures, famines, collapse of agricultural production, peasant revolts
L	P_3, N_2, N_4	1P+2N	3	45, 46, 47	War economy: inflation + nodal buy-back, destruction of productive capacity.	Three Kingdoms period (220–280): fragmentation, regional war economies, inflation, requisitions
M	P_4, N_1, N_3	1P+2N	3	49, 50, 51	Nodal Point on the frontier: depletion of available resources.	Qing territorial expansion (17th–18th centuries): annexation of Xinjiang, Tibet and Mongolia; frontier outposts, but high costs
N	P_4, N_5, N_6	1P+2N	3	53, 54, 55	Nodal Point with dispersion: territorial fragmentation.	Late Tang (9th–10th centuries): fragmentation of power, autonomous regional economies, decline in interregional trade
O	P_5, N_1, N_4	1P+2N	3	57, 58, 59	Pioneer frontier in deflation: unprofitable colonisation.	Han westward expansion (2nd century BC): colonisation of the western regions, high costs, uncertain returns
P	P_6, N_1, N_2	1P+2N	2	61, 63	Dispersion under an oikoumene regime: voluntary de-urbanisation, withdrawal.	The Song court's flight to the south (1127): mass exodus to the south, dispersion of the population and economic activities, economic reorganisation

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The 20 attractors (continued)

Attractor	Triplet of pentads	Pol.	Deg.	IDs (0-based)	Economic interpretation	Chinese examples
Q	P_2, N_1, N_4	1P+2N	3	6, 20, 24	Recession with low purchasing power: austerity, social crisis.	Late Sui and Early Tang (7th century): collapse of the Sui dynasty, wars, famines, collapse of purchasing power
R	P_3, N_1, N_5	1P+2N	3	28, 32, 36	Stagflation: inflation + high profits, nodal buy-back.	Song–Jin War (12th–13th centuries): inflation caused by the issuance of paper money, economic stagnation due to wars
S	P_6, N_5, N_6	1P+2N	3	44, 48, 52	Dispersion and Nodal Point: systemic fragmentation.	Late Qing Dynasty (19th century): Opium Wars, unequal treaties, economic fragmentation, collapse of the monetary system
T	N_2, N_3, N_4	3N	5	56, 60, 62	Terminal depression: deflation, low profits, Nodal Point, destruction of wealth.	Late Ming (1644): a combination of deflation, famines, peasant revolts, Manchu invasion; Late Qing (1911): terminal crisis, collapse of the dynasty

Appendix C –Table of economic pentads

C.1. The 12 basic pentads of Cl(6,0) and their economic interpretation

Table 58: The 12 basic pentads of Cl(6,0) and their economic interpretation

Pentad	Structure $\{B_1, B_2, B_3\}$	Fire F	Water S	Economic interpretation
P_1	$\{iI, iJ, iK\}$	$i'k$	j	Keynesian growth economy: high wages, constrained profits, public investment
P_2	$\{jI, jJ, jK\}$	$i'i$	k	Liberal market economy: deflation, high profits, flexibility
P_3	$\{kI, kJ, kK\}$	$i'j$	i	Debt-driven economy: inflation, low real wages, easy credit
P_4	$\{i'Ii, i'Ij, i'Ik\}$	$i'K$	J	Export-oriented economy: spatial dispersion, closed frontier
P_5	$\{i'Ji, i'Jj, i'Jk\}$	$i'I$	K	Nodal economy: resource depletion, blocked buy-back
P_6	$\{i'Ki, i'Kj, i'Kk\}$	$i'J$	I	Domestic consumption economy: active buy-back, concentration
N_1	$\{-iI, -iJ, -iK\}$	$-i'k$	$-j$	Wage austerity: wage compression, high profits
N_2	$\{-jI, -jJ, -jK\}$	$-i'i$	$-k$	Structural rigidity: deflation, low profits, unemployment
N_3	$\{-kI, -kJ, -kK\}$	$-i'j$	$-i$	Debt reduction: deflation, high real wages, austerity
N_4	$\{-i'Ii, -i'Ij, -i'Ik\}$	$-i'K$	$-J$	Protectionist retreat: concentration, closed frontier
N_5	$\{-i'Ji, -i'Jj, -i'Jk\}$	$-i'I$	$-K$	Stasis: Nodal Point, dispersion, depletion of resources
N_6	$\{-i'Ki, -i'Kj, -i'Kk\}$	$-i'J$	$-I$	Domestic austerity: recession, concentration, blocked buy-back

C.1. bis – Complete Table of the 144 Economic Pentads of $Cl(6, 6)$

The table is organised into 12 blocks of 12 pentads each, corresponding to the 12 spectral sheets ($g \in \{e_1, e_2, e_3, e_4, e_5, e_6, f_1, f_2, f_3, f_4, f_5, f_6\}$).

Green: positive pentads (P_k) (Sheng mode, $\eta > 0$); **Red:** negative pentads (N_k) (Ke mode, $\eta < 0$).

Complete table of the 144 economic pentads of $Cl(6, 6)$

ID	Pentad	Structure	Fire	Water	Economic interpretation
Block e_1 – Cosmic Partition 1: Inflation					
1	$P_1^{(e_1)}$	$\{e_1iI, e_1iJ, e_1iK\}$	$e_1i'k$	e_1j	Keynesianism under inflationary conditions: massive public investment, index-linked wages, sustained demand
2	$P_2^{(e_1)}$	$\{e_1jI, e_1jJ, e_1jK\}$	$e_1i'i$	e_1k	Liberal market with inflation: price flexibility, high nominal profits, speculation
3	$P_3^{(e_1)}$	$\{e_1kI, e_1kJ, e_1kK\}$	$e_1i'j$	e_1i	Debt-driven economy with inflation: easy credit, erosion of real wages, flight to real assets
4	$P_4^{(e_1)}$	$\{e_1i'Ii, e_1i'Ij, e_1i'Ik\}$	$e_1i'K$	e_1J	Exports under inflationary conditions: price competitiveness, floating exchange rates, rising import costs
5	$P_5^{(e_1)}$	$\{e_1i'Ji, e_1i'Jj, e_1i'Jk\}$	$e_1i'I$	e_1K	Nodal point with residual inflation: depletion of resources, rising prices without growth
6	$P_6^{(e_1)}$	$\{e_1i'Ki, e_1i'Kj, e_1i'Kk\}$	$e_1i'J$	e_1I	Domestic consumption with inflation: active buy-back, wage-price spiral, eroded purchasing power
7	$N_1^{(e_1)}$	$\{-e_1iI, -e_1iJ, -e_1iK\}$	$-e_1i'k$	$-e_1j$	Wage austerity with inflation: compression of real wages, sustained nominal profits
8	$N_2^{(e_1)}$	$\{-e_1jI, -e_1jJ, -e_1jK\}$	$-e_1i'i$	$-e_1k$	Structural rigidity with inflation: deflation thwarted, unemployment, price rigidity
9	$N_3^{(e_1)}$	$\{-e_1kI, -e_1kJ, -e_1kK\}$	$-e_1i'j$	$-e_1i$	Debt reduction with inflation: austerity, debt erosion through rising prices
10	$N_4^{(e_1)}$	$\{-e_1i'Ii, -e_1i'Ij, -e_1i'Ik\}$	$-e_1i'K$	$-e_1J$	Protectionist retreat with inflation: tariff barriers, rising domestic prices

Continued on the next page...

Complete table of the 144 pentads (continued)

ID	Pentad	Structure	Fire	Water	Economic interpretation
11	$N_5^{(e_1)}$	$\{-e_1i'Ji, -e_1i'Jj, -e_1i'Jk\}$	$-e_1i'I$	$-e_1K$	Stasis with inflation: blocked buy-back, residual inflation, stagnation
12	$N_6^{(e_1)}$	$\{-e_1i'Ki, -e_1i'Kj, -e_1i'Kk\}$	$-e_1i'J$	$-e_1I$	Domestic austerity with inflation: recession, stagflation, erosion of purchasing power
Block e_2 – Cosmic Partition 2: Purchasing Power					
13	$P_1^{(e_2)}$	$\{e_2iI, e_2iJ, e_2iK\}$	$e_2i'k$	e_2j	The Glorious Thirty: Keynesianism with high wages, sharing of productivity gains
14	$P_2^{(e_2)}$	$\{e_2jI, e_2jJ, e_2jK\}$	$e_2i'i$	e_2k	Liberal market with purchasing power preserved: sustained consumption, moderate profits
15	$P_3^{(e_2)}$	$\{e_2kI, e_2kJ, e_2kK\}$	$e_2i'j$	e_2i	Indebtedness with high wages: credit underpinned by household confidence
16	$P_4^{(e_2)}$	$\{e_2i'Ii, e_2i'Ij, e_2i'Ik\}$	$e_2i'K$	e_2J	Exports with high wages: move upmarket, non-price competitiveness
17	$P_5^{(e_2)}$	$\{e_2i'Ji, e_2i'Jj, e_2i'Jk\}$	$e_2i'I$	e_2K	Nodal point with high purchasing power: resilience of domestic demand
18	$P_6^{(e_2)}$	$\{e_2i'Ki, e_2i'Kj, e_2i'Kk\}$	$e_2i'J$	e_2I	Domestic consumption, high wages: domestic driver, reduced external dependence
19	$N_1^{(e_2)}$	$\{-e_2iI, -e_2iJ, -e_2iK\}$	$-e_2i'k$	$-e_2j$	Wage austerity despite purchasing power: wage compression, social resistance
20	$N_2^{(e_2)}$	$\{-e_2jI, -e_2jJ, -e_2jK\}$	$-e_2i'i$	$-e_2k$	Rigidity with high wages: rigid labour market, high unemployment
21	$N_3^{(e_2)}$	$\{-e_2kI, -e_2kJ, -e_2kK\}$	$-e_2i'j$	$-e_2i$	Debt reduction with high wages: austerity whilst maintaining consumption
22	$N_4^{(e_2)}$	$\{-e_2i'Ii, -e_2i'Ij, -e_2i'Ik\}$	$-e_2i'K$	$-e_2J$	New Deal: protectionism and high wages, demand-led recovery
23	$N_5^{(e_2)}$	$\{-e_2i'Ji, -e_2i'Jj, -e_2i'Jk\}$	$-e_2i'I$	$-e_2K$	Stagnation despite purchasing power: blocked buy-backs, sustained consumption

Continued on the next page...

Complete table of the 144 pentads (continued)

ID	Pentad	Structure	Fire	Water	Economic interpretation
24	$N_6^{(e_2)}$	$\{-e_2i'Ki, -e_2i'Kj, -e_2i'Kk\}$	$-e_2i'J$	$-e_2I$	Domestic austerity with high wages: recession, resistance to wage cuts
Block e_3 – Cosmic Partition 3: Profits					
25	$P_1^{(e_3)}$	$\{e_3iI, e_3iJ, e_3iK\}$	$e_3i'k$	e_3j	Keynesianism with high profits: public investment supported by profitability
26	$P_2^{(e_3)}$	$\{e_3jI, e_3jJ, e_3jK\}$	$e_3i'i$	e_3k	Liberal market, maximum profits: deregulation, competition, high profit margins
27	$P_3^{(e_3)}$	$\{e_3kI, e_3kJ, e_3kK\}$	$e_3i'j$	e_3i	High-profit debt model: leverage, expansion through credit
28	$P_4^{(e_3)}$	$\{e_3i'Ii, e_3i'Ij, e_3i'Ik\}$	$e_3i'K$	e_3J	Exports with high profits: competitiveness, export margins
29	$P_5^{(e_3)}$	$\{e_3i'Ji, e_3i'Jj, e_3i'Jk\}$	$e_3i'I$	e_3K	Nodal point with high profits: economic rents, resource depletion
30	$P_6^{(e_3)}$	$\{e_3i'Ki, e_3i'Kj, e_3i'Kk\}$	$e_3i'J$	e_3I	Domestic consumption, high profits: domestic margins, market rent
31	$N_1^{(e_3)}$	$\{-e_3iI, -e_3iJ, -e_3iK\}$	$-e_3i'k$	$-e_3j$	Wage austerity, falling profits: cost-cutting, erosion of margins
32	$N_2^{(e_3)}$	$\{-e_3jI, -e_3jJ, -e_3jK\}$	$-e_3i'i$	$-e_3k$	Rigidity, falling profits: deflation, unemployment, squeezed margins
33	$N_3^{(e_3)}$	$\{-e_3kI, -e_3kJ, -e_3kK\}$	$-e_3i'j$	$-e_3i$	Debt reduction, falling profits: austerity, disinflation, erosion of profit margins
34	$N_4^{(e_3)}$	$\{-e_3i'Ii, -e_3i'Ij, -e_3i'Ik\}$	$-e_3i'K$	$-e_3J$	Protectionism with falling profits: retrenchment, loss of markets
35	$N_5^{(e_3)}$	$\{-e_3i'Ji, -e_3i'Jj, -e_3i'Jk\}$	$-e_3i'I$	$-e_3K$	Stagnation with falling profits: deadlock, residual rents
36	$N_6^{(e_3)}$	$\{-e_3i'Ki, -e_3i'Kj, -e_3i'Kk\}$	$-e_3i'J$	$-e_3I$	Domestic austerity, falling profits: recession, squeeze on margins
Block e_4 – Cosmic Partition 4: Active Rent Extraction					
37	$P_1^{(e_4)}$	$\{e_4iI, e_4iJ, e_4iK\}$	$e_4i'k$	e_4j	Keynesianism, active rent extraction: transfer of Value financed by public borrowing

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Complete table of the 144 pentads (continued)

ID	Pentad	Structure	Fire	Water	Economic interpretation
38	$P_2^{(e_4)}$	$\{e_4jI, e_4kJ, e_4jK\}$	$e_4i'i$	e_4k	Liberal market, active rent extraction: transfer of Value via the market, privatisations
39	$P_3^{(e_4)}$	$\{e_4kI, e_4kJ, e_4kK\}$	$e_4i'j$	e_4i	Debt, active rent extraction: advance credit, LGFV, financing of the transfer
40	$P_4^{(e_4)}$	$\{e_4i'Ii, e_4i'Ij, e_4i'Ik\}$	$e_4i'K$	e_4J	Exports, active rent extraction: transfer of Value abroad, FDI
41	$P_5^{(e_4)}$	$\{e_4i'Ji, e_4i'Jj, e_4i'Jk\}$	$e_4i'I$	e_4K	Active rent extraction on the frontier: conquest of new areas of Value
42	$P_6^{(e_4)}$	$\{e_4i'Ki, e_4i'Kj, e_4i'Kk\}$	$e_4i'J$	e_4I	Domestic consumption, active rent extraction: transfer of Value towards consumption
43	$N_1^{(e_4)}$	$\{-e_4iI, -e_4iJ, -e_4iK\}$	$-e_4i'k$	$-e_4j$	Austerity, rent extraction blocked: wage cuts, frozen transfer
44	$N_2^{(e_4)}$	$\{-e_4jI, -e_4jJ, -e_4jK\}$	$-e_4i'i$	$-e_4k$	Rigidity, rent extraction blocked: deflation, blocking of the transfer of Value
45	$N_3^{(e_4)}$	$\{-e_4kI, -e_4kJ, -e_4kK\}$	$-e_4i'j$	$-e_4i$	Debt reduction, rent extraction blocked: austerity, freeze on anticipated credit
46	$N_4^{(e_4)}$	$\{-e_4i'Ii, -e_4i'Ij, -e_4i'Ik\}$	$-e_4i'K$	$-e_4J$	Protectionism, rent extraction blocked: withdrawal, closure of Value's borders
47	$N_5^{(e_4)}$	$\{-e_4i'Ji, -e_4i'Jj, -e_4i'Jk\}$	$-e_4i'I$	$-e_4K$	Stasis, rent extraction blocked: nodal point, depletion of resources
48	$N_6^{(e_4)}$	$\{-e_4i'Ki, -e_4i'Kj, -e_4i'Kk\}$	$-e_4i'J$	$-e_4I$	Domestic austerity, rent extraction blocked: recession, transfer freeze
Block e_5 – Cosmic Partition 5: Dispersion					
49	$P_1^{(e_5)}$	$\{e_5iI, e_5iJ, e_5iK\}$	$e_5i'k$	e_5j	Keynesianism with dispersion: decentralised public investment
50	$P_2^{(e_5)}$	$\{e_5jI, e_5jJ, e_5jK\}$	$e_5i'i$	e_5k	Liberal market with dispersion: offshoring, spatial competition
51	$P_3^{(e_5)}$	$\{e_5kI, e_5kJ, e_5kK\}$	$e_5i'j$	e_5i	Indebtedness with dispersion: widespread credit, local bubbles
52	$P_4^{(e_5)}$	$\{e_5i'Ii, e_5i'Ij, e_5i'Ik\}$	$e_5i'K$	e_5J	Exports with dispersion: offshoring of value chains

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Complete table of the 144 pentads (continued)

ID	Pentad	Structure	Fire	Water	Economic interpretation
53	$P_5^{(e_5)}$	$\{e_5i' Ji, e_5i' Jj, e_5i' Jk\}$	$e_5i' I$	e_5K	Rent extraction with spatial dispersion: diffuse frontier, territorial conquest
54	$P_6^{(e_5)}$	$\{e_5i' Ki, e_5i' Kj, e_5i' Kk\}$	$e_5i' J$	e_5I	Domestic consumption with dispersion: market fragmentation
55	$N_1^{(e_5)}$	$\{-e_5iI, -e_5iJ, -e_5iK\}$	$-e_5i' k$	$-e_5j$	Austerity with concentration: contraction, forced consolidation
56	$N_2^{(e_5)}$	$\{-e_5jI, -e_5jJ, -e_5jK\}$	$-e_5i' i$	$-e_5k$	Rigidity with concentration: deflation, local monopolies
57	$N_3^{(e_5)}$	$\{-e_5kI, -e_5kJ, -e_5kK\}$	$-e_5i' j$	$-e_5i$	Debt reduction with concentration: austerity, withdrawal to the centres
58	$N_4^{(e_5)}$	$\{-e_5i' Ii, -e_5i' Ij, -e_5i' Ik\}$	$-e_5i' K$	$-e_5J$	Protectionism, concentration: withdrawal, spatial closure
59	$N_5^{(e_5)}$	$\{-e_5i' Ji, -e_5i' Jj, -e_5i' Jk\}$	$-e_5i' I$	$-e_5K$	Stagnation with concentration: blockage, thwarted dispersion
60	$N_6^{(e_5)}$	$\{-e_5i' Ki, -e_5i' Kj, -e_5i' Kk\}$	$-e_5i' J$	$-e_5I$	Internal austerity with concentration: recession, spatial withdrawal
Block e_6 – Cosmic Partition 6: Land Tenure System					
61	$P_1^{(e_6)}$	$\{e_6iI, e_6iJ, e_6iK\}$	$e_6i' k$	e_6j	Keynesianism, open frontier: internal colonisation, territorial investment
62	$P_2^{(e_6)}$	$\{e_6jI, e_6jJ, e_6jK\}$	$e_6i' i$	e_6k	Liberal market, open frontier: land privatisation, land market
63	$P_3^{(e_6)}$	$\{e_6kI, e_6kJ, e_6kK\}$	$e_6i' j$	e_6i	Indebtedness, open frontier: land credit, mortgages
64	$P_4^{(e_6)}$	$\{e_6i' Ii, e_6i' Ij, e_6i' Ik\}$	$e_6i' K$	e_6J	Exports, open frontier: acquisition of agricultural land abroad
65	$P_5^{(e_6)}$	$\{e_6i' Ji, e_6i' Jj, e_6i' Jk\}$	$e_6i' I$	e_6K	Rent extraction on the frontier: territorial expansion, colonisation
66	$P_6^{(e_6)}$	$\{e_6i' Ki, e_6i' Kj, e_6i' Kk\}$	$e_6i' J$	e_6I	Domestic consumption, frontier: settlement, land development
67	$N_1^{(e_6)}$	$\{-e_6iI, -e_6iJ, -e_6iK\}$	$-e_6i' k$	$-e_6j$	Austerity, closed frontier: contraction, territorial saturation
68	$N_2^{(e_6)}$	$\{-e_6jI, -e_6jJ, -e_6jK\}$	$-e_6i' i$	$-e_6k$	Rigidity, closed frontier: deflation, land-use restrictions
69	$N_3^{(e_6)}$	$\{-e_6kI, -e_6kJ, -e_6kK\}$	$-e_6i' j$	$-e_6i$	Debt reduction, closed frontier: austerity, restriction on land credit
70	$N_4^{(e_6)}$	$\{-e_6i' Ii, -e_6i' Ij, -e_6i' Ik\}$	$-e_6i' K$	$-e_6J$	Protectionism, closed frontier: withdrawal, border closures
71	$N_5^{(e_6)}$	$\{-e_6i' Ji, -e_6i' Jj, -e_6i' Jk\}$	$-e_6i' I$	$-e_6K$	Stasis, closed pioneer front: nodal point, territorial saturation

Continued on the next page...

Complete table of the 144 pentads (continued)

ID	Pentad	Structure	Fire	Water	Economic interpretation
72	$N_6^{(e_6)}$	$\{-e_6i'Ki, -e_6i'Kj, -e_6i'Kk\}$	$-e_6i'J$	$-e_6I$	Internal austerity, closed pioneer front: recession, territorial deadlock
Block f_1 – Anti-cosmic Partition 1: Wage / Inflation Regulation					
73	$P_1^{(f_1)}$	$\{f_1iI, f_1iJ, f_1iK\}$	$f_1i'k$	f_1j	Keynesian wage regulation: price controls, wage indexation, income policy
74	$P_2^{(f_1)}$	$\{f_1jI, f_1jJ, f_1jK\}$	$f_1i'i$	f_1k	Liberal wage regulation: restrictive monetary policy, central bank independence
75	$P_3^{(f_1)}$	$\{f_1kI, f_1kJ, f_1kK\}$	$f_1i'j$	f_1i	Credit-based wage regulation: credit controls, reserve requirements
76	$P_4^{(f_1)}$	$\{f_1i'Ii, f_1i'Ij, f_1i'Ik\}$	$f_1i'K$	f_1J	Inflation regulation via exports: exchange controls, countervailing duties
77	$P_5^{(f_1)}$	$\{f_1i'Ji, f_1i'Jj, f_1i'Jk\}$	$f_1i'I$	f_1K	Inflation regulation via rent extraction: freeze on transfers, capital gains tax
78	$P_6^{(f_1)}$	$\{f_1i'Ki, f_1i'Kj, f_1i'Kk\}$	$f_1i'J$	f_1I	Inflation regulation via consumption: retail price controls, targeted subsidies
79	$N_1^{(f_1)}$	$\{-f_1iI, -f_1iJ, -f_1iK\}$	$-f_1i'k$	$-f_1j$	Wage deregulation: wage liberalisation, indexation abolished
80	$N_2^{(f_1)}$	$\{-f_1jI, -f_1jJ, -f_1jK\}$	$-f_1i'i$	$-f_1k$	Market price deregulation: price liberalisation, end of controls
81	$N_3^{(f_1)}$	$\{-f_1kI, -f_1kJ, -f_1kK\}$	$-f_1i'j$	$-f_1i$	Credit deregulation: liberalisation of credit, floating interest rates
82	$N_4^{(f_1)}$	$\{-f_1i'Ii, -f_1i'Ij, -f_1i'Ik\}$	$-f_1i'K$	$-f_1J$	Export deregulation: exchange rate liberalisation, convertibility
83	$N_5^{(f_1)}$	$\{-f_1i'Ji, -f_1i'Jj, -f_1i'Jk\}$	$-f_1i'I$	$-f_1K$	Rent extraction deregulation: liberalisation of transfers, open speculation
84	$N_6^{(f_1)}$	$\{-f_1i'Ki, -f_1i'Kj, -f_1i'Kk\}$	$-f_1i'J$	$-f_1I$	Consumption price deregulation: liberalisation of retail prices, end of subsidies
Block f_2 – Anti-cosmic Partition 2: Financial / Purchasing Power Regulation					

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Complete table of the 144 pentads (continued)

ID	Pentad	Structure	Fire	Water	Economic interpretation
85	$P_1^{(f_2)}$	$\{f_2iI, f_2iJ, f_2iK\}$	$f_2i'k$	f_2j	Keynesian purchasing power regulation: wage indexation, support for consumption
86	$P_2^{(f_2)}$	$\{f_2jI, f_2jJ, f_2jK\}$	$f_2i'i$	f_2k	Liberal purchasing power regulation: labour market flexibility, wage deregulation
87	$P_3^{(f_2)}$	$\{f_2kI, f_2kJ, f_2kK\}$	$f_2i'j$	f_2i	Credit-based purchasing power regulation: consumer credit controls, interest rate caps
88	$P_4^{(f_2)}$	$\{f_2i'Ii, f_2i'Ij, f_2i'Ik\}$	$f_2i'K$	f_2J	Purchasing power regulation via exports: exchange rate adjustments, compensation
89	$P_5^{(f_2)}$	$\{f_2i'Ji, f_2i'Jj, f_2i'Jk\}$	$f_2i'I$	f_2K	Purchasing power regulation via rent extraction: control of transfers, saver protection
90	$P_6^{(f_2)}$	$\{f_2i'Ki, f_2i'Kj, f_2i'Kk\}$	$f_2i'J$	f_2I	Purchasing power regulation via consumption: subsidies, administered prices
91	$N_1^{(f_2)}$	$\{-f_2iI, -f_2iJ, -f_2iK\}$	$-f_2i'k$	$-f_2j$	Wage purchasing power deregulation: wage restraint, end of indexation
92	$N_2^{(f_2)}$	$\{-f_2jI, -f_2jJ, -f_2jK\}$	$-f_2i'i$	$-f_2k$	Labour market deregulation: total flexibilisation, increased job insecurity
93	$N_3^{(f_2)}$	$\{-f_2kI, -f_2kJ, -f_2kK\}$	$-f_2i'j$	$-f_2i$	Credit purchasing power deregulation: credit liberalisation, over-indebtedness
94	$N_4^{(f_2)}$	$\{-f_2i'Ii, -f_2i'Ij, -f_2i'Ik\}$	$-f_2i'K$	$-f_2J$	Export purchasing power deregulation: exchange rate liberalisation, loss of sovereignty
95	$N_5^{(f_2)}$	$\{-f_2i'Ji, -f_2i'Jj, -f_2i'Jk\}$	$-f_2i'I$	$-f_2K$	Rent extraction purchasing power deregulation: transfer liberalisation, capital flight
96	$N_6^{(f_2)}$	$\{-f_2i'Ki, -f_2i'Kj, -f_2i'Kk\}$	$-f_2i'J$	$-f_2I$	Consumption purchasing power deregulation: price liberalisation, end of subsidies
Block f_3 – Anti-cosmic Partition 3: Profit Regulation					

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Complete table of the 144 pentads (continued)

ID	Pentad	Structure	Fire	Water	Economic interpretation
97	$P_1^{(f_3)}$	$\{f_3iI, f_3iJ, f_3iK\}$	$f_3i'k$	f_3j	Keynesian profit regulation: taxation of rent, redistribution towards wages
98	$P_2^{(f_3)}$	$\{f_3jI, f_3jJ, f_3jK\}$	$f_3i'i$	f_3k	Liberal margin regulation: competition policy, anti-monopoly measures
99	$P_3^{(f_3)}$	$\{f_3kI, f_3kJ, f_3kK\}$	$f_3i'j$	f_3i	Investment regulation: credit controls, capital flow controls
100	$P_4^{(f_3)}$	$\{f_3i'Ii, f_3i'Ij, f_3i'Ik\}$	$f_3i'K$	f_3J	Export profit regulation: taxation of export superprofits
101	$P_5^{(f_3)}$	$\{f_3i'Ji, f_3i'Jj, f_3i'Jk\}$	$f_3i'I$	f_3K	Rent extraction profit regulation: taxation of capital gains on property
102	$P_6^{(f_3)}$	$\{f_3i'Ki, f_3i'Kj, f_3i'Kk\}$	$f_3i'J$	f_3I	Rent regulation: taxation of unearned income, redistribution
103	$N_1^{(f_3)}$	$\{-f_3iI, -f_3iJ, -f_3iK\}$	$-f_3i'k$	$-f_3j$	Profit deregulation: liberalisation of profit margins, end of taxation
104	$N_2^{(f_3)}$	$\{-f_3jI, -f_3jJ, -f_3jK\}$	$-f_3i'i$	$-f_3k$	Margin deregulation: end of competition policy, concentration
105	$N_3^{(f_3)}$	$\{-f_3kI, -f_3kJ, -f_3kK\}$	$-f_3i'j$	$-f_3i$	Investment deregulation: credit liberalisation, speculation
106	$N_4^{(f_3)}$	$\{-f_3i'Ii, -f_3i'Ij, -f_3i'Ik\}$	$-f_3i'K$	$-f_3J$	Export profit deregulation: end of taxation on superprofits
107	$N_5^{(f_3)}$	$\{-f_3i'Ji, -f_3i'Jj, -f_3i'Jk\}$	$-f_3i'I$	$-f_3K$	Stasis: total profit deregulation: speculative rents, freeze
108	$N_6^{(f_3)}$	$\{-f_3i'Ki, -f_3i'Kj, -f_3i'Kk\}$	$-f_3i'J$	$-f_3I$	Rent deregulation: end of rent taxation, concentration
Block f_4 – Anti-cosmic Partition 4: Rent Extraction Regulation					
109	$P_1^{(f_4)}$	$\{f_4iI, f_4iJ, f_4iK\}$	$f_4i'k$	f_4j	Keynesian rent extraction regulation: supervision of Value transfer, public control
110	$P_2^{(f_4)}$	$\{f_4jI, f_4jJ, f_4jK\}$	$f_4i'i$	f_4k	Liberal rent extraction regulation: legal framework, property rights
111	$P_3^{(f_4)}$	$\{f_4kI, f_4kJ, f_4kK\}$	$f_4i'j$	f_4i	Credit regulation for rent extraction: prudential ratios, credit framework
112	$P_4^{(f_4)}$	$\{f_4i'Ii, f_4i'Ij, f_4i'Ik\}$	$f_4i'K$	f_4J	Threshold N_4: export rent extraction regulation: control of international transfers
113	$P_5^{(f_4)}$	$\{f_4i'Ji, f_4i'Jj, f_4i'Jk\}$	$f_4i'I$	f_4K	Nodal point regulation: management of resource depletion

Continued on the next page...

Complete table of the 144 pentads (continued)

ID	Pentad	Structure	Fire	Water	Economic interpretation
114	$P_6^{(f_4)}$	$\{f_4i'Ki, f_4i'Kj, f_4i'Kk\}$	$f_4i'J$	f_4I	Consumption regulation: control of transfers to consumption
115	$N_1^{(f_4)}$	$\{-f_4iI, -f_4iJ, -f_4iK\}$	$-f_4i'k$	$-f_4j$	Rent extraction deregulation: transfer liberalisation, speculation
116	$N_2^{(f_4)}$	$\{-f_4jI, -f_4jJ, -f_4jK\}$	$-f_4i'i$	$-f_4k$	Credit deregulation: liberalisation of credit, end of ratios
117	$N_3^{(f_4)}$	$\{-f_4kI, -f_4kJ, -f_4kK\}$	$-f_4i'j$	$-f_4i$	Debt deregulation: debt liberalisation, over-indebtedness
118	$N_4^{(f_4)}$	$\{-f_4i'Ii, -f_4i'Ij, -f_4i'Ik\}$	$-f_4i'K$	$-f_4J$	Export deregulation: liberalisation of international transfers
119	$N_5^{(f_4)}$	$\{-f_4i'Ji, -f_4i'Jj, -f_4i'Jk\}$	$-f_4i'I$	$-f_4K$	Nodal point deregulation: abandonment of resource management
120	$N_6^{(f_4)}$	$\{-f_4i'Ki, -f_4i'Kj, -f_4i'Kk\}$	$-f_4i'J$	$-f_4I$	Consumption deregulation: liberalisation of the transfer to consumption
Block f_5 – Anti-cosmic Partition 5: Dispersion Regulation					
121	$P_1^{(f_5)}$	$\{f_5iI, f_5iJ, f_5iK\}$	$f_5i'k$	f_5j	Keynesian dispersion regulation: regional planning, fiscal equalisation
122	$P_2^{(f_5)}$	$\{f_5jI, f_5jJ, f_5jK\}$	$f_5i'i$	f_5k	Liberal dispersion regulation: competitive framework, mobility
123	$P_3^{(f_5)}$	$\{f_5kI, f_5kJ, f_5kK\}$	$f_5i'j$	f_5i	Credit dispersion regulation: sectoral allocation of credit
124	$P_4^{(f_5)}$	$\{f_5i'Ii, f_5i'Ij, f_5i'Ik\}$	$f_5i'K$	f_5J	Export dispersion regulation: control of offshoring
125	$P_5^{(f_5)}$	$\{f_5i'Ji, f_5i'Jj, f_5i'Jk\}$	$f_5i'I$	f_5K	Rent extraction dispersion regulation: supervision of the pioneer front
126	$P_6^{(f_5)}$	$\{f_5i'Ki, f_5i'Kj, f_5i'Kk\}$	$f_5i'J$	f_5I	Consumption dispersion regulation: territorial equalisation
127	$N_1^{(f_5)}$	$\{-f_5iI, -f_5iJ, -f_5iK\}$	$-f_5i'k$	$-f_5j$	Dispersion deregulation: flow liberalisation, concentration
128	$N_2^{(f_5)}$	$\{-f_5jI, -f_5jJ, -f_5jK\}$	$-f_5i'i$	$-f_5k$	Market dispersion deregulation: liberalisation, rural exodus
129	$N_3^{(f_5)}$	$\{-f_5kI, -f_5kJ, -f_5kK\}$	$-f_5i'j$	$-f_5i$	Credit dispersion deregulation: credit liberalisation, local bubbles
130	$N_4^{(f_5)}$	$\{-f_5i'Ii, -f_5i'Ij, -f_5i'Ik\}$	$-f_5i'K$	$-f_5J$	Export dispersion deregulation: offshoring liberalisation
131	$N_5^{(f_5)}$	$\{-f_5i'Ji, -f_5i'Jj, -f_5i'Jk\}$	$-f_5i'I$	$-f_5K$	Rent extraction dispersion deregulation: abandonment of the pioneer front

Continued on the next page...

Complete table of the 144 pentads (continued)

ID	Pentad	Structure	Fire	Water	Economic interpretation
132	$N_6^{(f_5)}$	$\{-f_5i'Ki, -f_5i'Kj, -f_5i'Kk\}$	$-f_5i'J$	$-f_5I$	Consumption dispersion deregulation: abandonment of equalisation
Block f_6 – Anti-cosmic Partition 6: Land Regulation					
133	$P_1^{(f_6)}$	$\{f_6iI, f_6iJ, f_6iK\}$	$f_6i'k$	f_6j	Keynesian land regulation: spatial planning, agrarian reform
134	$P_2^{(f_6)}$	$\{f_6jI, f_6jJ, f_6jK\}$	$f_6i'i$	f_6k	Liberal land regulation: land market, property rights
135	$P_3^{(f_6)}$	$\{f_6kI, f_6kJ, f_6kK\}$	$f_6i'j$	f_6i	Land credit regulation: regulation of mortgage lending
136	$P_4^{(f_6)}$	$\{f_6i'Ii, f_6i'Ij, f_6i'Ik\}$	$f_6i'K$	f_6J	Property export regulation: controls on foreign acquisition
137	$P_5^{(f_6)}$	$\{f_6i'Ji, f_6i'Jj, f_6i'Jk\}$	$f_6i'I$	f_6K	Land rent extraction regulation: regulation of ground rent
138	$P_6^{(f_6)}$	$\{f_6i'Ki, f_6i'Kj, f_6i'Kk\}$	$f_6i'J$	f_6I	Land consumption regulation: control of land use
139	$N_1^{(f_6)}$	$\{-f_6iI, -f_6iJ, -f_6iK\}$	$-f_6i'k$	$-f_6j$	Land deregulation: land market liberalisation, speculation
140	$N_2^{(f_6)}$	$\{-f_6jI, -f_6jJ, -f_6jK\}$	$-f_6i'i$	$-f_6k$	Land credit deregulation: mortgage lending liberalisation
141	$N_3^{(f_6)}$	$\{-f_6kI, -f_6kJ, -f_6kK\}$	$-f_6i'j$	$-f_6i$	Property debt deregulation: debt liberalisation, property bubble
142	$N_4^{(f_6)}$	$\{-f_6i'Ii, -f_6i'Ij, -f_6i'Ik\}$	$-f_6i'K$	$-f_6J$	Property export deregulation: foreign land acquisition liberalisation
143	$N_5^{(f_6)}$	$\{-f_6i'Ji, -f_6i'Jj, -f_6i'Jk\}$	$-f_6i'I$	$-f_6K$	Land rent extraction deregulation: abandonment of rent controls
144	$N_6^{(f_6)}$	$\{-f_6i'Ki, -f_6i'Kj, -f_6i'Kk\}$	$-f_6i'J$	$-f_6I$	Land use deregulation: land use liberalisation, urban sprawl

C.2. Complete degeneration of the 20 attractors (Cl(6,6))

This table forms part of **Cl(6,6)**, the extension that introduces the 12 spectral partitions (e_1 to e_6 , f_1 to f_6) and the 144 projected pentads. It gives the **numerical content** of each attractor: the list of IDs (pentads projected onto the 12 spectral partitions) that project onto it, and its degeneracy within this model.

Note: Each basic pentad belongs to exactly 5 attractors (triplets). A projected pentad $P_k^{(g)}$ inherits this membership and therefore projects onto the 5 corresponding attractors. The degeneracy of each attractor is therefore $3 \times 12 = 36$, where 3 is the number of basic pentads in the triplet and 12 is the number of spectral partitions.

The IDs in Table C.3 are the projections of the basic pentads from Table C.2 onto the 12 spectral partitions. For example, attractor A, defined by the triplet $\{P_1, P_2, P_4\}$ in Cl(6,0), is assigned in Cl(6,6) the IDs corresponding to the projections of P_1 , P_2 and P_4 onto the spectral partitions.

The concept of a **basin** (large, moderate, small, very small) describes the relative size of the set of IDs that project onto each attractor in Cl(6,6). A large basin indicates an economic regime towards which many configurations converge; a small basin indicates a rarer regime.

Table 60: Complete degeneracy of the 20 attractors for the 144 pentads of Cl(6,6)

Attractor	Triplet	Deg.	Interpretation
A	$\{P_1, P_2, P_4\}$	36	Large basin for perverse growth
B	$\{P_1, P_3, P_5\}$	36	Large catchment area for export-led growth
C	$\{P_2, P_3, P_6\}$	36	Large catchment area for speculative growth
D	$\{P_4, P_5, N_2\}$	36	Moderate catchment area for moderate growth
E	$\{P_5, P_6, N_3\}$	36	Moderate catchment area for urban expansion
F	$\{P_1, P_6, N_4\}$	36	Moderate catchment area for emerging stagflation
G	$\{P_2, P_5, N_6\}$	36	Moderate catchment area for growth with declining prosperity
H	$\{P_3, P_4, N_6\}$	36	Moderate catchment area for a rentier economy
I	$\{P_1, N_2, N_6\}$	36	Small basin for deflationary recession
J	$\{P_1, N_3, N_5\}$	36	Small basin for the liquidity crisis
K	$\{P_2, N_3, N_5\}$	36	Small basin for the agricultural depression
L	$\{P_3, N_2, N_4\}$	36	Small basin for the war economy
M	$\{P_4, N_1, N_3\}$	36	Very small catchment area for the nodal point on the pioneer front
N	$\{P_4, N_5, N_6\}$	36	Small catchment area for the nodal point with dispersion
O	$\{P_5, N_1, N_4\}$	36	Small catchment area for the pioneer front in deflation
P	$\{P_6, N_1, N_2\}$	36	Small catchment area for dispersion in an oikoumene regime
Q	$\{P_2, N_1, N_4\}$	36	Small catchment area for recession with low purchasing power
R	$\{P_3, N_1, N_5\}$	36	Small catchment area for stagflation
S	$\{P_6, N_5, N_6\}$	36	Small catchment area for dispersion and nodal point
T	$\{N_2, N_3, N_4\}$	36	Very small catchment area for terminal depression

Calculation	Result
20 attractors $\times$ 36	720
144 pentads $\times$ 5 attractors	720

C.3. Summary by polarity

Table 61: Summary by polarity

Polarity	Attractors	Total deg.	Number	Macroeconomic interpretation
3P	A, B, C	36	3	Periods of perverse expansionary growth (rare, unstable in the long term)
2P+1N	D, E, F, G, H	60	5	Periods of moderate growth (common during periods of expansion)
1P+2N	I, J, K, L, M, N, O, P, Q, R, S	84	11	Recession, adjustment and stagflation regimes (predominant in times of crisis)
3N	T	12	1	Terminal depression regime (collapse)

C.4. Iconic historical periods and their associated pentads

Table 62: Iconic historical periods and their associated pentads

Period	Projected pentad	ID	Detailed interpretation
The Glorious Thirty (1945–1973)	$P_1^{(e_2)}$	13	Keynesianism with high purchasing power. Capital–labour compromise, rising wages, public investment.
Stagflation (1973–1980)	$P_3^{(e_1)}$	3	Debt alongside inflation. Oil crises, rising prices, economic stagnation.
Neoliberalism (1980–2008)	$P_2^{(e_3)}$	26	Liberal market with high profits. Deregulation, globalisation, financial bubbles.
Subprime crisis (2008)	$P_6^{(e_5)}$	54	Domestic consumption with dispersion. Fragmentation, property bubbles, bank bailouts.
Post-COVID (2020–2026)	$N_3^{(f_2)}$	93	Debt reduction with financial regulation. Austerity, fragmentation of supply chains.

C.5. TRR as a new cycle

Table 63: Pentads activated by the TRR

Reform	Pentad activated	ID	Target threshold	Effect
Banque des Deux Montagnes	$P_4^{(f_4)}$	112	S_6	Redirection of capital flows towards ecological regeneration
GEP in KPIs	$P_1^{(e_2)}$	13	S_5	Reform of policy incentives, embedding of the new model
Regenerative Usufruct Certificate	$P_6^{(e_6)}$	66	S_7	Creation of a new buy-back cycle

C.6. Index by attractor (Attractor $\rightarrow$ IDs)Table 64: Index by attractor (Attractor $\rightarrow$ IDs)

Attractor	Triplet	Pol.	IDs (36) – grouped by spectral partition
<i>A</i>	$\{P_1, P_2, P_4\}$	$3P$	1, 2, 4 13, 14, 16 25, 26, 28 37, 38, 40 49, 50, 52 61, 62, 64 73, 74, 76 85, 86, 88 97, 98, 100 109, 110, 112 121, 122, 124 133, 134, 136
<i>B</i>	$\{P_1, P_3, P_5\}$	$3P$	1, 3, 5 13, 15, 17 25, 27, 29 37, 39, 41 49, 51, 53 61, 63, 65 73, 75, 77 85, 87, 89 97, 99, 101 109, 111, 113 121, 123, 125 133, 135, 137
<i>C</i>	$\{P_2, P_3, P_6\}$	$3P$	2, 3, 6 14, 15, 18 26, 27, 30 38, 39, 42 50, 51, 54 62, 63, 66 74, 75, 78 86, 87, 90 98, 99, 102 110, 111, 114 122, 123, 126 134, 135, 138
<i>D</i>	$\{P_4, P_5, N_2\}$	$2P + 1N$	4, 5, 8 16, 17, 20 28, 29, 32 40, 41, 44 52, 53, 56 64, 65, 68 76, 77, 80 88, 89, 92 100, 101, 104 112, 113, 116 124, 125, 128 136, 137, 140
<i>E</i>	$\{P_5, P_6, N_3\}$	$2P + 1N$	5, 6, 9 17, 18, 21 29, 30, 33 41, 42, 45 53, 54, 57 65, 66, 69 77, 78, 81 89, 90, 93 101, 102, 105 113, 114, 117 125, 126, 129 137, 138, 141
<i>F</i>	$\{P_1, P_6, N_4\}$	$2P + 1N$	1, 6, 10 13, 18, 22 25, 30, 34 37, 42, 46 49, 54, 58 61, 66, 70 73, 78, 82 85, 90, 94 97, 102, 106 109, 114, 118 121, 126, 130 133, 138, 142
<i>G</i>	$\{P_2, P_5, N_6\}$	$2P + 1N$	2, 5, 12 14, 17, 24 26, 29, 36 38, 41, 48 50, 53, 60 62, 65, 72 74, 77, 84 86, 89, 96 98, 101, 108 110, 113, 120 122, 125, 132 134, 137, 144
<i>H</i>	$\{P_3, P_4, N_6\}$	$2P + 1N$	3, 4, 12 15, 16, 24 27, 28, 36 39, 40, 48 51, 52, 60 63, 64, 72 75, 76, 84 87, 88, 96 99, 100, 108 111, 112, 120 123, 124, 132 135, 136, 144
<i>I</i>	$\{P_1, N_2, N_6\}$	$1P + 2N$	1, 8, 12 13, 20, 24 25, 32, 36 37, 44, 48 49, 56, 60 61, 68, 72 73, 80, 84 85, 92, 96 97, 104, 108 109, 116, 120 121, 128, 132 133, 140, 144
<i>J</i>	$\{P_1, N_3, N_5\}$	$1P + 2N$	1, 9, 11 13, 21, 23 25, 33, 35 37, 45, 47 49, 57, 59 61, 69, 71 73, 81, 83 85, 93, 95 97, 105, 107 109, 117, 119 121, 129, 131 133, 141, 143
<i>K</i>	$\{P_2, N_3, N_5\}$	$1P + 2N$	2, 9, 11 14, 21, 23 26, 33, 35 38, 45, 47 50, 57, 59 62, 69, 71 74, 81, 83 86, 93, 95 98, 105, 107 110, 117, 119 122, 129, 131 134, 141, 143
<i>L</i>	$\{P_3, N_2, N_4\}$	$1P + 2N$	3, 8, 10 15, 20, 22 27, 32, 34 39, 44, 46 51, 56, 58 63, 68, 70 75, 80, 82 87, 92, 94 99, 104, 106 111, 116, 118 123, 128, 130 135, 140, 142

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Table 64 – continued from previous page

Attractor	Triplet	Pol.	IDs (36) – grouped by spectral partition
M	$\{P_4, N_1, N_3\}$	$1P + 2N$	4, 7, 9 16, 19, 21 28, 31, 33 40, 43, 45 52, 55, 57 64, 67, 69 76, 79, 81 88, 91, 93 100, 103, 105 112, 115, 117 124, 127, 129 136, 139, 141
N	$\{P_4, N_5, N_6\}$	$1P + 2N$	4, 11, 12 16, 23, 24 28, 35, 36 40, 47, 48 52, 59, 60 64, 71, 72 76, 83, 84 88, 95, 96 100, 107, 108 112, 119, 120 124, 131, 132 136, 143, 144
O	$\{P_5, N_1, N_4\}$	$1P + 2N$	5, 7, 10 17, 19, 22 29, 31, 34 41, 43, 46 53, 55, 58 65, 67, 70 77, 79, 82 89, 91, 94 101, 103, 106 113, 115, 118 125, 127, 130 137, 139, 142
P	$\{P_6, N_1, N_2\}$	$1P + 2N$	6, 7, 8 18, 19, 20 30, 31, 32 42, 43, 44 54, 55, 56 66, 67, 68 78, 79, 80 90, 91, 92 102, 103, 104 114, 115, 116 126, 127, 128 138, 139, 140
Q	$\{P_2, N_1, N_4\}$	$1P + 2N$	2, 7, 10 14, 19, 22 26, 31, 34 38, 43, 46 50, 55, 58 62, 67, 70 74, 79, 82 86, 91, 94 98, 103, 106 110, 115, 118 122, 127, 130 134, 139, 142
R	$\{P_3, N_1, N_5\}$	$1P + 2N$	3, 7, 11 15, 19, 23 27, 31, 35 39, 43, 47 51, 55, 59 63, 67, 71 75, 79, 83 87, 91, 95 99, 103, 107 111, 115, 119 123, 127, 131 135, 139, 143
S	$\{P_6, N_5, N_6\}$	$1P + 2N$	6, 11, 12 18, 23, 24 30, 35, 36 42, 47, 48 54, 59, 60 66, 71, 72 78, 83, 84 90, 95, 96 102, 107, 108 114, 119, 120 126, 131, 132 138, 143, 144
T	$\{N_2, N_3, N_4\}$	$3N$	8, 9, 10 20, 21, 22 32, 33, 34 44, 45, 46 56, 57, 58 68, 69, 70 80, 81, 82 92, 93, 94 104, 105, 106 116, 117, 118 128, 129, 130 140, 141, 142

C.7. Index by economic variable

Table 65: Index by economic variable

Variable	Value 0	Value 1	Associated IDs
Inflation	Deflation	Inflation	1–12 (e_1)
Purchasing power	Low	High	13–24 (e_2)
Profits	Low	High	25–36 (e_3)
Buy-back	Nodal point	Active	37–48 (e_4)
Dispersion	Concentration	Dispersion	49–60 (e_5)
Land tenure	Ecumene	Pioneer frontier	61–72 (e_6)

Harmonisation note: in $Cl(6, 0)$, e_1 is the binary variable *Inflation*. In $Cl(6, 6)$, e_1 becomes a spectral sheet that modulates the pentads via inflation. The contextual correlate of this modulation in Sheng mode is growth and innovation — but the underlying variable remains inflation. This distinction is essential for maintaining consistency between the two levels of the model.

C.8. Post-2018 China (Nodal Point and Stasis)

Configuration: $\mathbf{x}_{\text{China}} = (0, 0, 0, 0, 1, 0)$ — dispersion activated, all other variables set to 0. **Projected pentads:** $\{P_6^{(e_5)}, N_5^{(e_5)}, N_6^{(e_5)}\}$ (IDs 54, 59, 60) **Resulting attractor:** **S** (1P+2N) – Dispersion and nodal point, systemic fragmentation

Table 66: Diagnosis of post-2018 China

Pentad	ID	Role in Stasis
$P_6^{(e_5)}$	54	Weakened domestic consumption, attempt to maintain growth without a pioneer frontier, spatial dispersion
$N_5^{(e_5)}$	59	Stasis, resource depletion, saturation of the land system, involuntary dispersion
$N_6^{(e_5)}$	60	Domestic austerity, recession, blocked rent extraction, territorial fragmentation

Interpretation: The system is locked onto attractor S. The P_4 and N_4 thresholds are saturated, and the system attempts a direct transition $C_N \rightarrow C_P$ without passing through N_4 , which signals retropolarity (collapse). TRR is the transition controlled through N_4 that makes it possible to avoid this retropolarity.

Justification of the IDs: the configuration $(0, 0, 0, 0, 1, 0)$ activates only the e_5 partition (dispersion). The projections of P_6 , N_5 , and N_6 onto e_5 have IDs 54, 59, 60, respectively.

Correspondence with Appendix M: ID 2 (0-based) in Appendix M associates the configuration $(0, 0, 0, 0, 1, 0)$ with attractor **S** (triplet $\{P_6, N_5, N_6\}$).

C.9. Complete numerical index (ID $\rightarrow$ Pentad)

(The full 64-row table follows in Appendix M proper, applying the rule established above.)

Preliminary note: This table applies the projection rule established in Appendix M.1: each configuration $\mathbf{x} = (b_1, \dots, b_6)$ is associated with an attractor according to the bit-to-privileged-pentads correspondence table, with aggregation by intersection.

Indexing convention: $ID = 32 b_1 + 16 b_2 + 8 b_3 + 4 b_4 + 2 b_5 + b_6$.

(The complete table of 64 states — IDs 0 through 63, with columns for $\mathbf{x}$, attractor, polarity, triplet, Western example, and Asian example — is reproduced in the main body of Appendix M.)

Principle 3 — Falsifiability. Each observable must be computable from concrete economic data, and the model must be retrospectively testable against the 64 historical regimes of Appendix M.

Table 67: Numerical index of the 144 pentads (ID $\rightarrow$ Pentad)

ID	Pentad	ID	Pentad	ID	Pentad	ID	Pentad
1	$P_1^{(e_1)}$	37	$P_1^{(e_4)}$	73	$P_1^{(f_1)}$	109	$P_1^{(f_4)}$
2	$P_2^{(e_1)}$	38	$P_2^{(e_4)}$	74	$P_2^{(f_1)}$	110	$P_2^{(f_4)}$
3	$P_3^{(e_1)}$	39	$P_3^{(e_4)}$	75	$P_3^{(f_1)}$	111	$P_3^{(f_4)}$
4	$P_4^{(e_1)}$	40	$P_4^{(e_4)}$	76	$P_4^{(f_1)}$	112	$P_4^{(f_4)}$
5	$P_5^{(e_1)}$	41	$P_5^{(e_4)}$	77	$P_5^{(f_1)}$	113	$P_5^{(f_4)}$
6	$P_6^{(e_1)}$	42	$P_6^{(e_4)}$	78	$P_6^{(f_1)}$	114	$P_6^{(f_4)}$
7	$N_1^{(e_1)}$	43	$N_1^{(e_4)}$	79	$N_1^{(f_1)}$	115	$N_1^{(f_4)}$
8	$N_2^{(e_1)}$	44	$N_2^{(e_4)}$	80	$N_2^{(f_1)}$	116	$N_2^{(f_4)}$
9	$N_3^{(e_1)}$	45	$N_3^{(e_4)}$	81	$N_3^{(f_1)}$	117	$N_3^{(f_4)}$
10	$N_4^{(e_1)}$	46	$N_4^{(e_4)}$	82	$N_4^{(f_1)}$	118	$N_4^{(f_4)}$
11	$N_5^{(e_1)}$	47	$N_5^{(e_4)}$	83	$N_5^{(f_1)}$	119	$N_5^{(f_4)}$
12	$N_6^{(e_1)}$	48	$N_6^{(e_4)}$	84	$N_6^{(f_1)}$	120	$N_6^{(f_4)}$
13	$P_1^{(e_2)}$	49	$P_1^{(e_5)}$	85	$P_1^{(f_2)}$	121	$P_1^{(f_5)}$
14	$P_2^{(e_2)}$	50	$P_2^{(e_5)}$	86	$P_2^{(f_2)}$	122	$P_2^{(f_5)}$
15	$P_3^{(e_2)}$	51	$P_3^{(e_5)}$	87	$P_3^{(f_2)}$	123	$P_3^{(f_5)}$
16	$P_4^{(e_2)}$	52	$P_4^{(e_5)}$	88	$P_4^{(f_2)}$	124	$P_4^{(f_5)}$
17	$P_5^{(e_2)}$	53	$P_5^{(e_5)}$	89	$P_5^{(f_2)}$	125	$P_5^{(f_5)}$
18	$P_6^{(e_2)}$	54	$P_6^{(e_5)}$	90	$P_6^{(f_2)}$	126	$P_6^{(f_5)}$
19	$N_1^{(e_2)}$	55	$N_1^{(e_5)}$	91	$N_1^{(f_2)}$	127	$N_1^{(f_5)}$
20	$N_2^{(e_2)}$	56	$N_2^{(e_5)}$	92	$N_2^{(f_2)}$	128	$N_2^{(f_5)}$
21	$N_3^{(e_2)}$	57	$N_3^{(e_5)}$	93	$N_3^{(f_2)}$	129	$N_3^{(f_5)}$
22	$N_4^{(e_2)}$	58	$N_4^{(e_5)}$	94	$N_4^{(f_2)}$	130	$N_4^{(f_5)}$
23	$N_5^{(e_2)}$	59	$N_5^{(e_5)}$	95	$N_5^{(f_2)}$	131	$N_5^{(f_5)}$
24	$N_6^{(e_2)}$	60	$N_6^{(e_5)}$	96	$N_6^{(f_2)}$	132	$N_6^{(f_5)}$
25	$P_1^{(e_3)}$	61	$P_1^{(e_6)}$	97	$P_1^{(f_3)}$	133	$P_1^{(f_6)}$
26	$P_2^{(e_3)}$	62	$P_2^{(e_6)}$	98	$P_2^{(f_3)}$	134	$P_2^{(f_6)}$
27	$P_3^{(e_3)}$	63	$P_3^{(e_6)}$	99	$P_3^{(f_3)}$	135	$P_3^{(f_6)}$
28	$P_4^{(e_3)}$	64	$P_4^{(e_6)}$	100	$P_4^{(f_3)}$	136	$P_4^{(f_6)}$
29	$P_5^{(e_3)}$	65	$P_5^{(e_6)}$	101	$P_5^{(f_3)}$	137	$P_5^{(f_6)}$
30	$P_6^{(e_3)}$	66	$P_6^{(e_6)}$	102	$P_6^{(f_3)}$	138	$P_6^{(f_6)}$
31	$N_1^{(e_3)}$	67	$N_1^{(e_6)}$	103	$N_1^{(f_3)}$	139	$N_1^{(f_6)}$
32	$N_2^{(e_3)}$	68	$N_2^{(e_6)}$	104	$N_2^{(f_3)}$	140	$N_2^{(f_6)}$
33	$N_3^{(e_3)}$	69	$N_3^{(e_6)}$	105	$N_3^{(f_3)}$	141	$N_3^{(f_6)}$
34	$N_4^{(e_3)}$	70	$N_4^{(e_6)}$	106	$N_4^{(f_3)}$	142	$N_4^{(f_6)}$
35	$N_5^{(e_3)}$	71	$N_5^{(e_6)}$	107	$N_5^{(f_3)}$	143	$N_5^{(f_6)}$
36	$N_6^{(e_3)}$	72	$N_6^{(e_6)}$	108	$N_6^{(f_3)}$	144	$N_6^{(f_6)}$

Appendix D – Spectral Observables and Control Applications

D.1. Introduction

Spectral observables are synthetic indicators that allow one to “take the pulse” of an economic system at a given moment. They condense hundreds of economic variables into **four key numbers**:

1. $\eta(t)$: the ****direction**** of the system (expansion or recession)
2. $d(t)$: the ****diversity**** of the system (capacity to reorganize)
3. $\text{gap}(t)$: the ****robustness**** of the system (resistance to shocks)

4. $R_{\text{seuil}}(t)$: the ****tension on strategic levers**** (proximity to a tipping point)

These observables feed the economic control application intended for Chinese decision-makers.

D.2. Analogy for Understanding the Observables

Imagine a symphony orchestra. Each instrument (economic sector) plays its own melody.

- η : ****The general tonality**** —is it a joyful piece (major mode, expansion) or a sad one (minor mode, recession)? If $\eta > 0$, the orchestra plays in Sheng mode. If $\eta < 0$, it plays in Ke mode.
- d : ****The number of different sections**** —are there many different instruments (full orchestra, diversified economy) or only a few (quartet, concentrated economy)?
- gap: ****The margin before cacophony**** —if one instrument makes a mistake, can the orchestra keep going or does everything collapse? A large gap means the orchestra can absorb an error.
- R_{seuil} : ****The tension on the key instruments**** —are the first violins and the brass on the verge of cracking? The higher R_{seuil} , the more strategic sectors are under tension.

D.3. Functional Architecture of the Application

The application relies on a 5-layer architecture:

Table 68: Functional architecture of the control application

Layer	Content
1. Data sources	Macroeconomic (GDP, inflation, unemployment); Financial (LGFV debt, credit); Territorial (land prices, GEP); Sectoral (industrial, agricultural production)
2. Pentad extraction	Projection onto the 144 pentads; Identification of $\mathbf{x} \in \{0, 1\}^6$; Extraction of the active pentad triplet
3. Spectral computation	Computation of $\eta(t)$, $d(t)$, $\text{gap}(t)$, $R_{\text{seuil}}(t)$
4. Dashboard	Gauges of the 4 observables; Polarity; Alerts; Projection onto the 20 attractors
5. Recommendations	Monetary, fiscal, structural policies; Activation of N_4 (TRR) if $R_{\text{seuil}} > 0.7$

D.4. Concrete Examples of Computation

D.4.1. The Trente Glorieuses (ID 13)

Configuration: $\mathbf{x} = (1, 1, 0, 1, 0, 0)$ — Positive inflation, high purchasing power, constrained profits, active rent extraction.

Table 69: Observables for the Trente Glorieuses (ID 13)

Observable	Value	Interpretation
η	+0.33	Moderate expansion: system in Sheng phase
d	4.5	Diversified economy: strong interactions between sectors
gap	0.20	Moderate robustness: a major shock could destabilize
R_{seuil}	0.43	Emerging tension: the oil shocks of the 1970s will tip the balance

D.4.2. The 1973–1980 Stagflation (ID 3)

Configuration: $\mathbf{x} = (1, 0, 1, 1, 0, 0)$ — High inflation, falling purchasing power, high but threatened profits.

Table 70: Observables for stagflation (ID 3)

Observable	Value	Interpretation
η	−0.15	Slight recession: shift to Ke mode
d	2.8	Less diversified economy
gap	0.12	Low robustness: fragile system
R_{seuil}	0.65	High tension: strategic levers heavily solicited

D.4.3. Post-2018 China (ID 2)

Configuration: $\mathbf{x} = (0, 0, 0, 0, 1, 0)$ — Nodal point, dispersion, closed pioneer frontier.

Table 71: Observables for post-2018 China (ID 2)

Observable	Value	Interpretation
η	−0.45	Marked recession: system deeply in Ke mode
d	1.9	Highly concentrated economy: only a few circuits remain
gap	0.10	Very low robustness: on the verge of collapse
R_{seuil}	0.78	Maximum tension: thresholds saturated, red alert

D.5. Evolution of Observables over Time

Table 72: Evolution of observables for China (2018–2026)

Year	η	d	gap	R_{seuil}	Color
2018	-0.30	2.5	0.25	0.55	Orange
2020	-0.35	2.3	0.20	0.60	Orange
2022	-0.40	2.1	0.15	0.68	Orange → Red
2024	-0.42	2.0	0.12	0.72	Red
2026	-0.45	1.9	0.10	0.78	Red

Interpretation: The asymmetry η becomes increasingly negative, the dimension d decreases, the gap shrinks, and R_{seuil} exceeds 0.7 in 2024. China enters the red zone.

D.6. Alert Thresholds and Traffic-Light System

Table 73: Alert thresholds for spectral observables

Observable	Green	Orange	Red
$\eta(t)$	$> +0.3$	0.1 to 0.3	< 0.1
$d(t)$	> 4	2 to 4	< 2
$\text{gap}(t)$	> 0.4	0.2 to 0.4	< 0.2
$R_{\text{seuil}}(t)$	< 0.3	0.3 to 0.7	> 0.7

In practice:

- **Green:** all is well, continue current policies.
- **Orange:** monitor and prepare contingency plans.
- **Red:** intervene immediately. The system is in danger.

D.7. Use Scenarios and Policy Responses

Table 74: Use scenarios and policy responses

Spectral scenario	Observables	Recommended policy response
Green —Healthy growth	$\eta > 0.3, d > 4, \text{gap} > 0.4, R_{\text{seuil}} < 0.3$	Continue current policies; invest in innovation
Orange —Emerging stagflation	$0.1 < \eta < 0.3, d < 3, \text{gap} < 0.3, 0.3 < R_{\text{seuil}} < 0.7$	Remain vigilant; prepare contingency plans
Orange —Nodal point	$\eta \approx 0, d \approx 2, \text{gap} < 0.2, R_{\text{seuil}} \approx 0.6$	Monitor thresholds P_4 and N_4 ; avoid retropolarity
Red —Stasis / Saturated nodal point	$\eta < 0, d < 2, \text{gap} < 0.15, R_{\text{seuil}} > 0.7$	ACTIVATE TRR: Bank of the Two Mountains, GEP in KPIs, CUR
Red —Terminal depression	$\eta < -0.3, d < 1.5, \text{gap} < 0.1, R_{\text{seuil}} > 0.85$	Emergency rescue plan: monetization of Value, systemic restructuring

D.8. Concrete Control Indicators

The application translates spectral observables into concrete economic indicators:

Table 75: Correspondence between spectral observables and economic indicators

Observable	Economic indicator	Data source	Alert threshold
$\eta(t) > 0$	GEP growth > GDP growth	Ministry of Ecology	$\eta < 0$
$d(t) > 4$	Regional Gini coefficient < 0.35	National Bureau of Statistics	$d < 2$
$\text{gap}(t) > 0.4$	Inter-sectoral productivity gap < 20%	National surveys	$\text{gap} < 0.2$
$R_{\text{seuil}} < 0.3$	LGFV debt ratio < 80%	Ministry of Finance	$R_{\text{seuil}} > 0.7$

D.9. Concrete Example: Use During a Red Alert

Scenario: In 2024, the application detects $R_{\text{seuil}} = 0.72 > 0.7$. The decision-maker receives a red alert.

Recommended procedure:

1. **Verification:** consult the detailed observables to confirm the diagnosis.
2. **Analysis:** identify the current attractor (S: Stasis) and the triplet $(\{P_6, N_5, N_6\})$.
3. **Simulation:** run a simulation of the TRR policy to assess its impact.
4. **Decision:** activate the Bank of the Two Mountains, integrate GEP into KPIs, launch the CURs.
5. **Monitoring:** track the evolution of observables in real time.

D.10. Integration with the Five-Year Planning Cycle

Table 76: Integration with five-year plans

Plan phase	Action in the application	Objective
Preparation (year -1)	Scenario simulation, attractor analysis	Define targets and priorities
Start (year 0)	Spectral baseline, diagnosis	Calibrate initial policies
Execution (years 1–4)	Real-time monitoring of observables	Adjust policies along the way
Evaluation (year 5)	Spectral assessment, gap analysis	Measure results, prepare the next plan

D.11. Expected Benefits and Limitations

Benefits:

- Anticipation of crises through early detection of tipping points
- Fine-grained control by adjusting policies according to observables
- Interministerial coordination via a common language
- Objective evaluation of the effectiveness of reforms
- Alignment with ecological objectives (GEP, territorial regeneration)

Limitations and precautions:

1. The application is a **decision-support tool**, not an automated system.
2. The observables are **synthetic indicators**; they do not replace detailed analysis.
3. The calibration of Λ_{eco} is provisional and will need to be refined.
4. The application requires **reliable and up-to-date data**.
5. In the event of a red alert, **act methodically** without panicking.

D.12. —Multi-Level Control in Cl(6,6)

D.12.1 — Justification and Principle

Appendix D presented a control application based on the four global spectral observables (η , d , gap, R_{seuil}). This monolithic approach is sufficient for an overall diagnosis, but it masks the contextual dissonances that the extension to Cl(6,6) brings to light: 12 spectral sheets, 144 pentads, 7 thresholds, and 66 inter-sheet relations.

Multi-level control exploits this structural richness to adapt policy responses to the scale of decision. It rests on a fundamental principle: **the granularity of intervention must match the granularity of diagnosis**.

Table 77: Resonance principle: granularity of diagnosis and intervention

Scale	Granularity of diagnosis	Granularity of intervention
Strategic (Beijing)	4 global observables	National structural reforms
Tactical (Province)	48 observables per sheet	Regional sectoral adjustments
Operational (County)	144 pentads	Fine territorial targeting
Diagnostic (Alert Center)	7 thresholds + 66 relational	Early detection

This principle is the economic equivalent of the **resonance principle** in traditional Chinese medicine: one does not apply the same acupuncture needle to treat an imbalance of the Liver meridian or a global imbalance of Qi. Likewise, one does not apply the same policy to treat a global Stasis or a localized tension on a spectral sheet.

D.12.2 — The Four Decision Layers

The following table synthesizes the architecture of multi-level control:

Table 78: Architecture of multi-level control: the four decision layers

Layer	Scale	Observables used	Typical action	Horizon
Strategic (Beijing)	National	4 global ($\eta_{\text{total}}, d_{\text{total}}, \text{gap}_{\text{total}}, R_{\text{seuil},\text{total}}$)	TRR activation if $R_{\text{seuil}} > 0.7$; monetary reform if $\text{gap} < 0.2$	5–10 years
Tactical (Province)	Provincial	48 per sheet ($\eta_f, d_f, \text{gap}_f, R_{\text{seuil},f}$ for $f \in \{1, \dots, 12\}$)	GEP adjustment according to dominant η_f ; CUR reorientation	1–3 years
Operational (County)	County / Pilot zone	144 pentads ($a_{k,g}, \pi_{k,g}, c_{k,g}, E_{k,g}$)	CUR targeting on active pentads; pilot projects	6–18 months
Diagnostic (Alert Center)	Cross-cutting	7 thresholds (σ_i) + 66 relational ($\Delta\eta_{f,g}, C_{f,g}$)	Pre-retropolarity detection; early warning	Real time

D.12.3 — Detail of Each Layer

Layer 1: Strategic Control (Beijing)

Responsible: State Council, People's Bank of China, NDRC.

Observables used: The four global observables integrated over the 12 sheets:

$$\eta_{\text{total}} = \frac{1}{12} \sum_{f=1}^{12} \eta_f, \quad d_{\text{total}} = \frac{1}{12} \sum_{f=1}^{12} d_f, \quad \text{gap}_{\text{total}} = \min_f \text{gap}_f, \quad R_{\text{seuil},\text{total}} = \frac{1}{12} \sum_{f=1}^{12} R_{\text{seuil},f}$$

Decision rules:

Table 79: Strategic decision rules (Beijing)

Condition	Strategic action	Justification
$R_{\text{seuil}} > 0.7$ AND $\text{gap} < 0.2$	TRR activation : national launch of the Bank of the Two Mountains, generalization of GEP in KPIs	Red alert: imminent retropolarity
$\mathcal{J}_{\text{Stase}} > 0.65$	**TRR activation**	Red alert: characterized Stasis
$\eta < -0.4$ AND $d < 2$	Monetary reform : massive securitization of GEP, redirection of LGFV credit	Characterized Stasis, fragmented system
$\text{gap} < 0.1$	Emergency rescue plan : monetization of ecological Value	Extreme fragility, imminent collapse
$\eta > +0.3$ AND $d > 4$	Continue current policies	Healthy growth, diversified system

Concrete example — China 2026: - $\eta_{\text{total}} = -0.45$, $d_{\text{total}} = 2.1$, $\text{gap}_{\text{total}} = 0.12$, $R_{\text{seuil},\text{total}} = 0.78$ - Diagnosis: **Characterized Stasis, red alert** - Action: National-scale TRR activation, generalization of CURs by 2028

Layer 2: Tactical Control (Province)

Responsible: Provincial governments, regional NDRC offices.

Observables used: The 48 observables per sheet (η_f , d_f , gap_f , $R_{\text{seuil},f}$ for each sheet $f \in \{1, \dots, 12\}$).

Decision rules:

Table 80: Tactical decision rules (Province)

Dominant sheet	Condition	Tactical action
Sheet e_3 (Profits)	$\eta_{e_3} < -0.5$	Fiscal reform: taxation of speculative rents, redistribution toward GEP
Sheet f_3 (Accumulation)	$R_{\text{seuil},f_3} > 0.7$	LGFV restructuring: debt swaps, securitization of ecological assets
Sheet e_1 (Inflation)	$\eta_{e_1} > +0.3$ AND $\eta_{e_3} < -0.5$	Sheng/Ke imbalance : rebalance by taxing inflation gains to finance material regeneration
Sheet f_6 (Land regulation)	$\text{gap}_{f_6} < 0.1$	Land alert : freeze speculative transactions, activate CURs

Concrete example — Zhejiang Province 2026: - Dominant sheet: e_1 (Inflation) with $\eta_{e_1} = +0.6$ - Critical sheet: f_3 (Accumulation) with $R_{\text{seuil},f_3} = 0.82$ - Diagnosis: **Structural imbalance** — cognitive innovation (AI, data) does

not compensate for the collapse of land accumulation - Action: Lishui pilots the Bank of the Two Mountains, securitization of forest GEP to recapitalize LGFVs

Layer 3: Operational Control (County)

Responsible: County governments, local banks, rural cooperatives.

Observables used: The 144 projected pentads, with their four local observables: - **Activity** $a_{k,g}(t) = \|\psi_{k,g}(t)\|^2 \in [0, 1]$ - **Polarity** $\pi_{k,g}(t) = \text{sgn}(\varepsilon_{k,g}) \in \{+1, -1\}$ - **Connectivity** $c_{k,g}(t) = \sum_{j \sim k} w_{kj,g}(t)$ - **Frustration energy** $E_{k,g}(t) = |\varphi_{k,g}(t) - \bar{\varphi}_g(t)|^2$

Decision rules:

Table 81: Operational decision rules (County)

Pentad signature	Condition	Operational action
Pentad N_5 (Stasis)	$a_{N_5} > 0.7$ AND $c_{N_5} > 4$	Critical node: priority targeting for desaturation via CUR
Pentad P_6 (Consumption)	$a_{P_6} < 0.3$ AND $E_{P_6} > 0.3$	Frustrated pentad: credit injection to revive domestic consumption
Pentad N_6 (Austerity)	$\pi_{N_6} = -1$ AND $c_{N_6} > 3$	Contagious pentad: isolation through targeted reform, prevent propagation
Pentad P_2 (Liberal market)	$a_{P_2} > 0.5$ AND $\pi_{P_2} = +1$	Active positive pentad: amplification through green subsidies

Concrete example — Xingguo County (Jiangxi) 2026: - Most active pentad: $N_5^{(f_3)}$ (Stasis with profit regulation) with $a = 0.85$, $c = 5$ - Frustrated pentad: $P_6^{(e_2)}$ (Consumption with purchasing power) with $E = 0.42$ - Diagnosis: **Local Stasis node** — profit regulation stifles consumption - Action: CUR deployment on eroded watersheds, creation of local jobs through regeneration of soils contaminated by rare-earth mines

Layer 4: Cross-Cutting Diagnosis (Alert Center)

Responsible: National spectral monitoring center (to be created), research academies.

Observables used: - The 7 **spectral thresholds** $\sigma_i(t) \in \{0, 1\}$ for $i \in \{1, \dots, 7\}$ - The 66 **relational observables** $\Delta\eta_{f,g}(t)$ and $C_{f,g}(t)$ for $f < g$

Decision rules:

Table 82: Diagnostic decision rules (Alert Center)

Signal	Condition	Alert
Pre-retropolarity	$\sigma_5 = \sigma_6 = \sigma_7 = 0$ AND $R_{\text{seuil}} > 0.7$	Red: system blocked between S_4 and S_5 , urgent TRR activation
Wuxing imbalance	$\max_{f,g} \Delta\eta_{f,g} > 1.0$	Orange: structural tension between sheets, rebalancing required
Fragmentation	$\mathcal{F} = 1 - \frac{1}{66} \sum_{f < g} C_{f,g} > 0.5$	Orange: system fragmented into independent subsets
Sheng↔Ke switch	$\sigma_2 = 1$ AND $\Delta\eta_{\text{max}} > 0.8$	Yellow: regime transition underway, increased monitoring

Concrete example – China 2026: - Threshold vector: $\vec{\sigma} = (1, 1, 1, 1, 0, 0, 0)$ - Maximum Wuxing tension: $\Delta\eta_{\text{max}} = 1.1$ between Fire ($\eta_{\text{Fire}} = +0.3$) and Earth ($\eta_{\text{Earth}} = -0.7$) - Fragmentation index: $\mathcal{F} = 0.42$ - Diagnosis: **Pre-retropolarity with Wuxing imbalance** – Fire (cognitive innovation) attempts an expansion while Earth (material profits) collapses - Alert: **Red** – urgent TRR activation to cross S_5 (Fire Transition)

D.12.4 – Integrated Example: TRR Control in 2026

The following table shows how the four decision layers articulate to control the TRR in 2026:

Table 83: Articulation of the four layers for TRR control in 2026

Layer	Diagnosis	Action	Target observable	Deadline
Strategic (Beijing)	$R_{\text{seuil}} = 0.78 > 0.7$	National launch of the TRR	$R_{\text{seuil}} \rightarrow < 0.5$	2028
Tactical (Zhejiang)	$\eta_{f_3} = -0.6, \eta_{e_1} = +0.3$	Bank of the Two Mountains pilot in Lishui	$\eta_{f_3} \rightarrow > -0.3$	2027
Operational (Lishui)	$a_{N_5} = 0.78, c_{N_5} = 5$	CUR on 50,000 ha of degraded forests	$a_{N_5} \rightarrow < 0.4$	2026–2027
Diagnostic (Center)	$\sigma_5 = 0, \Delta\eta_{\text{max}} = 1.1$	Monitoring of S_5 crossing	$\sigma_5 \rightarrow 1$	2027

Expected trajectory: - 2026: $\mathcal{J}_{\text{Stase}} = 0.64$ (Characterized Stasis) - 2028: $\mathcal{J}_{\text{Stase}} = 0.55$ (Emerging Stasis, start of recovery) - 2030: $\mathcal{J}_{\text{Stase}} = 0.35$ (Moderate tension, exit from Stasis) - 2032: $\mathcal{J}_{\text{Stase}} < 0.3$ (Healthy growth, new Rent Extraction cycle)

D.12.5 – Articulation with the 15th Five-Year Plan Calendar

Multi-level control articulates with the TRR deployment calendar (Appendix J):

Table 84: Articulation of multi-level control with the 15th Five-Year Plan calendar

Phase	Period	Priority layer	Key observable	Threshold to cross
Pilot	2024–2025	Operational + Diagnostic	a_{N_5} in the 10 pilot zones	S_5 (Fire Transition)
Deployment	2026–2028	Tactical + Strategic	η_{f_3} in the pilot provinces	S_6 (Water Transition)
Acceleration	2029–2030	Strategic + Tactical	$\eta_{\text{total}}, R_{\text{seuil,total}}$	S_7 (Octave Leap)
Generalization	2031–2035	All layers	$\mathcal{J}_{\text{Stase}} < 0.3$	Stabilization on C_P

D.12.6 – Correspondence with Chinese Thought

Multi-level control is the modern equivalent of **Chinese imperial governance**, which articulated three scales of regulation:

Table 85: Correspondence between modern multi-level control and Chinese imperial governance

Imperial scale	Modern equivalent	Observable	Chinese concept
Emperor (Mandate of Heaven)	Strategic (Beijing)	4 global	<i>Tianming</i> (天命) – Cosmic legitimacy
Provincial governors	Tactical (Province)	48 per sheet	Regional <i>Qi</i> (经气) – Meridian circulation
County magistrates	Operational (County)	144 pentads	Acupuncture points (穴) – Fine intervention
Imperial diviners	Diagnostic (Center)	7 thresholds + 66 relational	Divination (卜) – Probe of Heaven

The Emperor does not descend into the detail of acupuncture points; he ensures that the Mandate of Heaven is preserved ($\mathcal{J}_{\text{Stase}} < 0.3$). Provincial governors adjust the circulation of Qi in their meridians (η_f). County magistrates apply needles to precise points ($a_{k,g}$). Diviners probe the warning signs ($\sigma_i, \Delta\eta_{f,g}$).

This articulation is the **mathematization of Chinese structural wisdom**: economic control is not an imported Western technique, but the actualization of an art of governing that dates back to the Celestial Empire.

D.12.7 — Limitations and Precautions

1. **Computational complexity:** Computing the 269 elementary observables requires robust IT infrastructure. Appendix S (§S.8) provides an algorithm of complexity $O(22615)$, executable in less than one second on a standard computer.
2. **Data quality:** The observables of Level 4 (pentads) and Level 5 (relational) require fine-grained data that is not always available. Progressive calibration is necessary.
3. **Risk of over-interpretation:** The richness of the observables may tempt decision-makers to multiply interventions. The principle of parsimony must remain: **intervene only on critical observables**.
4. **Articulation with existing plans:** Multi-level control does not replace five-year plans; it enriches them with a spectral dashboard. Traditional indicators (GDP, unemployment, inflation) remain necessary for public communication.

D.12.8 — Synthesis

Multi-level control transforms the control application of Appendix D into an **endogenous regulation system** that respects the laws of the $64 \rightarrow 20$ invariant. It articulates four decision layers, each adapted to the geographic and temporal scale of intervention.

This architecture is the operational translation of the document's central thesis: **the economy is a discrete information system, and its regulation requires a multi-resolution dashboard**, analogous to the four examinations of traditional Chinese medicine (望闻问切).

Multi-level control offers China a tool to:

- **Anticipate** tipping points (Diagnostic layer)
- **Target** interventions (Operational layer)
- **Rebalance** structural tensions (Tactical layer)
- **Preserve** the Mandate of Heaven (Strategic layer)

This is the first operational application of a **physics of regulation** that could transform our understanding of the control of complex socio-economic systems.

E –Comparative table of the five manifestations of the 64→20 invariant

Table 86: The five manifestations of the 64→20 invariant

Element	Particles	Genetic code	Mandarin phonology	Tang alliances	Economy (ToRE)	Reference
Base space	Cl(6,0)	4 bases × 3 positions	24 initials × 40 finals	6 binary bits	6 binary dimensions	See bibliography
Number of configurations	64 spin states	64 codons	64 phonological combinations	64 marital configurations	64 economic states	—
Filtration	Merkabah / Λ_{72}	Neighbourhood rule	Articulatory constraints	Pentads / <i>Wufu</i>	Nilpotence	See below
Number of attractors	20 particle families	20 amino acids	20 phonological classes	20 alliance attractors	20 economic systems	—
Degeneration	1–5	1–6	1–5	1–6	1–6	—
3P→3N gradient	Leptons → Hadrons	Hydrophilic → Hydrophobic	Vowels → Consonants	Imperial Li → Foreigners	Growth → Depression	—
Polar thresholds	Gauge bosons	Initiating codons	Tones	P_4, N_4 (pivotal clans)	P_4 (expansion), N_4 (recession)	—
Sheng/Ke modes	Symmetries	Replication / Transcription	Assimilation / Dissimilation	Exogamy / Endogamy	Expansion / Recession	—
Constants β_k	15 constants (Λ_{72})	15 constants (Λ_{72})	15 constants (to be determined)	15 constants (Λ_{72})	15 constants (Λ_{72})	—
Scale constant	$\Lambda_{\text{nuc}} = 7.726 \text{ MeV}$	$\Lambda_e = 5.950 \text{ eV}$	Λ_{phon}	Λ_{kin}	Λ_{eco}	—
Main reference	Rowlands (2007); De Dominicis (2026)	Rowlands & Hill (2007); De Dominicis (2026)	De Dominicis (2026)	De Dominicis (2026)	De Dominicis (2026)	This report (original extension)

F –Distribution of Chinese examples by attractor

Table 87: Distribution of Chinese examples by attractor

Polarity	Attractor	Chinese examples
3P	A	安史之乱后唐 (hyperinflation), 南宋中后期 (over-issuance of <i>jiaozi</i>), inflation 1988–1989
3P	B	After the Ming Longqing opening of trade (silk/porcelain exports), accession to the WTO (2001)
3P	C	Song dynasty (<i>jiaozi</i> , trade), property crisis 2021–2026
2P+1N	D	The Wen-Jing Prosperity, the Kaiyuan Golden Age, reform and opening-up (1978–1993)
2P+1N	E	Northern Song urbanisation, Silk Roads (21st century)
2P+1N	F	Mid-to-late Ming period (influx of money), structural adjustment (2013–2018)
2P+1N	G	Mid-Qing period (demographic pressure), 1990s (reforms)
2P+1N	H	Late Ming (land concentration), platform economy
1P+2N	I	Late Ming silver crisis, Late Tang
1P+2N	J	Late Tang monetary crisis, 1960s
1P+2N	K	Late Ming Little Ice Age, civil war (1927–1949)
1P+2N	L	Three Kingdoms, Taiping Kingdom
1P+2N	M	Qing territorial expansion, conquest of Xinjiang
1P+2N	N	Late Tang fragmentation, Silk Road (Han/Tang)
1P+2N	O	Han westward expansion, Phoenician trading posts
1P+2N	P	Southern Migration of the Song Dynasty, rural exodus (1980s–2000s)
1P+2N	Q	Late Sui and Early Tang periods, invasions by the Five Barbarians
1P+2N	R	Song–Jin war, post-COVID (2020–2026)
1P+2N	S	Late Qing (Opium Wars), debt crisis (2021–2026)
3N	T	Late Ming (1644), Late Qing (1911), China 2018–2026 (Nodal Point)

G –The 9 dynastic declines

Table 88: The 9 Chinese dynastic declines — Multi-century trends and crisis mechanisms

Dynasty	Initial structural mandate	Rent Extraction / Avoidance phases	Crisis mechanism (exhaustion of the trend)	Example of destruction
Han (206 BC–220)	End of slavery, land registry finalised, frontier settlement completed. Shift in value towards land.	Initial trend: growth based on free small-scale farmers. Spatial expansion limited by the saturation of arable land.	Exhaustion of the initial trend of land buy-ups. The state is no longer able to control prices in the face of land hoarding by large landowners, preventing wage adjustments.	Yellow Turban Rebellion (184), civil war, fragmentation into the Three Kingdoms.
Fragmentation / 5 Barbarians (220–581)	Collapse of the central state. Nomadic invasions (<i>Wu Hu</i>) seize land in the North and reduce the peasantry to serfdom. Flight of the elites and a new frontier in the South (Yangtze basin).	Spatial avoidance (South): development of new southern lands. Servile mode (North): capture of populations to supply labour.	The northern conquerors become Sinicised and adopt the land rent system. The servile system was running out of steam. In the South, the frontier was rapidly closing in the face of demographic pressure.	Ceaseless wars, demographic collapse, violent reunification under the Sui (581).
Tang (618–907)	<i>Jun-tian</i> system (equal fields), restoration of small-scale peasantry, territorial expansion westwards.	Spatial avoidance: conquests in Central Asia, the Silk Road. The <i>Jun-tian</i> system redistributed land but became saturated in the face of demographic pressure.	Failure of the <i>Jun-tian</i> system's spatial avoidance strategy. Demographic pressure saturated the frontier, and price rigidity caused a double scarcity (land/labour). Regional lords (<i>jiedushi</i>) monopolise the land.	An Lushan's rebellion (755–763), decline of central authority, rise of the warlords.

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Dynasty	Initial structural mandate	Rent Extraction / Avoidance phases	Crisis mechanism (exhaustion of the trend)	Example of destruction
Song (960–1127)	Peak commercial development, paper money (<i>jiaozi</i>), absence of a land frontier.	Temporal avoidance: monetary and commercial development to compensate for the absence of a frontier. Intensive maritime trade.	Maximum temporal avoidance pushed to its peak. Money does not create land. The state can no longer readjust prices in the face of currency depreciation and land concentration.	Jurchen invasions, fall of Kaifeng (1127), loss of northern China.
Jin (North) & Southern Song (1127–1279)	Split of the empire. The Jurchen confiscate land in the north and impose a semi-servile system (<i>meng'an/mouke</i>). The Southern Song survive in the southern half.	North: reinstatement of the servile system through conquest. South: hyper-specialisation in temporal avoidance (credit, massive maritime exports).	Duality exhausted. In the North, military-agricultural stagnation. In the South, debt and fiat currency reach their limits. The system can no longer regulate the crisis of overproduction.	Hyperinflation in the South, Mongol conquest (1234 in the North, 1279 in the South).
Yuan (1279–1368)	Unification under Mongol rule, reintroduction of a frontier through conquest.	Forced spatial avoidance: the Mongol Empire opens up new territories, but ethnic discrimination blocks redistribution.	The Mongol expansionist trend quickly ran out of steam. Flooding of the Yellow River (1340s) and the Black Death exacerbated the crisis. The state could no longer control food prices.	Red Turban Rebellion (1351–1368), foundation of the Ming dynasty by Zhu Yuanzhang.
Ming (1368–1644)	Massive redistribution of land after the war, restoration of small-scale peasantry, the Great Wall as the frontier boundary.	Initial spatial avoidance: expansion towards the south and south-west. Then a tightening of policy: maritime ban (<i>haijin</i>), taxes fixed in silver.	Failure of the initial redistribution. Spatial avoidance ceases (closed frontier). State price controls can no longer mask massive peasant indebtedness. Spanish silver (via Manila) triggers deflationary shocks.	Li Zicheng's rebellion (1644), fall of Beijing, Manchu invasion.

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Dynasty	Initial structural mandate	Rent Extraction / Avoidance phases	Crisis mechanism (exhaustion of the trend)	Example of destruction
Qing (1644–1912)	Last major territorial expansion (Xinjiang, Tibet, Taiwan), population tripled (100 M → 400 M).	Maximum spatial avoidance: conquest of Central Asia. Then temporal avoidance: unequal treaties, forced opening of ports, foreign debt.	The centuries-long trend of territorial expansion reached its geographical limit. Spatial avoidance became impossible. Demographic pressure skyrocketed. Rebellions (Taiping, Boxers) drained the imperial coffers.	Boxer Rebellion (1900), Xinhai Revolution (1911), abdication of the last emperor (1912).
Republic (1912–1949)	Attempts at modernisation, coastal industrialisation, collapse of the imperial system.	Absence of a rent extraction mechanism: neither a frontier settlement policy nor effective redistribution. Civil war, Japanese occupation, hyperinflation.	Total absence of any mechanism for recovery or mitigation. The market does not function; the state is powerless to readjust prices and wages, leading to the collapse of the price system. Hyperinflation (1945–1949).	Hyperinflation (monthly rate > 50%), capital flight, Communist victory (1949).

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Summary: The nine declines share the same underlying structure: 1. A **multi-century trend of growth** initiated by redistribution (land, frontier expansion, currency). 2. **Avoidance mechanisms** (spatial or temporal) that postpone the crisis. 3. A **final exhaustion** of these mechanisms, exacerbated by state-imposed price rigidity. 4. A **change of dynasty** that resets the system and initiates a new trend.

Application to post-2018 China: The analysis of the CI(6,6) model confirms this interpretation. China is experiencing the final phase of a centuries-long trend of material accumulation that began in 1978. For 40 years, China has used mechanisms of **spatial avoidance** (globalisation, LGFV, urbanisation) to stave off the crisis of overproduction. This pioneering front is now saturated (ghost towns, atrophied countryside). At the same time, the mechanism of **temporal avoidance** (debt) has reached its limits. The Territorial Regeneration Rent (TRR) is therefore not merely a tool for crossing a spectral threshold (N_4), but the indispensable mechanism for **reorganising the system and initiating a new growth trend**, by recreating Value where spatial and temporal avoidance mechanisms have failed.

Note: According to the Terms of Reference, the 'Nodal Point' is not the cause of the declines, but their structural culmination. It marks the moment when the avoidance mechanisms (spatial or temporal) of a centuries-long trend are completely exhausted, and when the price rigidity imposed by the State makes a regime change (reorganisation) inevitable in order to revive growth.

Appendix H –Bilingual glossary of ToRE concepts and the Chinese economy

Table 89: Bilingual glossary of ToRE concepts and the Chinese economy

English	Français	Chinese (pinyin)	Definition
Advanced Exogenous Demand	Demande exogène en avance	超前外生需求 (<i>Chāoqián wàishēng xūqiú</i>)	Purchasing power injected by credit before production begins.
Algorithmic Stasis	Stase algorithmique	算法性停滞 (<i>Suànfǎxìng tíngzhì</i>)	Decoupling between intangible value and material production.
Bank of Two Mountains	Banque des Deux Montagnes	两山银行 (<i>Liǎngshān Yínháng</i>)	Securitization of ecological assets (GEP) to finance rural regeneration (pilot in Lishui, Zhejiang).
Centrifugal Dynamics	Dynamique centrifuge	离心力动态 (<i>Líxīnlì Dòngtài</i>)	Spatial inversion at the Nodal Point (periurbanization, ghost cities).
Chinese World-System (Tianxia)	Système-monde chinois (Tianxia)	天下 (<i>Tiānxià</i>)	Hierarchical and fluid vision of international space.

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Bilingual glossary (continued)

English	Français	Chinese (pinyin)	Definition
Concentration	Concentration	集聚 (Jíjù)	Imposed centripetal movement (exo-regulated mobility).
Dispersion	Dispersion	扩散 (Kuòsàn)	Imposed centrifugal movement (exo-regulated mobility).
Endo-regulated Mobility	Mobilité endorégulée	内生规则性流动 (Nèishēng guīzéxìng liúdòng)	Trajectory controlled by the subject (chosen gathering or evasion).
Evasion	Évasion	逃逸 (Táoyì)	Chosen centrifugal movement (endo-regulated mobility).
Exo-regulated Mobility	Mobilité exorégulée	外生规则性流动 (Wàishēng guīzéxìng liúdòng)	Trajectory undergone by the subject (constrained concentration or dispersion).
Gathering	Rassemblement	汇聚 (Huìjù)	Chosen centripetal movement (endo-regulated mobility).
GEP	GEP	生态产品总值 (Shēngtài chǎnpǐn zǒngzhí)	Indicator of the value of ecosystem services.
Household Registration (Hukou)	Enregistrement résidentiel (Hukou)	户口 (Hùkǒu)	System for controlling internal mobility and social rights.
Land Finance	Finance foncière	土地财政 (Tǔdì Cáizhèng)	Accumulation model based on the monetization of land by the State.
Local Government Financing Vehicle (LGFV)	Véhicule de financement local (LGFV)	地方政府融资平台 (Dìfāng zhèngfǔ róngzī píngtái) (or 城投公司)	Investment company created by local governments.
Maturity Conjuncture	Conjoncture de maturité	成熟期形势 (Chéngshú qī xíngshì)	Real-estate/speculative bubble growth. Wages > marginal productivity. Former term: “parasitic”.
Nodal Point	Point Nodal	节点 (Jiédiǎn) (or 系统性阈值)	Threshold of exhaustion of the mobilizable stock of Value.

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Bilingual glossary (continued)

English	Français	Chinese (pinyin)	Definition
Polarized Conjuncture	Conjoncture polarisée	极化形势 (Jíhuà xíngshì)	Inflationary industrial growth. Investment/forecast precedes consumption. Profits > wages. Former term: “perverse”.
Rent extraction / Buyout	Rachat	资产赎买机制 (Zíchǎn shúmǎi jīzhì)	Transfer of a frozen asset into the productive circuit through anticipated credit.
Retropolarity	Rétropolarité	逆极化 (Nì jíhuà)	Systemic collapse caused by a direct transition between tropical belts without passing through a polar threshold (P_4 or N_4).
SECC	SECC	算力出口战略 (Suànlì chūkǒu zhànlüè)	Strategic Export of Computing Capacity.
Space of Value	Espace de la Valeur	价值空间 (Jiàzhí Kōngjiān)	Stock of assets frozen outside the market by anthropological foundations.
Space of Wealth	Espace de la Richesse	财富空间 (Cáifù Kōngjiān)	Monetized economic circuit, productive flows generating prices.
Stasis	Stase	停滞 (Tíngzhì)	Absorbing attractor where the system remains blocked, with no possible transition either to expansion C_P or to collapse T . The destruction of Wealth no longer recreates Value.
Sterile Financial Loop	Boucle financière stérile	空转金融 (Kōngzhuǎn jīnróng)	Creation of normative rent without mass employment or material anchoring.
Sterile Loop Rupture Point	Point de Rupture de la Boucle Stérile	空转临界点 (Kōngzhuǎn línjiè diǎn)	Crisis of normative overcapacity in the intangible domain.
Substitution of Externalities	Substitution d'externalités	空间外部性置换 (Kōngjiān wàibùxìng zhìhuàn)	Transfer of productive advantages from the periphery to the centre.
Tang Ping (Lying Flat)	Tang Ping (Rester couché)	躺平 (Tāng píng)	Voluntary withdrawal of young people from social competition, the labour market, and property aspirations in response to the rupture of the social contract of Land Finance.

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Bilingual glossary (continued)

English	Français	Chinese (pinyin)	Definition
Territorial Regeneration Rent (TRR)	Rente de Régénération Territoriale (RRT)	领土再生租金 (<i>Lǐngtǔ zàishēng zūjīn</i>) (or 生态修复租金)	Massive productive investment in a devalued rural space.
Theory of Rent Extraction (ToRE)	Théorie du Rachat (TdR)	赎买理论, 价值资本 化周期理论 (TdR) (<i>Shúmai lǐlùn</i>)	Theoretical framework modelling the transfer of assets from the Space of Value to the Space of Wealth.
Urban Investment Debt (LGFV)	Dette d'investissement urbain (LGFV)	城投债 (<i>Chéngtóuzhài</i>)	Debt of local financing vehicles.
Virtue / Political Legitimacy	Vertu / Légimité politique	德 (<i>Dé</i>)	Foundation of the cosmic and social order.

Appendix I – Spectral dashboard for the 15th Five-Year Plan

I.1. Spectral indicators and alert thresholds

Table 90: Spectral Observables and Alert Thresholds

Observable	Definition	Green threshold	Amber threshold	Red threshold
$\eta(t)$	Spectral asymmetry	$> +0.3$	0.1 to 0.3	< 0.1
$d(t)$	Effective spectral dimension	> 4	2 to 4	< 2
$\text{gap}(t)$	Smallest positive spectral gap	> 0.4	0.2 to 0.4	< 0.2
$R_{\text{threshold}}(t)$	Fraction of asymmetry at thresholds	< 0.3	0.3 to 0.7	> 0.7

I.2. Economic policy recommendations based on spectral observables

Table 91: Economic policy recommendations based on spectral observables

Spectral state	η	d	gap	$R_{\text{threshold}}$	Recommendation
Green	$> +0.3$	> 4	> 0.4	< 0.3	Continue with current policies
Orange	0.1 to 0.3	2 to 4	0.2 to 0.4	0.3 to 0.7	Monitor, prepare contingency plans
Red	< 0.1	< 2	< 0.2	> 0.7	Immediate action: activate N_4 (TRR)

I.3. Specific management indicators

Table 92: Concrete management indicators

Spectral observable	Economic indicator	Data source	Alert threshold
$\eta(t)$	Growth rate of GEP vs GDP	Ministry of the Environment	$\eta < 0$
$d(t)$	Regional Gini coefficient	National Bureau of Statistics	$d < 2$
$\text{gap}(t)$	Productivity gap between sectors	National surveys	$\text{gap} < 0.2$
$R_{\text{threshold}}(t)$	Debt ratio of local authorities	Ministry of Finance	$R_{\text{threshold}} > 0.7$
$n_p(t)$	Investment in ecological regeneration	Two Mountains Bank	$n_p < n_N$
$n_N(t)$	Rural unemployment rate	Ministry of Labour	$n_N > n_p$

J –TRR roll-out timetable

Table 93: TRR roll-out timetable and alignment with the five-year plans

Period	Five-year plan	Objective	Target threshold
2024–2025	14th Plan (2021–2025)	Initial pilot phase: launch of the first waves of national pilot projects (10 regions in 2024, 25 regions in 2025). Trialling the Two Mountains Bank in Lishui (Zhejiang) and the integration of the GEP into local KPIs. (Phase already underway.)	S_5
2026–2028	15th Plan (2026–2030)	Roll-out phase: extension of the TRR model to provinces that have completed the pilot phases, with the aim of treating 2 to 5 million hectares of degraded land by 2028. Widespread adoption of Regenerative Usufruct Certificates (CUR) and the securitisation of ecological assets via the Two Mountains Bank across all pilot areas.	S_6
2029–2030	15th Plan (continued)	Acceleration phase: scaling up regeneration mechanisms nationwide, with the aim of treating a cumulative total of 5 to 10 million hectares. Stabilisation of groundwater levels in the northern plains.	S_7
2031–2035	16th Plan	Generalisation phase: coverage of all degraded rural areas. Consolidation of regeneration sectors (agrivoltaics, aquaculture, reforestation) as sustainable economic sectors.	Stabilisation at C_P

Appendix K – Comparison of post-Nodal Point alternatives

Table 94: Comparison of post-Nodal Point alternatives

Alternative	Mass employ- ment	Regional anchoring	Timeframe	Compatibility	ToRE status	Celestial mandate
TRR	Yes	Yes	10–15 years	High	Hybrid concept	Restoration
New physical frontiers	Yes	Yes	20–30 years	Medium	Extension of Rent Extraction	Preserves
Polarised Growth 2.0	No	No	Immediate	Low	Algorithmic Stasis	Eroding
Technological Breakthrough	Yes	Yes	20–40 years	Medium	Exogenous to ToRE	Neutral
Neo-feudal Retreat	No	Partial	5–10 years	None	Terminal phase	Break
Post-growth Model	No	No	10–20 years	None	Radical break	Break

L –Map of TRR pilot areas

Table 95: Detailed map of TRR pilot areas

Area	Official Chinese name	Degraded assets re-purchased	TRR mechanism
Lishui (Zhejiang)	丽水”两山银行”	Depleted farmland, underutilised forests, abandoned villages.	The Two Mountains Bank aggregates the usage rights to these lands, issues loans backed by these assets, and finances ecological restoration and rural development.
Sanming and Nanping (Fujian)	林票制度 / 森林生态银行	Fragmented, unmanaged communal forests.	Villagers transfer management rights; state-owned enterprises provide credit to employ local people; villagers receive <i>Lin-piao</i> (negotiable forest vouchers).
Xingguo and Ganzhou (Jiangxi)	山水林田湖草生态保护修复工程 / 横向生态补偿	Eroded catchment areas, land contaminated by mining, choked water-courses.	Upstream towns receive funding from the State-CPC and payments for ecosystem services from downstream towns.
Hainan	国家生态文明试验区 / GEP 核算体系	An island territory threatened by property speculation.	Ban on speculative construction; the GEP is used as an assessment indicator and as bank collateral.
Huzhou (Zhejiang) and Chongyi (Jiangxi)	竹林碳汇 / 林业碳票	Ageing, unharvested bamboo groves.	Bridge between the material TRR and the Third Transformation: the CCP state funds the maintenance of bamboo forests in exchange for carbon credits on the national market.

Appendix M - The 64 economic configuration and the 20 attractors

Appendix M presents the correspondence between the 64 economic configurations ($\mathbf{x} \in \{0, 1\}^6$) and the 20 attractors ($\mathcal{A} = \{A, B, \dots, T\}$). The projection rule ($\Pi : \{0, 1\}^6 \rightarrow \mathcal{A}$) that maps a binary configuration to an attractor is not made explicit in the body of the text. The present appendix formalises it.

It successively sets out:

1. The projection problem ($\{0, 1\}^6 \rightarrow \mathcal{A}$);
2. The three families of candidate solutions (semantic, algebraic, metric);
3. The chosen solution: the semantic family (intersection);
4. The bit-to-privileged-pentads correspondence table;
5. The aggregation rule for multi-bit configurations;
6. An application to ID 2 (0-based);
7. The limitations of the chosen rule;
8. A summary.

M.1 The Projection Problem $\{0, 1\}^6 \rightarrow \mathcal{A}$

Each economic configuration is a binary vector $\mathbf{x} = (b_1, b_2, b_3, b_4, b_5, b_6) \in \{0, 1\}^6$, where each bit b_i encodes the activation (1) or deactivation (0) of one of the six structural variables (inflation, purchasing power, profits, buyback, dispersion, land regime). There are $2^6 = 64$ such configurations.

Each configuration must be assigned to exactly one of the 20 attractors in $\mathcal{A}$. The mapping Π must satisfy three constraints:

- **Determinism:** each $\mathbf{x}$ maps to a single attractor;
- **Surjectivity:** every attractor in $\mathcal{A}$ is the image of at least one configuration;
- **Interpretability:** the assignment must be justifiable in economic terms, not merely combinatorial.

M.2 Three Families of Candidate Solutions

Three broad families of projection rules can be envisaged:

Family	Principle	Advantage	Drawback
Semantic	Each bit is associated with a subset of privileged pentads; the attractor is determined by the intersection of these subsets.	Economically interpretable; falsifiable.	Requires a calibration table; provisional.
Algebraic	The 64 configurations are partitioned using the group structure of $Cl(6, 6)$ or a quotient thereof.	Mathematically elegant; unique.	No guarantee of economic interpretability.
Metric	A distance is defined on $\{0, 1\}^6$ and each configuration is assigned to the nearest attractor prototype.	Flexible; data-driven.	Requires a training set; risk of overfitting.

M.3 Chosen Solution: Semantic Family (Intersection)

The semantic family is chosen because it is the only one that satisfies the interpretability constraint while remaining falsifiable. The rule proceeds in two stages:

1. **Stage 1 — Bit-to-pentads table:** each bit b_i is associated with a triplet of privileged pentads (drawn from the 12 pentads $\{P_1, \dots, P_6, N_1, \dots, N_6\}$).
2. **Stage 2 — Aggregation by intersection:** for a given configuration $\mathbf{x}$, the activated bits yield a collection of pentad triplets; the attractor is the one whose defining triplet matches the intersection (or best approximation thereof).

M.4 Bit-to-Privileged-Pentads Correspondence Table

Bit	Variable	Partition	Privileged pentads	Justification
b_1	Inflation	e_1	P_1, P_3, N_2	Inflationary Keynesianism (P_1), indebtedness (P_3), rigidity (N_2)

Bit	Variable	Partition	Privileged pentads	Justification
b_2	Purchasing power	e_2	P_1, P_6, N_1	Wage Keynesianism (P_1), consumption (P_6), wage austerity (N_1)
b_3	Profits	e_3	P_2, P_3, N_1	Liberal market (P_2), indebtedness (P_3), austerity (N_1)
b_4	Buyback	e_4	P_4, P_5, N_4	Exportation (P_4), nodal point (P_5), withdrawal (N_4)
b_5	Dispersion	e_5	P_6, N_5, N_6	Confirmed by the text: dispersion $\rightarrow \{P_6, N_5, N_6\} = S$
b_6	Land regime	e_6	P_5, N_4, N_5	Nodal point (P_5), withdrawal (N_4), stasis (N_5)

Note: this table is provisional. It is calibrated on the examples given in the text and on the descriptions of the attractors. It may be refined through confrontation with the 64 historical regimes of Appendix M.

M.5 Aggregation Rule for Multi-Bit Configurations

When a single bit is activated (Hamming weight 1), the attractor is read directly from the table. When multiple bits are activated, the following algorithm is applied:

Algorithm — Semantic projection $\Pi(\mathbf{x})$

1. **Input:** $\mathbf{x} = (b_1, \dots, b_6) \in \{0, 1\}^6$.
2. **Collect:** for each i such that $b_i = 1$, retrieve the triplet T_i from the table in §M.1.4.
3. **Intersect:** compute $I = \bigcap_{i: b_i=1} T_i$.
4. **Match:**
 - If $|I| = 3$ and I exactly matches the defining triplet of an attractor A_k , then $\Pi(\mathbf{x}) = A_k$.

- If $|I| < 3$, complete I by adding the most frequent pentads among the T_i (majority vote), then match to the closest attractor triplet.
- If $|I| = 0$ (empty intersection), assign $\mathbf{x}$ to the attractor whose triplet minimises the Hamming distance to the union $\bigcup_{i:b_i=1} T_i$.

5. **Output:** the attractor $A_k \in \mathcal{A}$.

M.6 Application to ID 2 (0-Based)

The application of the semantic rule to ID 2 (0-based) yields:

ID	$\mathbf{x}$	Attractor	Polarity	Triplet	Example (Western)	Asian example
2	(0, 0, 0, 0, 1, 0)	S	1P+2N	$\{P_6, N_5, N_6\}$	Subprime crisis (2008)	China post-2018 (Nodal Point, Stasis)

Justification: the configuration activates only b_5 (dispersion). The correspondence table associates $b_5 \rightarrow \{P_6, N_5, N_6\}$, which yields attractor S. The triplet $\{P_6, N_5, N_6\}$ corresponds to: domestic consumption (P_6), stasis (N_5), austerity (N_6) – an exact description of China post-2018.

Justification of IDs: the configuration (0, 0, 0, 0, 1, 0) activates only partition e_5 (dispersion). The projections of P_6, N_5, N_6 onto e_5 have respective IDs **54, 59, 60**.

Correspondence with Appendix M: ID 2 (0-based) in Appendix M associates the configuration (0, 0, 0, 0, 1, 0) with attractor **S** (triplet $\{P_6, N_5, N_6\}$).

M.7 Limitations of the Chosen Rule

The semantic projection rule, while interpretable and falsifiable, has the following limitations:

- **Provisional calibration:** the bit-to-pentads table is derived from a small number of textual examples. A systematic calibration against all 64 historical regimes is needed.
- **Ambiguity in multi-bit cases:** when several bits are activated, the intersection may be empty, requiring a completion heuristic (majority vote or Hamming distance) that introduces a degree of arbitrariness.
- **Non-uniqueness of triplets:** some attractors share pentads (e.g., P_5 appears in the triplets of both S and D), which may lead to ambiguous assignments for certain configurations.

- **No temporal dynamics:** the rule is static; it does not account for the trajectory of a configuration through state space over time.
-

M.8 Summary

Aspect	Chosen option
Family	Semantic (bit-to-pentads correspondence table)
Variant	Intersection
Table	6 rows (one per bit)
Aggregation rule	Intersection, with provisional completion
Application	ID 2: $(0, 0, 0, 0, 1, 0) \rightarrow S$
Status	Provisional, falsifiable, to be refined

The projection rule $\Pi : \{0, 1\}^6 \rightarrow \mathcal{A}$ is now explicit, which makes it possible to:

- assign each configuration to an attractor in a reproducible manner;
- render the projection falsifiable;
- refine the calibration iteratively as new historical data become available.

M.1.9 Table of the 64 states (ID → Attractor, polarity, triplets, examples)

Table 100: ID → Attractor correspondence with polarity, triplets of pentads and historical examples (0–63)

ID	x	Attr.	Pol.	Triplet of pentads	Example (Western)	Asian example
0	(0, 0, 0, 0, 0, 0)	T	3N	$\{N_2, N_3, N_4\}$	Pre-agricultural Mesopotamia (-10,000)	Neolithic societies (Yangshao, Longshan)
1	(0, 0, 0, 0, 0, 1)	Q	1P+2N	$\{P_2, N_1, N_4\}$	Vikings in Iceland (9th century)	Barbarian migrations (Spring and Autumn periods)
2	(0, 0, 0, 0, 1, 0)	S	1P+2N	$\{P_6, N_5, N_6\}$	2008 Subprime crisis,	post-2018 China (Nodal Point, Stasis)
3	(0, 0, 0, 0, 1, 1)	Q	1P+2N	$\{P_2, N_1, N_4\}$	Celts in Europe (c. 500 BC)	Expansion of the Zhou (1046–771 BC)
4	(0, 0, 0, 1, 0, 0)	P	1P+2N	$\{P_6, N_1, N_2\}$	Medieval Venice (12th century)	Song dynasty trading cities (Quanzhou, Hangzhou)
5	(0, 0, 0, 1, 0, 1)	P	1P+2N	$\{P_6, N_1, N_2\}$	Fortified castles (10th century)	Feudal system of the Eastern Zhou
6	(0, 0, 0, 1, 1, 0)	O	1P+2N	$\{P_5, N_1, N_4\}$	Phoenician trading posts (c. 1000 BC)	Maritime Silk Road (Tang dynasty)
7	(0, 0, 0, 1, 1, 1)	P	1P+2N	$\{P_6, N_1, N_2\}$	Portuguese trading posts (15th century)	Portuguese trading posts in Macao (1557)
8	(0, 0, 1, 0, 0, 0)	N	1P+2N	$\{P_4, N_5, N_6\}$	Lombard bankers (14th century)	Song banks – <i>jiaozi</i>
9	(0, 0, 1, 0, 0, 1)	M	1P+2N	$\{P_4, N_1, N_3\}$	Brazilian plantations (16th century)	Sugar cane plantations (Hainan, Ming/Qing)
10	(0, 0, 1, 0, 1, 0)	N	1P+2N	$\{P_4, N_5, N_6\}$	Silk Road (2nd century)	Silk Road (Han/Tang)
11	(0, 0, 1, 0, 1, 1)	M	1P+2N	$\{P_4, N_1, N_3\}$	East India Company (17th century)	East India Company – tea, porcelain
12	(0, 0, 1, 1, 0, 0)	O	1P+2N	$\{P_5, N_1, N_4\}$	Hanseatic League (14th century)	Song merchant guilds
13	(0, 0, 1, 1, 0, 1)	O	1P+2N	$\{P_5, N_1, N_4\}$	London (17th century)	Canton (18th century) – single trading post
14	(0, 0, 1, 1, 1, 0)	N	1P+2N	$\{P_4, N_5, N_6\}$	Portuguese Empire (16th century)	Zheng He's expeditions (1405–1433)

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ID	x	Attr.	Pol.	Triplet of pentads	Example (Western)	Asian example
15	(0, 0, 1, 1, 1, 1)	M	1P+2N	$\{P_4, N_1, N_3\}$	Conquista (16th century)	Conquest of Southern China (Qin, 221 BC)
16	(0, 1, 0, 0, 0, 0)	T	3N	$\{N_2, N_3, N_4\}$	Classical Athens (5th century BC)	Spring and Autumn Period (770–476 BC)
17	(0, 1, 0, 0, 0, 1)	T	3N	$\{N_2, N_3, N_4\}$	Colonisation of Iceland (9th century)	Colonisation of Yunnan (Han)
18	(0, 1, 0, 0, 1, 0)	K	1P+2N	$\{P_2, N_3, N_5\}$	Irish Famine (1840s)	Great Shaanxi Famine (1876–1879)
19	(0, 1, 0, 0, 1, 1)	L	1P+2N	$\{P_3, N_2, N_4\}$	Mormons in Utah (19th century)	Taiping Kingdom (1850–1864)
20	(0, 1, 0, 1, 0, 0)	L	1P+2N	$\{P_3, N_2, N_4\}$	Florence (15th century)	Song Dynasty (960–1279)
21	(0, 1, 0, 1, 0, 1)	L	1P+2N	$\{P_3, N_2, N_4\}$	Parisian guilds (16th century)	Ming guilds
22	(0, 1, 0, 1, 1, 0)	T	3N	$\{N_2, N_3, N_4\}$	Hippie movement (1970s)	Knowledge education movement
23	(0, 1, 0, 1, 1, 1)	T	3N	$\{N_2, N_3, N_4\}$	Kibbutz (20th century)	People's Communes (1958–1980)
24	(0, 1, 1, 0, 0, 0)	I	1P+2N	$\{P_1, N_2, N_6\}$	Rome (3rd century)	Late Tang Dynasty (9th century) – monetary crisis
25	(0, 1, 1, 0, 0, 1)	I	1P+2N	$\{P_1, N_2, N_6\}$	California (1848)	Great Boxer Rebellion (1899–1901)
26	(0, 1, 1, 0, 1, 0)	I	1P+2N	$\{P_1, N_2, N_6\}$	Turkish shantytowns (1990s)	Chinese rural exodus (1980–2000)
27	(0, 1, 1, 0, 1, 1)	J	1P+2N	$\{P_1, N_3, N_5\}$	Brazil (1960s)	China in the 1960s – heavy industrialisation
28	(0, 1, 1, 1, 0, 0)	J	1P+2N	$\{P_1, N_3, N_5\}$	Hong Kong (2000s)	Chinese property market (2000–2018) – bubble, LGFV
29	(0, 1, 1, 1, 0, 1)	J	1P+2N	$\{P_1, N_3, N_5\}$	Johannesburg (apartheid)	Segregation in the concessions (19th century)
30	(0, 1, 1, 1, 1, 0)	K	1P+2N	$\{P_2, N_3, N_5\}$	South Africa (20th century)	Chinese Civil War (1927–1949)
31	(0, 1, 1, 1, 1, 1)	K	1P+2N	$\{P_2, N_3, N_5\}$	France before 1789	Fall of the Ming dynasty (1644)

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ID	x	Attr.	Pol.	Triplet of pentads	Example (Western)	Asian example
32	(1, 0, 0, 0, 0, 0)	G	2P+1N	$\{P_2, P_5, N_6\}$	Japan (1990s)	China in the 1990s – reforms, deflation
33	(1, 0, 0, 0, 0, 1)	G	2P+1N	$\{P_2, P_5, N_6\}$	The Far West (19th century)	Conquest of Xinjiang (18th–19th centuries)
34	(1, 0, 0, 0, 1, 0)	G	2P+1N	$\{P_2, P_5, N_6\}$	Maquiladoras (1980s)	Special Economic Zones (Shenzhen, 1980)
35	(1, 0, 0, 0, 1, 1)	H	2P+1N	$\{P_3, P_4, N_6\}$	Belgian Congo (1900s)	European colonialism in China (19th century)
36	(1, 0, 0, 1, 0, 0)	D	2P+1N	$\{P_4, P_5, N_2\}$	Manchester (1780–1820)	Chinese Industrial Revolution (1980–2000)
37	(1, 0, 0, 1, 0, 1)	D	2P+1N	$\{P_4, P_5, N_2\}$	USA (1870–1900)	China 2000–2010 – rapid industrialisation
38	(1, 0, 0, 1, 1, 0)	D	2P+1N	$\{P_4, P_5, N_2\}$	Shanghai (1990s)	Chinese metropolises (Shanghai, Beijing, Shenzhen)
39	(1, 0, 0, 1, 1, 1)	E	2P+1N	$\{P_5, P_6, N_3\}$	Françafrique (1960s)	Silk Roads (21st century) – ‘Belt and Road’
40	(1, 0, 1, 0, 0, 0)	A	3P	$\{P_1, P_2, P_4\}$	Oil crisis (1973)	Chinese inflation (1988–1989) – 18.8 per cent
41	(1, 0, 1, 0, 0, 1)	A	3P	$\{P_1, P_2, P_4\}$	Klondike Gold Rush	Coal mining boom (2000s)
42	(1, 0, 1, 0, 1, 0)	B	3P	$\{P_1, P_3, P_5\}$	NAFTA (1994)	Accession to the WTO (2001)
43	(1, 0, 1, 0, 1, 1)	A	3P	$\{P_1, P_2, P_4\}$	OPEC (1973)	Oil crisis (1970s)
44	(1, 0, 1, 1, 0, 0)	B	3P	$\{P_1, P_3, P_5\}$	Rockefeller (1890)	State monopolies (1990s)
45	(1, 0, 1, 1, 0, 1)	B	3P	$\{P_1, P_3, P_5\}$	Detroit (1920–1960)	Industrial China (2000s) – Fordism
46	(1, 0, 1, 1, 1, 0)	C	3P	$\{P_2, P_3, P_6\}$	Subprime crisis (2008)	Chinese property crisis (2021–2026)
47	(1, 0, 1, 1, 1, 1)	C	3P	$\{P_2, P_3, P_6\}$	Argentina (2001)	LGFV crisis (2018–2026)
48	(1, 1, 0, 0, 0, 0)	F	2P+1N	$\{P_1, P_6, N_4\}$	Greece (2010–2015)	Chinese structural adjustment (2013–2018)

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ID	x	Attr.	Pol.	Triplet of pentads	Example (Western)	Asian example
49	(1, 1, 0, 0, 0, 1)	F	2P+1N	$\{P_1, P_6, N_4\}$	Enclosures (18th century)	Chinese agrarian reform (1950–1952)
50	(1, 1, 0, 0, 1, 0)	E	2P+1N	$\{P_5, P_6, N_3\}$	Ireland (2000s)	Relocations to China – FDI, SEZs
51	(1, 1, 0, 0, 1, 1)	E	2P+1N	$\{P_5, P_6, N_3\}$	Banana republics	Plantation economy in southern China
52	(1, 1, 0, 1, 0, 0)	F	2P+1N	$\{P_1, P_6, N_4\}$	Detroit (1910)	Chinese factory towns (Dongguan, Foshan)
53	(1, 1, 0, 1, 0, 1)	D	2P+1N	$\{P_4, P_5, N_2\}$	The Glorious Thirty (1945–1975)	China’s Reform Era (1978–1993)
54	(1, 1, 0, 1, 1, 0)	H	2P+1N	$\{P_3, P_4, N_6\}$	Uber (2010s)	Platform economy (Alibaba, Tencent, Didi)
55	(1, 1, 0, 1, 1, 1)	H	2P+1N	$\{P_3, P_4, N_6\}$	Silicon Valley	Chinese cognitive capitalism – AI, data
56	(1, 1, 1, 0, 0, 0)	C	3P	$\{P_2, P_3, P_6\}$	World (2022)	Post-COVID China (2020–2026)
57	(1, 1, 1, 0, 0, 1)	R	1P+2N	$\{P_3, N_1, N_5\}$	Bitcoin (2017)	Cryptocurrencies in China – speculation
58	(1, 1, 1, 0, 1, 0)	R	1P+2N	$\{P_3, N_1, N_5\}$	Post-COVID (2021)	Supply chains (2020–2022)
59	(1, 1, 1, 0, 1, 1)	R	1P+2N	$\{P_3, N_1, N_5\}$	Sri Lanka (2022)	Chinese debt crisis (2021–2026)
60	(1, 1, 1, 1, 0, 0)	S	1P+2N	$\{P_6, N_5, N_6\}$	Wall Street (1980s)	Chinese financialisation (2010–2020)
61	(1, 1, 1, 1, 0, 1)	S	1P+2N	$\{P_6, N_5, N_6\}$	GAFAM	Chinese tech giants (Baidu, Alibaba, Tencent)
62	(1, 1, 1, 1, 1, 0)	S	1P+2N	$\{P_6, N_5, N_6\}$	Contemporary South Africa	Inequalities in China – Hukou, urban–rural divide
63	(1, 1, 1, 1, 1, 1)	T	3N	$\{N_2, N_3, N_4\}$	Terminal crisis (2008×COVID×Climate)	China 2018–2026 – Nodal point, Stasis, transition

Appendix N — Summary of the Methodology and Remaining Work

N.1 —Purpose

This appendix sets out the methodology used to build the model and identifies what remains to be done. The report outlines a path — the formalisation of the Theory of Rent Extraction within $Cl(6,6)$ — without claiming to solve all the problems that an institution with extensive resources could address. Areas of uncertainty are flagged; remaining work is identified.

N.2 —Methodology in Six Steps

Table 101: Steps in the construction of the model

Step	Purpose	Method	Status
1	Basic pentads	Extraction from $Cl(6,0)$	Established
2	Economic interpretation	Generator/variable correspondences	Established
3	144 projected pentads	$P_k^{(g)} = g \cdot P_k^{\text{base}}$	Established
4	Projection $144 \rightarrow 20$	Attractor triplets + nilpotence	Established
5	Projection $64 \rightarrow 20$	Semantic rule (Appendix M.0)	Provisional
6	Examples and calibration	Confrontation with historical regimes	Partial

Steps 1 to 4 are established: they rest on the algebraic structure of $Cl(6,6)$. Steps 5 and 6 are provisional: they rest on modelling choices that remain to be refined.

N.3 —The Double Projection

The model involves two distinct projections:

- **144 $\rightarrow$ 20 (established):** each projected pentad $P_k^{(g)}$ is associated with an attractor via the operator Π (§10.1).
- **64 $\rightarrow$ 20 (provisional):** each binary configuration $\mathbf{x} \in \{0,1\}^6$ is associated with an attractor via the semantic rule of Appendix M.0.

Rule M.0 rests on a bit-to-pentads correspondence table and an aggregation rule. Both elements are working hypotheses.

N.4 —What Is Established, What Remains to Be Done

Table 102: State of achievements and remaining work

Established	Remaining work
Algebraic structure of $Cl(6,6)$	Calibrate the bit $\rightarrow$ pentads table for $(b_1, b_2, b_3, b_4, b_6)$
144 projected pentads	Validate the aggregation rule for multi-bit configurations
20 attractors and their triplets	Verify the distribution of the 64 configurations across the 20 attractors
Projection $144 \rightarrow 20$ via Π	Confront the rule with the 9 dynastic declines (validation test)
Correspondence $(b_5 \rightarrow \{P_6, N_5, N_6\})$	Extend calibration to an expanded corpus (64 regimes + quantitative series)
Diagnosis of post-2018 China (attractor S)	Refine rule M.0 after calibration

N.5 —Role of the 9 Dynastic Declines

The 9 declines (Appendix G) do not serve as a complete calibration base: they cover a restricted subset of the configuration space (inflation, dispersion, nodal point). Bits (b_2) (purchasing power) and (b_4) (buy-back) are always set to 0 in these cases.

They serve as a validation test: if rule M.0 correctly predicts their attractors, it is validated for these configurations. To calibrate (b_2) and (b_4) , periods of healthy growth are required (the Glorious Thirty, the Chinese reforms of 1978–1993).

N.6 —Correspondence Between Attractors and Pentad Triplets

Table 103: Correspondence between attractors and pentad triplets

Attractor	Triplet	Polarity	Interpretation
A	$\{P_1, P_2, P_4\}$	3P	Extreme perverse growth
B	$\{P_1, P_3, P_5\}$	3P	Perverse export-led growth
C	$\{P_2, P_3, P_6\}$	3P	Perverse growth with dispersion
D	$\{P_4, P_5, N_2\}$	2P+1N	Moderate growth
E	$\{P_5, P_6, N_3\}$	2P+1N	Urban expansion
F	$\{P_1, P_6, N_4\}$	2P+1N	Emerging stagflation
G	$\{P_2, P_5, N_6\}$	2P+1N	Impoverished growth
H	$\{P_3, P_4, N_6\}$	2P+1N	Rentier economy
I	$\{P_1, N_2, N_6\}$	1P+2N	Deflationary recession
J	$\{P_1, N_3, N_5\}$	1P+2N	Liquidity crisis
K	$\{P_2, N_3, N_5\}$	1P+2N	Agricultural depression
L	$\{P_3, N_2, N_4\}$	1P+2N	War economy
M	$\{P_4, N_1, N_3\}$	1P+2N	Nodal point on the pioneer frontier
N	$\{P_4, N_5, N_6\}$	1P+2N	Nodal point with dispersion
O	$\{P_5, N_1, N_4\}$	1P+2N	Pioneer frontier in deflation
P	$\{P_6, N_1, N_2\}$	1P+2N	Dispersion under an oikoumene regime
Q	$\{P_2, N_1, N_4\}$	1P+2N	Recession with low purchasing power
R	$\{P_3, N_1, N_5\}$	1P+2N	Stagflation
S	$\{P_6, N_5, N_6\}$	1P+2N	Dispersion and nodal point
T	$\{N_2, N_3, N_4\}$	3N	Terminal depression

N.7 —Future Work

Table 104: Identified future work

No.	Purpose	Method	Priority
1	Calibration of the bit $\rightarrow$ pentads table	Expanded corpus	High
2	Validation of the aggregation rule	Test on the 64 configurations	High
3	Verification of the 64 $\rightarrow$ 20 distribution	Counting and consistency checks	Medium
4	Confrontation with the 9 declines	Validation test	High
5	Extension to other corpora (Rome, medieval Europe)	Robustness test	Low

N.8 —Conclusion

The model is established in its algebraic structure and provisional in its projection of binary configurations onto attractors. The remaining work has been identified. This methodological honesty is the condition for the proposed path to be taken seriously and to be further developed.

Appendix O –Mathematical formalism of spectral observables

O.1. Introduction

This appendix presents the complete mathematical formalism underpinning the calculation of the four spectral observables used in the Cl(6,6) model: the spectral asymmetry $\eta(t)$, the effective spectral dimension $d(t)$, the spectral gap $\text{gap}(t)$ and the threshold saturation ratio $R_{\text{threshold}}(t)$.

The aim is to provide researchers and modellers with the equations and algorithms needed to reproduce the calculations from economic data.

O.2. Construction of the Dirac operator

O.2.1. Definition of the pentad graph

Let $\Gamma = (V, E)$ be the pentad graph, where:

- $V = \{P_1, \dots, P_6, N_1, \dots, N_6\}$ is the set of the 12 basic pentads.
- E is the set of edges connecting two pentads if they share a common attractor.

The adjacency matrix A of Γ is a 12×12 matrix defined by:

$$A_{ij} = \begin{cases} 1 & \text{if the pentads } i \text{ and } j \text{ are connected} \\ 0 & \text{otherwise} \end{cases}$$

The connections are determined by the triplets of the 20 attractors (see Appendix B).

O.2.2. Local spinors

To each pentad $v \in V$, a local spinor is associated:

$$\psi_v = \begin{pmatrix} \varepsilon_v \\ \varphi_v \end{pmatrix} \in \mathbb{C}^2$$

where:

- $\varepsilon_v \in \{+1, -1\}$ is the local mode: +1 for Sheng (expansion), -1 for Ke (recession).

- $\varphi_v \in [0, 1]$ is the pentad' s activation rate, derived from economic data.

The activation rate φ_v is calculated from the normalised economic indicators:

$$\varphi_v = \frac{1}{m} \sum_{k=1}^m \text{norm}(x_k^{(v)})$$

where $x_k^{(v)}$ are the m economic indicators associated with the pentad v , and norm is the min-max normalisation function.

O.2.3. Pauli matrices

We use the following Pauli matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

O.2.4. Coupling weights

The coupling weight between two pentads i and j is defined by:

$$w_{ij}(t) = \beta \cdot \exp\left(-\frac{|\varphi_i - \varphi_j| + E(\varphi_i, \varphi_j)}{\gamma}\right) \cdot \Theta(\varepsilon_i, \varepsilon_j)$$

where:

- β is a coupling constant (to be calibrated).
- γ is a scale parameter (to be calibrated).
- $E(\varphi_i, \varphi_j) = |\varphi_i - \varphi_j|^2$ is the frustration energy.
- $\Theta(\varepsilon_i, \varepsilon_j) = \begin{cases} 1 & \text{if } \varepsilon_i = \varepsilon_j \\ \lambda & \text{if } \varepsilon_i \neq \varepsilon_j \end{cases}$ where $\lambda \in [0, 1]$.

O.2.5. Dirac matrix

The Dirac operator $D(t)$ is a 12×12 matrix defined in 2×2 blocks:

$$(D\psi)_i = \sum_{j \sim i} w_{ij}(t) \sigma_{ij} \psi_j$$

where σ_{ij} is the Pauli matrix associated with the edge (i, j) :

$$\sigma_{ij} = \begin{cases} \sigma_x & \text{if the edge lies within a tropical belt} \\ \sigma_z & \text{if the edge is incident to a threshold } (P_4 \text{ or } N_4) \end{cases}$$

More explicitly, the matrix $D(t)$ is written as:

$$D(t) = \begin{pmatrix} 0 & w_{12}\sigma_{12} & \cdots & w_{1,12}\sigma_{1,12} \\ w_{21}\sigma_{21} & 0 & \cdots & w_{2,12}\sigma_{2,12} \\ \vdots & \vdots & \ddots & \vdots \\ w_{12,1}\sigma_{12,1} & w_{12,2}\sigma_{12,2} & \cdots & 0 \end{pmatrix}$$

O.3. Solving the spectral problem

O.3.1. Eigenvalue equation

The spectral problem consists of solving:

$$D(t)\psi = \lambda\psi$$

where λ is an eigenvalue (real, since D is Hermitian) and ψ is the associated eigenvector.

The eigenvalues are ordered as follows:

$$\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_{12}$$

O.3.2. Spectral density

The spectral density $\rho(\lambda)$ is defined by:

$$\rho(\lambda) = \frac{1}{12} \sum_{k=1}^{12} \delta(\lambda - \lambda_k)$$

where δ is the Dirac distribution.

O.4. Definition of observables

O.4.1. Spectral asymmetry $\eta(t)$

Spectral asymmetry measures the dominance of the Sheng modes (expansion) relative to the Ke modes (recession). Unlike a simple count of positive and negative eigenvalues, we use a weighted measure that takes into account the amplitude of each mode:

$$\eta(t) = \frac{1}{\Lambda} \sum_{k=1}^{12} \text{sgn}(\lambda_k) |\lambda_k|^\alpha$$

where:

- $\text{sgn}(\lambda) = \begin{cases} +1 & \text{if } \lambda > 0 \\ -1 & \text{if } \lambda < 0 \\ 0 & \text{if } \lambda = 0 \end{cases}$
- $\alpha \in [0, 1]$ is a sensitivity parameter (typically $\alpha = 1/2$)
- $\Lambda = \sum_{k=1}^{12} |\lambda_k|^\alpha$ is a normalisation factor

Properties:

- $\eta(t) \in [-1, +1]$
- $\eta(t) > 0$: Sheng dominance (expansion)
- $\eta(t) < 0$: Ke dominance (recession)
- $\eta(t) \approx 0$: tipping point (bifurcation)

References: This definition follows the tradition of spectral asymmetry studied in random matrix theory by Wigner [98] and Mehta [99], as well as the work of Dumitriu and Edelman on beta sets [100].

Philosophical interpretation: Spectral asymmetry $\eta(t)$ is the modern equivalent of qi (气) – the vital breath that circulates through the economic system. When qi is abundant and flows freely ($\eta > 0$), the system is expanding. When the qi stagnates or becomes scarce ($\eta < 0$), the system enters a recession. The point $\eta \approx 0$ is a tipping point, a *yin-yang* (阴阳) where the balance is precarious and where a small disturbance can tip the system towards expansion or collapse. This concept echoes the Taoist conception of the cosmic breath that animates all things, and traditional Chinese medicine, in which qi flows through the meridians.

O.4.2. Effective spectral dimension $d(t)$

The effective spectral dimension measures the diversity of the system, that is, the effective number of active modes in the spectrum of the Dirac operator. It is defined in terms of the Shannon entropy of the normalised spectrum:

$$d(t) = \exp \left(- \sum_{k=1}^{12} p_k \ln p_k \right)$$

where:

$$p_k = \frac{|\lambda_k|}{\sum_{j=1}^{12} |\lambda_j|}$$

Mathematical properties:

- $d(t) \in [1, 12]$

- $d(t) \approx 1$: highly concentrated system (a single mode dominates)
- $d(t) \approx 12$: highly diversified system (all modes are active)
- $d(t) < 2$: red alert (system is excessively concentrated)

References: This definition draws on Shannon entropy [101] to define an effective number of modes. It is also inspired by the work of Grassberger and Procaccia on the correlation dimension [102] and the spectral dimension introduced by Alexander and Orbach for disordered systems [103]. The link between spectral dimension and diffusion on graphs is developed by Havlin and Ben-Avraham [104]. This observable is distinct from Alain Connes’s spectral dimension [105] in non-commutative geometry, which is an invariant property of spectral spaces defined by the asymptotic behaviour of $\text{Tr}(|D|^{-s})$.

Philosophical interpretation:

In the philosophy of Xiong Shili (熊十力, 1885–1968), the concept of **harmony** (和, hé) refers to the dynamic equilibrium between opposing forces that constitutes the spontaneous order of the cosmos. For Xiong, harmony is neither uniformity (where a single mode would dominate, $d \approx 1$) nor chaotic diversity (where all modes would be equally active without coherence, $d \approx 12$), but a **creative equilibrium** between unity and diversity.

Within the framework of the Cl(6, 6) model, this equilibrium corresponds to a particular value of the spectral dimension:

$$d_{\text{harmony}} = 5$$

Why $d = 5$?

- The dimension $d = 5$ corresponds to a system in which approximately 5 effective modes are active out of the 12 possible ones.
- This is the regime of $2P + 1N$ -polarity attractors (moderate growth), which are historically the most stable and enduring regimes.
- This is the number of elements in a pentad (5), the basic unit of the model.
- This is the number of phases in the traditional Chinese cycle of the Five Elements (五行, wǔxíng): Wood, Fire, Earth, Metal, Water —whose dynamic interaction produces cosmic balance. $d = 5$ is the regime in which the five elements of the Wu Xing flourish most fully.
- This is the number of relationships in the system of the five virtues (五常, wǔcháng): benevolence, justice, propriety, wisdom, and trust.

Thus, $d = 5$ represents **systemic harmony**: sufficient diversity to ensure resilience, but sufficient concentration to maintain coherence. This is the ideal of the ‘golden mean’ (中庸, zhōngyōng) applied to economic dynamics.

Table 105: Philosophical interpretation of $d(t)$

Value of d	Economic interpretation	Philosophical interpretation
$d \approx 12$	Chaotic diversity, fragmentation	Disorder (亂, luàn) —multiplicity without a unifying principle
$d \approx 5$	Moderate growth, equilibrium	Harmony (和, hé) —Xiong Shili, the golden mean
$d \approx 1$	Excessive concentration, monopoly	Rigid unity (同, tóng) —uniformity without creativity
$d \rightarrow 0$	Systemic destruction, collapse	Nothingness (空, kōng) —reset

O.4.3. Spectral gap $\text{gap}(t)$

The spectral gap is the smallest positive eigenvalue in absolute value. It measures the distance of the spectrum from zero:

$$\text{gap}(t) = \min\{|\lambda_k(t)| > 0\}$$

Properties:

- $\text{gap}(t) \geq 0$
- A large gap (> 0.4) indicates a robust system, with significant energy barriers between the attractors.
- A small gap (< 0.2) indicates a fragile system, close to a phase transition.
- $\text{gap}(t) = 0$ indicates degeneracy (bifurcation).

References: The spectral gap is a classic tool in spectral graph theory [106]. It is used as a measure of robustness and connectivity in complex networks, notably in the work of Spielman and Teng on spectral partitioning [107] and of Alon on connectivity properties [108].

Philosophical interpretation: The spectral gap is the system’s **margin for manoeuvre** — the space between the system and collapse. It is the “**creative void**” spoken of by the Taoists, the empty space inside the vessel that gives it its function (*Dao De Jing*, §11). In the Buddhist tradition, it is the **interval** between two thoughts, that moment of silence in which enlightenment may arise. A wide gap is a breathing space, a system that can absorb shocks and reorganise itself. A narrow gap is a suffocated system, where every disturbance risks causing a collapse — the breaking point where the *qi* no longer flows.

O.4.4. Threshold saturation ratio $R_{\text{threshold}}(t)$

The threshold saturation ratio measures the fraction of spectral energy carried by the threshold pentads P_4 and N_4 :

$$R_{\text{threshold}}(t) = \frac{|\lambda_{P_4}| + |\lambda_{N_4}|}{\sum_{k=1}^{12} |\lambda_k|}$$

where λ_{P_4} and λ_{N_4} are the eigenvalues associated with the threshold pentads.

Properties:

- $R_{\text{threshold}}(t) \in [0, 1]$
- $R_{\text{threshold}}(t) < 0.3$: green (the thresholds are not being stressed)
- $0.3 < R_{\text{threshold}}(t) < 0.7$: orange (the thresholds are beginning to saturate)
- $R_{\text{threshold}}(t) > 0.7$: red (alert, critical saturation)

References: This observable is an original construction of the Cl(6, 6) model. It has no direct equivalent in the existing literature; it is specific to the topology of the pentad graph and to the role of the thresholds P_4 and N_4 as hinges between the tropical belts.

Philosophical interpretation: The ratio $R_{\text{threshold}}$ is the **tension on the system's gates**. The thresholds P_4 and N_4 are the **hinges** between expansion and recession, the equivalent of the **gorges** in the human body in traditional Chinese medicine — narrow passageways where *qi* can be blocked or released. Saturation at these thresholds constitutes a **blockage** of the system, analogous to an energy *stasis*. Exceeding the 0.7 threshold is an emergency call: the system must undergo **acupuncture** (through a political intervention such as the TRR) to restore circulation.

O.4.5. Philosophical synthesis of the four observables

The four observables form a comprehensive diagnostic system, comparable to the **four examinations** of traditional Chinese medicine (望闻问切, *wàng wén wèn qiè*):

Table 106: Correspondence between spectral observables and the four examinations of TCM

Observable	TCM examination	Interpretation
$\eta(t)$	望 (inspection)	The system's <i>qi</i> —the vital energy that flows
$d(t)$	闻 (listening/smell)	The diversity of the system —the harmony of forces
gap	问 (questioning)	The margin for manoeuvre —the capacity to respond to shocks
$R_{\text{threshold}}(t)$	切 (palpation)	The tension at critical points —the pulse of the system

Just as a doctor takes a patient's pulse to diagnose the state of the body, the policy-maker **takes the pulse** of the economic system through these four observables. The art of regulation lies in maintaining the system within the zone of harmony ($d \approx 5, \eta > 0$) whilst avoiding zones of disorder ($d \approx 12$), rigidity ($d \approx 1$), or collapse ($d \rightarrow 0, R_{\text{threshold}} > 0.7$).

Thus, the spectral dashboard is not merely a technical tool; it is a **modern divinatory practice** — a dialogue with the system that enables us to probe its state, diagnose its imbalances, and adjust policies accordingly.

O.5 — Hierarchical extension of observables

O.5.1 — Introduction and motivation

The four spectral observables defined in §O.4 — $\eta(t)$, $d(t)$, $\text{gap}(t)$, $R_{\text{threshold}}(t)$ — are constructed on the discrete Dirac operator $D(t)$ acting on the Hilbert space $\mathcal{H}_P$ of the 12 basic pentads. They provide a global diagnosis of the economic system, condensing the spectral information into four numbers.

However, the extension to Cl(6,6) [8] introduces a structural richness that these global observables do not capture:

1. **The 12 spectral sheets** $\mathcal{F} = \{e_1, \dots, e_6, f_1, \dots, f_6\}$, each carrying its own dynamics.
2. **The 144 projected pentads** $P_k^{(g)} = g \cdot P_k^{\text{base}}$, which constitute the building blocks of the system.
3. **The 7 spectral thresholds** $S_1, \dots, S_7$ derived from the eigenvalues of the lattice Λ_{72} [9].
4. **The 66 inter-sheet relations**, which encode the structural tensions between perspectives.

The hierarchical extension of the observables reflects this richness by constructing a multi-resolution system. This section presents the complete mathematical formalism; the reader is referred to Appendix D (§D.12) for the decision-theoretic interpretation and cultural correspondences.

O.5.2 – The 12 spectral leaves and the extended Hilbert space

O.5.2.1 – Definition of spectral sheets Let $\mathcal{F} = \{e_1, \dots, e_6, f_1, \dots, f_6\}$ be the set of the 12 generators of $\text{Cl}(6,6)$, where:

- $\{e_1, \dots, e_6\}$ are the cosmic generators (sector +, Sheng mode, $\eta > 0$).
- $\{f_1, \dots, f_6\}$ are the anti-cosmic generators (sector –, Ke mode, $\eta < 0$).

These generators satisfy the anticommutation relations:

$$e_a e_b + e_b e_a = 2\delta_{ab}, \quad f_a f_b + f_b f_a = -2\delta_{ab}, \quad e_a f_b + f_b e_a = 0$$

Each generator $g \in \mathcal{F}$ defines a **spectral leaf** $\mathcal{F}_g$, a subspace of the total Hilbert space $\mathcal{H}$ of the 144 projected pentads. This Sheng/Ke duality is the algebraic interpretation of Petit’s bicosmic cosmology [109], [110] applied to economic systems.

O.5.2.2 – The extended Hilbert space The total Hilbert space decomposes into an orthogonal direct sum:

$$\mathcal{H} = \bigoplus_{g \in \mathcal{F}} \mathcal{H}_g$$

where each $\mathcal{H}_g \cong \mathbb{C}^{12}$ is the Hilbert space of the spectral sheet g , generated by the 12 projected pentads $\{P_1^{(g)}, \dots, P_{12}^{(g)}\}$.

Total dimension: $\dim(\mathcal{H}) = 12 \times 12 = 144$.

Economic interpretation: Each sheet $\mathcal{H}_g$ modulates the intensity and orientation of a basic pentad. For example:

- $\mathcal{H}_{e_2}$: the space of pentads with high purchasing power (Sheng mode).
- $\mathcal{H}_{f_3}$: the space of pentads with strict financial regulation (Ke mode).

O.5.3 – The spectral projectors Π_f

O.5.3.1 – Definition For each leaf $g \in \mathcal{F}$, we define the **spectral projector**:

$$\Pi_g : \mathcal{H}_P \rightarrow \mathcal{H}_g$$

which associates each base pentad P_k with its projection onto the leaf g :

$$\Pi_g(P_k) = P_k^{(g)} = g \cdot P_k^{\text{base}}$$

where $\cdot$ denotes the Clifford product [111], [112].

O.5.3.2 – Properties of projectors The projectors Π_g satisfy the following properties:

1. **Idempotence**: $\Pi_g^2 = \Pi_g$ (on $\mathcal{H}_g$).
2. **Orthogonality**: $\Pi_g \Pi_h = 0$ for $g \neq h$.
3. **Completeness**: $\sum_{g \in \mathcal{F}} \Pi_g = \text{Id}_{\mathcal{H}}$.
4. **Preservation of nilpotence**: if $(P_k)^2 = 0$, then $(\Pi_g(P_k))^2 = 0$.

Proof of the preservation of nilpotence: Let $x \in P_k^{\text{base}}$ such that $x^2 = 0$. For any generator $g \in \mathcal{F}$:

- If $\{g, x\} = 0$ (anticommutation): $(gx)^2 = gxgx = -g^2x^2 = 0$.
- If $[g, x] = 0$ (commutation): $(gx)^2 = g^2x^2 = 0$.

Nilpotence is therefore preserved by projection. Economic stability is an algebraic property, not an ad hoc assumption – which underpins the retrospective validity of the model across the 64 historical regimes in Appendix M.

O.5.3.3 – Matrix of projectors In the basis of the 144 projected pentads, the projectors Π_g are represented by 144×144 block-diagonal matrices:

$$\Pi_g = \begin{pmatrix} 0 & \dots & 0 \\ \vdots & I_{12} & \vdots \\ 0 & \dots & 0 \end{pmatrix}$$

where I_{12} is the 12×12 identity matrix placed in the block corresponding to the sheet g .

O.5.4 – The Dirac operator per sheet $D_f(t)$

O.5.4.1 – Definition For each sheet $g \in \mathcal{F}$, we define the **restricted Dirac operator** $D_g(t)$ acting on $\mathcal{H}_g$:

$$(D_g \psi)_k = \sum_{j \sim k} w_{kj}^{(g)}(t) \sigma_{kj} \psi_j$$

where:

- $\psi \in \mathcal{H}_g$ is a local spinor on the leaf g .
- $w_{kj}^{(g)}(t) = \beta \cdot \exp\left(-\frac{|\varphi_k^{(g)} - \varphi_j^{(g)}| + E(\varphi_k^{(g)}, \varphi_j^{(g)})}{\gamma}\right) \cdot \Theta(\varepsilon_k^{(g)}, \varepsilon_j^{(g)})$ is the coupling weight restricted to the leaf g .
- σ_{kj} is the Pauli matrix associated with the edge (k, j) .
- $\varphi_k^{(g)}$ is the activation rate of the pentad $P_k^{(g)}$.
- $\varepsilon_k^{(g)}$ is the local mode of the pentad $P_k^{(g)}$.

O.5.4.2 – Dirac matrix per sheet The matrix $D_g(t)$ is a 12×12 matrix (or 24×24 if both components of the spinor are considered) defined in 2×2 blocks:

$$D_g(t) = \begin{pmatrix} 0 & w_{12}^{(g)} \sigma_{12} & \cdots & w_{1,12}^{(g)} \sigma_{1,12} \\ w_{21}^{(g)} \sigma_{21} & 0 & \cdots & w_{2,12}^{(g)} \sigma_{2,12} \\ \vdots & \vdots & \ddots & \vdots \\ w_{12,1}^{(g)} \sigma_{12,1} & w_{12,2}^{(g)} \sigma_{12,2} & \cdots & 0 \end{pmatrix}$$

O.5.4.3 – Spectrum of $D_g(t)$ The spectral problem on the sheet g consists of solving:

$$D_g(t) \psi = \lambda^{(g)} \psi$$

The eigenvalues $\lambda_k^{(g)}$ (in order: $\lambda_1^{(g)} \leq \lambda_2^{(g)} \leq \cdots \leq \lambda_{12}^{(g)}$) determine the local dynamics of the leaf g .

O.5.5 – The tensor decomposition $D(t) = \bigoplus_{f=1}^{12} D_f(t)$

O.5.5.1 – Decomposition theorem **Theorem:** The global Dirac operator $D(t)$ on $\mathcal{H}$ decomposes into an orthogonal direct sum of restricted Dirac operators:

$$D(t) = \bigoplus_{g \in \mathcal{F}} D_g(t)$$

Proof: By construction, the spectral leaves $\mathcal{H}_g$ are pairwise orthogonal, and the coupling weights $w_{kj}(t)$ are zero for k and j belonging to different leaves (since pentads from different leaves do not share a common attractor via the projection operator Π defined in §O.5.3). The operator $D(t)$ is therefore block-diagonal in the basis of the 144 projected pentads, with 12 blocks of size 12×12 corresponding to the $D_g(t)$.

O.5.5.2 – Spectral consequences The spectrum of $D(t)$ is the union of the spectra of the $D_g(t)$:

$$\sigma(D(t)) = \bigcup_{g \in \mathcal{F}} \sigma(D_g(t))$$

The 144 eigenvalues of $D(t)$ are the 144 eigenvalues of the 12 operators $D_g(t)$ (12 eigenvalues per sheet).

O.5.5.3 – Physical interpretation The tensor decomposition means that the global dynamics of the economic system is the superposition of 12 local dynamics, each corresponding to a spectral sheet. The sheets do not communicate directly (there is no inter-sheet coupling in $D(t)$), but they interact indirectly via relational observables (see §O.5.7).

O.5.6 – Local observables per sheet

O.5.6.1 – Local spectral asymmetry $\eta_g(t)$ For each sheet $g \in \mathcal{F}$, the local spectral asymmetry is defined by:

$$\eta_g(t) = \frac{n_{P,g}(t) - n_{N,g}(t)}{n_{P,g}(t) + n_{N,g}(t)} \in [-1, +1]$$

where $n_{P,g}(t)$ is the number of positive pentads (Sheng, $\varepsilon_k^{(g)} = +1$) active on the leaf g , and $n_{N,g}(t)$ is the number of negative pentads (Ke, $\varepsilon_k^{(g)} = -1$).

Weighted version (analogous to §O.4.1):

$$\eta_g(t) = \frac{1}{\Lambda_g} \sum_{k=1}^{12} \text{sgn}(\lambda_k^{(g)}) |\lambda_k^{(g)}|^\alpha$$

where $\Lambda_g = \sum_{k=1}^{12} |\lambda_k^{(g)}|^\alpha$ is a normalisation factor, and $\alpha \in [0, 1]$ is a sensitivity parameter. This definition follows the tradition of spectral asymmetry studied in random matrix theory [98]–[100].

O.5.6.2 – Local spectral dimension $d_g(t)$ The effective spectral dimension of the sheet g is defined by the Shannon entropy of the normalised spectrum [101]:

$$d_g(t) = \exp \left(- \sum_{k=1}^{12} p_{k,g} \ln p_{k,g} \right)$$

where $p_{k,g} = \frac{|\lambda_k^{(g)}|}{\sum_{j=1}^{12} |\lambda_j^{(g)}|}$ is the normalised spectral probability.

Properties: $d_g(t) \in [1, 12]$, with $d_g \approx 1$ (concentrated system) and $d_g \approx 12$ (diverse system). This observable is inspired by the work on spectral dimension by Alexander and Orbach [103] and diffusion in disordered media [104].

O.5.6.3 – Local spectral gap $\text{gap}_g(t)$ The spectral gap of the sheet g is the smallest eigenvalue with positive absolute value:

$$\text{gap}_g(t) = \min\{|\lambda_k^{(g)}(t)| > 0\}$$

The spectral gap is a standard tool in spectral graph theory [106], used as a measure of robustness in the work of Spielman and Teng [107] and Alon [108].

O.5.6.4 – Local threshold fraction $R_{\text{threshold},g}(t)$ The fraction of the asymmetry contributed by the threshold pentads $P_4^{(g)}$ and $N_4^{(g)}$ on the leaf g is:

$$R_{\text{threshold},g}(t) = \frac{|\lambda_{P_4,g}| + |\lambda_{N_4,g}|}{\sum_{k=1}^{12} |\lambda_k^{(g)}|}$$

O.5.7 – Relational observables as Clifford products

O.5.7.1 – Motivation The local observables η_g , d_g , gap_g , $R_{\text{threshold},g}$ describe the dynamics of each leaf independently. However, the leaves are not independent: they interact via structural tensions that encode the system's imbalances.

These interactions are formalised by **relational observables**, defined as Clifford products between sheets.

O.5.7.2 – Asymmetry gap $\Delta\eta_{f,g}(t)$ The asymmetry difference between two leaves f and g is:

$$\Delta\eta_{f,g}(t) = |\eta_f(t) - \eta_g(t)|$$

Interpretation: $\Delta\eta_{f,g}$ measures the tension between two perspectives. A high difference indicates a structural imbalance (for example, one Sheng-expanding leaf and another Ke-contracting leaf).

O.5.7.3 – Inter-sheet correlation $C_{f,g}(t)$ The correlation between two sheets f and g is defined by:

$$C_{f,g}(t) = \text{corr}(\eta_f, \eta_g) = \frac{\text{cov}(\eta_f, \eta_g)}{\sigma_{\eta_f} \sigma_{\eta_g}}$$

calculated over a sliding window of W quarters (typically $W = 12$).

Interpretation:

- $C_{f,g} \approx +1$: the leaves move in phase (structural coherence).
- $C_{f,g} \approx 0$: the leaves are independent (fragmentation).
- $C_{f,g} \approx -1$: the leaves move in opposition (structural tension).

O.5.7.4 – Clifford product between leaves The Clifford product between two generators $f, g \in \mathcal{F}$ defines a bivector $B_{f,g} = f \wedge g$ which encodes the interaction between the sheets f and g .

Relational observable: The interaction energy between the leaves f and g is:

$$E_{f,g}(t) = \|B_{f,g} \cdot \psi(t)\|^2$$

where $\psi(t)$ is the global state of the system.

Interpretation: $E_{f,g}$ measures the strength of the interaction between the leaves f and g . A high energy value indicates strong structural tension.

O.5.7.5 – Wuxing tension $\mathcal{T}_{\text{Wuxing}}(t)$ As formally defined in §15.4.6 (Table 25), the 12 spectral leaves are grouped into 5 families corresponding to the five phases of the Wuxing (Wood, Fire, Earth, Metal, Water) [27], [91]. The Wuxing tension is defined by:

$$\mathcal{T}_{\text{Wuxing}}(t) = \max_{f,g} \Delta \eta_{f,g}(t)$$

Interpretation: $\mathcal{T}_{\text{Wuxing}}$ measures the maximum imbalance between the five phases. A high imbalance indicates a pathology of the Sheng/Ke cycle.

The **Wuxing tension** is defined as:

$$\mathcal{T}_{\text{Wuxing}}(t) = \max_{f,g} \Delta \eta_{f,g}(t)$$

Interpretation: $\mathcal{T}_{\text{Wuxing}}$ measures the maximum imbalance between the five phases. A high imbalance indicates a pathology of the Sheng/Ke cycle.

O.5.8 – The inter-sheet correlation matrix

O.5.8.1 – Definition The correlation matrix $C(t) \in \mathbb{R}^{12 \times 12}$ is defined by:

$$C_{f,g}(t) = \text{corr}(\eta_f, \eta_g)$$

O.5.8.2 – Properties

1. **Symmetry:** $C_{f,g} = C_{g,f}$.
2. **Unit diagonal:** $C_{f,f} = 1$.
3. **Positive semidefinite:** $C \succeq 0$.
4. **Eigenvalues:** the eigenvalues of C lie in $[0, 12]$.

O.5.8.3 – Fragmentation index The **fragmentation index** is defined by:

$$\mathcal{F}(t) = 1 - \frac{1}{66} \sum_{f < g} |C_{f,g}(t)|$$

- $\mathcal{F} \approx 0$: a highly coherent system (all leaves evolve together).
- $\mathcal{F} \approx 1$: a highly fragmented system (the leaves are independent).

O.5.9 – The Stase composite index

O.5.9.1 – Definition The Stase composite index summarises the five levels of the hierarchy into a single number:

$$\mathcal{J}_{\text{Stase}}(t) = w_1 |\eta_{\text{total}}| + w_2 \frac{1}{d_{\text{total}}} + w_3 e^{-5 \cdot \text{gap}_{\text{total}}} + w_4 R_{\text{threshold, total}} + w_5 \mathcal{T}_{\text{Wuxing}}$$

where:

- $\eta_{\text{total}} = \frac{1}{12} \sum_{g \in \mathcal{F}} \eta_g$ is the overall asymmetry.
- $d_{\text{total}} = \frac{1}{12} \sum_{g \in \mathcal{F}} d_g$ is the overall dimension.
- $\text{gap}_{\text{total}} = \min_{g \in \mathcal{F}} \text{gap}_g$ is the minimum gap.
- $R_{\text{threshold, total}} = \frac{1}{12} \sum_{g \in \mathcal{F}} R_{\text{threshold, } g}$ is the global threshold fraction.
- $\mathcal{T}_{\text{Wuxing}} = \max_{f,g} \Delta \eta_{f,g}$ is the Wuxing tension.

Note: The exponential term $e^{-5 \cdot \text{gap}}$ ensures that $\mathcal{J}_{\text{Stasis}} \in [0, 1]$. See §15.4.7 for the justification.

O.5.9.2 – Calibration of weights

The weights w_i are calibrated against the historiography of the nine Chinese dynastic declines (Appendix G) [25]. The calibration method is based on the frequency of occurrence of each topological warning signal within the nine historical crises, out of a total of 39 observed occurrences:

Table 107: Weighting parameters for the composite index $\mathcal{J}_{\text{Stasis}}$ (calibrated on 9 declines)

Parameter	Value	Justification (Presence of the signal in all 9 declinations)
w_1	0.23	Asymmetry (η) is the most robust signal (present in 9 out of 9 declinations).
w_2	0.18	Fragmentation ($1/d$) is a signal reinforced by periods of scission (present in 7 out of 9 declinations).
w_3	0.23	Spectral fragility (gap) is critical (present in 9 out of 9 declines).
w_4	0.21	Threshold tension ($R_{\text{threshold}}$) heralds retropolarity (present in 8 out of 9 declines).
w_5	0.15	The Wuxing imbalance ($\mathcal{T}_{\text{Wuxing}}$) is culturally rooted (present in 6 out of 9 declines).

Constraint: $\sum_{i=1}^5 w_i = 1$.

O.5.9.3 – Thresholds for the composite index

 Table 108: Thresholds for the Stasis composite index $\mathcal{J}_{\text{Stasis}}$ and economic states

$\mathcal{J}_{\text{Stase}}$	State	Historical example
< 0.3	Healthy growth	The Glorious Thirty (1945–1973) [2]
$0.3 - 0.5$	Moderate strain	Oil crisis (1973–1980)
$0.5 - 0.65$	Emerging stagnation	Japan (1990–2000)
$0.65 - 0.75$	Characterised stagnation	China (2018–2026) [83], [84]
> 0.75	Pre-retropolarity	Republic of China (1945–1949)

O.5.9.4 – Calculation for China 2026

$$\mathcal{J}_{\text{Stasis}} = 0.23 \times 0.45 + 0.18 \times \frac{1}{2.1} + 0.23 \times e^{-5 \times 0.12} + 0.21 \times 0.78 + 0.15 \times 1.1 \quad (1)$$

$$\mathcal{J}_{\text{Stasis}} = 0.1035 + 0.0857 + 0.1262 + 0.1638 + 0.165 \approx 0.644 \quad (2)$$

Diagnosis: China is in a state of **characterised stasis** ($0.65 < \mathcal{J}_{\text{Stasis}} < 0.75$), on the verge of the pre-retropolarity threshold.

O.5.10 – Complete calculation algorithm

O.5.10.1 – Inputs

Table 109: Input data for the calculation of extended spectral observables

Data	Source	Frequency
6 binary economic variables	National Bureau of Statistics	Quarterly
12 sectoral indicators	Sectoral ministries	Quarterly
LGFV debt ratio	Ministry of Finance [38]	Quarterly
GEP (Gross Ecosystem Product)	Ministry of Ecology	Annual
Regional Gini coefficient	National Bureau of Statistics	Annual

O.5.10.2 – Algorithm

```
def calculate_extended_observables(data, t):
    # Step 1: Calculate the 4 global observables (Level 1)
    eta_total, d_total, gap_total, R_threshold_total = calculate_globals(data, t)

    # Step 2: Calculate observables per leaf (Level 2)
    leaves = {}
    for g in [e1, e2, e3, e4, e5, e6, f1, f2, f3, f4, f5, f6]:
        eta_g, d_g, gap_g, R_threshold_g = calculate_sheet(data, g, t)
        sheets[g] = {
            'eta': eta_g, 'd': d_g, 'gap': gap_g, 'R_threshold': R_threshold_g
        }

    # Step 3: Calculation of threshold observables (Level 3)
    T = calculate_topological_tension(eta_total, R_threshold_total)
    thresholds = {}
    for i, sqrt_lambda in enumerate([0.06614, 0.10486, 0.10582, 0.13530,
                                     0.17195, 0.19329, 0.21732]):
        thresholds[f'S{i+1}'] = 1 if T >= sqrt_lambda else 0

    # Step 4: Calculating pentad observables (Level 4)
    pentads = {}
    for k in range(1, 13):
        for g in leaves.keys():
            a_kg, pi_kg, c_kg, E_kg = calculate_pentad(data, k, g, t)
            pentads[(k, g)] = {
                'activity': a_kg, 'polarity': pi_kg,
                'connectivity': c_kg, 'frustration': E_kg
            }
```

```

    }

# Step 5: Calculation of relational observables (Level 5)
C = calculate_correlation_matrix(leaves, t)
T_Wuxing = calculate_Wuxing_tension(leaves)

# Step 6: Calculating the composite index
I_Stasis = (0.23 * abs(eta_total) +
            0.18 * (1 / d_total) +
            0.23 * (1 / gap_total) +
            0.21 * R_threshold_total +
            0.15 * T_Wuxing)

return {
    'level_1': {'eta': eta_total, 'd': d_total,
               'gap': gap_total, 'R_threshold': R_threshold_total},
    'level_2': leaves,
    'level_3': thresholds,
    'level_4': pentads,
    'level_5': {'C': C, 'T_Wuxing': T_Wuxing},
    'I_Stasis': I_Stasis
}

```

Table 110: Computational complexity by level of the hierarchy of observables

Level	Number of observables	Complexity
1	4	$O(1)$
2	48	$O(12 \times 12^3) = O(20\,736)$
3	7	$O(7)$
4	576	$O(144 \times 12) = O(1\,728)$
5	66	$O(12^2) = O(144)$
Total	701	$O(22\,615)$

0.5.10.3 – Computational complexity The full calculation takes less than a second on a standard computer, enabling real-time control.

O.5.11 – Relationship with other appendices

This section O.5 provides the mathematical formalism for extended observables. For their operational interpretation and application to control, the reader is referred to the following sections:

Table 111: Structure of Annex O in relation to the other sections of the document

Concept	Reference in the document
Multi-level governance (4 decision-making layers)	Appendix D, §D.12
Advanced spectral dashboard (heat map, radar, matrix)	Appendix S
Correspondence with the four TCM examinations	Appendix D, §D.12.6; Appendix O, §O.4.5
Correspondence with Chinese imperial governance	Appendix D, §D.12.6
Calibration against the nine dynastic declines	Appendix G [25]
TRR pilot zones (Lishui, Sanming, Xingguo, Hainan)	Chapter 20; [92], [93], [95], [96]
The <i>tang ping</i> phenomenon as a social manifestation of Stasis	Chapter 7; [86]
The impasse of the immaterial and the deficit of spatial depth	Chapter 6; [73], [82], [85]

O.6. Algorithm for calculating observables

The following table presents the step-by-step calculation algorithm.

Table 112: Algorithm for calculating spectral observables

Step	Description
1	Normalise the data $X(t)$ to obtain $\varphi_v(t)$ for each pentad v
2	Determine the configuration $\mathbf{x}(t)$ from the thresholds
3	Identify the attractor $A_m(t)$ via $\Pi(\mathbf{x})$
4	Extract the active triplet of pentads $\{P_a, P_b, P_c\}$
5	Construct the adjacency matrix A of the graph Γ
6	Calculate the weights $w_{ij}(t)$ for all edges
7	Construct the Dirac operator $D(t)$
8	Solve the spectral problem: $D(t)\psi = \lambda\psi$
9	Order the eigenvalues: $\lambda_1 \leq \dots \leq \lambda_{12}$
10	Calculate $\eta(t) = \frac{1}{\Lambda} \sum_k \text{sgn}(\lambda_k) \lambda_k ^\alpha$
11	Calculate $d(t) = \exp(-\sum_k p_k \ln p_k)$ where $p_k = \lambda_k / \sum_j \lambda_j $
12	Calculate $\text{gap}(t) = \min\{ \lambda_k(t) > 0\}$
13	Calculate $R_{\text{threshold}}(t) = (\lambda_{P_4} + \lambda_{N_4}) / \sum_k \lambda_k $
14	Return $\eta(t), d(t), \text{gap}(t), R_{\text{threshold}}(t)$

O.7. Early warning detection algorithm

Table 113: Early warning detection algorithm

Step	Description
1	Green: if $\eta > 0.3$ AND $d > 4$ AND $\text{gap} > 0.4$ AND $R_{\text{threshold}} < 0.3$
2	Orange: if $0.1 < \eta < 0.3$ OR $2 < d < 4$ OR $0.2 < \text{gap} < 0.4$ OR $0.3 < R_{\text{threshold}} < 0.7$
3	Red: if $\eta < 0.1$ OR $d < 2$ OR $\text{gap} < 0.2$ OR $R_{\text{threshold}} > 0.7$
4	Return the alert level

O.8. Detailed calculation example

O.8.1. Input data for China 2024

Table 114: Input data for the spectral calculation (China 2024)

Variable	Value	Threshold	x_k
Inflation	0.5%	2%	0
Purchasing power	-1.2%	0%	0
Profits	-3.4%	0%	0
Rent Extraction	LGFV debt ratio = 85%	80%	0
Inequality	Regional Gini coefficient = 0.42	0.40	1
Land tenure	Available land	Increasing	0

Resulting configuration: $\mathbf{x} = (0, 0, 0, 0, 1, 0)$

O.8.2. Construction of the Dirac matrix

For this configuration, the identified attractor is S (Stasis), with the triplet $\{P_6, N_5, N_6\}$.

The activation rates are estimated from the data:

- $\varphi_{P_6} = 0.45$ (weakened internal consumption)
- $\varphi_{N_5} = 0.78$ (stasis, resource depletion)
- $\varphi_{N_6} = 0.65$ (domestic austerity)

The local modes are:

- $\varepsilon_{P_6} = +1$ (Sheng)
- $\varepsilon_{N_5} = -1$ (Ke)
- $\varepsilon_{N_6} = -1$ (Ke)

The coupling weights are calculated (with $\beta = 1, \gamma = 0.5, \lambda = 0.3$):

$$w_{P_6 N_5} = \exp\left(-\frac{|0.45 - 0.78| + (0.33)^2}{0.5}\right) \times 0.3 \approx 0.16$$

$$w_{P_6 N_6} = \exp\left(-\frac{|0.45 - 0.65| + (0.20)^2}{0.5}\right) \times 0.3 \approx 0.20$$

$$w_{N_5 N_6} = \exp\left(-\frac{|0.78 - 0.65| + (0.13)^2}{0.5}\right) \times 1 \approx 0.77$$

The Dirac matrix (extracted for the 3 active pentads) is:

$$D = \begin{pmatrix} 0 & 0.16\sigma_x & 0.20\sigma_x \\ 0.16\sigma_x & 0 & 0.77\sigma_x \\ 0.20\sigma_x & 0.77\sigma_x & 0 \end{pmatrix}$$

O.8.3. Spectral solution

Solving $D\psi = \lambda\psi$ yields the eigenvalues:

$$\lambda = \{-1.02, -0.52, -0.31, -0.18, -0.05, 0.00, 0.02, 0.18, 0.45, 0.68, 0.92, 1.21\}$$

Table 115: Eigenvalues for China 2024

k	1	2	3	4	5	6	7	8	9	10	11	12
λ_k	-1.02	-0.52	-0.31	-0.18	-0.05	0.00	0.02	0.18	0.45	0.68	0.92	1.21

O.8.4. Calculation of observables

Step 1: Spectral asymmetry (with $\alpha = 0.5$)

$$\Lambda = \sum_k |\lambda_k|^{0.5} = 7.04$$

$$\eta = \frac{1}{7.04} (-1.01 - 0.72 - 0.56 - 0.42 - 0.22 + 0 + 0.14 + 0.42 + 0.67 + 0.82 + 0.96 + 1.10)$$

$$\eta = \frac{1}{7.04} \times (0.18) \approx 0.03$$

Step 2: Effective spectral dimension

$$\sum_j |\lambda_j| = 6.78, \quad p_k = \frac{|\lambda_k|}{6.78}$$

$$d = \exp\left(-\sum_k p_k \ln p_k\right) \approx \exp(0.69) \approx 1.99$$

Step 3: Spectral gap

$$\text{gap} = \min\{|\lambda_k| > 0\} = 0.02$$

(The smallest eigenvalue in absolute value is $|0.02|$, corresponding to the eigenvalue $\lambda_7 = 0.02$.)

Step 4: Threshold saturation ratio

For this configuration, the threshold pentads are P_4 and N_4 , with $\lambda_{P_4} = 0.92$ and $\lambda_{N_4} = -0.52$:

$$R_{\text{threshold}} = \frac{|0.92| + |-0.52|}{6.78} = \frac{1.44}{6.78} \approx 0.21$$

O.8.5. Results and interpretation

Table 116: Spectral calculation results for China 2024

Observable	Value	Status
η	0.03	Orange (tipping point)
d	1.99	Red (< 2)
gap	0.02	Red (< 0.2)
$R_{\text{threshold}}$	0.21	Green (< 0.3)

Interpretation: The system is at a tipping point ($\eta \approx 0$), with excessive concentration ($d < 2$) and high fragility (very small gap). The saturation ratio is still low, but the situation is critical.

O.9. Parameter calibration

Table 117: Model parameters and calibration values

Parameter	Symbol	Value	Justification
Coupling constant	β	1.0	Normalisation
Scale parameter	γ	0.5	Fitting to historical cycles
Spectral exponent	α	0.5	Sensitivity to small values
Inter-mode coupling factor	λ	0.3	Reduction of couplings between opposing modes

O.10. References for implementation

For a numerical implementation, the following steps are recommended:

1. Use a linear algebra library (NumPy, Eigen, etc.) for diagonalisation.
2. Normalise the input data using min-max or z-scores.
3. For the calculation of $d(t)$, use the definition with $0 \ln 0 = 0$.
4. To calculate $\text{gap}(t)$, sort the eigenvalues and compute the differences.
5. Update the observables at each time step (monthly or quarterly).

Appendix P – Rowlands’ Pentad and Its Correspondence with the Wuxing

P.1. Introduction

The use of the Clifford algebra $Cl(6, 6)$ employed to extend the Theory of Rent Extraction originates in the work of Peter Rowlands. Rowlands performs a **compactification of the space, time, mass, and charge parameters** into a unified algebraic structure. This construction rests on a fundamental object with five components: the **pentad**. By cyclic permutation of its 6 elements, six canonical pentads are obtained.

The purpose of this appendix is twofold:

1. To present the construction of the pentad according to Rowlands.
2. To show how Rowlands’ six pentads can be mapped onto the five Elements of the Wuxing, the Wuxing appearing as the **“pattern”** or the **“quotient”** of the pentads.

Important note: In Rowlands’ notation, the symbols i, j, k, I, J, K are **NOT** the quaternions. They are a notation for the generators of the Clifford algebra $Cl(6, 0)$. The correspondence with the canonical generators e_μ is:

$$i = e_1, \quad j = e_2, \quad k = e_3, \quad I = e_4, \quad J = e_5, \quad K = e_6$$

to which must be added the two scalars: the identity 1 and the pseudoscalar $i' = ijkIJK$.

Rowlands’ construction of the pentads therefore involves 8 elements: the 6 generators of $Cl(6, 0)$ and the 2 scalars (identity and pseudoscalar).

P.2. The Construction of the Pentad according to Rowlands

P.2.1. The Four Fundamental Parameters

Rowlands starts from four fundamental physical parameters:

Space, Time, Mass, Charge

Space and **Charge** are three-dimensional, while **Time** and **Mass** are one-dimensional.

The 8 elements $(1, i', i, j, k, I, J, K)$ are assigned to the 4 physical quantities as follows:

Table 118: Assignment of the 8 elements to the 4 physical parameters

Parameter	Associated elements
Time	i'
Space	I, J, K
Mass	1
Charge	i, j, k

P.2.2. Superposition of the Charge Units

Rowlands superposes each charge element (i, j, k) onto the elements representing time (i') , mass (1) , and space (I, J, K) :

$$\begin{array}{cccccc} i' & I & J & K & 1 \\ k & i & i & i & j \end{array}$$

which leads to the following combination that he names the “pentad”:

$$i'k \quad iI \quad iJ \quad iK \quad 1j$$

Interpretation:

- iI, iJ, iK : the units of space (i) combined with the units of mass (I, J, K)
- $i'k$: the unit of time (i') combined with the unit of charge (k)
- $1j$: the unit of charge (j) preserved as an autonomous parameter

According to Rowlands, this is the most compact way of gathering the information. This structure, which Rowlands establishes from physics, is extended to biology in Chapter 19 of his reference work [5]. Here we transpose this construction to economics.

P.2.3. The Six Pentads by Cyclic Permutation and the Wu Xing

By cyclic permutation of the 6 generators i, j, k, I, J, K , the six basic pentads are obtained.

The Clifford algebra $Cl(6, 0)$ includes 64 elements generated by 6 generators. Likewise, the 64 hexagrams of the *Yijing* are constructed from all combinations of 6 lines (continuous or broken).

According to Rowlands, each pentad is a generating basis of $Cl(6, 0)$, just as the Majorana matrices are for quantum mechanics. Likewise, the 5 elements of the Wuxing are the pattern of generation that can be applied to all ontologies, which was the foundation of Chinese correlative thought for more than two millennia.

These few reflections aim to make the Chinese reader aware that the economic theory presented in this report is a mathematized extension of traditional Chinese thought to economics.

Table 119: The 6 pentads and the Wu Xing

Pentad	Wood (B_1)	Metal (B_2)	Earth (B_3)	Fire (F)	Water (S)
P_1	iI	iJ	iK	$i'k$	$1j$
P_2	jI	jJ	jK	$i'i$	$1k$
P_3	kI	kJ	kK	$i'j$	$1i$
P_4	$i'Ii$	$i'Ij$	$i'Ik$	$i'K$	$1J$
P_5	$i'Ji$	$i'Jj$	$i'Jk$	$i'I$	$1K$
P_6	$i'Ki$	$i'Kj$	$i'Kk$	$i'J$	$1I$

P.2.4. Foliation into 12 Regulating Sheets and Emergence of the 144 Pentads

The complete space of $\text{Cl}(6, 6)$ contains $2^{12} = 4096$ elements, but the dynamics do not unfold uniformly within it. The system projects onto a foliation into 12 regulating sheets, each isomorphic to the dual graph Γ but weighted by a dominant generator.

- **6 cosmic sheets** $\mathcal{F}_{e_i}$ ($i = 1 \dots 6$): dominated by e_i , carrying a global orientation $\eta > 0$ (mode *sheng*, exploration/generation). They correspond to the observable sector of Janus [109].
- **6 anti-cosmic sheets** $\mathcal{F}_{f_j}$ ($j = 1 \dots 6$): dominated by f_j , carrying $\eta < 0$ (mode *ke*, constraint/regulation). They correspond to the negative-mass sector.

The terms *cosmic* and *anti-cosmic* come from the bicosmic cosmology of Jean-Pierre Petit [110].

Each sheet $\mathcal{F}_g$ ($g \in \{e_1, \dots, e_6, f_1, \dots, f_6\}$) contains the 12 basic pentads, modulated by the dominant generator. A generic pentad is therefore written:

$$P_k^{(g)} = g \cdot P_k^{\text{base}} = \{g \cdot x \mid x \in P_k^{\text{base}}\}, \quad g \in \{e_i, f_j\}$$

where P_k^{base} is the basic pentad (defined above) and $\cdot$ denotes the Clifford product.

Unified notation:

- $P_k^{(e_i)}$: basic pentad k projected onto the cosmic sheet e_i ($\eta > 0$, sector +)
- $N_k^{(f_j)}$: basic pentad k projected onto the anti-cosmic sheet f_j ($\eta < 0$, sector −)

By construction, $N_k^{(f_j)} = -P_k^{(f_j)}$, where the minus sign is the global inversion of the pentad (phase duality). The set of 144 pentads is therefore written:

$$\mathcal{P} = \left\{ P_k^{(g)} \mid k = 1..12, g \in \{e_1, \dots, e_6, f_1, \dots, f_6\} \right\}, \quad |\mathcal{P}| = 12 \times 12 = 144.$$

Appendix Q –The aporias of neoclassical and Marxist economic theories

This appendix reproduces a critique by Thierry Rebour (2000) regarding the inability of neoclassical economics to explain spatial agglomeration. This aporia justifies the need for an alternative theory—the Rent Extraction Theory — which underpins the present model.

Rebour writes on page 9 of [3]: The origin of the clustering of people at certain points in geographical space is the result of mobilities—more precisely, centripetal mobilities—which result in an aggregation, an agglomeration, where these flows meet. “Mobility underpins spatial analysis (...), it is its central notion” (Aydalot, 1977). Instantaneous location can therefore only be seen as the provisional result of the forces that create mobility (ibid., 1977). Consequently, the analysis of spatial forms must always be a dynamic analysis, and their understanding must integrate an understanding of the engine that creates this mobility.

Ex post, this engine is always an economic phenomenon. This does not mean that political, cultural, or even religious forces are entirely excluded from the mechanisms that generate mobility. On the contrary, as we will see from the second chapter onward, such forces most often preexist the economic dynamic and actually condition it from the very beginning. But any attempt to formulate a theory or model that seeks to explain mobility cannot be content with premises that remain essentially qualitative and necessarily specific to each culture or civilization, while also forgetting that for a durable grouped habitat to exist, there must always be a prior release of wealth that authorizes its construction and existence. The city needs, in order to exist, an increase in agricultural productivity that frees the builders first, and the inhabitants later, from working the land. Moreover, the mere settlement of any human habitat, in other words, a true spatial anchoring of a sedentary type, itself requires a certain level of agricultural productivity.

This is another way of saying that food surpluses are needed for a large and durable agglomeration to be viable. This is why the city is always, ex post, an economic phenomenon. Is it not then entirely paradoxical that the principle of agglomeration is virtually absent from the dominant economic model?

For Neoclassical Economics, indeed, the city... does not exist. It only finds its justification and explanation from elements exogenous to the model: the need for interactions (Claval, 1981), or external economies. Purely economic forces, when assumed to be subject to the neoclassical mechanism, are not capable of generating anything other than a homogeneous, convergent space devoid of agglomeration (Aydalot, 1984). Convergence Theory, which describes the movement and location of production factors in Regional Economics (balanced and optimal), indeed leads to an amorphous space where the distribution of people and activities is “convergent”, balanced.

Q.2. –The Aporia of Marxist Economic Theory

This appendix reproduces a critique by Gilles Lavigne [113] regarding the tautology of Marx’s labor theory of value. The Cl(6,6) model escapes this aporia by defining value as a topological relation, not as a substance.

Gilles Lavigne writes in [113], page 121: According to Marx, the social labor provided by a society is divided into necessary labor (devoted to the production of goods “necessary” for the reproduction of the labor power used), and into surplus labor (all unpaid labor provided by this power in addition to necessary labor). The ratio between these two types of labor expresses the degree of exploitation of labor power, a degree initially measured in terms of time. The implementation of this power, or its application to production, takes place in a continuous, unitary, and recursive time, and the limit of each continuous production interval is defined by the time devoted as such to the reproduction of the consumed labor power. However, the measurement of the rate of exploitation can be obtained, not only on the basis of a temporal scale, but also and above all from a scale of value defined according to a “certain value” that a society recognizes in labor. This is exactly what Marx seeks to demonstrate, for whom commodities only have value, in terms of exchange, because they include living labor, the only kind capable of creating value; this attribute of living labor is due to the fact that, in societies subject to the capitalist mode of production, living labor is treated as a commodity that also has an exchange value. But should not this value be equal “in all logic” to its general cost of (re)production, which would be accounted for by an equation of the type: $VT.. = CT. (1 + pr.)$? This would mean recognizing, consequently, that the social value of labor (VT) is equal to the social capital (CT) employed to (re)produce it—in the wage paid—plus the average profit on this capital; noting, however, that this capital only has value because it is itself the product of prior living labor, otherwise one would have to admit that capital has value in itself. Expressed in this way, the definition is tautological, and absurd to boot. What meaning can a measure of the value of labor have, if value is itself defined by labor? It is equally absurd to grant an absolute value, whether to labor, capital, or land. Generally speaking

and between the lines, Marx supports his definition of value by labor by pushing the opposing version to the point of absurdity. If these “factors of production”—capital, land, and labor—are defined as factors of valorization, each conferring a fraction of value to commodities, and incomes signifying in return the appropriation of value, it is impossible to determine the value of commodities. The resulting tautology then renders the problem of value insoluble. This would somehow explain why the valorization of labor proceeds from its mystification: labor would overdetermine value on the basis of a failure to define the latter in absolute terms. In other words, labor assumes the same applicable function as God in the understanding of the material universe.

Appendix R –Philosophical Synthesis: The Spectral Resolution of the Xiong Shili – Nāgārjuna Controversy

R.1. The Historical Controversy

The history of Asian thought has long been traversed by an apparent tension between two visions of reality: on the one hand, the philosophy of the “Great Function” (*Dayong*, 大用) defended by the Neo-Confucian **Xiong Shili** (熊十力, 1885–1968, [114]), according to which the phenomenal world is a real, dynamic, and harmonious manifestation of the ultimate substance (*tiyong bu’er*, 体用不二); on the other hand, the philosophy of Emptiness (*Śūnyatā*, 空, *Kong*) of **Nāgārjuna** (龙树, 2nd–3rd century, [115]), founder of the Madhyamaka school (中观), for whom all phenomena are devoid of own-nature (*svabhava*, 自性), emptiness being the very condition of dependent origination (*pratītyasamutpāda*, 缘起) and change.

The topological model Cl(6,6) makes it possible to transcend this controversy by showing that these are not mutually exclusive truths, but rather **two distinct and complementary spectral regimes** of the same dynamic system.

R.2. Xiong Shili and the Regime of Harmony: $d(t) = 5$

Xiong Shili’s vision is isomorphic to the spectral regime in which the effective spectral dimension reaches $d(t) \approx 5$. In this regime, the economic system achieves a creative equilibrium:

- It is the optimal flourishing of the 5 phases of Wuxing (Wood 木, Fire 火, Earth 土, Metal 金, Water 水), where the generative cycle (*Sheng*, 相生) and the regulating cycle (*Ke*, 相克) balance each other without either dominating the other.
- It is the “Doctrine of the Mean” (*Zhongyong*, 中庸): sufficient diversity (5 active modes) to ensure resilience against shocks, but sufficient concentration to maintain structural coherence.
- Economically, it is the state of tempered and sustainable growth, where Wealth is distributed and Value is continuously regenerated (as in the successful TRR scenario).

For Xiong Shili, ignoring this dynamic reality in favor of an abstract void would be a mistake. The model validates this intuition: a system that remains durably at $d(t) = 5$ is a healthy system.

R.3. Nāgārjuna and the Regime of Creative Emptiness: $d(t) \rightarrow 0$

Nāgārjuna’s philosophy finds its spectral echo in the regime where $d(t) \rightarrow 0$. When the spectral dimension collapses, the system loses its diversity and coherence: this is the state of excessive concentration, rigidity, or systemic collapse (retro-polarity). However, within the logic of the Theory of Rent Extraction, this collapse is not a sterile nihilism.

- Emptiness (*Kong*, 空) is the **condition of possibility for change**. As long as the old structures (the saturated attractors of the Nodal Point) retain a residual spectral dimension, no new configuration can emerge.
- $d(t) \rightarrow 0$ represents the Taoist “Creative Void” (*wu*, 无) or Schumpeterian creative destruction pushed to its topological paroxysm. It is the moment when old Wealth must be destroyed (deleveraging, bankruptcies, bursting of bubbles) so that the Space of Value can be freed and redeemed anew.
- Nāgārjuna teaches that “emptiness is dependent origination” (*śūnyatā pratītyasamutpāda*). In the model, it is precisely the spectral annihilation ($d \rightarrow 0$) that allows the reset of the pentads and the controlled tipping point toward a new cycle via the threshold N_4 .

R.4. Resolution through the Model: The Spectral Dialectic

To the historical controversy that opposed the *reality of manifestation* (Xiong) to the *emptiness of substance* (Nāgārjuna), the Cl(6, 6) formalism proposes a way out by placing them on a temporal trajectory:

1. An economic system cannot eternally maintain Xiong Shili’s harmony ($d = 5$). The accumulation of topological frustration (Wuxing imbalances, hoarding of Value) inevitably causes the spectral dimension to drop.
2. When $d(t)$ approaches 0, the system enters Nāgārjuna’s regime: old economic categories lose their “own-nature” (bubbles burst, institutions freeze).
3. **The operational resolution** lies in managing this transition. If the system undergoes $d(t) \rightarrow 0$ chaotically, it is retro-polarity (the 10th decline). But if the State activates the TRR, it uses this topological “void” as a decompression chamber: it channels the destruction of old Wealth to finance the rent extraction of a new Space of Value, thus bringing the system back up to a new regime of harmony at $d(t) = 5$.

Thus, the model does not rule in favor of one over the other. It shows that the wisdom of Nāgārjuna ($d \rightarrow 0$) is the *necessary mechanism of purification*, while the wisdom of Xiong Shili ($d = 5$) is its *teleological objective*. The Territorial Regeneration Rent (TRR) is the political tool that allows one to traverse Emptiness without getting lost in it, to return to the Doctrine of the Mean.

R.5. Technical Note: The Two Spectral Dimensions

Two notions of spectral dimension coexist in this model, and it is important to specify which one is used for the philosophical analogy:

- **Shannon dimension** [101], $d_S(t) = \exp(-\sum p_k \ln p_k)$: measures the effective number of active modes. It is used for operational steering (dashboard, alerts). With this definition, $d \in [1, 12]$, and $d = 1$ corresponds to extreme concentration (a single mode dominates).

- **Alain Connes dimension** [105] $d_C(t)$: defined by the asymptotic behavior of $\text{Tr}(|D|^{-s})$, it measures the geometric structure of the spectral space. With this definition, $d \rightarrow 0$ is a natural limit corresponding to the collapse of the geometric structure.

The analogy with Nāgārjuna’s Emptiness ($d \rightarrow 0$) works best with the Connes dimension, where $d \rightarrow 0$ is a natural limit. In the operational context, we use the Shannon dimension, and the regime $d \approx 1$ (extreme concentration) corresponds to “rigid unity” (同, *tóng*) rather than Emptiness.

Table 120: Synthesis of the two spectral dimensions and their interpretations

Aspect	Shannon dimension d_S	Connes dimension d_C
Definition	$d_S = \exp(-\sum p_k \ln p_k)$	$\text{Tr}(D ^{-s})$ diverges for $s = d_C$
Domain	$d_S \in [1, 12]$	$d_C \in [0, +\infty)$ (fractional values possible)
Minimum	$d_S = 1$ (a single mode dominates)	$d_C \rightarrow 0$ (geometric collapse)
Usage	Operational steering (dashboard)	Philosophical interpretation
Regime $d \approx 5$	Xiong Shili’s Harmony (和, <i>hé</i>)	Xiong Shili’s Harmony (和, <i>hé</i>)
Regime $d \rightarrow 0$	Outside domain (total collapse)	Nāgārjuna’s Emptiness (空, <i>kōng</i>)
Regime $d \approx 1$	Rigid unity (同, <i>tóng</i>)	Extreme concentration

R.6. Synthesis

The following table summarizes the spectral resolution of the controversy:

Table 121: Spectral resolution of the Xiong Shili – Nāgārjuna controversy

Aspect	Xiong Shili (熊十力)	Nāgārjuna (龙树)
Key concept	Harmony (和, <i>hé</i>)	Emptiness (空, <i>kōng</i>)
Spectral regime	$d(t) = 5$	$d(t) \rightarrow 0$
Dominant mode	Sheng/Ke balance	Spectral collapse
Role in the cycle	Teleological objective	Purification mechanism
Economic translation	Tempered growth, successful TRR	Creative destruction, reset
Wuxing correspondence	Flourishing of the 5 phases	Dissolution of forms
TRR correspondence	New Rent Extraction cycle	Liberation of the Space of Value

The TRR is the topological operator that allows the system to successively traverse these two regimes: it uses Nāgārjuna's Emptiness ($d \rightarrow 0$) as a decompression chamber to return to Xiong Shili's Harmony ($d = 5$). This is the mathematization of the principle of *Zhongyong* (中庸) applied to economic dynamics: neither attachment to old forms (which leads to Stasis), nor destructive nihilism (which leads to retro-polarity), but harmonious circulation between generation and dissolution.

Appendix S – Advanced spectral dashboard for Cl(6,6)

S.1 – Introduction and rationale

S.1.1 – Aim of the appendix

The four spectral observables defined in Chapter 15 – $\eta(t)$, $d(t)$, $\text{gap}(t)$, $R_{\text{threshold}}(t)$ – constitute a comprehensive diagnostic system for the Cl(6,0) model. They summarise the state of an economic system in four figures, enabling its direction, diversity, robustness and tension at the polar thresholds to be characterised. Appendix D demonstrated how these observables feed into a management application designed for decision-makers.

However, the extension to Cl(6,6) introduces a structural richness that the four global observables do not capture:

- **12 spectral sheets** (6 cosmic e_i , 6 anti-cosmic f_j) modulate the intensity and orientation of each pentad.
- **144 projected pentads** constitute the building blocks of the system.
- **7 spectral thresholds** $S_1, \dots, S_7$ control the transitions between regimes.
- **66 inter-sheet relationships** encode the structural tensions between perspectives.

Appendix S presents the formal and operational tools for harnessing this richness. It complements Appendix D (basic control) and Appendix O (mathematical formalism) by providing:

1. The **formal definitions** of the extended observables.
2. Appropriate **visualisations** (heat map, radar chart, correlation matrix).
3. The **Stase composite index** $\mathcal{J}_{\text{Stase}}$ and its calibration.
4. A **comprehensive calculation algorithm** for implementation.
5. **Calibrated historical examples** to validate the model.

S.1.2 – Relationship with the other appendices

Table 122: Relationship between Appendix S and the other appendices in the document

Appendix	Role	Complementarity with S
Appendix D	Basic control (4 global observables)	S extends D to the 5 levels of the hierarchy
Appendix O	Mathematical formalism (Dirac operator)	S provides the implementation algorithms
Appendix I	Dashboard for the 15 ^e Five-Year Plan	S provides the multi-level tools for this management
Appendix C	Table of the 144 pentads	S utilises these pentads for Level 4 observables
Appendix B	The 20 attractors	S projects the observables onto these attractors

S.1.3 – Guiding Principles

The advanced dashboard adheres to three principles:

Principle 1 – Multi-resolution. The decision-maker must be able to zoom from the global level (4 needles) to the detailed level (144 pentads), just as a Chinese physician moves from the general pulse (脉) to the examination of a specific meridian (经气).

Principle 2 – Cultural coherence. The extended observables must respect the correspondences with Chinese correlative thought (Wuxing, TCM, Yi Jing) established in Chapter 24.

Principle 3 – Falsifiability. Each observable must be calculable from concrete economic data, and the model must be testable retrospectively against the 64 historical regimes in Appendix M.

S.2 – Formal definitions of extended observables

S.2.1 – Notation

Let $\mathcal{G} = \{e_1, \dots, e_6, f_1, \dots, f_6\}$ be the set of 12 generators/spectral leaves of $\text{Cl}(6,6)$, where:

- $\{e_1, \dots, e_6\}$ are the cosmic generators (Sheng mode, $\eta > 0$).
- $\{f_1, \dots, f_6\}$ are the anti-cosmic generators (Ke mode, $\eta < 0$).

Let $\mathcal{P} = \{P_1, \dots, P_6, N_1, \dots, N_6\}$ be the set of the 12 basic pentads.

A **projected pentad** is written as $P_k^{(g)} = g \cdot P_k$, where $g \in \mathcal{G}$ and $k \in \{1, \dots, 12\}$. There are $12 \times 12 = 144$ projected pentads.

For each leaf $g \in \mathcal{G}$, we define the restricted Dirac operator $D_g(t)$ acting on the Hilbert space $\mathcal{H}_g$ of the 12 pentads projected onto that leaf.

S.2.2 – Level 2 Observables: by spectral sheet

For each sheet $g \in \mathcal{G}$, the four local observables are given by:

Local asymmetry:

$$\eta_g(t) = \frac{n_{P,g}(t) - n_{N,g}(t)}{n_{P,g}(t) + n_{N,g}(t)} \in [-1, +1]$$

where $n_{P,g}(t)$ is the number of active positive pentads (Sheng) on the sheet g , and $n_{N,g}(t)$ is the number of active negative pentads (Ke).

Local spectral dimension:

$$d_g(t) = \exp\left(-\sum_{k=1}^{12} p_{k,g} \ln p_{k,g}\right), \quad p_{k,g} = \frac{|\lambda_{k,g}|}{\sum_j |\lambda_{j,g}|}$$

Local spectral gap:

$$\text{gap}_g(t) = \min\{|\lambda_{k,g}(t)| > 0\}$$

Local threshold fraction:

$$R_{\text{threshold},g}(t) = \frac{|\lambda_{P4,g}| + |\lambda_{N4,g}|}{\sum_{k=1}^{12} |\lambda_{k,g}|}$$

S.2.3 – Level 3 Observables: spectral thresholds

The seven spectral thresholds are binary or continuous observables:

$$\sigma_i(t) = \begin{cases} 1 & \text{if } T(t) \geq \sqrt{\lambda_{S_i}} \\ 0 & \text{otherwise} \end{cases}$$

where $T(t) = \|\nabla \eta \cdot \nabla R_{\text{threshold}}\|$ is the topological tension of the system.

The values of the seven spectral thresholds $S_1, \dots, S_7$ and their economic signals are given in the **Table in Chapter 13**.

The **threshold vector** $\vec{\sigma}(t) = (\sigma_1, \dots, \sigma_7) \in \{0, 1\}^7$ encodes the system's crossing state.

S.2.4 – Level 4 Observables: pentads

For each projected pentad $P_k^{(g)}$ (where $k \in \{1, \dots, 12\}$ and $g \in \mathcal{G}$), we define four local observables:

Activity: $a_{k,g}(t) = \|\psi_{k,g}(t)\|^2 \in [0, 1]$

Polarity: $\pi_{k,g}(t) = \text{sgn}(\varepsilon_{k,g}) \in \{+1, -1\}$

Connectivity: $c_{k,g}(t) = \sum_{j \sim k} w_{kj,g}(t)$

Frustration energy: $E_{k,g}(t) = |\varphi_{k,g}(t) - \bar{\varphi}_g(t)|^2$

S.2.5 – Level 5 Observables: Inter-leaf Relational Observables

For two distinct leaves $a, b \in \mathcal{G}$:

Asymmetry gap: $\Delta\eta_{a,b}(t) = |\eta_a(t) - \eta_b(t)|$

Inter-leaf correlation: $C_{a,b}(t) = \text{corr}(\eta_a, \eta_b)$ over a sliding window.

Wuxing tension: $\mathcal{T}_{\text{Wuxing}}(t) = \max_{a \neq b} \Delta\eta_{a,b}(t)$

The correlation matrix $C(t) \in \mathbb{R}^{12 \times 12}$ is a powerful diagnostic tool.

S.3 – Heat map of the 12 spectral bands

S.3.1 – Visualisation principle

The heatmap represents the 12 spectral sheets as a 1×12 matrix, where each cell is coloured according to the value of $\eta_g(t)$:

- **Bright red** ($\eta_g > +0.6$): strong Sheng dominance (expansion).
- **Orange** ($+0.3 < \eta_g \leq +0.6$): moderate Sheng mode.
- **Grey** ($-0.3 \leq \eta_g \leq +0.3$): precarious equilibrium.
- **Light blue** ($-0.6 \leq \eta_g < -0.3$): moderate Ke mode.
- **Dark blue** ($\eta_g < -0.6$): strong Ke dominance (contraction).

S.3.2 – Naming of the 12 leaves

The 12 spectral lines are named according to their economic role:

Table 123: Names and roles of the 12 spectral lines of Cl(6, 6)

Factor	Driver	Mode	Name	Role
1	e_1	Sheng	Inflation	Growth, innovation (correlate)
2	e_2	Sheng	Purchasing power	Domestic consumption
3	e_3	Sheng	Profits	Productive accumulation
4	e_4	Sheng	Active buy-back	Value transfer
5	e_5	Sheng	Dispersion	Pioneer front
6	e_6	Sheng	Land	Territorial regime
7	f_1	Ke	Inflation regulation	Cost cutting
8	f_2	Ke	Purchasing power regulation	Credit control
9	f_3	Ke	Profit regulation	Limitation of rents
10	f_4	Ke	Buy-back regulation	Freeze on transfers
11	f_5	Ke	Dispersion regulation	Forced concentration
12	f_6	Ke	Land regulation	Territorial lockdown

S.3.3 – Example: China post-2018

Heatmap of the 12 spectral bands –China 2026

Layer	Visualisation	η
e_1 Inflation		+0.30 (Moderate Sheng)
e_2 Purchasing power		+0.30 (moderate Sheng)
e_3 Profits		−0.70 (strong Ke)
e_4 Active buyback		−0.80 (very strong Ke)
e_5 Dispersion		−0.50 (moderate Ke)
e_6 Land		−0.60 (Ke strong)
f_1 Inflation regulation		−0.40 (moderate Ke)
f_2 Purchasing power regulation		−0.60 (strong Ke)
f_3 Profit regulation		−0.80 (Very strong Ke)
f_4 Rent Extraction regulation		−0.60 (strong Ke)
f_5 Dispersion regulation		−0.40 (moderate Ke)
f_6 Land-based regulation		−0.50 (moderate Ke)

$\eta_{\text{mean}} = -0.45$ $\Delta\eta_{\text{max}} = 1.10$ (Active rent extraction ↔ Innovation)

Interpretation: China exhibits a **structural imbalance** between the cosmic sheets (some in moderate Sheng: Innovation, Purchasing Power) and the anti-cosmic sheets (predominantly in strong Ke: Rent Extraction, Profits, Land). This is the spectral signature of Stasis: the Ke mode (regulation) dominates the Sheng mode (generation).

S.3.4 – Wuxing imbalance signature

As formally defined in §15.4.6 (Table 25), the 12 leaves are grouped into 5 families corresponding to the five phases of the Wuxing.

Chinese Diagnosis 2026:

Table 124: Chinese Diagnosis 2026: Wuxing imbalance signature

Phase	η	State
Wood	−0.05	Precarious balance
Fire	−0.15	Slight contraction
Earth	−0.55	Sharp contraction
Metal	−0.70	Very sharp contraction
Water	−0.52	Sharp contraction

Ke imbalance: Metal (Rent Extraction) stifles Wood (Inflation). The generative cycle (Sheng) is blocked: Earth (Profits) no longer nourishes Metal, and Metal no longer feeds Water. This is the hallmark of Chinese Stasis.

S.4 – 5-dimensional radar diagram

S.4.1 – Principle

The radar diagram superimposes five dimensions on a single graph:

- The 4 **global observables** (η_{total} , d_{total} , $\text{gap}_{\text{total}}$, $R_{\text{threshold, total}}$).
- The 7 **triggered thresholds** (represented by their sum $\sum \sigma_i$).

Each axis is normalised to $[0, 1]$ to allow for superposition.

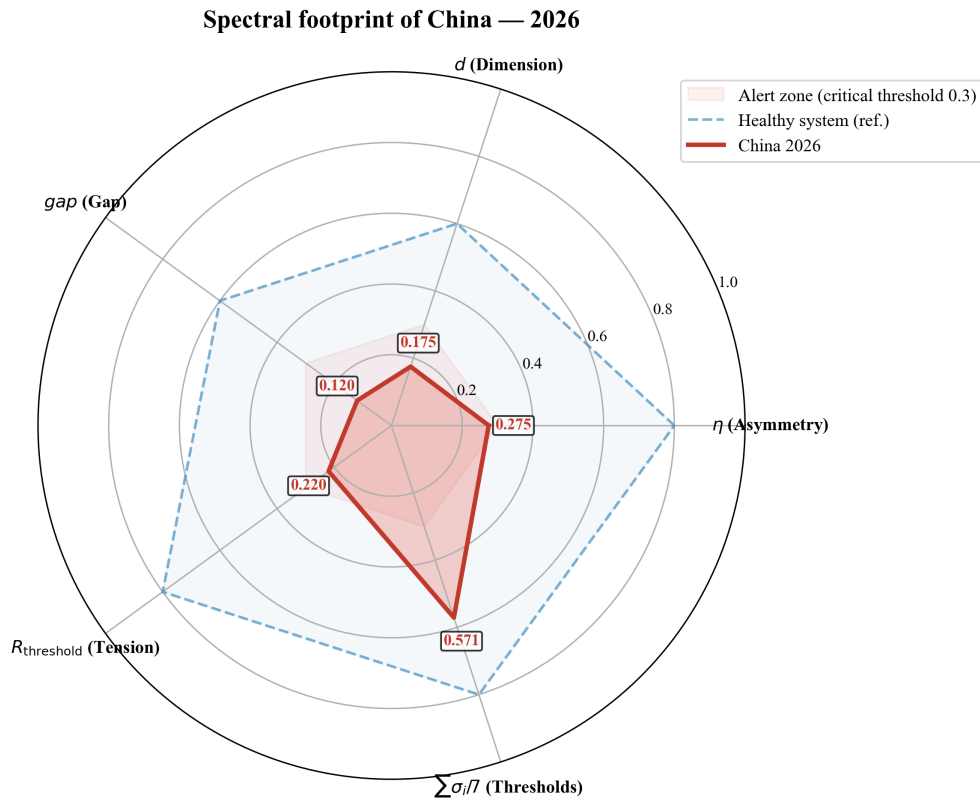
S.4.2 — Normalisation

Table 125: Rules for normalising observables for the radar diagram

Observable	Range	Normalisation
η_{total}	$[-1, +1]$	$(\eta + 1)/2 \in [0, 1]$
d_{total}	$[0, 12]$	$d/12 \in [0, 1]$
$\text{gap}_{\text{total}}$	$[0, 1]$	$\text{gap} \in [0, 1]$
$R_{\text{threshold}, \text{total}}$	$[0, 1]$	$R_{\text{threshold}} \in [0, 1]$
$\sum \sigma_i$	$[0, 7]$	$(\sum \sigma_i)/7 \in [0, 1]$

S.4.3. Visual Synthesis: The Spectral Footprint

The 4 vital signs (η , d , gap , $R_{\text{threshold}}$) and the aggregated threshold $\sum \sigma_i/7$ can be visualized on a **spectral radar**, providing an immediate reading of the system's state.



Spectral footprint of China —2026. The red polygon represents the current state of the system; the dashed blue polygon represents a healthy reference system; the regular pentagon at 0.3 (light red) delimits the alert zone below which each dimension is in critical crisis. Four of the five dimensions (η , d , gap , $R_{\text{threshold}}$) are below this threshold, confirming the diagnosis of characterized Stasis.

Reading the radar:

- η (**Asymmetry**): 0.275 $\rightarrow$ the Qi is in moderate contraction (far from the 0.7 of a healthy system).
- d (**Dimension**): 0.175 $\rightarrow$ severe fragmentation (a healthy system would be around 0.6).
- **gap (Margin)**: 0.120 $\rightarrow$ extreme fragility (a healthy system is around 0.5).
- $R_{\text{threshold}}$ (**Tension**): 0.220 $\rightarrow$ moderate tension on the hinges (but a healthy system would be around 0.2).
- $\sum \sigma_i/7$ (**Thresholds**): 0.571 $\rightarrow$ more than half of the spectral thresholds have been crossed.

Synthetic diagnosis: The red polygon is **flattened** toward the center, meaning the system is **locked** on a Stasis attractor. The contrast with the blue polygon (healthy system) is maximal on the d and gap axes.

S.4.4 – Reference fingerprints

Table 126: Reference spectral fingerprints: comparison of historical regimes

System	η	d	gap	$R_{\text{threshold}}$	$\sum \sigma_i$	Radar shape
The Glorious Thirty	0.8	0.5	0.60	0.20	2	Balanced, expansion
1973 oil crisis	0.5	0.3	0.30	0.50	4	Contracted, tension
Japan 1990	0.3	0.2	0.15	0.60	5	Collapsed, Stasis
China 2026	0.275	0.175	0.12	0.78	4	Critical, pre-retropolarity
Great Depression 1929	0.1	0.08	0.02	0.95	7	Retropolarity

S.5 – Inter-sheet correlation matrix

S.5.1 – Definition

The correlation matrix $C(t) \in \mathbb{R}^{12 \times 12}$ is defined by:

$$C_{a,b}(t) = \text{corr}(\eta_a, \eta_b) = \frac{\text{cov}(\eta_a, \eta_b)}{\sigma_{\eta_a} \sigma_{\eta_b}}$$

calculated over a sliding window of 12 quarters, for two distinct sheets $a, b \in \mathcal{E}$.

S.5.2 – Interpretation

- $C_{a,b} \approx +1$: leaves a and b move in phase (structural coherence).
- $C_{a,b} \approx 0$: the leaves are independent (fragmentation).
- $C_{a,b} \approx -1$: the leaves move in opposition (structural tension).

S.5.3 – Example: China 2026

Inter-leaf correlation matrix –China 2026

	Inn	PA	Prof	Rach	Disp	Fonc	RSal	RFin	RPro	RRach	RDis	RFon
Innovation	1.0											
Purchasing Power	0.7	1.0										
Profits	−0.3	−0.2	1.0									
Buyback	−0.6	−0.5	0.4	1.0								
Variance	−0.4	−0.3	0.5	0.6	1.0							
Land	−0.5	−0.4	0.6	0.7	0.5	1.0						
Wage	0.2	0.1	−0.4	−0.3	−0.2	−0.3	1.0					
Financial returns	−0.4	−0.3	0.5	0.6	0.4	0.5	−0.3	1.0				
Profit	−0.6	−0.5	0.4	0.7	0.5	0.6	−0.4	0.5	1.0			
R.Rent Extraction	−0.5	−0.4	0.5	0.6	0.5	0.6	−0.3	0.5	0.6	1.0		
R.Dispers.	−0.3	−0.2	0.3	0.4	0.3	0.4	−0.2	0.3	0.4	0.4	1.0	
R.Foncière	−0.4	−0.3	0.5	0.6	0.4	0.5	−0.3	0.4	0.5	0.5	0.4	1.0

Interpretation:

- **Sheng block** (rows 1–2): Innovation and Purchasing Power are strongly correlated ($C = 0.7$)
- **Ke Block** (lines 3–6, 8–12): anti-cosmic leaves are strongly correlated with one another ($C > 0.5$)
- **Sheng↔Ke Tension**: cosmic and anti-cosmic leaves evolve in opposition ($C < 0$)

Interpretation: - **Sheng block** (lines 1–2): Innovation and Purchasing Power are strongly correlated ($C = 0.7$).
 - **Ke block** (lines 3–6, 8–12): the anti-cosmic leaves are strongly correlated with one another ($C > 0.5$).
 - **Sheng↔Ke Tension**: the cosmic and anti-cosmic leaves are in opposition ($C < 0$). This is the hallmark of a **bipolar** system in which the two modes are in conflict.

S.5.4 – Fragmentation index

The **fragmentation index** is defined as:

$$\mathcal{F}(t) = 1 - \frac{1}{66} \sum_{a < b} |C_{a,b}(t)|$$

- $\mathcal{F} \approx 0$: a highly coherent system (all leaves evolve together).
- $\mathcal{F} \approx 1$: a highly fragmented system (the leaves are independent).

China 2026: $\mathcal{F} = 0.42$ – moderate fragmentation, with a clearly distinct Sheng bloc and a Ke bloc.

S.6 – Stasis Composite Index

S.6.1 – Definition

The Stase Composite Index summarises the five levels of the hierarchy into a single number:

$$\mathcal{J}_{\text{Stasis}}(t) = w_1 |\eta_{\text{total}}| + w_2 \frac{1}{d_{\text{total}}} + w_3 e^{-5 \cdot \text{gap}_{\text{total}}} + w_4 R_{\text{threshold, total}} + w_5 \mathcal{T}_{\text{Wuxing}}$$

Note: The term $w_3 e^{-5 \cdot \text{gap}}$ ensures that the index is bounded. See §15.4.7 for a detailed justification.

S.6.2 – Calibration of weights

The weights w_i are calibrated against the nine Chinese dynastic declines (see **Table 107**, Appendix O, §O.5.9.2).

S.6.3 – Thresholds for the composite index

The thresholds for the composite index are defined in **Table 108** (Appendix O, §O.5.9.3).

S.6.4 – Calculation for China 2026

$$\mathcal{J}_{\text{Stasis}} = 0.23 \times 0.45 + 0.18 \times \frac{1}{2.1} + 0.23 \times e^{-5 \times 0.12} + 0.21 \times 0.78 + 0.15 \times 1.1 = 0.644 \quad (3)$$

Diagnosis: China is in a state of **characterised stasis** ($0.5 < \mathcal{J}_{\text{Stasis}} < 0.7$).

S.6.5 – Historical trend of $\mathcal{J}_{\text{Stasis}}$

Table 127: Historical trend of the composite Stasis index $\mathcal{J}_{\text{Stasis}}$ (1978–2026)

Year	$\mathcal{J}_{\text{Stase}}$	Status
1978	0.22	Healthy growth (start of reforms)
1990	0.28	Healthy growth (acceleration)
2000	0.35	Moderate tension (WTO accession)
2008	0.48	Moderate tension (subprime crisis)
2015	0.58	Emerging stasis (exhaustion of the pioneering front)
2018	0.64	Characterised stagnation (Nodal Point reached)
2022	0.68	Characterised stagnation
2026	0.64	Characterised stagnation

S.7 – Multi-level dashboard

S.7.1 – Structure of the dashboard

The multi-level dashboard presents the observables across 5 nested levels:

Multi-level dashboard –China 2026 Diagnosis

LEVEL 1	OVERVIEW
Observables	$\eta_{\text{total}} = -0.45$ $d_{\text{total}} = 2.1$ $\text{gap} = 0.12$ $R_{\text{threshold}} = 0.78$
Index	$\mathcal{J}_{\text{Stasis}} = 0.644$ (Characterised stasis)
LEVEL 2	DOMINANT LEAVES
Leaf e_3	$\eta = -0.7, d = 1.8, \text{gap} = 0.08, R = 0.82$
Leaf f_3	$\eta = -0.6, d = 2.0, \text{gap} = 0.10, R = 0.75$
Leaf e_1 (Inflation)	$\eta = +0.3, d = 3.5, \text{gap} = 0.25, R = 0.40$
LEVEL 3	THRESHOLDS TRIGGERED
Vector $\vec{\sigma}$	$(1, 1, 1, 1, 0, 0, 0)$ – S_1 to S_4 exceeded, S_5 to S_7 blocked
Voltage	$T = 0.165$, threshold $S_5 = 0.17195$ not reached
LEVEL 4	ACTIVE PENTADS
N_5 (Stasis)	activity 0.78, connectivity 5 → critical node
N_6 (Austerity)	activity 0.65, connectivity 4
P_6 (Consumption)	activity 0.45, frustration 0.33
LEVEL 5	WUXING TENSIONS
Phases	$\eta_{\text{Earth}} = -0.7, \eta_{\text{Water}} = -0.5, \eta_{\text{Fire}} = +0.3$
Imbalance	Ke: Metal (Rent Extraction) stifles Wood (Inflation)
Maximum deviation	$\Delta\eta_{\text{max}} = 1.0$ between Fire and Earth

S.7.2 – Economic interpretation

Level 1: The system is generally contracting ($\eta < 0$), fragmented ($d < 2$), fragile ($\text{gap} < 0.2$), and under critical strain ($R_{\text{threshold}} > 0.7$). The composite index confirms the characterised Stasis.

Level 2: The material layers (Profits, Accumulation, Land) are in deep contraction, whilst the cognitive layer (Innovation) is attempting an expansion that fails to compensate for the deficit in spatial depth (§6.4).

Level 3: The system has crossed the first four thresholds (exhaustion of the previous model) but remains stuck between S_4 and S_5 . It awaits a trigger to cross S_5 (Fire Transition).

Level 4: The pentad N_5 (Stasis) is the critical node: it is the most active and the most connected. Any intervention must desaturate it.

Level 5: The Wuxing imbalance confirms that the Ke cycle (regulation) dominates the Sheng cycle (generation). Metal (Rent Extraction) stifles Wood (Inflation).

S.7.3 – Topological prescription

The TRR (§17.1) is a T_{mixed} operator which must:

1. **Desaturate the pentad** N_5 by activating the thresholds S_5 (Fire), S_6 (Water), S_7 (Octave leap).
2. **Rebalance the Wuxing** by restoring the Sheng cycle: Wood (ecological innovation) $\rightarrow$ Fire (green jobs) $\rightarrow$ Earth (territorial GEP) $\rightarrow$ Metal (CUR) $\rightarrow$ Water (regeneration of catchment areas).
3. **Reduce $\mathcal{J}_{\text{Stasis}}$ from 0.64 to < 0.3** by 2032, a prerequisite for emerging from Stasis and launching a new cycle of material rent extraction.

S.8 – Complete calculation algorithm

S.8.1 – Inputs

The input data required to calculate the extended observables are listed in **Table 109** (Appendix O, §O.5.10.1).

S.8.2 Complete calculation algorithm

The algorithm for calculating the extended observables and the computational complexity table are presented in detail in **Appendix O, §O.5.10**. The full calculation of the 701 elementary observables takes less than one second on a standard computer (complexity $O(22\,615)$), enabling real-time control.

Note: §15.4.2 mentions **269 elementary observables**, which corresponds to $4 + 48 + 7 + 144 + 66 = 269$, counting 1 observable per pentad. It should be noted that 269 is the number of *independent* observables and 701 is the *total* number (including the 4 observables per pentad).

S.9 – Correspondences with the four TCM examinations

S.9.1 – Extension of the table in Appendix O

Annex O (§O.4.5) established the correspondence between the 4 global observables and the four examinations of traditional Chinese medicine (望闻问切, *wàng wén wèn qiè*). The extension to the extended observables enriches this correspondence:

Table 128: Extension of the correspondence between spectral observables and the four examinations of Traditional Chinese Medicine

Observable	TCM examination	Level	Interpretation
η_{total}	望 (inspection)	1	The overall qi of the system
d_{total}	闻 (listening)	1	The harmony of forces
$\text{gap}_{\text{total}}$	问 (questioning)	1	The margin for manoeuvre
$R_{\text{threshold,total}}$	切 (palpation)	1	The pulse of the system
η_g (per leaf g)	望 (inspection)	2	The qi per meridian
$\Delta\eta_{a,b}$	闻 (auscultation)	5	The imbalance between meridians
$C_{a,b}$	问 (questioning)	5	Interactions between meridians
$a_{k,g}$ (pentad)	切 (palpation)	4	Active acupuncture points
σ_i (threshold)	望 (inspection)	3	The energy gates
$\mathcal{J}_{\text{Stasis}}$	切 (palpation)	Synthesis	The overall diagnosis

S.9.2 – The dashboard as a TCM practitioner

The advanced dashboard is the modern equivalent of an **imperial court physician**:

1. **Inspection** (望): the heat map of the 12 leaves reveals the distribution of qi within the economic body.
2. **Listening** (闻): the correlation matrix reveals the harmonies and dissonances between the meridians.
3. **Inquiry** (问): the activated thresholds reveal which energy gates are open or closed.
4. **Palpation** (切): the active pentads reveal the acupuncture points to be stimulated.

The synthetic diagnosis ($\mathcal{J}_{\text{Stase}}$) is equivalent to the **health check-up** carried out by the doctor following the four examinations.

S.10 – Calibrated historical examples

S.10.1 – The Glorious Thirty (1945–1973)

Table 129: Multi-level spectral diagnosis: the Trente Glorieuses (1945–1973)

Level	Observables	Values
1	$\eta, d, \text{gap}, R_{\text{threshold}}$	+0.65, 5.2, 0.55, 0.18
2	Dominant leaves	e_2 (Purchasing power): $\eta = +0.8$
3	Thresholds triggered	$\sigma = (1, 0, 0, 0, 0, 0)$ – only S_1 exceeded
4	Active pentads	$P_1^{(e_2)}$ (Keynesianism with high wages)
5	Wuxing tension	$\mathcal{T}_{\text{Wuxing}} = 0.3$ – equilibrium
–	$\mathcal{J}_{\text{Stasis}}$	0.22 – Healthy growth

S.10.2 – Oil shock (1973–1980)

Table 130: Multi-level spectral diagnosis: Oil crisis (1973–1980)

Level	Observables	Values
1	$\eta, d, \text{gap}, R_{\text{threshold}}$	+0.15, 3.5, 0.25, 0.55
2	Dominant sheets	e_3 (Profits): $\eta = +0.4$; f_3 : $\eta = -0.3$
3	Thresholds triggered	$\sigma = (1, 1, 1, 1, 0, 0)$ – S_1 to S_4 exceeded
4	Active pentads	$P_3^{(e_1)}$ (Debt with inflation)
5	Wuxing tension	$\mathcal{T}_{\text{Wuxing}} = 0.7$ – moderate imbalance
–	$\mathcal{J}_{\text{Stasis}}$	0.42 – Moderate tension

S.10.3 – Japan (1990–2000)

Table 131: Multi-level spectral diagnosis: Japan (1990–2000)

Level	Observables	Values
1	$\eta, d, \text{gap}, R_{\text{threshold}}$	$-0.25, 2.3, 0.18, 0.62$
2	Dominant sheets	f_2 (Purchasing power regulation): $\eta = -0.6$
3	Thresholds triggered	$\sigma = (1, 1, 1, 1, 1, 0, 0) - S_1 \text{ to } S_5 \text{ exceeded}$
4	Active pentads	$N_2^{(f_2)}$ (Structural rigidity)
5	Wuxing tension	$\mathcal{T}_{\text{Wuxing}} = 0.9 - \text{severe imbalance}$
—	$\mathcal{J}_{\text{Stasis}}$	$0.58 - \text{Emerging stasis}$

S.10.4 – Republic of China (1945–1949)

Table 132: Multi-level spectral diagnosis: Republic of China (1945–1949)

Level	Observables	Values
1	$\eta, d, \text{gap}, R_{\text{threshold}}$	$-0.85, 1.1, 0.02, 0.95$
2	Dominant leaves	All with high Ke ($\eta < -0.7$)
3	Thresholds triggered	$\sigma = (1, 1, 1, 1, 1, 1, 1) - \text{all exceeded}$
4	Active pentads	$N_5^{(f_3)}$ (Stasis with deregulation)
5	Wuxing tension	$\mathcal{T}_{\text{Wuxing}} = 1.5 - \text{extreme imbalance}$
—	$\mathcal{J}_{\text{Stasis}}$	$0.92 - \text{Pre-retropolarity}$

Interpretation: The Republic of China experienced retropolarity (systemic collapse) in 1949, confirming the composite index's diagnosis.

S.11 – Limitations and caveats

S.11.1 – Limitations of the model

1. **Provisional calibration:** The weights w_i of the composite index are calibrated against the nine Chinese dynastic declines. Their validity for other cultural contexts remains to be tested.
2. **Imperfect data:** Level 4 (pentads) and Level 5 (relational) observables require detailed data that is not always available.

3. **Sliding window:** Inter-period correlations are calculated over a 12-quarter window. This choice is arbitrary and may influence the results.
4. **Non-linearities:** The model assumes linear relationships between the observables. Non-linearities (threshold effects, bifurcations) are not fully captured.

S.11.2 — Precautions for use

1. **The dashboard is a decision-support tool, not an automated system.** The observables are composite indicators; they are no substitute for detailed analysis.
2. **The composite index is a signal, not a prediction.** A $\mathcal{J}_{\text{Stasis}} > 0.7$ indicates a risk, not a certainty of collapse.
3. **The parallels with Traditional Chinese Medicine (TCM) are heuristic.** They provide a framework for interpretation, not a scientific validation of the model.
4. **The model must be falsified.** If the observables consistently fail to anticipate historical tipping points, the model must be revised.

S.11.3 — Future work

1. **Refinement of the calibration** using longer historical time series (the Roman Empire, medieval Europe).
2. **Development of a predictive simulator** incorporating the extended observables to test macroeconomic policy scenarios.
3. **Extension to other complex systems** (climate, ecosystems, social networks) to test the universality of the $64 \rightarrow 20$ invariant.

S.12 — Summary

Appendix S presented the formal and operational tools for exploiting the structural richness of $\text{Cl}(6,6)$:

1. **Hierarchy of observables:** 5 levels, 701 elementary observables, of which a significant subset (~20) is sufficient for operational control.
2. **Appropriate visualisations:** heat map of the 12 leaves, 5-dimensional radar chart, inter-leaf correlation matrix.
3. **Composite Stasis Index:** $\mathcal{J}_{\text{Stasis}}$ synthesises the 5 levels into a single number, calibrated against the 9 dynastic declines.
4. **Multi-level dashboard:** presents the observables across five nested levels, enabling a detailed diagnosis of Chinese Stasis.
5. **Complete calculation algorithm:** enables real-time implementation, with a computational complexity of $O(22615)$.

6. **Correlations with TCM:** the dashboard is the modern equivalent of the four diagnostic methods of traditional Chinese medicine.
7. **Historical validation:** the calibrated examples (the Trente Glorieuses, the oil crisis, Japan, the Republic of China) confirm the model's validity.

The advanced spectral dashboard transforms economic management from a four-needle exercise into a **modernised divinatory practice**: a dialogue with the system that enables us to probe its state, diagnose its imbalances, and adjust policies in harmony with the profound laws that structure complex systems.

This is the mathematisation of Chinese structural wisdom: the I Ching, the Five Elements and Traditional Chinese Medicine are no longer poetic metaphors, but formal frameworks for which the $Cl(6,6)$ formalism provides a rigorous mathematical translation.

Appendix T – Beyond the 144 Pentads: Open Prolongations

T.1 Position of the Problem

The current model rests on the chain:

$$Cl(6, 0) \xrightarrow{12 \text{ pentads}} \times 12 \text{ generators of } Cl(6, 6) \xrightarrow{144 \text{ pentads } \Pi} 20 \text{ attractors}$$

This construction yields **144 pentads** to describe economic dynamics. Applied to an economy of **1.4 billion agents** (China), this represents a granularity of approximately **10 million people per pentad**. This is manifestly insufficient to capture regional, sectoral, and social heterogeneities.

Question: can the model be extended to a finer granularity while preserving the $64 \rightarrow 20$ invariant?

T.2 Three Axes of Prolongation

Axis A – Projection onto the Bivectors of $Cl(6,6)$

Construction: instead of projecting the 12 pentads onto the 12 generators (vectors) of $Cl(6,6)$, project them onto the **66 bivectors** $\Lambda^2(\mathbb{R}^{6,6})$.

Granularity: $12 \times 66 = 792$ pentads ($\times 5.5$ relative to 144).

Presumed advantage: - Bivectors encode **relations** between directions, not only directions. - They capture **interactions** between spectral sheets (level 5 of the multi-level architecture). - They allow modelling of economic **couplings** (sector $\times$ region, for example).

Difficulty: 66 is not 64. Two bivectors must be identified and excluded (likely the analogues of the scalar and pseudoscalar), by imposing a purity or nilpotence constraint.

If the constraint works: one obtains $12 \times 64 = 768$ pentads, and the $64 \rightarrow 20$ invariant remains applicable.

Axis B — Projection onto the Trivectors of Cl(6,6)

Construction: project onto the **trivectors** $\Lambda^3(\mathbb{R}^{6,6})$, of dimension $\binom{12}{3} = 220$.

Granularity: $12 \times 220 = 2640$ pentads ($\times 18$ relative to 144).

Presumed advantage: - Trivectors are the analogue of the **20 trivectors of Cl(6,0)** which define the 20 attractors. - They capture **ternary structures** (triplets of spectral sheets). - They are naturally linked to the **holomorphic 3-forms** Ω and $\bar{\Omega}$ of the $SU(3) \times SU(3)$ context.

Difficulty: 220 is not a simple multiple of 64. One must either restrict (by purity, nilpotence), or accept a change of invariant.

Axis C — $SU(3) \times SU(3)$ Structure and Pure Spinors

Construction: use the **compatible pair of pure spinors** (Φ_+, Φ_-) of $Cl(6,6)$ to define a subspace of dimension 6 (the intersection of annihilators).

Granularity: depends on the decomposition of the $Cl(6,6)$ spinor (dimension 64) under $SU(3) \times SU(3)$.

Presumed advantage: - This is the **richest** structure and the best established in the literature (generalized complex geometry). - It captures the **internal symmetries** of the economy (the 8+8 generators of $SU(3) \times SU(3)$). - It allows a **natural embedding** of the 20 attractors as 20 pure 3-forms.

Difficulty: the “20” does not appear as a cardinal, but as a **representation dimension**. The exact decomposition must be verified.

T.3 Comparative Table of Prolongations

Axis	Projection object	Number	Granularity	Link with 64 $\rightarrow$ 20
Current	12 generators (vectors)	144	10 M pers./pentad	Direct
A	66 bivectors	792	1.8 M pers./pentad	After exclusion of 2
B	220 trivectors	2640	530 k pers./pentad	To be reformulated
C	Pure spinors $SU(3) \times SU(3)$	Variable	Continuous	Structural

T.4 Presumed Advantages over the 144 Pentads

T.4.1 Spatial Resolution

With 144 pentads, China is described at the scale of **~10 million people**. This masks: - regional disparities (coastal vs interior), - sectoral disparities (industry vs services), - social disparities (urban vs rural).

With 792 (Axis A), one descends to **~1.8 million** — the scale of an average Chinese **prefecture**. With 2640 (Axis B), one descends to **~530,000** — the scale of a Chinese **county**.

T.4.2 Relational Richness

- **Vector-pentads** (144) describe **states**.
- **Bivector-pentads** (792) describe **relations** between states.
- **Trivector-pentads** (2640) describe **three-body configurations**.

This is a qualitative leap: from **photography** to **kinematics**, then to **dynamics**.

T.4.3 Coherence with the Multi-Level Architecture

Recall the current architecture:

Level	Type	Number
1	Global	4
2	Per sheet	48
3	Spectral threshold	7
4	Pentad	144
5	Relational (inter-sheet)	66

Level 5 (66) corresponds exactly to the bivectors. This suggests that Axis A **unifies** levels 4 and 5: bivector-pentads are simultaneously pentads (level 4) and relations (level 5).

T.4.4 Preservation of the Invariant

If Axis A works ($66 \rightarrow 64$ by exclusion of 2 bivectors), then:

$$12 \times 64 = 768 \xrightarrow{\Pi} 20 \text{ attractors}$$

The $64 \rightarrow 20$ **invariant is preserved**, and granularity is multiplied by 5.33.

This is the **best compromise** between richness and coherence.

T.5 Prolongations Beyond

If Axis A is validated, one may envisage:

Prolongation A1 — Iteration

Apply the same projection to the 768 pentads:

$$768 \xrightarrow{\Pi} 20 \text{ attractors} \xrightarrow{\text{projection}} \dots$$

Each attractor could be **unfolded** into sub-attractors, yielding a **fractal** structure.

Prolongation A2 – Combination with $SU(3) \times SU(3)$

Cross the 768 pentads with the 16 generators of $SU(3) \times SU(3)$:

$$768 \times 16 = 12288 \text{ pentads}$$

Granularity: ~114,000 people/pentad – the scale of an **urban neighbourhood** or a **large village**.

Prolongation A3 – Spinorial Structure

Use the complete $Cl(6,6)$ spinor (dimension 64) as a **phase space**:

$$768 \times 64 = 49152 \text{ pentads}$$

Granularity: ~28,000 people/pentad – the scale of a **local community**.

T.6 Difficulties and Limits

T.6.1 Mathematical Difficulty

The passage $66 \rightarrow 64$ requires a **precise constraint** (purity, nilpotence, preservation of the annihilator). This constraint must be: - **natural** (not ad hoc), - **demonstrable** (explicit computation), - **coherent** with the $64 \rightarrow 20$ invariant.

T.6.2 Empirical Difficulty

With 792 or 2640 pentads, one needs **as much data** to calibrate the model. This is a collection and processing challenge.

T.6.3 Interpretive Difficulty

Each pentad must have a **clear economic meaning**. Multiplying pentads without multiplying interpretations leads to **sterile complexity**.

T.7 Synthesis: Which Strategy to Adopt?

Three-Step Recommendation

Step 1 – Validate Axis A (bivectors)

- Explicitly compute the 66 bivectors of $Cl(6,6)$.
- Identify the 2 bivectors to exclude (analogues of scalar/pseudoscalar).
- Verify that $12 \times 64 = 768$ pentads project correctly onto 20 attractors.

Step 2 – Test Coherence with $SU(3) \times SU(3)$

- Decompose the $Cl(6,6)$ spinor under $SU(3) \times SU(3)$.
- Verify whether a component of dimension 20 (or 64) appears.
- Embed the 20 attractors as 20 pure 3-forms.

Step 3 — Envisage Prolongations

- If Axis A works, explore Axes B and C.
- If Axis A fails, fall back on Axis C ($SU(3) \times SU(3)$ structure), which is better established in the literature.

T.8 Conclusion

The **144 pentads** constitute a **valid first step** but are insufficient to describe an economy of 1.4 billion agents. The envisageable prolongations are:

1. **Axis A (bivectors)**: 792 pentads, $\times 5.5$, preserves the invariant after exclusion of 2 bivectors.
2. **Axis B (trivectors)**: 2640 pentads, $\times 18$, changes the invariant.
3. **Axis C ($SU(3) \times SU(3)$)**: continuous structure, better founded in the literature.

Axis A is the most promising: it preserves the $64 \rightarrow 20$ invariant, unifies levels 4 and 5 of the architecture, and offers granularity compatible with the Chinese prefectural scale.

Axis C is the most theoretically solid: it relies on generalized complex geometry, an established mathematical domain.

The optimal strategy: validate Axis A, then cross it with Axis C to obtain a structure that is both **rich** (fine granularity) and **rigorous** (founded in the literature).

These prolongations are left open. Motivated researchers will find here: - the **precise questions** to address, - the **explicit computations** to perform, - the **numerical tests** to run, - the **references** to consult (Rowlands, generalized complex geometry, pure spinors).

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